

Quantum Mechanics

$$\widehat{A} \left| \psi_n \right\rangle = \lambda \left| \psi_n \right\rangle$$

HONG YI CHEN

Hong Yi Chen

National Taiwan Normal University
Department of Physics

88 TING-CHOU RD. SEC. 4, TAIPEI 116, TAIWAN

—Classical Mechanics—

1. Particle Mechanics

- 1-1 Newtonian Mechanics
- 1-2 Lagrangian Mechanics
- 1-3 Hamiltonian Mechanics
- 1-4 Symmetry and Transformations

—Quantum Theory of Probability—

2. Matter Wave

- 2-1 Matter Wave
- 2-2 The Probability Interpretation
- 2-3 From Classical to Quantum Theory

—Quantum Mechanics—

3. Schrödinger Wave Mechanics

- 3-1 The Schrödinger Equation
- 3-2 The Infinite Square Potential Well
- 3-3 The δ -function Potential Well
- 3-4 The Parabolic Potential Well – Harmonic Oscillator
- 3-5 The Hydrogen Atom

4. Heisenberg Matrix Mechanics

- 4-1 The Matrix
- 4-2 Linear Algebra and Dirac Notation
- 4-3 Postulates of Quantum Mechanics
- 4-4 The Ladder Operator Method for Harmonic Oscillator
- 4-5 Time-Evolution Operators

5. Feynman Path Integral

5-1 Path Integral

5-2 Propagator of Free Particle and Harmonic Oscillator

5-3 Stationary Phase Approximation

5-4 Gauge Transformation

Appendix – Math background

6. Orbital and Spin Operators

6-1 Quantization of Orbital Angular Momentum

6-2 Magnetic Moment

6-3 Algebra of Orbital Operators

6-4 Algebra of Spin Operators

6-5 Continuous Group $SO(3)$ and $SU(2)$

6-6 Addition of Two Spins

—Application of Quantum Mechanics—

7. Time-Independent Perturbation Theory

7-1 Nondegenerate Perturbation Theory

7-2 Degenerate Perturbation Theory

7-3 Spin-Orbit Coupling

7-4 Zeeman Effect

8. Time-Dependent Perturbation Theory

8-1 First-Order Perturbation Theory

8-2 Adiabatic Process

8-3 Fermi Golden Rule

9. Quantum Entanglement

9-1 Quantum States of Multi-Particle System

9-2 Bell's Inequality

9-3 Density Matrix

—Preface—

Basic Concepts

Single particle

Quantum probability: Chap. 2-2, Chap. 4-3

Linear algebra: Chap. 3-2, Chap. 4-1, Chap 4-2

Two particles

Quantum probability: Chap. 9-1

Linear algebra: Chap. 6-6, Chap. 9-3

Academic Years of Lecture

- 2017 Fall - 2018 Spring
- 2018 Fall - 2019 Spring
- 2019 Fall - 2020 Spring
- 2020 Fall - 2021 Spring
- 2021 Fall - 2022 Spring
- 2023 Fall - 2024 Spring

The latest notes are available at



- web.phy.ntnu.edu.tw/~hongyi/?url=notes

—Bibliography—

Graduate Level

Prerequisite Knowledge: Analytical Mechanics, Modern Physics

1. Claude Cohen-Tannoudji, *Quantum Mechanics* (Wiley-VCH).
2. J. J. Sakurai, *Modern Quantum Mechanics* (Addison-Wesley).
3. R. Shankar, *Principles of Quantum Mechanics* (Plenum).
4. Theoretical Physics Reference.
<http://www.theoretical-physics.net/dev/index.html>

Undergraduate Level

Prerequisite Knowledge: Analytical Mechanics, Modern Physics

1. David J. Griffiths, *Introduction to Quantum Mechanics* (Pearson).
2. David H. McIntyre, *Quantum Mechanics* (Pearson).
3. 姚珩, 量子力學 (滄海).

—Chapter 1—

Particle Mechanics

1-1 Newtonian Mechanics

A. NEWTON'S SECOND LAW (1687)

- (1) The matter is made of particles which are described by points $\vec{r}$. The motion of a particle of mass m is described by Newton's equation of motion,

$$m\ddot{\vec{r}} = \vec{F}^{(\text{ext})}$$

where $\vec{F}^{(\text{ext})}$ is the external force acting on the particle.

OS:

The differential equation which describes the relation between the dynamical observables is called equation of motion.

- (2) Explicit form of $\vec{F}$

- Conservative system:

$$\vec{F}^{(\text{ext})} = \vec{F}^{(\text{ext})}(\vec{r})$$

is independent of the velocity and time. The force can then be obtained from a potential $V(\vec{r})$ as

$$\vec{F}^{(\text{ext})} = -\nabla V(\vec{r})$$

- Nonconservative system:

$$\vec{F}^{(\text{ext})} = \vec{F}^{(\text{ext})}(\vec{r}, \vec{v}, t)$$

depends on the velocity of the particle and possibly on time.

EXAMPLES:

- One dimensional harmonic oscillator

$$V(x) = \frac{1}{2}kx^2$$

$$F^{(\text{ext})}(x) = -\nabla V(x) = -kx$$

Newton's equation of motion:

$$m\ddot{x} = -kx$$

$$\Rightarrow \ddot{x} + \frac{k}{m}x = 0$$

$$\Rightarrow x = A \sin \omega t + B \cos \omega t, \quad \omega = \sqrt{\frac{k}{m}}$$

- In an electromagnetic field, a moving electron is subject to the

Lorentz force

$$\vec{F}^{(\text{ext})}(\vec{r}, \vec{v}, t) = -e\vec{E} - \frac{e}{c}\vec{v} \times \vec{B}$$

B. DEGREES OF FREEDOM

- (1) For a system of N particles with masses $\{m_i\}$ and coordinates $\{\vec{r}_i\}$, the set of equations is

$$\sum_{i=1}^N m_i \ddot{\vec{r}}_i = \sum_{i=1}^N \vec{F}_i^{(\text{ext})}$$

The degrees of freedom are $3N$, i.e., we need to solve $3N$ independent equations with $3N$ unknown variables.

EXAMPLES:

1. A three-body system



2.

The degrees of freedom are $3N = 3 \times 3 = 9$.

$\Rightarrow$ Solve 9 independent equations with 9 unknown variables.

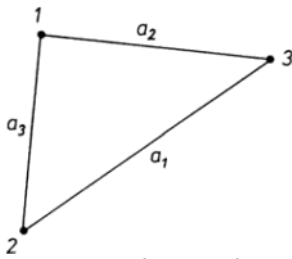
- (2) Holonomic constraints

Constraints on the position (configuration) of a system of particles can be expressed as a set of independent equations of the form

$$f_\lambda(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, t) = 0, \quad \lambda = 1, 2, \dots, \Lambda$$

EXAMPLES:

1. A three-body system at constant distances a_1, a_2, a_3



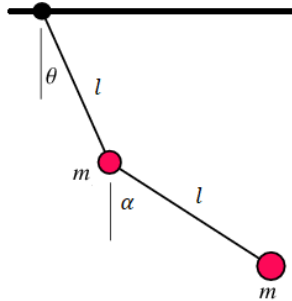
$$\left. \begin{aligned} f_1 &\equiv |\vec{r}_1 - \vec{r}_2| - a_3 = 0 \\ f_2 &\equiv |\vec{r}_2 - \vec{r}_3| - a_1 = 0 \\ f_3 &\equiv |\vec{r}_3 - \vec{r}_1| - a_2 = 0 \end{aligned} \right\} \Rightarrow \Lambda = 3$$

The constraints reduce the degrees of freedom to

$$f = 3N - \Lambda = 3 \times 3 - 3 = 9 - 3 = 6.$$

$\Rightarrow$ Solve 9 dependent equations with 6 unknown variables.

2. Write down the equation of motion of a double pendulum



ANSWER:

Newton's equation of motion:

$$m\ddot{\vec{r}}_1 = -\frac{T_1}{l}\vec{r}_1 + \frac{T_2}{l}(\vec{r}_2 - \vec{r}_1) + m\vec{a}$$

$$m\ddot{\vec{r}}_2 = -\frac{T_2}{l}(\vec{r}_2 - \vec{r}_1) + m\vec{a}$$

Since

$$\vec{r}_1 = l \sin \theta \hat{x} + l \cos \theta \hat{y}$$

$$\vec{r}_2 = \vec{r}_1 + l \sin \alpha \hat{x} + l \cos \alpha \hat{y}$$

$$\dot{\vec{r}}_1 = l\dot{\theta} \cos \theta \hat{x} - l\dot{\theta} \sin \theta \hat{y}$$

$$\dot{\vec{r}}_2 = l(\dot{\theta} \cos \theta + \dot{\alpha} \cos \alpha)\hat{x} - l(\dot{\theta} \sin \theta + \dot{\alpha} \sin \alpha)\hat{y}$$

$$\ddot{\vec{r}}_1 = l(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta)\hat{x} - l(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta)\hat{y}$$

$$\ddot{\vec{r}}_2 = l(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta + \ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha)\hat{x} - l(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta + \ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha)\hat{y}$$

$$\begin{cases} ml(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) = -T_1 \sin \theta + T_2 \sin \alpha \\ -ml(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta) = -T_1 \cos \theta + T_2 \cos \alpha + mg \\ ml(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta + \ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha) = -T_2 \sin \alpha \\ -ml(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta + \ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha) = -T_2 \cos \alpha + mg \end{cases}$$

Use trigonometric identities $\sin \alpha \cos \theta - \cos \alpha \sin \theta = \sin(\alpha - \theta)$

$$\begin{cases} l\ddot{\theta} = \frac{T_2}{m} \sin(\alpha - \theta) - g \sin \theta \dots\dots \textcircled{1} \\ l\dot{\theta}^2 = \frac{T_1}{m} - \frac{T_2}{m} \cos(\alpha - \theta) - g \cos \theta \dots\dots \textcircled{2} \\ l\ddot{\theta} \cos(\alpha - \theta) + l\dot{\theta}^2 \sin(\alpha - \theta) + l\ddot{\alpha} = -g \sin \alpha \dots\dots \textcircled{3} \\ -l\ddot{\theta} \sin(\alpha - \theta) + l\dot{\theta}^2 \cos(\alpha - \theta) + l\dot{\alpha}^2 = \frac{T_2}{m} - g \cos \alpha \dots\dots \textcircled{4} \end{cases}$$

Substituting $\textcircled{1}$ and $\textcircled{2}$ into $\textcircled{3}$ and $\textcircled{4}$, we obtain

$$\begin{cases} l\ddot{\theta} = \frac{T_2}{m} \sin(\alpha - \theta) - g \sin \theta \dots\dots \text{(a)} \\ l\dot{\theta}^2 = \frac{T_1}{m} - \frac{T_2}{m} \cos(\alpha - \theta) - g \cos \theta \\ l\ddot{\alpha} = -\frac{T_1}{m} \sin(\alpha - \theta) \dots\dots \text{(b)} \\ l\dot{\alpha}^2 = 2\frac{T_2}{m} - T_1 \cos(\alpha - \theta) \end{cases}$$

Equations (a) and (b) give

$$\begin{cases} T_1 = -m \frac{l\ddot{\alpha}}{\sin(\alpha - \theta)} \\ T_2 = m \frac{l\ddot{\theta}}{\sin(\alpha - \theta)} + m \frac{g \sin \theta}{\sin(\alpha - \theta)} \end{cases}$$

Substituting T_1 and T_2 into equations above, we obtain

$$\begin{cases} l\ddot{\theta} = l\ddot{\theta} \\ -l\dot{\theta}^2 \sin(\alpha - \theta) = l\ddot{\alpha} + l\ddot{\theta} \cos(\alpha - \theta) + g \sin \alpha \\ l\ddot{\alpha} = l\ddot{\alpha} \\ l\dot{\alpha}^2 \sin(\alpha - \theta) = 2l\ddot{\theta} + l\ddot{\alpha} \cos(\alpha - \theta) + 2g \sin \theta \end{cases}$$

$$\Rightarrow \begin{cases} 2l\ddot{\theta} + l\ddot{\alpha} \cos(\theta - \alpha) = -l\dot{\alpha}^2 \sin(\theta - \alpha) - 2g \sin \theta \\ l\ddot{\alpha} + l\ddot{\theta} \cos(\theta - \alpha) = l\dot{\theta}^2 \sin(\theta - \alpha) - g \sin \alpha \end{cases}$$

1-2 Lagrangian Mechanics

A. D'ALEMBERT'S PRINCIPLE (1743)

- (1) The equations of motion with constraints

$$\vec{F}_i = \vec{F}_i^{(\text{ext})} + \vec{f}_i = \dot{\vec{p}}_i$$

where $\vec{F}_i^{(\text{ext})}$ is the applied force and $\vec{f}_i$ is the force of constraint.

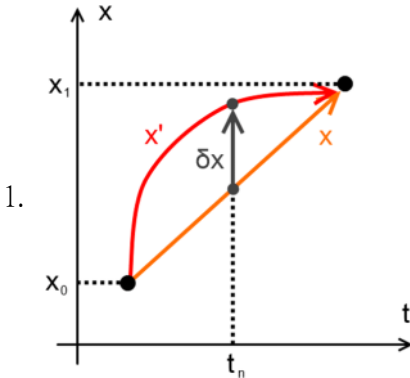
$$\vec{F}_i^{(\text{ext})} - \dot{\vec{p}}_i + \vec{f}_i = 0$$

- (2) Consider an infinitesimal change of coordinates occurring in a manner consistent with the constraints at a fixed time and called virtual displacement $\delta\vec{r}_i$.

OS:

It is called virtual rather than real since no actual displacement can take place without the passage of time.

EXAMPLES:



- (3) The virtual work of forces is

$$\sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta\vec{r}_i - \sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta\vec{r}_i + \sum_i \vec{f}_i \cdot \delta\vec{r}_i = 0$$

Here, $\delta\vec{r}_i$ are not completely independent but are connected by the constraints. Then, we restrict ourselves to systems for which the virtual work of the forces of constraint vanishes, and obtain

$$\sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta \vec{r}_i - \sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta \vec{r}_i = 0 \quad \text{D'Alembert's principle}$$

B. LAGRANGE'S EQUATIONS (1788)

- (1) A system of N particles subject to Λ holonomic constraints

$$f_\lambda(\vec{r}_1, \dots, \vec{r}_N, t) = 0, \quad \lambda = 1, \dots, \Lambda$$

The $3N$ space coordinates can be replaced by a set of $(3N - \Lambda)$ coordinates, i.e.,

$$\{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N\} \rightarrow \{q_1, q_2, \dots, q_f\}, \quad f = 3N - \Lambda$$

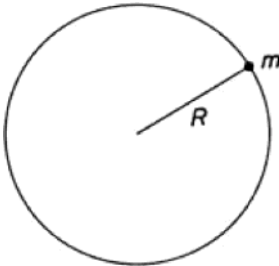
where the *generalized* coordinate transformations are

$$\vec{r}_i = \vec{r}_i(q_1, \dots, q_f, t), \quad i = 1, \dots, N \text{ and } f = 3N - \Lambda$$

and q_f is called *generalized* coordinate.

EXAMPLES:

1. A particle moving on the surface of a sphere of radius R



The constraint is $x^2 + y^2 + z^2 = R^2$ (holonomic), so that
 $f = 3N - 1 = 3 - 1 = 2$.

Instead of the dependent coordinates $\{x, y, z\}$, one introduces the independent coordinates $q_1 = \theta, q_2 = \phi$. The coordinate transformations are

$$\begin{aligned} x &= R \sin \theta \cos \phi \\ y &= R \sin \theta \sin \phi \\ z &= R \cos \theta \end{aligned}$$

- (2) According to D'Alembert's principle

$$\sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta \vec{r}_i - \sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta \vec{r}_i = 0$$

Define the virtual displacement as

$$\delta \vec{r}_i = \sum_{k=1}^f \frac{\partial \vec{r}_i}{\partial q_k} \delta q_k$$

The first term becomes

$$\sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta \vec{r}_i = \sum_i^N \sum_k^f \vec{F}_i^{(\text{ext})} \cdot \frac{\partial \vec{r}_i}{\partial q_k} \delta q_k$$

The second term becomes

$$\sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta \vec{r}_i = \sum_i^N m_i \ddot{\vec{r}}_i \cdot \sum_k^f \frac{\partial \vec{r}_i}{\partial q_k} \delta q_k$$

Thus, we obtain

$$\begin{aligned} & \sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta \vec{r}_i - \sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta \vec{r}_i \\ &= \sum_k^f \sum_i^N \vec{F}_i^{(\text{ext})} \cdot \frac{\partial \vec{r}_i}{\partial q_k} \delta q_k - \sum_k^f \sum_i^N m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \delta q_k \\ &= \sum_k^f \left[\sum_i^N \vec{F}_i^{(\text{ext})} \cdot \frac{\partial \vec{r}_i}{\partial q_k} - \sum_i^N m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \right] \delta q_k \\ &= 0 \end{aligned}$$

- (3) For the first term, we assume that the external forces are derivable from a scalar potential function $V(\vec{r}_1, \dots, \vec{r}_N, t)$, i.e.,

$$\vec{F}_i^{(\text{ext})} = -\nabla_i V(\vec{r}_1, \dots, \vec{r}_N, t) \text{ is the conservative force.}$$

Thus, we obtain

$$\sum_i^N \vec{F}_i^{(\text{ext})} \cdot \frac{\partial \vec{r}_i}{\partial q_k} = \sum_i^N -\nabla_i V(\vec{r}_1, \dots, \vec{r}_N, t) \cdot \frac{\partial \vec{r}_i}{\partial q_k} = -\frac{\partial}{\partial q_k} V(q_k, t)$$

where V the system's potential energy in term of generalized coordinates q_k .

For the second term, we use integration by parts and obtain

$$\begin{aligned}
\sum_i^N m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} &= \sum_{i=1}^N m_i \frac{d}{dt} \dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \\
&= \sum_{i=1}^N m_i \left[\frac{d}{dt} \left(\dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \right) - \dot{\vec{r}}_i \cdot \frac{d}{dt} \frac{\partial \vec{r}_i}{\partial q_k} \right] \\
&= \sum_{i=1}^N m_i \left[\frac{d}{dt} \left(\dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \right) - \dot{\vec{r}}_i \cdot \frac{\partial \dot{\vec{r}}_i}{\partial q_k} \right] \\
&= \sum_{i=1}^N m_i \left[\frac{d}{dt} \left(\dot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} \right) - \frac{1}{2} \frac{\partial}{\partial q_k} \dot{\vec{r}}_i^2 \right] \dots (a)
\end{aligned}$$

Then, we consider

$$\begin{aligned}
\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} &= \frac{\partial}{\partial \dot{q}_k} \left[\sum_{l=1}^f \left(\frac{\partial \vec{r}_i}{\partial q_l} \dot{q}_l \right) + \frac{\partial \vec{r}_i}{\partial t} \right] \\
&= \sum_{l=1}^f \left[\frac{\partial \dot{q}_l}{\partial \dot{q}_k} \left(\frac{\partial \vec{r}_i}{\partial q_l} \right) + \dot{q}_l \frac{\partial}{\partial q_l} \frac{\partial \vec{r}_i}{\underbrace{\partial \dot{q}_k}_{=0}} \right] + \frac{\partial}{\partial t} \frac{\partial \vec{r}_i}{\underbrace{\partial \dot{q}_k}_{=0}} \\
&= \sum_{l=1}^f \frac{\partial \dot{q}_l}{\partial \dot{q}_k} \left(\frac{\partial \vec{r}_i}{\partial q_l} \right) \\
&= \sum_{l=1}^f \delta_{kl} \frac{\partial \vec{r}_i}{\partial q_l} \\
&= \frac{\partial \vec{r}_i}{\partial q_k}
\end{aligned}$$

Thus, the equation (a) becomes

$$\begin{aligned}
\sum_i^N m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} &= \sum_{i=1}^N m_i \left[\frac{d}{dt} \left(\dot{\vec{r}}_i \cdot \frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} \right) - \frac{1}{2} \frac{\partial}{\partial q_k} \dot{\vec{r}}_i^2 \right] \\
&= \sum_{i=1}^N m_i \left[\frac{d}{dt} \left(\frac{1}{2} \frac{\partial}{\partial \dot{q}_k} \dot{\vec{r}}_i^2 \right) - \frac{1}{2} \frac{\partial}{\partial q_k} \dot{\vec{r}}_i^2 \right] \\
&= \sum_{i=1}^N \left[\frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} \left(\frac{1}{2} m_i \dot{\vec{r}}_i^2 \right) - \frac{\partial}{\partial q_k} \left(\frac{1}{2} m_i \dot{\vec{r}}_i^2 \right) \right]
\end{aligned}$$

Since

$$T = \sum_{i=1}^N \frac{1}{2} m_i \dot{\vec{r}}_i^2 = T(q_k, \dot{q}_k)$$

is the system's kinetic energy in term of generalized coordinates q_k and $\dot{q}_k$, thus, we obtain

$$\sum_i^N m_i \ddot{\vec{r}}_i \cdot \frac{\partial \vec{r}_i}{\partial q_k} = \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} T(q_k, \dot{q}_k) - \frac{\partial}{\partial q_k} T(q_k, \dot{q}_k)$$

(4) Then, D'Alembert's principle becomes

$$\begin{aligned} & \sum_{i=1}^N \vec{F}_i^{(\text{ext})} \cdot \delta \vec{r}_i - \sum_{i=1}^N \dot{\vec{p}}_i \cdot \delta \vec{r}_i \\ &= \sum_k^f \left[-\frac{\partial}{\partial q_k} V(q_k, t) - \left(\frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} T(q_k, \dot{q}_k) - \frac{\partial}{\partial q_k} T(q_k, \dot{q}_k) \right) \right] \delta q_k \\ &= \sum_k^f \left[\frac{\partial}{\partial q_k} (T(q_k, \dot{q}_k) - V(q_k, t)) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} T(q_k, \dot{q}_k) \right] \delta q_k \\ &= 0 \end{aligned}$$

Since the displacements δq_k are independent ($\delta \vec{r}_i$ are not), thus every term must vanish individually. We obtain

$$\frac{\partial}{\partial q_k} (T(q_k, \dot{q}_k) - V(q_k, t)) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} T(q_k, \dot{q}_k) = 0$$

Since

$$\frac{\partial}{\partial \dot{q}_k} V(q_k, t) = 0$$

we obtain

$$\frac{\partial}{\partial q_k} (T(q_k, \dot{q}_k) - V(q_k, t)) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} (T(q_k, \dot{q}_k) - V(q_k, t)) = 0$$

(5) Define the *Lagrangian* function

$$L(q_k, \dot{q}_k, t) = T(q_k, \dot{q}_k) - V(q_k, t)$$

as the difference between the system's kinetic energy and potential energy. Thus, we obtain Lagrange's equation of motion

$$\boxed{\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) = 0}$$

For a velocity-dependent potential (non-conservative force), we have

$$\frac{\partial}{\partial \dot{q}_k} V(q_k, \dot{q}_k, t) \neq 0$$

If the Lagrange's equation holds:

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) = 0$$

we can let the Lagrangian function as

$$L(q_k, \dot{q}_k, t) = T(q_k, \dot{q}_k) - V(q_k, \dot{q}_k, t)$$

COMMENTS:

1. The Lagrangian L is not a physical quantity but a mathematical quantity which provides a direct path to the equation of motion.
2. The Lagrangian denotes a way to mathematically encode the relationship between motion (kinetic energy) and forces (potential energy) in a system, allowing us to predict how things move.

EXAMPLES:

1. One dimensional harmonic oscillator

$$T = \frac{1}{2} m \dot{x}^2$$

$$V = \frac{1}{2} k x^2$$

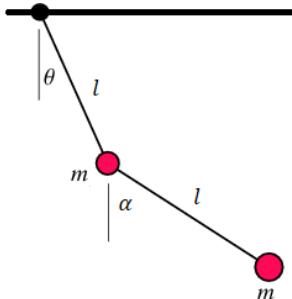
The Lagrangian is

$$L = T - V = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2$$

Lagrange's equation of motion:

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = -kx - m\ddot{x} = 0 \Rightarrow \ddot{x} + \frac{k}{m} x = 0$$

2. Write down the equation of motion of a double pendulum



ANSWER:

The kinetic energy:

$$\begin{aligned}
 T &= T_1 + T_2 = \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\
 x &= l \sin \theta + l \sin \alpha \Rightarrow \dot{x} = l\dot{\theta} \cos \theta + l\dot{\alpha} \cos \alpha \\
 y &= l \cos \theta + l \cos \alpha \Rightarrow \dot{y} = -l\dot{\theta} \sin \theta - l\dot{\alpha} \sin \alpha \\
 T &= \frac{1}{2}ml^2\dot{\theta}^2 \\
 &\quad + \frac{1}{2}ml^2(\dot{\theta}^2 + \dot{\alpha}^2 + 2\dot{\theta}\dot{\alpha} \cos \theta \cos \alpha + 2\dot{\theta}\dot{\alpha} \sin \theta \sin \alpha) \\
 &= \frac{1}{2}ml^2\dot{\theta}^2 + \frac{1}{2}ml^2(\dot{\theta}^2 + \dot{\alpha}^2 + 2\dot{\theta}\dot{\alpha} \cos(\theta - \alpha))
 \end{aligned}$$

The potential energy:

$$V = V_1 + V_2 = -mgl \cos \theta - mgl \cos \theta - mgl \cos \alpha$$

The Lagrangian is

$$\begin{aligned}
 L &= T - V \\
 &= ml^2\dot{\theta}^2 + \frac{1}{2}ml^2\dot{\alpha}^2 + ml^2\dot{\theta}\dot{\alpha} \cos(\theta - \alpha) \\
 &\quad + 2mgl \cos \theta + mgl \cos \alpha
 \end{aligned}$$

θ -term:

$$\begin{aligned}
 \frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} &= 0 \\
 \Rightarrow -ml^2\dot{\theta}\dot{\alpha} \sin(\theta - \alpha) - 2mgl \sin \theta \\
 &\quad - 2ml^2\ddot{\theta} - ml^2\ddot{\alpha} \cos(\theta - \alpha) + ml^2\dot{\theta}\dot{\alpha} \sin(\theta - \alpha) \\
 &\quad - ml^2\dot{\alpha}^2 \sin(\theta - \alpha) = 0 \\
 \Rightarrow 2ml^2\ddot{\theta} + ml^2\ddot{\alpha} \cos(\theta - \alpha) + ml^2\dot{\alpha}^2 \sin(\theta - \alpha) + 2mgl \sin \theta \\
 &= 0 \dots (a)
 \end{aligned}$$

α -term:

$$\begin{aligned}
 \frac{\partial L}{\partial \alpha} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\alpha}} &= 0 \\
 \Rightarrow ml^2\dot{\theta}\dot{\alpha} \sin(\theta - \alpha) - mgl \sin \alpha \\
 &\quad - ml^2\ddot{\alpha} - ml^2\ddot{\theta} \cos(\theta - \alpha) + ml^2\dot{\theta}^2 \sin(\theta - \alpha) \\
 &\quad - ml^2\dot{\theta}\dot{\alpha} \sin(\theta - \alpha) = 0 \\
 \Rightarrow ml^2\ddot{\alpha} + ml^2\ddot{\theta} \cos(\theta - \alpha) - ml^2\dot{\theta}^2 \sin(\theta - \alpha) + mgl \sin \alpha \\
 &= 0 \dots (b)
 \end{aligned}$$

We collect (a) and (b) and obtain the Lagrange's equations, which are the same ones we got from Newton's law:

$$\begin{cases} 2l\ddot{\theta} + l\ddot{\alpha} \cos(\theta - \alpha) = -l\dot{\alpha}^2 \sin(\theta - \alpha) - 2g \sin \theta \\ l\ddot{\alpha} + l\ddot{\theta} \cos(\theta - \alpha) = l\dot{\theta}^2 \sin(\theta - \alpha) - g \sin \alpha \end{cases}$$

3. An electron in an electromagnetic field moves according to the Lorentz force

$$\vec{F} = -e\vec{E} - \frac{e}{c}\vec{v} \times \vec{B}$$

The electric field $\vec{E}$ and magnetic field $\vec{B}$ can be written in terms of a scalar potential φ and a vector potential $\vec{A}$,

$$\vec{E} = -\nabla\varphi - \frac{1}{c}\frac{\partial\vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\begin{aligned}\vec{F} &= e\nabla\varphi + \frac{e}{c}\frac{\partial\vec{A}}{\partial t} - \frac{e}{c}\vec{v} \times \nabla \times \vec{A} \\ &= e\nabla\varphi + \frac{e}{c}\frac{\partial\vec{A}}{\partial t} - \frac{e}{c}(\nabla\vec{v} \cdot \vec{A} - \vec{v} \cdot \nabla\vec{A})\end{aligned}$$

Since $\frac{d\vec{A}}{dt} = \frac{\partial\vec{A}}{\partial t} + \vec{v} \cdot \nabla\vec{A}$

$$\begin{aligned}\vec{F} &= e\nabla\varphi + \frac{e}{c}\frac{\partial\vec{A}}{\partial t} - \frac{e}{c}\left(\nabla\vec{v} \cdot \vec{A} - \frac{d\vec{A}}{dt} + \frac{\partial\vec{A}}{\partial t}\right) \\ &= e\nabla\varphi - \frac{e}{c}\nabla\vec{v} \cdot \vec{A} + \frac{e}{c}\frac{d\vec{A}}{dt}\end{aligned}$$

$$\frac{d\vec{A}}{dt} = \frac{d}{dt}(\nabla_v\vec{v} \cdot \vec{A}), \text{ since } \nabla_v\vec{v} = 1$$

$$\begin{aligned}\vec{F} &= e\nabla\left(\varphi - \frac{1}{c}\vec{v} \cdot \vec{A}\right) + \frac{e}{c}\frac{d}{dt}\nabla_v\vec{v} \cdot \vec{A} \\ &= e\nabla\left(\varphi - \frac{1}{c}\vec{v} \cdot \vec{A}\right) - e\frac{d}{dt}\nabla_v\left(\varphi - \frac{1}{c}\vec{v} \cdot \vec{A}\right)\end{aligned}$$

Let $V = \varphi - \frac{1}{c}\vec{v} \cdot \vec{A}$ be the velocity-dependent potential

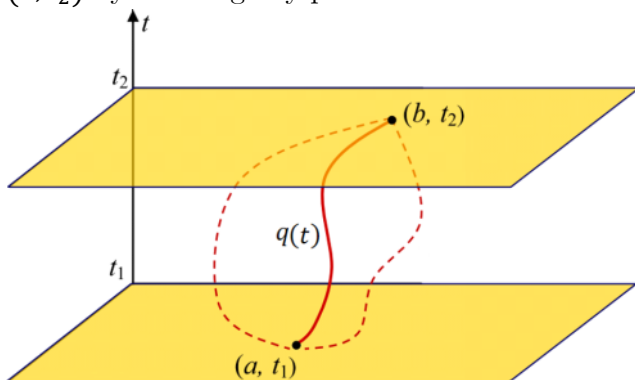
The Lagrangian is

$$\begin{aligned}L &= T - V \\ &= \frac{1}{2}mv^2 - (-e)\left(\varphi - \frac{1}{c}\vec{v} \cdot \vec{A}\right) \\ &= \frac{1}{2}mv^2 + e\varphi - \frac{e}{c}\vec{v} \cdot \vec{A}\end{aligned}$$

C. ACTION FUNCTIONAL

- (1) Consider a particle traverses a space between two points (a, t_1) and

(b, t_2) by choosing any path.



The Lagrangian function is given by

$$L = L(q(t), \dot{q}(t))$$

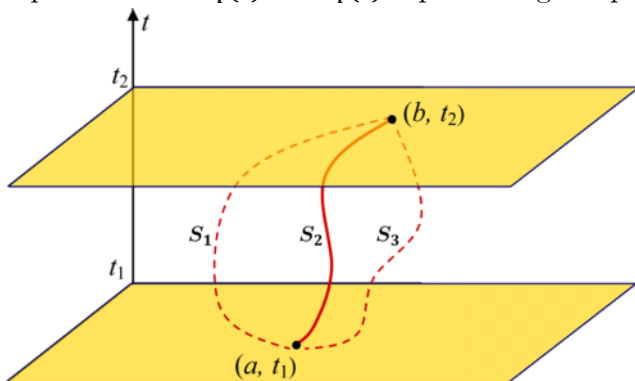
where $q(t)$ and $\dot{q}(t)$ are parametric functions describing the path.

Mathematically, the Lagrangian function will return a function instead of a number. Since a physical system is represented by a number, we thus need a mathematical object which can take in a function as its input and return a number. We shall then take in entire functions $q(t)$ and $\dot{q}(t)$ as its input and return a number by writing a *definite integral* of the Lagrangian function:

$$S = \int_{t_1}^{t_2} L(q(t), \dot{q}(t)) dt$$

where S is called action functional or simply action in physics.

The number describes the value of the action (functional) S for the input functions $q(t)$ and $\dot{q}(t)$ representing the path of the system.



(2) The differential or variation of an action (functional) S

Consider an infinitesimal change of $q(t)$ as

$$q(t) \rightarrow q(t) + \delta q(t)$$

Thus, we obtain

$$\begin{aligned}\delta S &= \sum_{k=1}^f \left(\int_{t_1}^{t_2} L(q_k + \delta q_k, \dot{q}_k + \delta \dot{q}_k) dt - \int_{t_1}^{t_2} L(q_k, \dot{q}_k) dt \right) \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(L(q_k, \dot{q}_k) + \frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} \delta q_k + \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \delta \dot{q}_k - L(q_k, \dot{q}_k) \right) dt \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} \delta q_k(t) + \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \delta \dot{q}_k(t) \right) dt\end{aligned}$$

Applying integration by parts to the second term, we obtain

$$\begin{aligned}\delta S &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} \delta q_k(t) - \frac{d}{dt} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \delta q_k(t) \right) dt \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} - \frac{d}{dt} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \right) \delta q_k(t) dt\end{aligned}$$

OS:

Alternatively,

$$\begin{aligned}\delta S &= \frac{\delta S}{\delta q_l(t)} \delta q_l(t) \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} - \frac{d}{dt'} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \right) \frac{\delta q_k(t')}{\delta q_l(t)} \delta q_l(t) dt' \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} - \frac{d}{dt'} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \right) \delta(t' - t) \delta_{kl} \delta q_l(t) dt'\end{aligned}$$

or simply

$$\begin{aligned}\frac{\delta S}{\delta q_l} &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} - \frac{d}{dt'} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \right) \delta(t' - t) \delta_{kl} dt' \\ &= \left(\frac{\partial L(q_l, \dot{q}_l)}{\partial q_l} - \frac{d}{dt} \frac{\partial L(q_l, \dot{q}_l)}{\partial \dot{q}_l} \right)\end{aligned}$$

Thus, the variation of S can be expressed as:

$$\delta S = \sum_{k=1}^f \int_{t_1}^{t_2} \frac{\delta S}{\delta q_k} \delta q_k(t') dt'$$

- (3) Hamilton (1833) states that out of all the possible paths connecting a and b , the physical path $q_k(t)$ would be the one for which the action has the least possible value, i.e.,

$$\delta S = 0 = \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} - \frac{d}{dt} \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \right) \delta q_k(t) dt$$

Since the displacements δq_k are independent, thus every term must vanish individually. We obtain

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} = 0$$

OS:

or simply

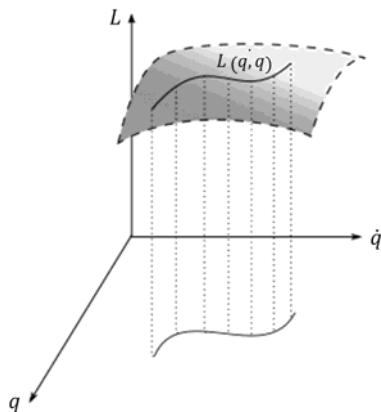
$$\frac{\delta S}{\delta q_k} = 0 \Rightarrow \frac{\partial L}{\partial q_k} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} = 0$$

[Calculus of Variations For Dummies: An Intuitive Introduction – Profound Physics](#)

1-3 Hamiltonian Mechanics

A. HAMILTON'S EQUATIONS (1833)

- (1) Mathematically, we assume that the generalized coordinates q_k and velocities $\dot{q}_k$ are independent variables. Thus, the Lagrangian is a function that describes the curve in space,



where the curve is not a path or trajectory mentioned in action, and the space spanned by $(q_k, \dot{q}_k)$ is called "phase space".

- (2) As we treat q_k and $\dot{q}_k$ are independent variables, we can have the differential of the Lagrangian function $L(q, \dot{q}, t)$ as

$$dL(q, \dot{q}, t) = \sum_k \frac{\partial L}{\partial q_k} dq_k + \sum_k \frac{\partial L}{\partial \dot{q}_k} d\dot{q}_k + \frac{\partial L}{\partial t} dt$$

Since

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) = 0 \Rightarrow \frac{\partial L}{\partial q_k} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right)$$

and

$$\frac{\partial L}{\partial q_k} = -\nabla_q V = \text{force} = \frac{d}{dt} p_k$$

thus, we can define a conjugate momentum

$$p_k = \frac{\partial L}{\partial \dot{q}_k}$$

and obtain

$$dL(q, \dot{q}, t) = \sum_k \dot{p}_k dq_k + \sum_k p_k d\dot{q}_k + \frac{\partial L}{\partial t} dt \dots (a)$$

- (3) Mathematically, we would prefer a function of (q, p, t) instead of a function of $(q, \dot{q}, t)$ because p is not an explicit differentiation of q with respect to time.

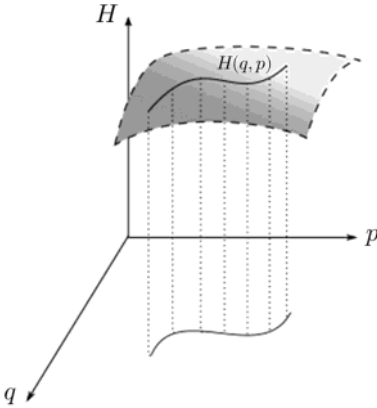
Since

$$p_k = \frac{\partial L}{\partial \dot{q}_k}$$

we then use the Legendre transformation (1787) to define a function from L ,

$$H(q, p, t) = \sum_k p_k \dot{q}_k - L(q, \dot{q}, t)$$

where $H(q, p, t)$ is called the Hamiltonian.



The differential of H gives

$$dH(q, p, t) = d\left(\sum_k p_k \dot{q}_k\right) - dL(q, \dot{q}, t)$$

Since $d(p\dot{q})$ appears in the first term of dH and $p(d\dot{q})$ appears in the second term of dL , we thus apply the integration by parts to the second term of equation (a),

$$\sum_k p_k d\dot{q}_k = d\left(\sum_k p_k \dot{q}_k\right) - \sum_k \dot{q}_k dp_k$$

Then we obtain

$$\begin{aligned}
dL &= \sum_k \dot{p}_k dq_k + d\left(\sum_k p_k \dot{q}_k\right) - \sum_k \dot{q}_k dp_k + \frac{\partial L}{\partial t} dt \\
&\Rightarrow d\left(\sum_k p_k \dot{q}_k\right) - dL = \sum_k \dot{q}_k dp_k - \sum_k \dot{p}_k dq_k - \frac{\partial L}{\partial t} dt \\
&\Rightarrow dH(q, p, t) = \sum_k \dot{q}_k dp_k - \sum_k \dot{p}_k dq_k - \frac{\partial L}{\partial t} dt \dots (b)
\end{aligned}$$

On the other hand,

$$dH(q, p, t) = \sum_k \frac{\partial H}{\partial q_k} dq_k + \sum_k \frac{\partial H}{\partial p_k} dp_k + \frac{\partial H}{\partial t} dt \dots (c)$$

Comparing equations (b) and (c), we obtain Hamilton's equations of motion:

$$\begin{aligned}
\dot{q}_k &= \frac{\partial H}{\partial p_k} \\
\dot{p}_k &= -\frac{\partial H}{\partial q_k} = \frac{\partial L}{\partial q_k} \\
\frac{\partial L}{\partial t} &= -\frac{\partial H}{\partial t}
\end{aligned}$$

(4) Physical interpretation of $H(q, p)$

Considering a system has no explicit dependence of time

$$L(q, \dot{q}) = T(\dot{q}) - V(q)$$

Thus, we obtain

$$\begin{aligned}
H(q, p) &= \sum_k p_k \dot{q}_k - L(q, \dot{q}) \\
&= \sum_k \frac{\partial L(q, \dot{q})}{\partial \dot{q}_k} \dot{q}_k - (T(\dot{q}) - V(q)) \\
&= \sum_k \frac{\partial T(\dot{q})}{\partial \dot{q}_k} \dot{q}_k - T(\dot{q}) + V(q)
\end{aligned}$$

For the time-independent constraint (holonomic constraint):

$$\begin{aligned}
T(\dot{q}) &= \sum_{i=1}^N \frac{1}{2} m_i \dot{\vec{r}}_i^2 \\
&= \sum_i \frac{1}{2} m_i \left(\sum_s \frac{\partial \vec{r}_i}{\partial q_s} \dot{q}_s + \underbrace{\frac{\partial \vec{r}_i}{\partial t}}_{=0} \right) \cdot \left(\sum_t \frac{\partial \vec{r}_i}{\partial q_t} \dot{q}_t + \underbrace{\frac{\partial \vec{r}_i}{\partial t}}_{=0} \right) \\
&= \sum_{st} \sum_i \underbrace{\frac{1}{2} m_i \frac{\partial \vec{r}_i}{\partial q_s} \cdot \frac{\partial \vec{r}_i}{\partial q_t}}_{=c(s,t)} \dot{q}_s \dot{q}_t
\end{aligned}$$

Thus, we obtain

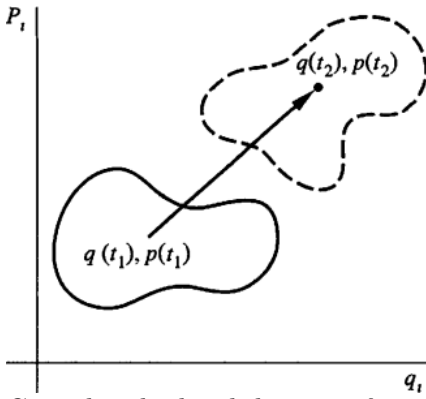
$$\begin{aligned}
\sum_k \frac{\partial T(\dot{q})}{\partial \dot{q}_k} \dot{q}_k &= \sum_k \frac{\partial}{\partial \dot{q}_k} \left(\sum_{st} c(s, t) \dot{q}_s \dot{q}_t \right) \dot{q}_k \\
&= \sum_k \left(\sum_{st} c(s, t) \delta_{sk} \dot{q}_t + \sum_{st} c(s, t) \dot{q}_s \delta_{tk} \right) \dot{q}_k \\
&= \sum_k \left(\sum_t c(k, t) \dot{q}_t + \sum_s c(s, k) \dot{q}_s \right) \dot{q}_k \\
&= \sum_k \sum_s c(s, k) \dot{q}_k \dot{q}_s \\
&= 2T(\dot{q})
\end{aligned}$$

The Hamiltonian becomes

$$H(q, p) = 2T(\dot{q}) - T(\dot{q}) + V(q) = T(\dot{q}) + V(q)$$

$\Rightarrow$ The time-independent Hamiltonian corresponds to the total energy of the system including both kinetic energy and potential energy. Thus, the time-independent Hamiltonian is a measurable physical quantity.

(5) Liouville's [lyoo-veel] theorem in phase space



Consider the local density of points in phase space as

$$\rho(q, p, t) = \frac{\Delta N}{\Delta \mathcal{V}}$$

where ΔN is the fixed number of points in the volume $\Delta \mathcal{V}$. The time evolution of $\rho(q, p, t)$ makes a curve, called "flow", in phase space.

Thus, the total time-derivative, of $\rho(q, p, t)$ is

$$\frac{d\rho}{dt} = \sum_{k=1} \left(\frac{\partial \rho}{\partial q_k} \dot{q}_k + \frac{\partial \rho}{\partial p_k} \dot{p}_k \right) + \frac{\partial \rho}{\partial t}$$

From the continuity equation:

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= - \sum_{k=1}^f \left(\frac{\partial}{\partial q_k} (\rho \dot{q}_k) + \frac{\partial}{\partial p_k} (\rho \dot{p}_k) \right) \\ &= - \sum_k^f \left(\frac{\partial \rho}{\partial q_k} \dot{q}_k + \rho \frac{\partial \dot{q}_k}{\partial q_k} + \frac{\partial \rho}{\partial p_k} \dot{p}_k + \rho \frac{\partial \dot{p}_k}{\partial p_k} \right) \end{aligned}$$

Thus, we obtain

$$\begin{aligned} \frac{d\rho}{dt} &= \sum_{k=1} \left(\frac{\partial \rho}{\partial q_k} \dot{q}_k + \frac{\partial \rho}{\partial p_k} \dot{p}_k \right) - \left(\frac{\partial \rho}{\partial q_k} \dot{q}_k + \rho \frac{\partial \dot{q}_k}{\partial q_k} + \frac{\partial \rho}{\partial p_k} \dot{p}_k + \rho \frac{\partial \dot{p}_k}{\partial p_k} \right) \\ &= \sum_{k=1} \left(-\rho \frac{\partial \dot{q}_k}{\partial q_k} - \rho \frac{\partial \dot{p}_k}{\partial p_k} \right) \\ &= - \sum_{k=1} \rho \left(\frac{\partial \dot{q}_k}{\partial q_k} + \frac{\partial \dot{p}_k}{\partial p_k} \right) \end{aligned}$$

Assume that the points move according to the Hamilton's equations,

$$\dot{q}_k = \frac{\partial H}{\partial p_k} \Rightarrow \frac{\partial \dot{q}_k}{\partial q_k} = \frac{\partial^2 H}{\partial q_k \partial p_k}$$

$$\dot{p}_k = -\frac{\partial H}{\partial q_k} \Rightarrow \frac{\partial \dot{p}_k}{\partial p_k} = -\frac{\partial^2 H}{\partial p_k \partial q_k}$$

We obtain

$$\frac{d\rho}{dt} = -\sum_{k=1} \rho \left(\frac{\partial^2 H}{\partial q_k \partial p_k} - \frac{\partial^2 H}{\partial p_k \partial q_k} \right) = 0 \Rightarrow \rho = \frac{\Delta N}{\Delta \mathcal{V}} = \text{constant}$$

Since ΔN is the fixed number, thus, $\Delta \mathcal{V}$ is invariant, which obeys Liouville's theorem.

- (6) Therefore, the time-independent Hamiltonian represents the total energy of the system and is more suitable for describing systems in terms of energy conservation and phase space (q, p) .

EXAMPLES:

1. One dimensional harmonic oscillator

The Lagrangian

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

The conjugate momentum

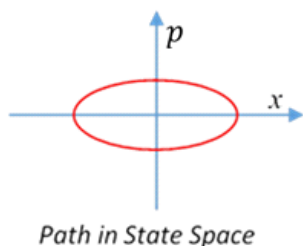
$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

The Hamiltonian

$$\begin{aligned} H &= \sum_k p_k \dot{q}_k - L \\ &= p\dot{x} - \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 \\ &= p \frac{p}{m} - \frac{1}{2}m \left(\frac{p}{m} \right)^2 + \frac{1}{2}kx^2 \\ &= \underbrace{\frac{p^2}{2m}}_T + \underbrace{\frac{1}{2}kx^2}_V \end{aligned}$$

Hamilton's equations of motion:

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m} \\ \dot{p} = -\frac{\partial H}{\partial x} = -kx \end{cases}$$



2. Two dimensional projectile motion

The Lagrangian

$$L = T - V = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy$$

The conjugate momentum

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y}$$

The Hamiltonian

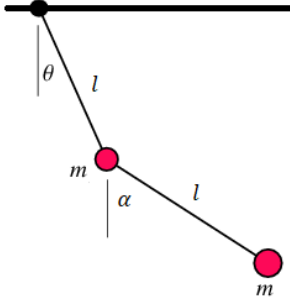
$$\begin{aligned} H &= \sum_k p_k \dot{q}_k - L \\ &= p_x \dot{x} + p_y \dot{y} - \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\dot{y}^2 + mgy \\ &= p_x \frac{p_x}{m} + p_y \frac{p_y}{m} - \frac{1}{2}m \left(\frac{p_x}{m} \right)^2 - \frac{1}{2}m \left(\frac{p_y}{m} \right)^2 + mgy \\ &= \underbrace{\frac{p_x^2}{2m} + \frac{p_y^2}{2m}}_T + \underbrace{mgy}_V \end{aligned}$$

Hamilton's equations of motion:

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial p_x} = \frac{p_x}{m} \\ \dot{p}_x = -\frac{\partial H}{\partial x} = 0 \end{cases}$$

$$\begin{cases} \dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{m} \\ \dot{p}_y = -\frac{\partial H}{\partial y} = -mg \end{cases}$$

3. Write down the equation of motion of a double pendulum



ANSWER:

The Lagrangian is given by

$$L = ml^2 \dot{\theta}^2 + \frac{1}{2} ml^2 \dot{\alpha}^2 + ml^2 \dot{\theta} \dot{\alpha} \cos(\theta - \alpha) + 2mgl \cos \theta + mgl \cos \alpha$$

The conjugate momenta

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = 2ml^2 \dot{\theta} + ml^2 \dot{\alpha} \cos(\theta - \alpha) \dots \dots \textcircled{1}$$

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = ml^2 \dot{\alpha} + ml^2 \dot{\theta} \cos(\theta - \alpha) \dots \dots \textcircled{2}$$

The Hamiltonian is given by,

$$H = p_\theta \dot{\theta} + p_\alpha \dot{\alpha} - L$$

Since $\dot{\theta}$, $\dot{\alpha}$, L should be written in terms of θ , α , p_θ , p_α , we need to rewrite equations $\textcircled{1}$ and $\textcircled{2}$ in the matrix form as

$$\begin{pmatrix} p_\theta \\ p_\alpha \end{pmatrix} = \begin{pmatrix} 2ml^2 & ml^2 \cos(\theta - \alpha) \\ ml^2 \cos(\theta - \alpha) & ml^2 \end{pmatrix} \begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix}$$

$$\begin{pmatrix} \dot{\theta} \\ \dot{\alpha} \end{pmatrix} = \begin{pmatrix} 2ml^2 & ml^2 \cos(\theta - \alpha) \\ ml^2 \cos(\theta - \alpha) & ml^2 \end{pmatrix}^{-1} \begin{pmatrix} p_\theta \\ p_\alpha \end{pmatrix}$$

$$= \frac{1}{\det \mathbf{K}} \cdot ml^2 \begin{pmatrix} 1 & -\cos(\theta - \alpha) \\ -\cos(\theta - \alpha) & 2 \end{pmatrix} \begin{pmatrix} p_\theta \\ p_\alpha \end{pmatrix}$$

where

$$\det \mathbf{K} = m^2 l^4 (2 - \cos^2(\theta - \alpha))$$

Thus, we obtain

$$\dot{\theta} = \frac{p_{\theta} - p_{\alpha} \cos(\theta - \alpha)}{ml^2(1 + \sin^2(\theta - \alpha))}$$

$$\dot{\alpha} = \frac{-p_{\theta} \cos(\theta - \alpha) + 2p_{\alpha}}{ml^2(1 + \sin^2(\theta - \alpha))}$$

The Hamiltonian in terms of $\theta, \alpha, p_{\theta}, p_{\alpha}$

$$H = \underbrace{\frac{p_{\theta}^2 + 2p_{\alpha}^2 - 2p_{\theta}p_{\alpha} \cos(\theta - \alpha)}{2ml^2(1 + \sin^2(\theta - \alpha))}}_{\bar{T}} + \underbrace{-2mgl \cos \theta - mgl \cos \alpha}_{\bar{V}}$$

Hamilton's equations are

$$\begin{cases} \dot{\theta} = \frac{\partial H}{\partial p_{\theta}} = \frac{p_{\theta} - p_{\alpha} \cos(\theta - \alpha)}{ml^2(1 + \sin^2(\theta - \alpha))} \\ \dot{p}_{\theta} = -\frac{\partial H}{\partial \theta} = -h_1 + h_2 \sin 2(\theta - \alpha) - 2mgl \sin \theta \\ \dot{\alpha} = \frac{\partial H}{\partial p_{\alpha}} = \frac{-p_{\theta} \cos(\theta - \alpha) + 2p_{\alpha}}{ml^2(1 + \sin^2(\theta - \alpha))} \\ \dot{p}_{\alpha} = -\frac{\partial H}{\partial \alpha} = h_1 - h_2 \sin 2(\theta - \alpha) - mgl \sin \alpha \end{cases}$$

where

$$h_1 = \frac{p_{\theta}p_{\alpha} \sin(\theta - \alpha)}{ml^2(1 + \sin^2(\theta - \alpha))}$$

$$h_2 = \frac{p_{\theta}^2 + 2p_{\alpha}^2 - 2p_{\theta}p_{\alpha} \cos(\theta - \alpha)}{2l^2(1 + \sin^2(\theta - \alpha))^2}$$

4. Consider an electron in an electromagnetic field

$$L = \underbrace{\frac{1}{2}mv^2}_{\bar{T}} - \underbrace{\left(-e\varphi + \frac{e}{c} \vec{v} \cdot \vec{A}\right)}_{\bar{V}}$$

Define the conjugate momentum,

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} - \frac{e}{c} \vec{A}$$

we obtain

$$\begin{aligned} H &= \vec{p} \cdot \vec{v} - L \\ &= mv^2 - \frac{e}{c} \vec{v} \cdot \vec{A} - \left(\frac{1}{2}mv^2 + e\varphi - \frac{e}{c} \vec{v} \cdot \vec{A}\right) \\ &= \frac{1}{2}mv^2 - e\varphi \end{aligned}$$

Since

$$m\vec{v} = \vec{p} + \frac{e}{c} \vec{A}$$

we obtain the Hamiltonian

$$H = \underbrace{\frac{1}{2m} \left(\vec{p} + \frac{e}{c} \vec{A} \right)^2}_T + \underbrace{-e\varphi}_V$$

B. POISSON BRACKET (1809) AND CANONICAL COORDINATES

- (1) The Poisson bracket of two continuous functions $f(q, p)$ and $g(q, p)$ is defined to be

$$[f, g]_{qp} \equiv \sum_{k=1} \left(\frac{\partial f}{\partial q_k} \frac{\partial g}{\partial p_k} - \frac{\partial f}{\partial p_k} \frac{\partial g}{\partial q_k} \right)$$

The definition of the Poisson bracket leads to the following identities:

Anticommutativity:

$$[f, g]_{qp} = -[g, f]_{qp}$$

Bilinear:

$$[af_1 + bf_2, g]_{qp} = a[f_1, g]_{qp} + b[f_2, g]_{qp}$$

Product identity:

$$[f_1 f_2, g]_{qp} = f_1 [f_2, g]_{qp} + [f_1, g]_{qp} f_2$$

- (2) The Poisson bracket of variables themselves are called the fundamental Poisson bracket,

$$[q_i, q_j]_{qp} = \sum_k \left(\frac{\partial q_i}{\partial q_k} \frac{\partial q_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \frac{\partial q_j}{\partial q_k} \right) = \sum_k (\delta_{ik} \cdot 0 - 0 \cdot \delta_{jk}) = 0$$

$$[p_i, p_j]_{qp} = \sum_k \left(\frac{\partial p_i}{\partial q_k} \frac{\partial p_j}{\partial p_k} - \frac{\partial p_i}{\partial p_k} \frac{\partial p_j}{\partial q_k} \right) = \sum_k (0 \cdot \delta_{jk} - \delta_{ik} \cdot 0) = 0$$

$$[q_i, p_j]_{qp} = \sum_k \left(\frac{\partial q_i}{\partial q_k} \frac{\partial p_j}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \frac{\partial p_j}{\partial q_k} \right) = \sum_k (\delta_{ik} \delta_{jk} - 0 \cdot 0) = \delta_{ij}$$

- (3) The Hamilton's equations in terms of Poisson brackets

$$[q_i, H]_{qp} = \sum_k \left(\frac{\partial q_i}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial q_i}{\partial p_k} \frac{\partial H}{\partial q_k} \right) = \sum_k \delta_{ik} \frac{\partial H}{\partial p_k} = \frac{\partial H}{\partial p_i}$$

$$[p_i, H]_{qp} = \sum_k \left(\frac{\partial p_i}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial p_i}{\partial p_k} \frac{\partial H}{\partial q_k} \right) = - \sum_k \delta_{ik} \frac{\partial H}{\partial q_k} = - \frac{\partial H}{\partial q_i}$$

Thus, we obtain

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = [q_i, H]_{qp}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} = [p_i, H]_{qp}$$

$\Rightarrow$ The variables q and p satisfy the Poisson bracket and obey the Hamilton's equations are called canonical coordinates of the phase space.

EXAMPLES:

1. One dimensional harmonic oscillator

$$\begin{aligned} H &= \frac{p^2}{2m} + \frac{1}{2}kx^2 \\ \dot{x} &= [x, H]_{xp} = \frac{1}{2m}[x, p^2] = \frac{1}{2m}(p[x, p] + [x, p]p) = \frac{p}{m} \\ \dot{p} &= [p, H]_{xp} = \frac{1}{2}k[p, x^2] = \frac{k}{2}(x[p, x] + [p, x]x) = -kx \end{aligned}$$

2. Two dimensional projectile motion

$$\begin{aligned} H &= \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + mgy \\ \dot{x} &= [x, H] = \frac{1}{2m}[x, p_x^2] = \frac{1}{2m}(p_x[x, p_x] + [x, p_x]p_x) = \frac{p_x}{m} \\ \dot{p}_x &= [p_x, H] = \frac{1}{2m}[p_x, p_x^2] = 0 \\ \dot{y} &= [y, H] = \frac{1}{2m}[y, p_y^2] = \frac{1}{2m}(p_y[y, p_y] + [y, p_y]p_y) = \frac{p_y}{m} \\ \dot{p}_y &= [p_y, H] = \frac{1}{2m}[p_y, p_y^2] + mg[p_y, y] = -mg \end{aligned}$$

C. HAMILTON-JACOBI EQUATION (1843)

- (1) Consider the time-dependent action

$$S(t) = \int_{t_1}^t L dt' = \int_{t_1}^t \left(\sum_k p_k \dot{q}_k - H \right) dt'$$

The differential of S is

$$dS = \sum_k \frac{\partial S}{\partial q_k} dq_k + \frac{\partial S}{\partial t} dt = \sum_k p_k dq_k - H dt$$

Comparing the L.H.S. and R.H.S., we obtain

$$\frac{\partial S(t)}{\partial q_k} = p_k$$

$$\frac{\partial S(t)}{\partial t} = -H(q, p) \dots (a)$$

From the equation (a), we have

$$\begin{aligned} \frac{\partial S(t)}{\partial t} &= -H(q, p) \\ \Rightarrow \frac{\partial S(t)}{\partial t} + H\left(q, \frac{\partial S}{\partial q}\right) &= 0 \end{aligned}$$

This equation is called Hamilton-Jacobi equation.

EXAMPLES:

1. One dimensional harmonic oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

The Hamilton-Jacobi equation is

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial x} \right)^2 + \frac{1}{2}m\omega^2 x^2 = 0$$

Since H has no explicit time dependence, $H = E$ which is the total energy. So the action has the form,

$$\begin{aligned} \frac{\partial S(x, t)}{\partial t} &= -H(x, p) \Rightarrow \frac{\partial S}{\partial t} = -E \Rightarrow S(x, t) = S_0(x) - Et \\ \frac{\partial S(x, t)}{\partial x} &= p \Rightarrow p = \frac{\partial S_0}{\partial x} \end{aligned}$$

The Hamilton-Jacobi equation becomes

$$\begin{aligned} \frac{1}{2m} \left(\frac{dS_0}{dx} \right)^2 + \frac{1}{2}m\omega^2 x^2 &= E \\ \frac{dS_0}{dx} &= \sqrt{2mE - m^2\omega^2 x^2} \end{aligned}$$

Integration of this equation gives

$$S_0 = \int \sqrt{2mE - m^2\omega^2 x^2} dx = \sqrt{2mE} \int \sqrt{1 - \frac{m\omega^2 x^2}{2E}} dx$$

That is,

$$S = \sqrt{2mE} \int \sqrt{1 - \frac{m\omega^2 x^2}{2E}} dx - Et$$

Note that,

$$\frac{\partial S}{\partial E} = \sqrt{\frac{2m}{E}} \int \frac{dx}{\sqrt{1 - \frac{m\omega^2 x^2}{2E}}} - t$$

This can be integrated to give

$$t = \frac{1}{\omega} \sin^{-1} \sqrt{\frac{m\omega^2}{2E}} x + t_0$$

That is,

$$\begin{aligned} x &= \sqrt{\frac{2E}{m\omega^2}} \sin \omega(t - t_0) \\ p &= \frac{\partial S_0}{\partial x} \\ &= \sqrt{2mE - m^2\omega^2 x^2} \\ &= \sqrt{2mE - m^2\omega^2 \frac{2E}{m\omega^2} \sin^2 \omega(t - t_0)} \\ &= \sqrt{2mE(1 - \sin^2 \omega(t - t_0))} \\ &= \sqrt{2mE} \cos \omega(t - t_0) \end{aligned}$$

1-4 Symmetry and Transformation

A. SYMMETRY IN LAGRANGIAN MECHANICS

(1) Symmetry

In physics, a symmetry of a physical system is any transformation that leaves a system physically unchanged, that is, a symmetry would be a transformation that leaves the equations of motion of a system unchanged.

Mathematically, changes the Lagrangian of the system up to a total time derivative, that is:

$$L'(q, \dot{q}, t) = L(q, \dot{q}, t) + \frac{d}{dt}F(q, t)$$

Since

$$\frac{dF}{dt} = \sum_l \frac{\partial F}{\partial q_l} \dot{q}_l + \frac{\partial F}{\partial t}$$

the Lagrange's equation of L' is

$$\begin{aligned} \frac{\partial L'}{\partial q_k} - \frac{d}{dt} \frac{\partial L'}{\partial \dot{q}_k} &= \frac{\partial}{\partial q_k} \left(L + \frac{dF}{dt} \right) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} \left(L + \frac{dF}{dt} \right) \\ &= \underbrace{\frac{\partial L}{\partial q_k} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k}}_{=0} + \frac{\partial}{\partial q_k} \left(\frac{dF}{dt} \right) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} \left(\frac{dF}{dt} \right) \\ &= \frac{\partial}{\partial q_k} \left(\frac{dF}{dt} \right) - \frac{d}{dt} \frac{\partial}{\partial \dot{q}_k} \left(\frac{dF}{dt} \right) \end{aligned}$$

Since

$$\begin{aligned} \frac{\partial}{\partial q_k} \left(\frac{dF}{dt} \right) &= \frac{\partial}{\partial q_k} \left(\sum_l \frac{\partial F}{\partial q_l} \dot{q}_l + \frac{\partial F}{\partial t} \right) = \sum_l \frac{\partial^2 F}{\partial q_k \partial q_l} \dot{q}_l + \frac{\partial^2 F}{\partial q_k \partial t} \\ \frac{\partial}{\partial \dot{q}_k} \left(\frac{dF}{dt} \right) &= \frac{\partial}{\partial \dot{q}_k} \left(\sum_l \frac{\partial F}{\partial q_l} \dot{q}_l + \frac{\partial F}{\partial t} \right) = \sum_l \frac{\partial F}{\partial q_l} \delta_{kl} = \frac{\partial F}{\partial q_k} \end{aligned}$$

we obtain

$$\frac{\partial L'}{\partial q_k} - \frac{d}{dt} \frac{\partial L'}{\partial \dot{q}_k} = \sum_l \frac{\partial^2 F}{\partial q_k \partial q_l} \dot{q}_l + \frac{\partial^2 F}{\partial q_k \partial t} - \frac{d}{dt} \frac{\partial F}{\partial q_k}$$

Since

$$\frac{d}{dt} \frac{\partial F}{\partial q_k} = \frac{\partial}{\partial q_k} \left(\sum_l \frac{\partial F}{\partial q_l} \dot{q}_l + \frac{\partial F}{\partial t} \right) = \sum_l \frac{\partial^2 F}{\partial q_k \partial q_l} \dot{q}_l + \frac{\partial^2 F}{\partial q_k \partial t}$$

we obtain

$$\frac{\partial L'}{\partial q_k} - \frac{d}{dt} \frac{\partial L'}{\partial \dot{q}_k} = 0$$

$\Rightarrow$ The transformation of the Lagrangian, i.e., $L' = L + dF/dt$, which leaves the equation of motion unchanged, is called the gauge transformation.

EXAMPLES:

1. Consider an electron in an electromagnetic field

$$L = \frac{1}{2}mv^2 + e\varphi - \frac{e}{c} \vec{v} \cdot \vec{A}$$

The scalar potential φ and vector potential $\vec{A}$ can be transformed from point to point (local) as

$$\varphi(\vec{r}, t) \rightarrow \varphi - \frac{1}{c} \frac{\partial f(\vec{r}, t)}{\partial t}, \quad \vec{A}(\vec{r}, t) \rightarrow \vec{A} + \nabla f(\vec{r}, t)$$

where $f(\vec{r}, t)$ is an arbitrary scalar function in space-time.

$$\begin{aligned} L' &= \frac{1}{2}mv^2 + e \left(\varphi - \frac{1}{c} \frac{\partial f(\vec{r}, t)}{\partial t} \right) - \frac{e}{c} \vec{v} \cdot (\vec{A} + \nabla f(\vec{r}, t)) \\ &= \underbrace{\frac{1}{2}mv^2 + e\varphi - \frac{e}{c} \vec{v} \cdot \vec{A}}_L - \frac{e}{c} \frac{\partial f(\vec{r}, t)}{\partial t} - \frac{e}{c} \vec{v} \cdot \nabla f(\vec{r}, t) \\ &= L - \frac{e}{c} \left(\frac{\partial f(\vec{r}, t)}{\partial t} + \vec{v} \cdot \nabla f(\vec{r}, t) \right) \end{aligned}$$

Since

$$\frac{d}{dt} f(\vec{r}, t) = \frac{\partial f(\vec{r}, t)}{\partial t} + \frac{\partial f(\vec{r}, t)}{\partial \vec{r}} \frac{\partial \vec{r}}{\partial t} = \frac{\partial f(\vec{r}, t)}{\partial t} + \vec{v} \cdot \nabla f(\vec{r}, t)$$

we obtain

$$L' = L + \frac{d}{dt} \underbrace{\left(-\frac{e}{c} f(\vec{r}, t) \right)}_{=F(\vec{r}, t)} \cdots \cdots \text{gauge transformation}$$

2. Gauge invariance:

$$\vec{B} = \nabla \times \vec{A}' = \nabla \times (\vec{A} + \nabla f) = \nabla \times \vec{A} + \underbrace{\nabla \times \nabla f}_{=0} = \nabla \times \vec{A}$$

$$\begin{aligned}
\vec{E} &= -\nabla\varphi' - \frac{1}{c} \frac{\partial \vec{A}'}{\partial t} \\
&= -\nabla \left(\varphi - \frac{1}{c} \frac{\partial f}{\partial t} \right) - \frac{1}{c} \frac{\partial}{\partial t} (\vec{A} + \nabla f) \\
&= -\nabla\varphi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} + \frac{1}{c} \nabla \frac{\partial f}{\partial t} - \frac{1}{c} \frac{\partial}{\partial t} \nabla f \\
&= -\nabla\varphi - \frac{\partial \vec{A}}{\partial t}
\end{aligned}$$

$\vec{B}$ and $\vec{E}$ are invariant under the gauge transformation.

(2) Noether [nũ'tər] theorem (1918)

Mathematically, a symmetry is any transformation that changes the Lagrangian of the system only by

$$\delta L = L' - L = \frac{dF}{dt}$$

The variation of the Lagrangian is

$$\delta L = \sum_k \left(\frac{\partial L}{\partial q_k} \delta q_k + \frac{\partial L}{\partial \dot{q}_k} \delta \dot{q}_k \right)$$

Since

$$p_k = \frac{\partial L}{\partial \dot{q}_k} \dots \text{conjugate momentum}$$

$$\frac{d}{dt} p_k = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} = \frac{\partial L}{\partial q_k} \dots \text{Lagrange's equation}$$

we obtain

$$\begin{aligned}
\delta L &= \sum_k (\dot{p}_k \delta q_k + p_k \delta \dot{q}_k) = \sum_k \frac{d}{dt} (p_k \delta q_k) = \frac{d}{dt} \left(\sum_k p_k \delta q_k \right) = \frac{dF}{dt} \\
&\Rightarrow \frac{d}{dt} \left(\sum_k p_k \delta q_k - F \right) = 0
\end{aligned}$$

Since the total time derivative equals to zero, we obtain a conservation law and the quantity

$$Q = \sum_k p_k \delta q_k - F$$

is a conserved quantity.

(3) Alternatively, a symmetry is also defined by the condition:

$$\delta S = 0$$

The differential or variation of S is

$$\begin{aligned}\delta S &= \sum_{k=1}^f \left(\int_{t_1}^{t_2} L(q_k + \delta q_k, \dot{q}_k + \delta \dot{q}_k) dt - \int_{t_1}^{t_2} L(q_k, \dot{q}_k) dt \right) \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(L(q_k, \dot{q}_k) + \frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} \delta q_k + \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \delta \dot{q}_k - L(q_k, \dot{q}_k) \right) dt \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{\partial L(q_k, \dot{q}_k)}{\partial q_k} \delta q_k(t) + \frac{\partial L(q_k, \dot{q}_k)}{\partial \dot{q}_k} \delta \dot{q}_k(t) \right) dt\end{aligned}$$

Using Lagrange's equations,

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} = 0$$

we obtain

$$\begin{aligned}\delta S &= \sum_{k=1}^f \int_{t_1}^{t_2} \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} \delta q_k(t) + \frac{\partial L}{\partial \dot{q}_k} \frac{d}{dt} \delta q_k(t) \right) dt \\ &= \sum_{k=1}^f \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \delta q_k(t) \right) dt\end{aligned}$$

As $\delta S = 0$, we have

$$\frac{d}{dt} \sum_{k=1}^f \frac{\partial L}{\partial \dot{q}_k} \delta q_k = 0 \Rightarrow Q = \sum_k \frac{\partial L}{\partial \dot{q}_k} \delta q_k = \sum_k p_k \delta q_k = \text{constant}$$

where Q is a conserved quantity.

OS:

[Noether' s Theorem: A Complete Guide With Examples – Profound Physics](#)

EXAMPLES:

1. As symmetries associated with translations of a system in space, show the conservation of momentum.

ANSWER:

Consider an infinitesimal transformation $\delta q_k = \text{constant}$. The transformation is given by

$$q'_k = q_k + \delta q_k$$

The total time derivative is

$$\frac{d}{dt}q'_k = \frac{d}{dt}(q_k + \delta q_k) = \frac{dq_k}{dt} + \underbrace{\frac{d(\delta q_k)}{dt}}_{=0} = \frac{dq_k}{dt}$$

Thus, the total time derivative of the generalized coordinate does not change in this infinitesimal transformation.

The variation in the Lagrangian is then given as:

$$\delta L = \sum_k \left(\frac{\partial L}{\partial q_k} \delta q_k + \frac{\partial L}{\partial \dot{q}_k} \delta \dot{q}_k \right) = \sum_k \frac{\partial L}{\partial q_k} \delta q_k$$

Suppose that the Lagrangian is independent of the generalized coordinate q_k :

$$\frac{\partial L}{\partial q_k} = 0 \Rightarrow \delta L = 0$$

$\Rightarrow$ Space translation is a symmetry of the system.

Then, according to Noether's theorem, we will find a conserved quantity of the form (in this case, $F = 0$):

$$Q = \sum_k p_k \delta q_k - F = \left(\sum_k p_k \right) \delta q_k$$

Since

$$\frac{dQ}{dt} = 0$$

we obtain

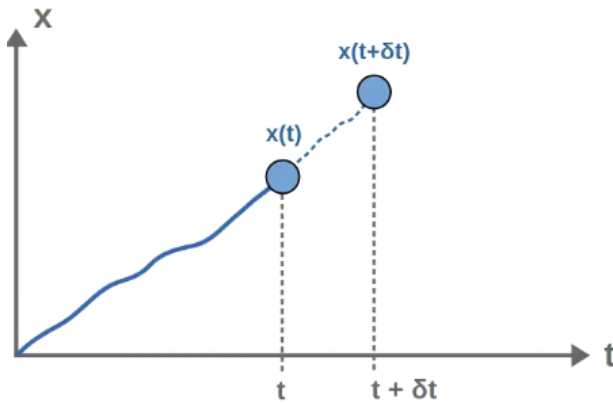
$$\sum_k p_k = \text{constant}$$

Translation symmetry in a given generalized coordinate results in the conservation of generalized momentum.

2. As symmetries associated with translations of a system in time, show the conservation of energy.

ANSWER:

For an infinitesimal change of time δt , we obtain



$$q_k(t + \delta t) \approx q_k(t) + \frac{dq_k(t)}{dt} \delta t = q_k(t) + \dot{q}_k(t) \delta t$$

$$\dot{q}_k(t + \delta t) \approx \dot{q}_k(t) + \frac{d\dot{q}_k(t)}{dt} \delta t = \dot{q}_k(t) + \ddot{q}_k(t) \delta t$$

$$\Rightarrow \delta q_k = \dot{q}_k(t) \delta t$$

$$\Rightarrow \delta \dot{q}_k = \ddot{q}_k(t) \delta t$$

The variation in the Lagrangian:

$$\delta L = \sum_k \left(\frac{\partial L}{\partial q_k} \delta q_k + \frac{\partial L}{\partial \dot{q}_k} \delta \dot{q}_k \right) = \sum_k \left(\frac{\partial L}{\partial q_k} \dot{q}_k \delta t + \frac{\partial L}{\partial \dot{q}_k} \ddot{q}_k \delta t \right)$$

Since

$$\frac{dL}{dt} = \sum_k \left(\frac{\partial L}{\partial q_k} \dot{q}_k + \frac{\partial L}{\partial \dot{q}_k} \ddot{q}_k \right) + \frac{\partial L}{\partial t}$$

we obtain

$$\delta L = \left(\frac{dL}{dt} - \frac{\partial L}{\partial t} \right) \delta t \equiv \frac{dF}{dt} \Rightarrow \frac{\partial L}{\partial t} = 0$$

Thus, the Lagrangian of the system is independent of time explicitly and

$$\delta L = \frac{d(L\delta t)}{dt} \Rightarrow F = L\delta t$$

$\Rightarrow$ Time translation is a symmetry of the system.

The conserved quantity is

$$Q = \sum_k p_k \delta q_k - F = \sum_k p_k \dot{q}_k \delta t - L \delta t = \underbrace{\left(\sum_k p_k \dot{q}_k - L \right)}_{\text{Hamiltonian}} \delta t$$

Time translation symmetry results in the conservation of energy.

B. SYMMETRY IN HAMILTONIAN MECHANICS

(1) Symmetry

In Hamiltonian mechanics, Hamilton's equations are expressed in terms of Poisson bracket with the Hamiltonian as

$$\begin{aligned}\dot{q}_l &= [q_l, H]_{qp} \\ \dot{p}_l &= [p_l, H]_{qp}\end{aligned}$$

A symmetry is a transformation preserves Poisson bracket.

Mathematically, a transformation of canonical coordinates are given as

$$\begin{aligned}Q_k &= Q_k(q_l, p_l) \\ P_k &= P_k(q_l, p_l)\end{aligned}$$

which satisfy the fundamental Poisson bracket:

$$\begin{aligned}[Q_k, P_{k'}]_{qp} &= \delta_{kk'} \\ [Q_k, Q_{k'}]_{qp} &= 0 \\ [P_k, P_{k'}]_{qp} &= 0\end{aligned}$$

The Poisson bracket for two different sets of canonical coordinates:

$$\begin{aligned}[f, g]_{qp} &= \sum_k \left(\frac{\partial f}{\partial q_k} \frac{\partial g}{\partial p_k} - \frac{\partial f}{\partial p_k} \frac{\partial g}{\partial q_k} \right) \\ &= \sum_{kl'l'} \left(\frac{\partial f}{\partial Q_l} \frac{\partial Q_l}{\partial q_k} + \frac{\partial f}{\partial P_l} \frac{\partial P_l}{\partial q_k} \right) \left(\frac{\partial g}{\partial Q_{l'}} \frac{\partial Q_{l'}}{\partial p_k} + \frac{\partial g}{\partial P_{l'}} \frac{\partial P_{l'}}{\partial p_k} \right) \\ &\quad - \left(\frac{\partial f}{\partial Q_l} \frac{\partial Q_l}{\partial p_k} + \frac{\partial f}{\partial P_l} \frac{\partial P_l}{\partial p_k} \right) \left(\frac{\partial g}{\partial Q_{l'}} \frac{\partial Q_{l'}}{\partial q_k} + \frac{\partial g}{\partial P_{l'}} \frac{\partial P_{l'}}{\partial q_k} \right) \\ &= \sum_{ll'} \left(\frac{\partial f}{\partial Q_l} \frac{\partial g}{\partial P_{l'}} [Q_l, P_{l'}]_{qp} - \frac{\partial f}{\partial P_l} \frac{\partial g}{\partial Q_{l'}} [Q_{l'}, P_l]_{qp} \right. \\ &\quad \left. + \frac{\partial f}{\partial Q_l} \frac{\partial g}{\partial Q_{l'}} [Q_l, Q_{l'}]_{qp} + \frac{\partial f}{\partial P_l} \frac{\partial g}{\partial P_{l'}} [P_l, P_{l'}]_{qp} \right) \\ &= \sum_{ll'} \left(\frac{\partial f}{\partial Q_l} \frac{\partial g}{\partial P_{l'}} \delta_{ll'} - \frac{\partial f}{\partial P_l} \frac{\partial g}{\partial Q_{l'}} \delta_{ll'} \right) \\ &= \sum_l \left(\frac{\partial f}{\partial Q_l} \frac{\partial g}{\partial P_l} - \frac{\partial f}{\partial P_l} \frac{\partial g}{\partial Q_l} \right) \\ &= [f, g]_{QP}\end{aligned}$$

The transformation of the canonical coordinates which leaves the Poisson bracket unchanged is called the canonical transformation.

(2) Conservation laws

Mathematically, we can define infinitesimal canonical transformations by changing the coordinates and momenta in a system as:

$$\begin{aligned}\delta q_k &= \frac{\partial Q}{\partial p_k} \delta \gamma \\ \delta p_k &= -\frac{\partial Q}{\partial q_k} \delta \gamma\end{aligned}$$

where $Q = Q(q_k, p_k)$ is called a generator of the transformation.

The variation of the Hamiltonian:

$$\begin{aligned}\delta H &= \sum_k \left(\frac{\partial H}{\partial q_k} \delta q_k + \frac{\partial H}{\partial p_k} \delta p_k \right) \\ &= \sum_k \left(\frac{\partial H}{\partial q_k} \frac{\partial Q}{\partial p_k} \delta \gamma - \frac{\partial H}{\partial p_k} \frac{\partial Q}{\partial q_k} \delta \gamma \right) \\ &= \sum_k \left(\frac{\partial H}{\partial q_k} \frac{\partial Q}{\partial p_k} - \frac{\partial H}{\partial p_k} \frac{\partial Q}{\partial q_k} \right) \delta \gamma \\ &= [H, Q]_{qp} \delta \gamma\end{aligned}$$

Since

$$\begin{aligned}[H, Q]_{qp} &= \sum_k \left(\frac{\partial H}{\partial q_k} \frac{\partial Q}{\partial p_k} - \frac{\partial H}{\partial p_k} \frac{\partial Q}{\partial q_k} \right) \\ &= \sum_k \left(-\dot{p}_k \frac{\partial Q}{\partial p_k} - \dot{q}_k \frac{\partial Q}{\partial q_k} \right) \\ &= -\frac{dQ(q, p)}{dt} \\ \Rightarrow \frac{dQ}{dt} &= [Q, H]_{qp} = -\delta H\end{aligned}$$

If this transformation is to be a symmetry, we should have $\delta H = 0$, i.e.,

$$[H, Q]_{qp} = 0 \text{ and } \frac{dQ}{dt} = 0$$

Thus, the quantity Q is a generator of the transformation and a conserved quantity.

So, in Hamiltonian mechanics, given that we know Q that is conserved, we can easily find the associated symmetry transformation.

EXAMPLES:

1. The Lagrangian of a moving particle has constant angular

momentum about the origin, so

$$L = \frac{1}{2} m r^2 \dot{\theta}^2$$

Define the conjugate momentum

$$\ell = \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta} \dots \text{angular momentum}$$

we obtain the Hamiltonian

$$H = \ell \dot{\theta} - \frac{1}{2} m r^2 \dot{\theta}^2 = \frac{\ell^2}{m r^2} - \frac{\ell^2}{2 m r^2} = \frac{\ell^2}{2 m r^2}$$

The total time derivative of ℓ is

$$\frac{d\ell}{dt} = [\ell, H]_{qp} = 0$$

$\Rightarrow \ell$ is a constant of motion, i.e., a conserved quantity.

(3) Generating function of canonical transformation

$$\text{Let } F = \int_{t_1}^{t_2} (L(q, \dot{q}, t) - L'(Q, \dot{Q}, t)) dt$$

$$\frac{dF}{dt} = L(q, \dot{q}, t) - L'(Q, \dot{Q}, t)$$

The Legendre transformation gives

$$L = \sum_k p_k \dot{q}_k - H(q, p)$$

$$L' = \sum_k P_k \dot{Q}_k - H'(Q, P)$$

$$\begin{aligned} \frac{dF}{dt} &= \left(\sum_k p_k \dot{q}_k - H \right) - \left(\sum_k P_k \dot{Q}_k - H' \right) \\ &= \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H) \end{aligned}$$

The generating function F has to be chosen such that the transformation from the initial variables (q, p) to the final variables (Q, P) is a canonical transformation.

There are four possible type of generating functions:

Type 1: $F_1 = F_1(q, Q, t)$

$$\begin{aligned}\frac{dF_1}{dt} &= \sum_k \left(\frac{\partial F_1}{\partial q_k} \dot{q}_k + \frac{\partial F_1}{\partial Q_k} \dot{Q}_k \right) + \frac{\partial F_1}{\partial t} \\ &= \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H)\end{aligned}$$

The generating function F_1 determines the canonical variables p and P

$$\Rightarrow p_k = \frac{\partial F_1}{\partial q_k}, \quad P_k = -\frac{\partial F_1}{\partial Q_k}, \quad H' = H + \frac{\partial F_1}{\partial t}$$

Type 2: $F_2 = F_2(q, P, t) - \sum_l Q_l P_l$

$$\begin{aligned}\frac{dF_2}{dt} &= \sum_k \left(\frac{\partial F_2}{\partial q_k} \dot{q}_k + \frac{\partial F_2}{\partial P_k} \dot{P}_k - \dot{Q}_k P_k - Q_k \dot{P}_k \right) + \frac{\partial F_2}{\partial t} \\ &= \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H) \\ \Rightarrow p_k &= \frac{\partial F_2}{\partial q_k}, \quad Q_k = \frac{\partial F_2}{\partial P_k}, \quad H' = H + \frac{\partial F_2}{\partial t}\end{aligned}$$

Type 3: $F_3 = F_3(p, Q, t) + \sum_l q_l p_l$

$$\begin{aligned}\frac{dF_3}{dt} &= \sum_k \left(\frac{\partial F_3}{\partial p_k} \dot{p}_k + \frac{\partial F_3}{\partial Q_k} \dot{Q}_k + \dot{q}_k p_k + q_k \dot{p}_k \right) + \frac{\partial F_3}{\partial t} \\ &= \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H) \\ \Rightarrow q_k &= -\frac{\partial F_3}{\partial p_k}, \quad P_k = -\frac{\partial F_3}{\partial Q_k}, \quad H' = H + \frac{\partial F_3}{\partial t}\end{aligned}$$

Type 4: $F_4 = F_4(p, P, t) + \sum_l q_l p_l - \sum_l Q_l P_l$

$$\begin{aligned}\frac{dF_4}{dt} &= \sum_k \left(\frac{\partial F_4}{\partial p_k} \dot{p}_k + \frac{\partial F_4}{\partial P_k} \dot{P}_k + \dot{q}_k p_k + q_k \dot{p}_k - \dot{Q}_k P_k - Q_k \dot{P}_k \right) + \frac{\partial F_4}{\partial t} \\ &= \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H) \\ \Rightarrow q_k &= -\frac{\partial F_4}{\partial p_k}, \quad Q_k = \frac{\partial F_4}{\partial P_k}, \quad H' = H + \frac{\partial F_4}{\partial t}\end{aligned}$$

In summary,

$$H'(Q, P) = H(q, p) + \frac{\partial F}{\partial t}$$

EXAMPLES:

1. We made a canonical transformation that has led to a zero Hamiltonian,

$$H'(Q, P) = H(q, p) + \frac{\partial F}{\partial t} = 0$$

which also leads to

$$\dot{Q}_k = \frac{\partial H'(Q, P)}{\partial P_k} = 0, \quad \dot{P}_k = -\frac{\partial H'(Q, P)}{\partial Q_k} = 0$$

The generating function is

$$\frac{dF}{dt} = \sum_k (p_k \dot{q}_k - P_k \dot{Q}_k) + (H' - H) = \sum_k p_k \dot{q}_k - H = L$$

Thus, F is the action S .

Since

$$\frac{\partial F}{\partial t} = \frac{\partial S}{\partial t}$$

we obtain the Hamilton-Jacobi equation

$$H + \frac{\partial S}{\partial t} = 0$$

2. Gauge transformations as canonical transformation

$$\begin{aligned} L'(q, \dot{q}, t) &= L(q, \dot{q}, t) + \frac{df(q, t)}{dt} \\ \Rightarrow \begin{cases} Q_k = q_k \\ P_k = \frac{\partial L'}{\partial \dot{q}_k} = p_k + \frac{\partial}{\partial \dot{q}_k} \frac{df}{dt} = p_k + \frac{\partial f}{\partial q_k} \end{cases} \\ H'(Q, P, t) &= \sum_k P_k \dot{Q}_k - L' \\ &= \sum_k \left[\left(p_k + \frac{\partial f}{\partial q_k} \right) \dot{q}_k \right] - \left(L + \frac{df}{dt} \right) \\ &= \left(\sum_k p_k \dot{q}_k - L \right) + \sum_k \frac{\partial f}{\partial q_k} \dot{q}_k - \frac{df}{dt} \end{aligned}$$

Since

$$\frac{df}{dt} = \sum_k \frac{\partial f}{\partial q_k} \dot{q}_k + \frac{\partial f}{\partial t}$$

we obtain

$$\begin{aligned}
H'(Q, P, t) &= H(Q, p, t) - \frac{\partial f(Q, t)}{\partial t} \\
\frac{\partial H'}{\partial P_k} &= \sum_l \frac{\partial H}{\partial p_l} \frac{\partial p_l}{\partial P_k} = \sum_l \frac{\partial H}{\partial p_l} \delta_{kl} = \frac{\partial H}{\partial p_k} = \dot{Q}_k \\
\frac{\partial H'}{\partial Q_k} &= \frac{\partial H}{\partial Q_k} + \sum_l \frac{\partial H}{\partial p_l} \frac{\partial p_l}{\partial Q_k} - \frac{\partial^2 f}{\partial Q_k \partial t}
\end{aligned}$$

Since

$$\begin{aligned}
p_k &= P_k - \frac{\partial f}{\partial q_k} \\
\frac{\partial p_l}{\partial Q_k} &= -\frac{\partial^2 f}{\partial Q_k \partial Q_l}
\end{aligned}$$

we obtain

$$\begin{aligned}
\frac{\partial H'}{\partial Q_k} &= -\dot{p}_k - \sum_l \dot{Q}_l \frac{\partial^2 f}{\partial Q_k \partial Q_l} - \frac{\partial^2 f}{\partial Q_k \partial t} \\
&= -\dot{p}_k - \frac{d}{dt} \frac{\partial f}{\partial Q_k} \\
&= -\frac{d}{dt} \left[p_k + \frac{\partial f}{\partial Q_k} \right] \\
&= -P_k
\end{aligned}$$

—Chapter 2—

Matter Wave

2-1 Matter Wave

A. WAVE PACKET

- (1) In 1926, Erwin Schrödinger wrote a partial differential equation that governs the wavefunction based on a postulate of de Broglie that all matter has an associated matter wave.

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi(x, t)$$

- (2) The description of the wavefunction (matter wave)

Consider a matter wave as a traveling wave with wave vector k ,

$$\Psi(x, t) = Ae^{i(kx - \omega t)}$$

Thus, we obtain the velocity

$$v = \frac{\omega}{k} = 2\pi\nu \cdot \frac{1}{2\pi/\lambda} = \underbrace{h\nu}_{=E} \cdot \underbrace{\frac{1}{h/\lambda}}_{=p} = \frac{E}{p} = \frac{\gamma mc^2}{\gamma mv} = \frac{c^2}{v} \geq c$$

$\Rightarrow$ Violate the Einstein's special relativity.

- (3) Consider a matter wave as the superposition of large number of traveling waves with different wave vectors k , called a wave packet.



$$A_1 e^{i(k_1 x - \omega_1 t)}$$



$$A_2 e^{i(k_2 x - \omega_2 t)}$$



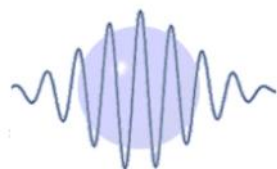
$$A_3 e^{i(k_3 x - \omega_3 t)}$$



$$A_4 e^{i(k_4 x - \omega_4 t)}$$

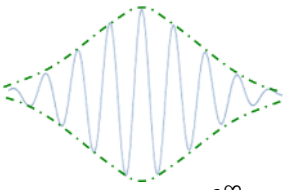


$$A_5 e^{i(k_5 x - \omega_5 t)}$$



$$\Psi(x, t) = \sum_n A_n e^{i(k_n x - \omega_n t)}$$

Suppose that the wave packet is a Gaussian function $A(k) = e^{-\alpha(k-k_0)^2}$



$$\Psi(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega_k t)} dk = \int_{-\infty}^{\infty} e^{-\alpha(k-k_0)^2} e^{i(kx - \omega_k t)} dk$$

The Taylor expansion of ω_k around ω_0

$$\begin{aligned} \omega_k &= \omega_0 + \left. \frac{d\omega_k}{dk} \right|_{k_0} (k - k_0) + \frac{1}{2} \left. \frac{d^2\omega_k}{dk^2} \right|_{k_0} (k - k_0)^2 \\ &= \omega_0 + v_g(k - k_0) + \beta(k - k_0)^2 \end{aligned}$$

$$\text{where } v_g = \left. \frac{d\omega_k}{dk} \right|_{k_0} \text{ and } \beta = \frac{1}{2} \left. \frac{d^2\omega_k}{dk^2} \right|_{k_0} = \frac{1}{2} \frac{dv_g}{dk}$$

Let $q = k - k_0$

$$\begin{aligned} \omega_k &= \omega_0 + v_g q + \beta q^2 \\ \Psi(x, t) &= \int_{-\infty}^{\infty} e^{-\alpha q^2} e^{i(k_0+q)x} e^{-i(\omega_0 + v_g q + \beta q^2)t} dq \\ &= e^{i(k_0 x - \omega_0 t)} \int_{-\infty}^{\infty} e^{-\alpha q^2} e^{iq(x - v_g t)} e^{-i\beta q^2 t} dq \\ &= e^{i(k_0 x - \omega_0 t)} \int_{-\infty}^{\infty} e^{-(\alpha + i\beta t)q^2 + i(x - v_g t)q} dq \\ &= e^{i(k_0 x - \omega_0 t)} \sqrt{\frac{\pi}{\alpha + i\beta t}} e^{-(x - v_g t)^2 / 4(\alpha + i\beta t)} \end{aligned}$$

The Taylor expansion up to the first order only, i.e., $\beta = 0$,

$$\begin{aligned} \omega_k &= \omega_0 + v_g q \\ \Psi(x, t) &= e^{i(k_0 x - \omega_0 t)} \sqrt{\frac{\pi}{\alpha}} e^{-(x - v_g t)^2 / 4\alpha} \\ &= e^{-i\omega_0 t} e^{ik_0(x - v_g t) + ik_0 v_g t} \sqrt{\frac{\pi}{\alpha}} e^{-(x - v_g t)^2 / 4\alpha} \\ &= e^{i(k_0 v_g t - \omega_0 t)} e^{ik_0(x - v_g t)} \sqrt{\frac{\pi}{\alpha}} e^{-(x - v_g t)^2 / 4\alpha} \\ &= e^{i(k_0 v_g t - \omega_0 t)} \Psi(x - v_g t, 0) \end{aligned}$$

Since $|\Psi(x, t)|^2 = |\Psi(x - v_g t, 0)|^2$

$$v_g = \left. \frac{d\omega_k}{dk} \right|_{k_0} \dots \text{group velocity}$$

For the case of an electron, the kinetic energy is

$$E = \frac{p^2}{2m} = \frac{1}{2m} \left(\frac{h}{\lambda} \right)^2 = \frac{1}{2m} \left(\underbrace{\frac{h}{2\pi}}_{=\hbar} \cdot \underbrace{\frac{2\pi}{\lambda}}_{=k} \right)^2 = \frac{\hbar^2 k^2}{2m}$$

Since

$$E = h\nu = \underbrace{\frac{h}{2\pi}}_{=\hbar} \cdot \underbrace{2\pi\nu}_{=\omega} = \hbar\omega$$

thus, we obtain

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} \Rightarrow \omega = \frac{\hbar k^2}{2m}$$

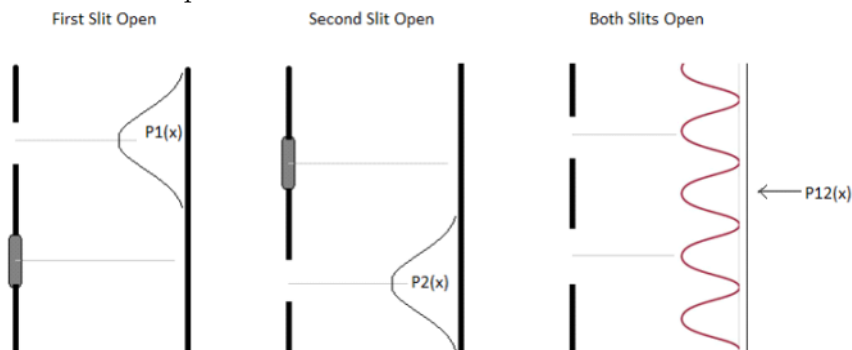
The group velocity of the wave packet is

$$v_g = \frac{d\omega_k}{dk} = \frac{\hbar k}{m} = \frac{p}{m} < c$$

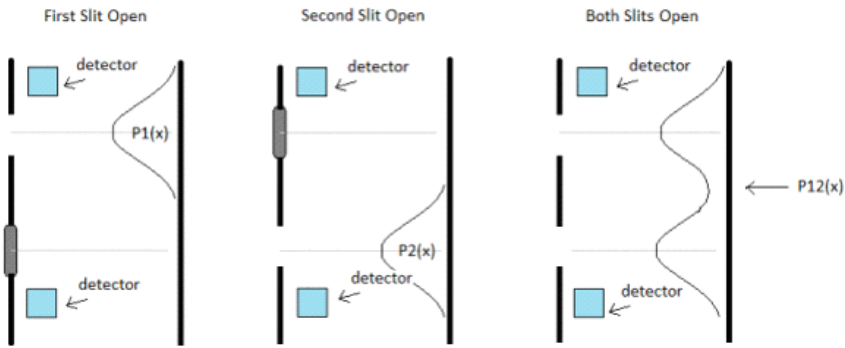
B. CHARACTERISTICS OF MATTER WAVE

- (1) The characteristics of de Broglie's matter wave
 - i) The matter wave must exhibit both wave and particle behaviors at the same time.
 - ii) The matter wave does not change with time.
- i) Thought-experiment of the simultaneous observation of wave and particle behaviors of electrons

Double-slit experiments of electrons



As soon as it gets detected at the slits, electrons start giving the expected particle pattern on the screen instead of the wave pattern.



Conclusion:

One cannot observe the wave and particle behaviors at the same time.

ii) The matter wave does not change with time

Since

$$\Psi(x, t) = e^{i(k_0 x - \omega_0 t)} \sqrt{\frac{\pi}{\alpha + i\beta t}} e^{-(x - v_g t)^2 / 4(\alpha + i\beta t)}$$

we obtain the width of the wave packet, or the size of a particle, as

$$|\Psi(x, t)|^2 = \frac{\pi}{\sqrt{\alpha^2 + \beta^2 t^2}} e^{-(x - v_g t)^2 / 4 \left(\frac{\alpha^2 + \beta^2 t^2}{2\alpha} \right)}$$

$$\Delta x(t) = 2 \left(\frac{\alpha^2 + \beta^2 t^2}{2\alpha} \right)^{1/2} = \sqrt{2\alpha} \left(1 + \frac{\beta^2 t^2}{\alpha^2} \right)^{1/2}$$

$$\Rightarrow \Delta x(0) = \sqrt{2\alpha}$$

For the case of an electron, we have

$$\beta = \frac{1}{2} \frac{d^2 \omega_k}{dk^2} = \frac{1}{2} \frac{dv_g}{dk} = \frac{1}{2} \frac{d}{dk} \left(\frac{\hbar k}{m} \right) = \frac{\hbar}{2m}$$

Thus, the width of the wave packet is

$$\Delta x(t) = \sqrt{2\alpha} \left(1 + \frac{\hbar^2}{4\alpha^2 m^2} t^2 \right)^{1/2}$$

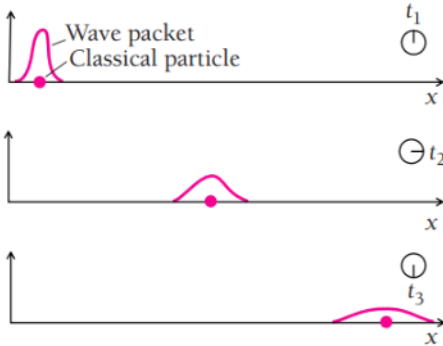
$$\text{Let } \frac{1}{\tau^2} = \frac{\hbar^2}{4\alpha^2 m^2} \Rightarrow \sqrt{2\alpha} = \sqrt{\frac{\hbar \tau}{m}}$$

We obtain

$$\Delta x(t) = \sqrt{\frac{\hbar \tau}{m}} \left(1 + \frac{t^2}{\tau^2} \right)^{1/2}$$

For $t \gg \tau$

$$\Delta x(t) \approx \sqrt{\frac{\hbar \tau}{m}} = \sqrt{\frac{\hbar}{m \tau}} t$$



$\Rightarrow \Delta x$ increases with time, i.e., the wave packet **spreads with time**.

$\Rightarrow$ For an electron, Δx of a wave packet does not mean the spatial extent of the electron itself.

(2) Conclusion

The matter wave cannot be a physically real wave associated with any moving particle.

EXAMPLES:

1. An electromagnetic wave consists of spatially localized, discrete wave-packets. Find the group velocity and the width of the wave packet.

ANSWER:

- (a) Since the speed of light is $c = \text{constant}$, i.e.,

$$v(k) = c$$

The linear dispersion relation is

$$\omega_k = v(k) \cdot k = ck$$

The group velocity is

$$v_g = \frac{d\omega_k}{dk} = c$$

- (b) Since

$$\beta = \frac{1}{2} \frac{d^2 \omega_k}{dk^2} \bigg|_{k_0} = \frac{1}{2} \frac{dv_g}{dk} = 0$$

The width of the wave packet is

$$\Delta x = \sqrt{2\alpha} = \text{constant}$$

$\Rightarrow$ The different frequency components of the wave packet move at the same speed. The wave packet retains the shape, i.e., do not disperse.

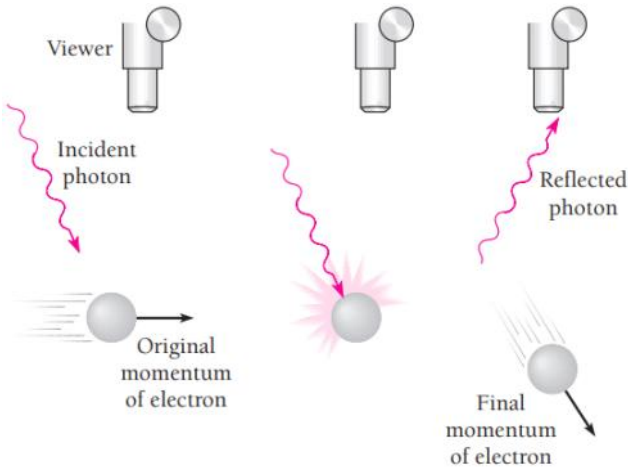
2-2 The Probability Interpretation

A. HEISENBERG UNCERTAINTY PRINCIPLE

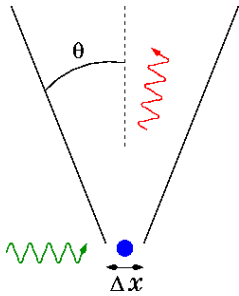
(1) Observer effect:

When you try to observe something, you will always change something about it, even if you do not want to. The observer effect only happens when you observe something in order to measure it.

Heisenberg gamma-ray microscope:



According to Rayleigh criterion (spatial resolution):



Δx indicates a region that a single particle is confined. According to Rayleigh criterion (spatial resolution),

$$\Delta x \approx \frac{f\lambda}{2R}$$

Since $\frac{R}{f} = \tan \theta \approx \theta$

$$\Delta x \approx \frac{\lambda}{2\theta}$$

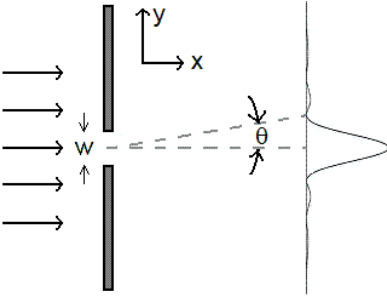
According to Compton scattering, the momentum change of the electron in x direction

$$\Delta p \approx p \sin \theta \approx \frac{h}{\lambda} \theta$$

$$\Delta x \Delta p \approx \frac{\lambda}{2\theta} \frac{h}{\lambda} \theta = \frac{h}{2}$$

$\Rightarrow$ If we increase the precision in measuring one quantity, we are forced to lose precision in measuring the other (Werner K. Heisenberg, 1925).

- (2) No observer effect:
Single slit diffraction



Δy indicates a region that a single particle is confined, i.e.,

$$\Delta y \approx w$$

$$\Delta p_y = p \sin \theta$$

Since $w \sin \theta = \lambda$

$$\Delta p_y = \frac{p\lambda}{w}$$

$$\Delta y \Delta p_y \approx w \times \frac{p\lambda}{w} = p\lambda = h$$

$\Rightarrow$ It is impossible to measure accurately both the position and the momentum of a particle at the same time, i.e., a particle cannot have arbitrarily precise position and momentum simultaneously (Werner K. Heisenberg, 1925)

B. BORN'S STATISTICAL DESCRIPTION

- (1) According to the dispersion of an electron:

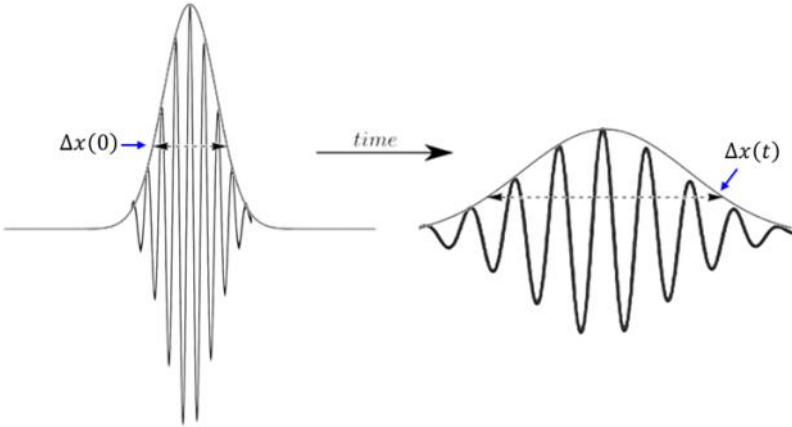
Assuming at $t = 0$, the size of the wave packet is equal to the electron's Compton wavelength.

$$\Delta x(0) = \frac{\hbar}{mc}$$

The time constant of the spreading of the wave packet

$$\tau = \frac{m}{\hbar} \left(\frac{\hbar}{mc} \right)^2 = \frac{\hbar}{mc^2} \sim 10^{-21} \text{ s}$$

$$\Delta x(t) = \Delta x(0) \frac{t}{\tau} = \frac{\hbar}{mc} \frac{mc^2}{\hbar} t = ct$$

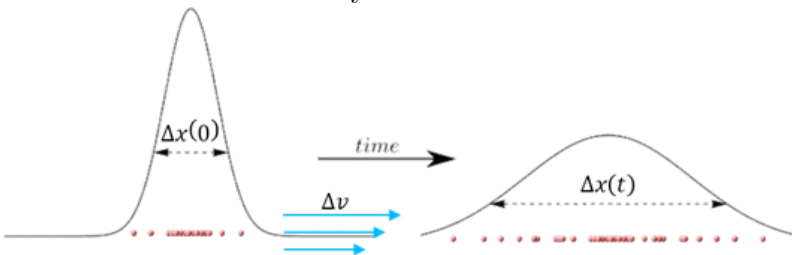


$\Rightarrow \Delta x(t)$ indicates the size of an electron at time t .

(2) According to the Heisenberg uncertainty principle:

$$\Delta p \sim \frac{\hbar}{\Delta x(0)} \Rightarrow \Delta v = \frac{\hbar}{m \Delta x(0)}$$

Δv indicates the uncertainty of the velocities of electrons.

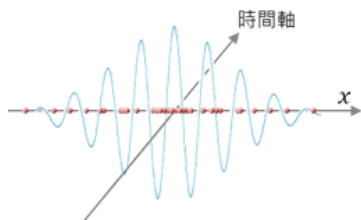


The *velocity fluctuation* (Δv) expands the extent of positions by a proportionate amount of

$$\Delta x(t) = \Delta v \cdot t = \frac{\hbar}{m \Delta x(0)} t = \frac{\hbar}{m \frac{\hbar}{mc}} t = ct$$

$\Rightarrow \Delta x(t)$ indicates the extent of possible positions of electrons at time t .

- (3) Since the size of a matter wave is equal to the extent of possible positions, i.e.,



such coincidence gives rise to two questions:

1. How to measure the possible positions of an electron?
2. What is the relation between the matter wave and the extent of possible positions?

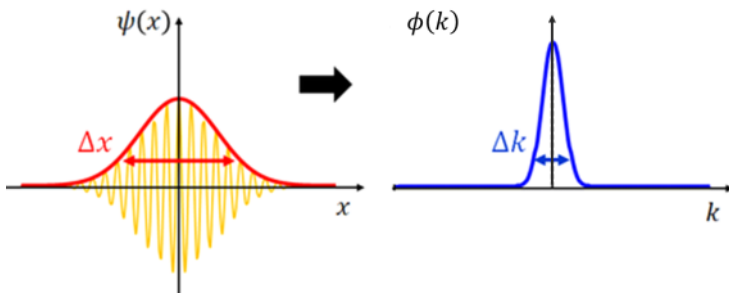
BORN'S IDEAS:

1. Image many virtual copies of a system, comprised of one electron, at once called ensemble. The electrons in each virtual system are at different positions. The extent of possible positions of an electron is the measurement of all copies of systems at the same time.
 2. The matter wave of an electron is a statistical function describing the ensemble of possible positions, called wavefunction $\Psi(x, t)$. Thus, Δx is obtained by the standard deviation of the wavefunction.
- (4) Since a statistical function associated the standard deviation is called the probability distribution, Born (1926) proposed that the square of the wavefunction, $|\Psi(x, t)|^2$ is the **probability distribution** of finding a particle at a particular place and time.

C. PROBABILITY INTERPRETATION

- (1) The uncertainty principle (1927):

COMMENT:



Suppose the distribution of the particle's position is a Gaussian distribution

$$\psi(x) = \left(\frac{1}{\pi a}\right)^{1/4} e^{-x^2/2a}$$

Fourier transform of $\psi(x)$

$$\begin{aligned}\phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\pi a}\right)^{1/4} e^{-x^2/2a} e^{-ikx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\pi a}\right)^{1/4} e^{-k^2 a/2} \int_{-\infty}^{\infty} e^{-(x+ika)^2/2a} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\pi a}\right)^{1/4} e^{-k^2 a/2} \int_{-\infty+ika}^{\infty+ika} e^{-x^2/2a} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{\pi a}\right)^{1/4} e^{-k^2 a/2} \sqrt{2\pi a} \\ &= \left(\frac{a}{\pi}\right)^{1/4} e^{-k^2 a/2}\end{aligned}$$

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$$

$$(\Delta k)^2 = \langle k^2 \rangle - \langle k \rangle^2$$

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi(x) x \psi(x) dx = \sqrt{\frac{1}{\pi a}} \int_{-\infty}^{\infty} x e^{-x^2/a} dx = 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \psi(x) x^2 \psi(x) dx = \sqrt{\frac{1}{\pi a}} \int_{-\infty}^{\infty} x^2 e^{-x^2/a} dx = \frac{a}{2}$$

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a}{2}$$

$$\langle k \rangle = \int_{-\infty}^{\infty} \phi(k) k \phi(k) dk = \sqrt{\frac{a}{\pi}} \int_{-\infty}^{\infty} k e^{-k^2 a/2} dk = 0$$

$$\begin{aligned}\langle k^2 \rangle &= \int_{-\infty}^{\infty} \phi(k) k^2 \phi(k) dk = \sqrt{\frac{a}{\pi}} \int_{-\infty}^{\infty} k^2 e^{-k^2 a} dk = \frac{1}{2a} \\ p &= \hbar k \\ (\Delta p)^2 &= \hbar^2 \langle k^2 \rangle - \hbar^2 \langle k \rangle^2 = \frac{\hbar^2}{2a} \\ (\Delta x)^2 (\Delta p)^2 &= \frac{a}{2} \times \frac{\hbar^2}{2a} = \frac{\hbar^2}{4} \\ \Delta x \Delta p &= \frac{\hbar}{2}\end{aligned}$$

■

⇒ Heisenberg Uncertainty principle is

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

and states a fundamental property of quantum systems, and is not a statement about the observational success of current technology.

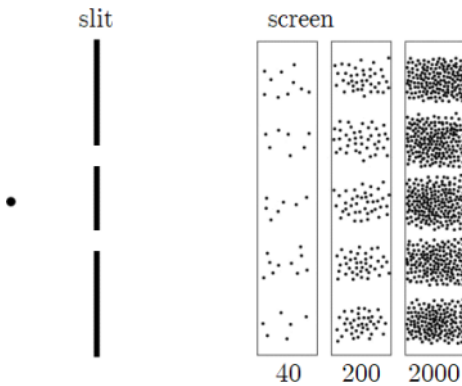
OS:

In the classical world, uncertainty is due to ignorance: the more you know, the better your predictions. In the quantum world, the uncertainty is inherent and can't be eliminated by gaining more knowledge.

(2) The wave-particle duality

Neils Bohr (1927) proposed that electrons can possess both wave-like and particle-like characters, yet cannot exhibit both wave-like and particle-like behaviors at the same time. In other word, experiments on physical systems can exhibit either particle or wave behavior but not both simultaneously which are called **complementary principle**.

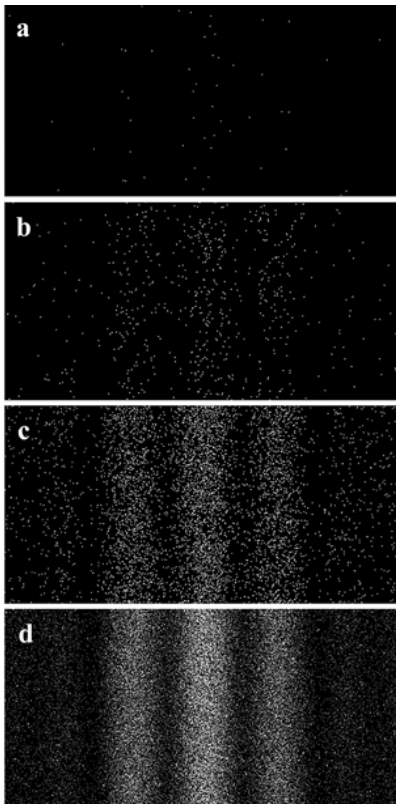
The double-slit experiment of electrons:



Send one electron at a time and it gets detected at one point-like flash on the screen. **Each electron goes through both slits at once (wave-like behavior).** The interference pattern, observed after the large number of electrons are arrived, indicates the probability where electrons can be found.

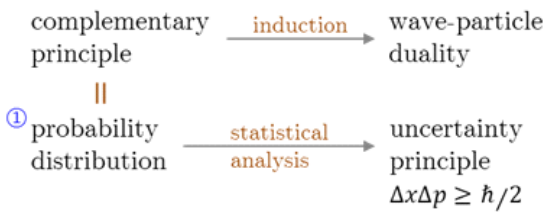
COMMENTS:

1. Claus Jönsson, Zeitschrift für Physik (1961): observed an interference pattern with electron beams.
2. Pier Giorgio Merli, Gian Franco Missiroli, and Giulio Pozzi, American Journal of Physics (1971): obtained an interference pattern with the continuous arrival of the electrons, one at a time.
3. Ken Harada, et. al., Scientific Reports (2018): obtained an interference pattern using field emission electron microscope providing coherent electron waves and a direct detection camera system enabling single-electron detections at a sub-second exposure time.



(a) 50 electrons (b) 500 electrons (c) 5,000 electrons (d) 50,000 electrons

⇒ Summary:



2-3 From Classical to Quantum Theory

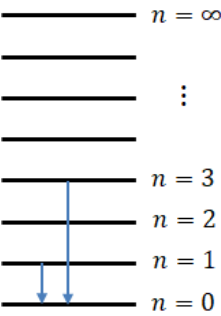
A. HEISENBERG MATRIX EQUATION

- (1) Heisenberg: all observable quantities are always related to a pair of Bohr orbits, but never just a single one.

A dipole transition between two orbitals n and m is determined by the frequency ω_{nm} and by the intensity x_{nm} .

$$\langle x \rangle = \sum_{nm} x_{nm} e^{-i\omega_{nm}t}, \text{ where } \omega_{nm} = \frac{E_n - E_m}{\hbar}$$

The energy levels of a simple harmonic oscillator is



The transition written in a table

$n \setminus m$	1	2	3	4	
1	0	1	0	0	$\xrightarrow{\text{Matrix form}} \mathbf{X} = \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & 0 & \dots \\ 1 & 0 & \sqrt{2} & 0 & \dots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \dots \\ 0 & 0 & \sqrt{3} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$
2	1	0	$\sqrt{2}$	0	
3	0	$\sqrt{2}$	0	$\sqrt{3}$	
4	0	0	$\sqrt{3}$	0	

- (2) Heisenberg conjectured that to each Newtonian observable, such as position q or momentum p , there corresponds an infinite matrix, and that the elements of these matrices would actually yield measurable quantities.

$$\mathbf{P} = i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & -1 & 0 & 0 & \dots \\ 1 & 0 & -\sqrt{2} & 0 & \dots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & \dots \\ 0 & 0 & \sqrt{3} & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Heisenberg found that for given two matrices $\mathbf{X}$ and $\mathbf{P}$ describing two

observables,

$$\mathbf{X} \cdot \mathbf{P} - \mathbf{P} \cdot \mathbf{X}$$

$$\begin{aligned} &= \frac{i\hbar}{2} \begin{pmatrix} 1 & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & 1 & 0 & -\sqrt{6} & \cdots \\ \sqrt{2} & 0 & 1 & 0 & \cdots \\ 0 & \sqrt{6} & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} - \frac{i\hbar}{2} \begin{pmatrix} -1 & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & -1 & 0 & -\sqrt{6} & \cdots \\ \sqrt{2} & 0 & -1 & 0 & \cdots \\ 0 & \sqrt{6} & 0 & -1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= i\hbar \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= i\hbar \mathbf{1} \end{aligned}$$

(3) Dirac canonical quantization (1925)

The Poisson bracket of two variables becomes the commutator of two operators, i.e.,

$$[f, H]_{q,p} \rightarrow \frac{1}{i\hbar} [\hat{f}, \hat{H}]$$

The canonical commutation relation becomes

$$\begin{aligned} [x, p]_{q,p} = 1 &\rightarrow \frac{1}{i\hbar} [\hat{x}, \hat{p}] = 1 \\ \Rightarrow [\hat{x}, \hat{p}] &= i\hbar \\ \Rightarrow \begin{cases} \mathbf{X} \rightarrow \hat{x} \\ \mathbf{P} \rightarrow \hat{p} \end{cases} \end{aligned}$$

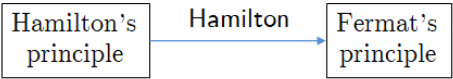
(4) Consider an observable $\mathcal{O}$ and its total time derivative is

$$\frac{d\hat{\mathcal{O}}}{dt} = \frac{1}{i\hbar} [\hat{\mathcal{O}}, \hat{H}] \cdots \cdots \text{Heisenberg's equation of motion}$$

B. SCHRÖDINGER [shroh-ding-er] **WAVE EQUATION**

(1) The analogy between the Hamiltonian formulation of classical mechanics and optics.

Classical Mechanics Geometrical optics

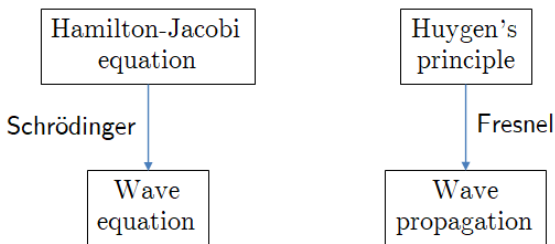


(2) Inherent in the Hamiltonian formulation of classical mechanics, there is

a picture of wave propagation.

Hamiltonian mechanics

Physical optics



(3) For a particle of mass m and momentum p

$$H(x, p, t) = \frac{p^2}{2m} + V(x, t)$$

The Hamilton-Jacobi equation

$$\frac{\partial S}{\partial t} + \frac{1}{2m} \left(\frac{\partial S}{\partial x} \right)^2 + V(x, t) = 0 \dots (a)$$

OS:

The motion of an action S can be thought of as a wave moving through space. S is analogous to a wavefront for electromagnetic waves in optics.

Let $\Psi(x, t) = \Psi_0 e^{iS/\hbar}$

$$S = -i\hbar (\ln \Psi - \ln \Psi_0)$$

Since

$$\frac{\partial S}{\partial t} = -i\hbar \frac{1}{\Psi} \frac{\partial \Psi}{\partial t} \Rightarrow \frac{\partial S}{\partial t} \Psi = -i\hbar \frac{\partial}{\partial t} \Psi$$

$$\frac{\partial S}{\partial x} = -i\hbar \frac{1}{\Psi} \frac{\partial \Psi}{\partial x} \Rightarrow \frac{\partial S}{\partial x} \Psi = -i\hbar \frac{\partial}{\partial x} \Psi$$

Substituting into (a), we obtain

$$\begin{aligned}
 -i\hbar \frac{\partial}{\partial t} \Psi - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi + V(x, t) \Psi &= 0 \\
 \Rightarrow i\hbar \frac{\partial}{\partial t} \Psi &= \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right] \Psi
 \end{aligned}$$

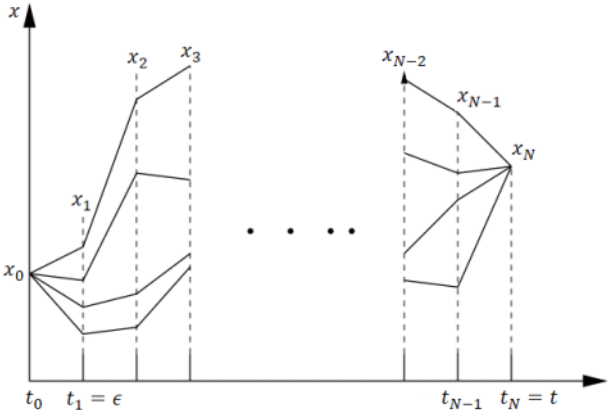
C. FEYNMAN PATH INTEGRALS

(1) Propagator $K(x, t; x_0, t_0)$: a function of two space-time points, a "final"

position and time (x, t) , and an "initial" position and time (x_0, t_0)

$$\Psi(x, t) = \int K(x, t; x_0, t_0) \Psi(x_0, t_0) dx$$

- (2) The propagation from (x_0, t_0) to (x, t) can be viewed as the sum of the paths that go through the intermediate states on each time slice.



$$\begin{aligned} K(x, t; x_0, t_0) &= \int dx_1 \cdots dx_{N-1} K(x, x; x_{N-1}, t_{N-1}) \cdots K(x_1, t_1; x_0, t_0) \\ &= \int D[x(t)] e^{iS[x(t)]/\hbar} \end{aligned}$$

$$S[x(t)] = \int_{t_1}^{t_2} L(x, \dot{x}, t) dt$$

Feynman's conjecture: The propagator is the sum of all paths connecting x_0 and x . Each path is assigned a weight $e^{iS[x(t)]/\hbar}$.

—Chapter 3—

Schrödinger Wave Mechanics

3-1 The Schrödinger Equation

A. STATISTICAL INTERPRETATION OF WAVEFUNCTION

- (1) Born's statistical interpretation of the wavefunction of the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi(x, t) + V(x, t) \Psi(x, t)$$

$|\Psi(x, t)|^2$ is the *probability density* of finding the particle at position x and time t .

- (2) The total probability of finding a particle at time t is

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1$$

Mathematically, this is also called the normalization.

Consequences of normalization:

1. Conservation of global probability

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 0$$

Using Leibniz integral rule

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} |\Psi(x, t)|^2 dx$$

Since

$$\frac{\partial}{\partial t} |\Psi(x, t)|^2 = \frac{\partial}{\partial t} \Psi^* \Psi = \Psi^* \frac{\partial \Psi}{\partial t} + \frac{\partial \Psi^*}{\partial t} \Psi$$

and

$$\begin{aligned} \frac{\partial}{\partial t} \Psi &= \frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi \right) \\ \frac{\partial}{\partial t} \Psi^* &= \frac{-1}{i\hbar} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + V\Psi^* \right) \end{aligned}$$

we obtain

$$\begin{aligned} \frac{\partial}{\partial t} |\Psi(x, t)|^2 &= -\frac{\hbar}{i2m} \left(\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right) \\ &= \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \end{aligned}$$

Thus, $\Psi(x, t)$ and $\partial \Psi(x, t) / \partial x$ must be *continuous*, *finite*, and *single valued*.

Since

$$\int_{-\infty}^{\infty} \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) dx = \frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \Big|_{-\infty}^{\infty} = 0$$

thus, we obtain

$$\Psi(x, t) \rightarrow 0 \text{ at } x \rightarrow \pm\infty.$$

2. Conservation of local probability

From the equation

$$\frac{\partial}{\partial t} |\Psi(x, t)|^2 = \frac{i\hbar}{2m} \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right)$$

$$\text{Let } \rho = |\Psi(x, t)|^2 \text{ and } j = -\frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right)$$

We obtain

$$\frac{\partial}{\partial x} j + \frac{\partial}{\partial t} \rho = 0 \dots \text{continuity equation}$$

(3) From the idea of the probability, we can find the probabilistic expected values (expectation value),

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx \\ \langle p \rangle &= m \frac{d}{dt} \langle x \rangle \\ &= m \int_{-\infty}^{\infty} x \frac{\partial}{\partial t} |\Psi(x, t)|^2 dx \\ &= \frac{i\hbar}{2} \int x \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx \end{aligned}$$

Using integration by parts

$$\langle p \rangle = -\frac{i\hbar}{2} \int \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx \langle x \rangle$$

Using integration by parts of the second term

$$\langle p \rangle = -i\hbar \int \Psi^* \frac{\partial}{\partial x} \Psi dx = \int \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx$$

(4) Define differential operators

$$\hat{x} = x, \quad \hat{p} = -i\hbar \frac{\partial}{\partial x}$$

The expectation value can be written as

$$\langle \hat{x} \rangle = \int \Psi^* \hat{x} \Psi dx$$

$$\langle \hat{p} \rangle = \int \Psi^* \hat{p} \Psi dx$$

- (5) The uncertainty, in the variance formulation, by the Cauchy-Schwarz inequality states that

$$(\Delta x)^2 (\Delta p)^2 \geq (\Delta x \Delta p)^2$$

where

$$\Delta \hat{x} = \hat{x} - \langle \hat{x} \rangle$$

$$\Delta \hat{p} = \hat{p} - \langle \hat{p} \rangle$$

$$(\Delta x)^2 = \langle (\Delta \hat{x})^2 \rangle = \langle (\hat{x} - \langle \hat{x} \rangle)^2 \rangle = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

$$(\Delta p)^2 = \langle (\Delta \hat{p})^2 \rangle = \langle (\hat{p} - \langle \hat{p} \rangle)^2 \rangle = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

While $\langle \Delta \hat{x} \Delta \hat{p} \rangle$ can be decomposed into a real and imaginary part,

$$\langle \Delta \hat{x} \Delta \hat{p} \rangle = \underbrace{\frac{1}{2} \langle \Delta \hat{x} \Delta \hat{p} + \Delta \hat{p} \Delta \hat{x} \rangle}_{\text{real}} + \underbrace{\frac{1}{2i} \langle \Delta \hat{x} \Delta \hat{p} - \Delta \hat{p} \Delta \hat{x} \rangle}_{\text{imaginary}}$$

thus, we obtain

$$\begin{aligned} (\Delta x \Delta p)^2 &= \langle \Delta \hat{x} \Delta \hat{p} \rangle^2 \\ &= \frac{1}{4} \langle \Delta \hat{x} \Delta \hat{p} + \Delta \hat{p} \Delta \hat{x} \rangle^2 + \left| \frac{1}{2i} \langle [\Delta \hat{x}, \Delta \hat{p}] \rangle \right|^2 \\ &\geq \left| \frac{1}{2i} \langle [\Delta \hat{x}, \Delta \hat{p}] \rangle \right|^2 \end{aligned}$$

So that,

$$(\Delta x)^2 (\Delta p)^2 \geq (\Delta x \Delta p)^2 \geq \left| \frac{1}{2i} \langle [\Delta \hat{x}, \Delta \hat{p}] \rangle \right|^2 = \left| \frac{1}{2i} \langle [\hat{x}, \hat{p}] \rangle \right|^2$$

The commutation relation of $\hat{x}$ and $\hat{p}$ is

$$\begin{aligned} [\hat{x}, \hat{p}] \Psi &= (\hat{x} \hat{p} - \hat{p} \hat{x}) \Psi \\ &= x \left(-i\hbar \frac{\partial}{\partial x} \Psi \right) - \left(-i\hbar \frac{\partial}{\partial x} \right) x \Psi \\ &= -i\hbar x \frac{\partial}{\partial x} \Psi + i\hbar x \frac{\partial}{\partial x} \Psi + i\hbar \Psi \\ &= i\hbar \Psi \\ \Rightarrow [\hat{x}, \hat{p}] &= i\hbar \end{aligned}$$

Thus, we obtain

$$(\Delta x)^2 (\Delta p)^2 \geq \left| \frac{1}{2i} \langle [\hat{x}, \hat{p}] \rangle \right|^2 = \left| \frac{1}{2i} i\hbar \right|^2 = \frac{\hbar^2}{4}$$

$$\Rightarrow \Delta x \Delta p \geq \frac{\hbar}{2}$$

The uncertainty principle between two observables is related to their commutator.

B. WAVEFUNCTION OF SCHRÖDINGER EQUATION

- (1) Consider the time-independent potential, $V(x, t) = V(x)$. The Schrödinger equation is,

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi(x, t) + V(x) \Psi(x, t)$$

Using separation of variables:

$$\Psi(x, t) = \psi(x) f(t)$$

We obtain

$$i\hbar \frac{\partial}{\partial t} \psi(x) f(t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) f(t) + V(x) \psi(x) f(t)$$

Divided by $\Psi(x, t)$

$$i\hbar \frac{1}{f(t)} \frac{d}{dt} f(t) = -\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)$$

Since the L.H.S. term depends only on t , and the R.H.S. term depends only on x , it follows that both sides must be a constant. Take the separation constant λ , we obtain two independent equations,

$$\begin{cases} i\hbar \frac{1}{f(t)} \frac{d}{dt} f(t) = \lambda \dots\dots (a) \\ -\frac{\hbar^2}{2m} \frac{1}{\psi(x)} \frac{\partial^2}{\partial x^2} \psi(x) + V(x) = \lambda \dots\dots (b) \end{cases}$$

- (2) Solution of the equation (a)

$$\frac{df(t)}{f(t)} = -\frac{i\lambda}{\hbar} dt$$

Integration of this equation gives

$$\ln f(t) = -\frac{i\lambda t}{\hbar}$$

$$\Rightarrow f(t) = e^{-i\lambda t/\hbar}$$

- (3) Solution of the equation (b)

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)\psi(x) = \lambda\psi(x)$$

Take expectation on both sides

$$\begin{aligned} \int \psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \psi(x) dx + \int \psi^*(x) V(x) \psi(x) dx \\ = \lambda \int \psi^*(x) \psi(x) dx \end{aligned}$$

$$\frac{1}{2m} \int \psi^*(x) \left(-i\hbar \frac{\partial}{\partial x} \right)^2 \psi(x) dx + \langle V(x) \rangle = \lambda$$

$$\frac{\langle \hat{p}^2 \rangle}{2m} + \langle V(x) \rangle = \lambda = E$$

We obtain the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)\psi(x) = E\psi(x)$$

(4) The total wavefunction of the Schrödinger equation is

$$\Psi(x, t) = \psi(x) e^{-iEt/\hbar}$$

EXAMPLES:

1. Consider a free particle $V(x) = 0$. The time-independent Schrödinger equation is

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) &= E\psi(x) \\ \Rightarrow \frac{d^2}{dx^2} \psi(x) + \frac{2mE}{\hbar^2} \psi(x) &= 0 \\ \Rightarrow \psi(x) &= Ae^{ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar} \end{aligned}$$

Using the normalization condition to determine A :

Since

$$\int_{-\infty}^{\infty} Ae^{-ikx} Ae^{ikx} dx = A^2 \int_{-\infty}^{\infty} dx = \infty$$

the wavefunction is not normalizable. Thus, we use so called the box normalization. Limits a plane wave in a box with length L

$$\begin{aligned} A^2 \int_0^L dx &= A^2 L = 1 \Rightarrow A = \frac{1}{\sqrt{L}} \\ \Rightarrow \psi(x) &= \frac{1}{\sqrt{L}} e^{ikx} \end{aligned}$$

The total energy of a free particle is,

$$\begin{aligned}
E &= \frac{1}{L} \int_0^L e^{-ikx} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) e^{ikx} dx \\
&= \frac{1}{L} \frac{\hbar^2 k^2}{2m} \int_0^L dx \\
&= \frac{\hbar^2 k^2}{2m} \\
&= \frac{p^2}{2m}
\end{aligned}$$

C. TIME-INDEPENDENT SCHRÖDINGER EQUATION

- (1) The time-independent Schrödinger equation is given by

$$\begin{aligned}
& -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)\psi(x) = E\psi(x) \\
\Rightarrow & \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x) = E\psi(x) \\
\Rightarrow & \left[\frac{1}{2m} \left(-i\hbar \frac{\partial}{\partial x} \right)^2 + V(x) \right] \psi(x) = E\psi(x) \\
\Rightarrow & \left[\frac{\hat{p}^2}{2m} + V(x) \right] \psi(x) = E\psi(x)
\end{aligned}$$

- (2) Define a differential operator

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

The time-independent Schrödinger equation can be written as an eigenvalue equation:

$$\hat{\mathcal{H}}\psi(x) = E\psi(x)$$

where E is the eigenvalue, $\psi(x)$ is the eigenfunction, and $\hat{\mathcal{H}}$ is the Hamiltonian differential operator.

1. For the terms of the position differential operator $\hat{x}$ in $\hat{\mathcal{H}}$

Take expectation

$$\int_{-\infty}^{\infty} \psi'^*(x) \hat{x} \psi(x) dx = \int_{-\infty}^{\infty} \psi'^*(x) x \psi(x) dx$$

Take the complex conjugate on both sides

$$\begin{aligned}\left(\int_{-\infty}^{\infty} \psi^*(x) \hat{x} \psi'(x) dx\right)^* &= \left(\int_{-\infty}^{\infty} \psi^*(x) x \psi'(x) dx\right)^* \\ &= \int_{-\infty}^{\infty} \psi'^*(x) x \psi(x) dx\end{aligned}$$

Since

$$\left(\int_{-\infty}^{\infty} \psi^*(x) \hat{x} \psi'(x) dx\right)^* = \int_{-\infty}^{\infty} \psi'^*(x) \hat{x}^* \psi(x) dx$$

we obtain

$$\int_{-\infty}^{\infty} \psi'^*(x) \hat{x}^* \psi(x) dx = \int_{-\infty}^{\infty} \psi'^*(x) \hat{x} \psi(x) dx$$

which indicates that $\hat{x}$ is a Hermitian operator

$$\hat{x}^* = \hat{x}$$

2. For the terms of the momentum differential operator $\hat{p}$ in $\hat{\mathcal{H}}$

Take expectation

$$\int_{-\infty}^{\infty} \psi'^*(x) \hat{p} \psi(x) dx = \int_{-\infty}^{\infty} \psi'^*(x) \left(-i\hbar \frac{\partial}{\partial x}\right) \psi(x) dx$$

Take the complex conjugate on both sides

$$\begin{aligned}\left(\int_{-\infty}^{\infty} \psi^*(x) \hat{p} \psi'(x) dx\right)^* &= \left(\int_{-\infty}^{\infty} \psi^*(x) \left(-i\hbar \frac{\partial}{\partial x}\right) \psi'(x) dx\right)^* \\ &= \int_{-\infty}^{\infty} \left(i\hbar \frac{\partial \psi'^*(x)}{\partial x}\right) \psi(x) dx\end{aligned}$$

Using integration by parts

$$\begin{aligned}&= - \int_{-\infty}^{\infty} \psi'^*(x) \left(i\hbar \frac{\partial \psi(x)}{\partial x}\right) dx \\ &= \int_{-\infty}^{\infty} \psi'^*(x) \hat{p} \psi(x) dx\end{aligned}$$

Since

$$\left(\int_{-\infty}^{\infty} \psi^*(x) \hat{p} \psi'(x) dx\right)^* = \int_{-\infty}^{\infty} \psi'^*(x) \hat{p}^* \psi(x) dx$$

we obtain

$$\int_{-\infty}^{\infty} \psi'^*(x) \hat{p}^* \psi(x) dx = \int_{-\infty}^{\infty} \psi'^*(x) \hat{p} \psi(x) dx$$

which indicates that $\hat{p}$ is a Hermitian operator.

$$\hat{p}^* = \hat{p}$$

Since $\hat{p}$ and $\hat{x}$ are Hermitian operators, thus $\hat{\mathcal{H}}$ is a Hermitian differential operator.

- (3) According to Sturm-Liouville [lyoo-veel] theory, as $\hat{\mathcal{H}}$ is a Hermitian differential operator (the operator is its own complex conjugate),

$$\hat{\mathcal{H}}^* = \hat{\mathcal{H}}$$

there are an infinite number of eigenvalues, which can be labelled by an integer number n . Thus, the eigenvalue equation is given by

$$\hat{\mathcal{H}}\psi_n(x) = E_n\psi_n(x)$$

1. Take expectation on both sides

$$\int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx = E_n \int \psi_n^*(x) \psi_n(x) dx = E_n$$

Take complex conjugate on both sides

$$\left(\int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx \right)^* = E_n^*$$

Since

$$\begin{aligned} \left(\int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx \right)^* &= \int \psi_n^*(x) \hat{\mathcal{H}}^* \psi_n(x) dx \\ &= \int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx \\ &= E_n \end{aligned}$$

thus **the eigenvalues are real:**

$$E_n^* = E_n$$

2. Take expectation on both sides for $k \neq n$

$$\int \psi_k^*(x) \hat{\mathcal{H}} \psi_n(x) dx = E_n \int \psi_k^*(x) \psi_n(x) dx$$

Take complex conjugate on both sides

$$\begin{aligned} \left(\int \psi_k^*(x) \hat{\mathcal{H}} \psi_n(x) dx \right)^* &= E_n^* \int \psi_n^*(x) \psi_k(x) dx \\ &= E_n \int \psi_n^*(x) \psi_k(x) dx \dots\dots (a) \end{aligned}$$

Since

$$\begin{aligned} \left(\int \psi_k^*(x) \hat{\mathcal{H}} \psi_n(x) dx \right)^* &= \int \left(\hat{\mathcal{H}} \psi_n(x) \right)^* \psi_k(x) dx \\ &= \int \psi_n^*(x) \hat{\mathcal{H}} \psi_k(x) dx \\ &= E_k \int \psi_n^*(x) \psi_k(x) dx \dots\dots (b) \end{aligned}$$

(a) - (b) gives

$$(E_n - E_k) \int \psi_n^*(x) \psi_k(x) dx = 0$$

thus **the eigenfunctions are orthonormal**:

$$\int \psi_n^*(x) \psi_k(x) dx = \delta_{nk}, \quad \delta_{nk} = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$

- (4) Sturm-Liouville theory states that as $\psi_n(x)$ are orthonormal, i.e., $\psi_n(x)$ are linear independent. Any wavefunction $\psi(x)$ in the system can be written as a linear combination of eigenfunctions $\psi_n(x)$, i.e.,

$$\psi(x) = \sum_n c_n \psi_n(x), \text{ where } c_n = \int \psi_n^*(x) \psi(x) dx$$

$\Rightarrow |\psi_n(x)|^2$ is the probability of finding a particle at position x associated with the eigenvalue E_n .

Since

$$\begin{aligned} \int \psi^*(x) \psi(x) dx &= \int \sum_m c_m^* \psi_m^*(x) \sum_n c_n \psi_n(x) dx \\ &= \sum_{nm} c_m^* c_n \int \psi_m^*(x) \psi_n(x) dx \\ &= \sum_{nm} c_m^* c_n \delta_{mn} \\ &= \sum_n |c_n|^2 \\ &= 1 \end{aligned}$$

$|c_n|^2$ is the probability of finding a particle associated with the eigenvalue E_n .

D. STATIONARY STATES

- (1) As wavefunctions are written as $\Psi_n(x, t) = \psi_n(x) e^{-iE_n t/\hbar}$, the state associated with the definite eigenvalue E_n is called the "stationary state".
- (2) Properties of stationary states:

1. The probability density does not depend on time

$$|\Psi_n(x, t)|^2 = \psi_n^*(x) e^{iE_n t/\hbar} \psi_n(x) e^{-iE_n t/\hbar} = |\psi_n(x)|^2$$

2. If $\hat{\mathcal{A}}$ does not explicitly depend on time, the expectation value is constant in time

$$\int \Psi_n^*(x, t) \hat{\mathcal{A}} \Psi_n(x, t) dx = \int \psi_n^*(x) \hat{\mathcal{A}} \psi_n(x) dx = \langle \hat{\mathcal{A}} \rangle$$

3. There is a definite energy

$$\langle \hat{\mathcal{H}} \rangle = \int \Psi_n^*(x, t) \hat{\mathcal{H}} \Psi_n(x, t) dx = \int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx = E_n$$

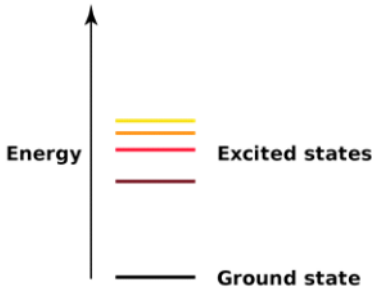
Moreover,

$$\langle \hat{\mathcal{H}}^2 \rangle = \int \psi_n^*(x) \hat{\mathcal{H}}^2 \psi_n(x) dx = E_n \int \psi_n^*(x) \hat{\mathcal{H}} \psi_n(x) dx = E_n^2$$

The variance of $\hat{\mathcal{H}}$ is

$$\langle \hat{\mathcal{H}}^2 \rangle - \langle \hat{\mathcal{H}} \rangle^2 = E_n^2 - E_n^2 = 0$$

- (3) E_n indicate the energy states hypothesised by Bohr (1913).



⇒ The lowest energy state is called the **ground state**.

⇒ Any state has a higher energy than the ground state is called **excited state**.

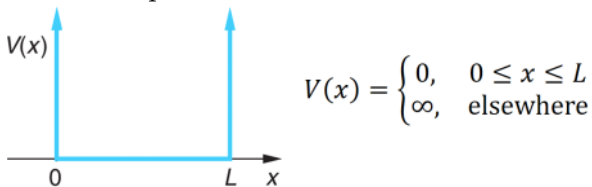
- (4) Time dependence of the stationary states

$$\Psi(x, t) = \sum_n c_n \Psi_n(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t / \hbar}$$

3-2 The Infinite Square Potential Well

A. EXACT SOLUTIONS

- (1) Consider a potential of the form



The Hamiltonian operator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

The Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x)$$

- (2) Solutions:

1. Outside the well, $V(x) = \infty$ as $x \leq 0$ or $x \geq L$

$$\psi^{\text{I}}(x) = 0$$

2. Inside the well, $V(x) = 0$ as $0 \leq x \leq L$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E\psi(x)$$

$$\Rightarrow \frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0, \text{ where } k = \frac{\sqrt{2mE}}{\hbar}$$

$$\Rightarrow \psi^{\text{II}}(x) = A \sin kx + B \cos kx$$

Three unknown variables: A , B , and k

- (3) Using boundary conditions to determine unknown variables:

Continuity of $\psi(x)$

$$\psi^{\text{I}}(0) = \psi^{\text{II}}(0) \Rightarrow 0 = A \times 0 + B \Rightarrow B = 0 \Rightarrow \psi^{\text{II}}(x) = A \sin kx$$

$$\psi^{\text{II}}(L) = \psi^{\text{I}}(L) \Rightarrow A \sin kL = 0 \Rightarrow \sin kL = 0 \Rightarrow kL = n\pi$$

$$k \rightarrow k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots, \infty$$

$$\psi^{\text{II}}(x) \rightarrow \psi_n(x) = A \sin \frac{n\pi}{L} x$$

Using the normalization condition:

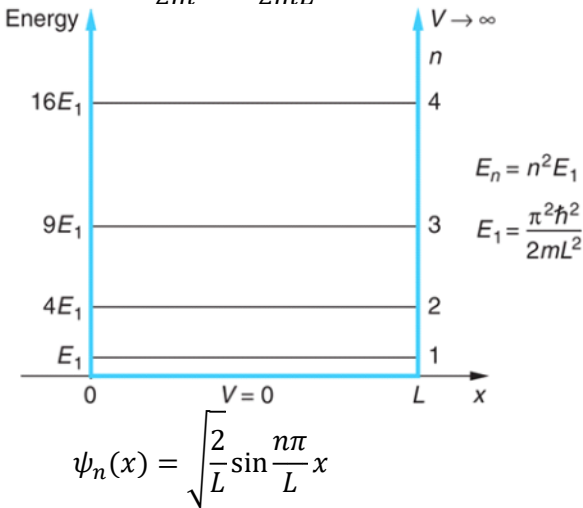
$$\begin{aligned}
 \int_{-\infty}^{\infty} \psi_n^*(x) \psi_n(x) dx &= A^2 \int_0^L \sin^2 k_n x dx \\
 &= A^2 \int_0^L \frac{1 - \cos 2k_n x}{2} dx \\
 &= A^2 \left[\frac{x}{2} - \frac{1}{4k_n} \sin 2k_n x \right]_0^L \\
 &= \frac{L}{2} A^2 \\
 &= 1
 \end{aligned}$$

$$\Rightarrow A = \sqrt{\frac{2}{L}} \dots \dots \text{normalization constant}$$

$$\Rightarrow \psi_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x$$

(4) Eigenenergy and eigenfunction:

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad n = 1, 2, 3, \dots, \infty \text{ (quantum number)}$$

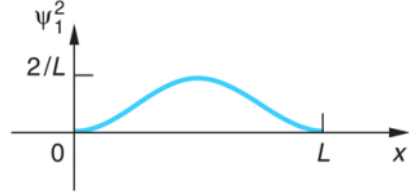
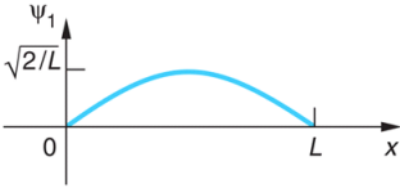


B. GROUND STATE PROPERTIES

(1) The ground state $n = 1$

$$E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$$

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$



(2) The average position and momentum of the particle

$$\begin{aligned} \langle \hat{x} \rangle &= \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \cdot x \cdot \sin\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2}{L} \int_0^L x \cdot \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2}{L} \int_0^L x \left(\frac{1 - \cos 2\pi x/L}{2} \right) dx \end{aligned}$$

Since

$$\begin{aligned} \int_0^L x \cos \frac{2\pi x}{L} dx &= \frac{L}{2\pi} \int_0^L x d\left(\sin \frac{2\pi x}{L}\right) \\ &= \frac{L}{2\pi} \left[x \sin \frac{2\pi x}{L} \Big|_0^L - \int_0^L \sin \frac{2\pi x}{L} dx \right] \\ &= \left(\frac{L}{2\pi} \right)^2 \cos \frac{2\pi x}{L} \Big|_0^L \\ &= 0 \end{aligned}$$

$$\langle \hat{x} \rangle = \frac{2}{L} \int_0^L \frac{x}{2} dx = \frac{2}{L} \frac{L^2}{4} = \frac{L}{2}$$

$$\begin{aligned} \langle \hat{p} \rangle &= \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(-i\hbar \frac{\partial}{\partial x} \right) \sin\left(\frac{\pi x}{L}\right) dx \\ &= -\frac{i2\pi\hbar}{L^2} \int_0^L \sin \frac{\pi x}{L} \cos \frac{\pi x}{L} dx \\ &= 0 \end{aligned}$$

The particle is moving back and forth, and so its average momentum is zero.

(3) The Heisenberg uncertainty relation

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

Calculate the uncertainty for position:

$$\begin{aligned}\langle \hat{x}^2 \rangle &= \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \cdot x^2 \cdot \sin\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2}{L} \int_0^L x^2 \left(\frac{1 - \cos 2\pi x/L}{2} \right) dx\end{aligned}$$

Since

$$\begin{aligned}\int_0^L x^2 \cos \frac{2\pi x}{L} dx &= \frac{L}{2\pi} \int_0^L x^2 d\left(\sin \frac{2\pi x}{L}\right) \\ &= \frac{L}{2\pi} \left[x^2 \sin \frac{2\pi x}{L} \Big|_0^L - \int_0^L 2x \sin \frac{2\pi x}{L} dx \right] \\ &= -2 \left(\frac{L}{2\pi} \right)^2 \int_0^L x d\left(-\cos \frac{2\pi x}{L}\right) \\ &= 2 \left(\frac{L}{2\pi} \right)^2 \left[x \cos \frac{2\pi x}{L} \Big|_0^L - \int_0^L \cos \frac{2\pi x}{L} dx \right] \\ &= 2 \left(\frac{L}{2\pi} \right)^2 L \\ &= \frac{L^3}{2\pi^2}\end{aligned}$$

$$\Rightarrow \langle \hat{x}^2 \rangle = \frac{2}{L} \left[\int_0^L \frac{x^2}{2} dx - \frac{1}{2} \times \frac{L^3}{2\pi^2} \right] = \frac{L^2}{3} - \frac{L^2}{2\pi^2}$$

$$\Rightarrow (\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = \left(\frac{L^2}{3} - \frac{L^2}{2\pi^2} \right) - \frac{L^2}{4} = \frac{L^2}{12} - \frac{L^2}{2\pi^2}$$

Calculate the uncertainty for momentum:

$$\begin{aligned}\langle \hat{p}^2 \rangle &= \frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \sin\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2\pi^2 \hbar^2}{L^3} \int_0^L \sin^2\left(\frac{\pi x}{L}\right) dx \\ &= \frac{2\pi^2 \hbar^2}{L^3} \times \frac{L}{2} \\ &= \frac{\pi^2 \hbar^2}{L^2}\end{aligned}$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 = \frac{\pi^2 \hbar^2}{L^2} - 0 = \frac{\pi^2 \hbar^2}{L^2}$$

$$\begin{aligned}
 (\Delta x)^2(\Delta p)^2 &= \left(\frac{L^2}{12} - \frac{L^2}{2\pi^2}\right) \frac{\pi^2 \hbar^2}{L^2} \\
 &= \left(\frac{\pi^2}{3} - 2\right) \frac{L^2}{4\pi^2} \frac{\pi^2 \hbar^2}{L^2} \\
 &= \left(\frac{\pi^2}{3} - 2\right) \frac{\hbar^2}{4}
 \end{aligned}$$

$$\Delta x \Delta p = \sqrt{\frac{\pi^2}{3} - 2} \frac{\hbar}{2} \approx 1.13 \frac{\hbar}{2} > \frac{\hbar}{2}$$

As a particle is confined in a box, so

$$\Delta x = \sqrt{\frac{L^2}{12} - \frac{L^2}{2\pi^2}} = \sqrt{\frac{\pi^2}{3} - 2} \frac{L}{2\pi}$$

The uncertainty for momentum is

$$\Delta p = \sqrt{\frac{\pi^2}{3} - 2} \frac{\hbar}{2} \frac{1}{\Delta x} = \frac{\pi \hbar}{L}$$

The minimum energy of the system is

$$E_{\min} = \frac{(\Delta p)^2}{2m} = \frac{\pi^2 \hbar^2}{2mL^2} = E_1 > V(x) = 0 \Big|_{0 \leq x \leq L}$$

The lowest energy $E_{\min}$ lying above zero is equal to the ground state energy E_1 and called the zero-point energy. The zero point energy is the smallest possible energy a physical system can possess, which is a direct consequence of the uncertainty principle.

$\underbrace{\Delta x \Delta p > \frac{\hbar}{2}}_{\text{uncertainty principle}} \iff \underbrace{E_{\min} = \frac{\pi^2 \hbar^2}{2mL^2} = E_1 > V(x) \Big _{0 \leq x \leq L}}_{\text{zero point energy}}$
--

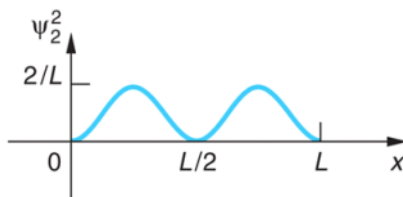
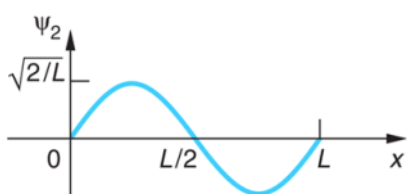
C. EXCITED STATE PROPERTIES

- (1) The first excited state $n = 2$

$$E_2 = \frac{4\pi^2 \hbar^2}{2mL^2} = \frac{2\pi^2 \hbar^2}{mL^2}$$

$$\psi_2(x) = \sqrt{\frac{2}{L}} \sin \frac{2\pi}{L} x$$





(2) Free expansion:

Assume there is a particle in the ground state

$$\psi(x) = \psi_1(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

Suddenly the well expands to twice its original size, i.e.,

$$0 \leq x \leq L \rightarrow 0 \leq x \leq 2L.$$

The new wavefunctions of the particle in a new infinite well is

$$\psi'_n(x) = \sqrt{\frac{1}{L}} \sin \frac{n\pi x}{2L}$$

1. What is the probability that the particle would be found in the ground state and the second state ($n = 2$) of the sudden expansion?
2. What is the expectation value of the energy of the particle before and after the sudden expansion?

ANSWERS:

1. Since $\int \psi_n'^*(x) \psi_m'(x) dx = \delta_{nm} \Rightarrow \psi_n'(x)$ forms the basis

The original ground state wavefunction can be expressed as the linear combination of the new normalized wavefunctions.

$$\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi}{L} x = \sum_n c'_n \psi'_n(x)$$

$$c'_1 = \int_0^L \psi_1'^*(x) \psi(x) dx = \frac{\sqrt{2}}{L} \int_0^L \sin \frac{\pi x}{2L} \sin \frac{\pi x}{L} dx = \frac{4\sqrt{2}}{3\pi}$$

$$|c_1'^2| = \frac{32}{9\pi^2}$$

$$c'_2 = \int_0^L \psi_2'^*(x) \psi(x) dx = \frac{\sqrt{2}}{L} \int_0^L \sin^2 \frac{\pi x}{L} dx = \frac{1}{\sqrt{2}}$$

$$|c_2'^2| = \frac{1}{2}$$

OS:

$$\begin{aligned}
 c'_n &= \int \psi_n'^*(x) \psi(x) dx \\
 c'_n &= \int_0^L \sqrt{\frac{1}{L}} \sin \frac{n\pi x}{2L} \cdot \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L} dx, \quad n \neq 2 \\
 &= \frac{\sqrt{2}}{L} \int_0^L \frac{1}{2} \left(\cos \left(\frac{n}{2} - 1 \right) \frac{\pi x}{L} - \cos \left(\frac{n}{2} + 1 \right) \frac{\pi x}{L} \right) dx \\
 &= \frac{\sqrt{2}}{2\pi} \left[\left(\frac{1}{\frac{n}{2} - 1} \right) \sin \left(\frac{n}{2} - 1 \right) \frac{\pi x}{L} - \left(\frac{1}{\frac{n}{2} + 1} \right) \sin \left(\frac{n}{2} + 1 \right) \frac{\pi x}{L} \right]_0^L \\
 &= \frac{1}{\sqrt{2}\pi} \left[\left(\frac{1}{\frac{n}{2} - 1} \right) \sin \left(\frac{n}{2} - 1 \right) \pi - \left(\frac{1}{\frac{n}{2} + 1} \right) \sin \left(\frac{n}{2} + 1 \right) \pi \right] \\
 &= \frac{1}{\sqrt{2}\pi} \left[\left(\frac{1}{\frac{n}{2} - 1} \right) - \left(\frac{1}{\frac{n}{2} + 1} \right) \right] \sin \left(\frac{n}{2} - 1 \right) \pi \\
 &= \frac{4\sqrt{2}}{\pi(n^2 - 4)} \sin \left(\frac{n}{2} - 1 \right) \pi \\
 c'_n &= \begin{cases} \frac{1}{2}, & n = 2 \\ \pm \frac{4\sqrt{2}}{\pi(n^2 - 4)}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}
 \end{aligned}$$

2. Before the sudden expansion, the particle is in the eigenstate (ground state). Therefore, the energy expectation is equal to the energy of the ground state,

$$\langle \hat{\mathcal{H}} \rangle = E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$$

After the sudden expansion,

$$\begin{aligned}
\langle \hat{H} \rangle &= \int_0^{2L} \psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right) \psi(x) dx \\
&= \frac{2}{L} \frac{\hbar^2}{2m} \int_0^L \frac{\pi^2}{L^2} \sin^2 \left(\frac{\pi x}{L} \right) dx \\
&= \frac{2}{L} \frac{\hbar^2}{2m} \times \frac{\pi^2 L}{L^2} \frac{1}{2} \\
&= \frac{\pi^2 \hbar^2}{2mL^2}
\end{aligned}$$

The expectation value of energy remains the same before and after the sudden change because the change is so quick that the wavefunction stays the same.

OS:

If you wanted to ensure that the particle has unity probability to be found in the ground state of the new potential, how would you change the potential (time scale)?

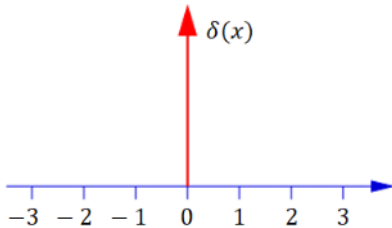
To ensure that the particle has unity probability to be found in the ground state, we need to change the potential slowly enough so that a new wavefunction is always in the ground state of a new well. It requires that the time scale of the change is $t \gg \hbar/\Delta E$, where ΔE is the difference between the ground state and an excited state. This change of the wavefunction is continuous, i.e. the eigenenergy of the ground state changes continuously from the old eigenenergy to a new eigenenergy. This does not violate energy conservation because the particle will interact with the walls and will lose some of its energy to the walls.

3-3 The δ -function Potential Well

A. DIRAC DELTA FUNCTION

- (1) A very “tall-and-thin” spike with unit area located at the origin

$$\delta(x) = 0 \text{ for } x \neq 0 \text{ and } \int_{-\infty}^{\infty} \delta(x) dx = 1$$



- (2) Limit representation

1. As the limit of $\epsilon \rightarrow 0$

Gaussian functions

$$f(x) = \frac{1}{\sqrt{2\pi}\epsilon} e^{-x^2/2\epsilon^2}$$

$$\delta(x) = \lim_{\epsilon \rightarrow 0} f(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{2\pi}\epsilon} e^{-x^2/2\epsilon^2}$$

Lorentzian functions

$$f(x) = \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2}$$

$$\delta(x) = \lim_{\epsilon \rightarrow 0} f(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2}$$

Sinc functions

$$f(x) = \frac{\sin(x/\epsilon)}{\pi x}$$

$$\delta(x) = \lim_{\epsilon \rightarrow 0} f(x) = \lim_{\epsilon \rightarrow 0} \frac{\sin(x/\epsilon)}{\pi x} = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \frac{\sin(x/\epsilon)}{\pi(x/\epsilon)}$$

2. As the limit of $\eta \rightarrow \infty$

Gaussian functions

$$f(x) = \frac{\eta}{\sqrt{2\pi}} e^{-x^2\eta^2/2}$$

$$\delta(x) = \lim_{\eta \rightarrow \infty} f(x) = \lim_{\eta \rightarrow \infty} \frac{\eta}{\sqrt{2\pi}} e^{-x^2\eta^2/2}$$

Lorentzian functions

$$f(x) = \frac{1}{\pi} \frac{\eta}{1 + x^2 \eta^2}$$

$$\delta(x) = \lim_{\eta \rightarrow \infty} f(x) = \lim_{\eta \rightarrow \infty} \frac{1}{\pi} \frac{\eta}{1 + x^2 \eta^2}$$

Sinc functions

$$f(x) = \frac{\sin(x\eta)}{\pi x}$$

$$\delta(x) = \lim_{\eta \rightarrow \infty} f(x) = \lim_{\eta \rightarrow \infty} \frac{\sin(x\eta)}{\pi x}$$

(3) Heaviside step function

$$\Theta(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} = \frac{1}{2} \left(1 + \frac{x}{|x|} \right)$$

$$\delta(x) = \frac{d}{dx} \Theta(x) = \frac{1}{2} \frac{d}{dx} \frac{x}{|x|} = \frac{1}{2} \lim_{\eta \rightarrow \infty} \frac{d}{dx} \frac{2}{\pi} \arctan x\eta = \lim_{\eta \rightarrow \infty} \frac{1}{\pi} \frac{\eta}{1 + x^2 \eta^2}$$

Let $\eta = 1/\epsilon$

$$\delta(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{1/\epsilon}{1 + x^2 / \epsilon^2} = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2}$$

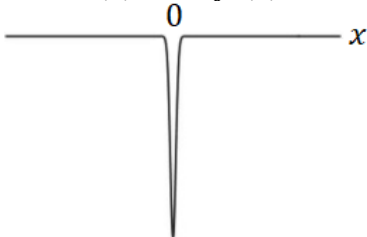
(4) Fourier transformation

$$\begin{aligned} \delta(x) &= \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{2\pi}\epsilon} e^{-x^2/2\epsilon^2} \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-k^2 \epsilon^2 / 2} e^{-ikx} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} dk \end{aligned}$$

B. EXACT SOLUTIONS

(1) Consider a potential of the form

$$V(x) = -V_0 \delta(x)$$



The Hamiltonian operator is

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - V_0 \delta(x)$$

The Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - V_0 \delta(x) \psi(x) = E \psi(x)$$

(2) Solutions for $E < 0$

1. In the region $x < 0, V(x) = 0$

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = -E \psi(x)$$

$$\frac{d^2}{dx^2} \psi(x) - \frac{2mE}{\hbar^2} \psi(x) = 0$$

$$\psi^{(-)}(x) = A e^{-\alpha x} + B e^{\alpha x}, \quad \alpha = \frac{\sqrt{-2mE}}{\hbar}$$

2. Similarly, in the region $x > 0, V(x) = 0$

$$\psi^{(+)}(x) = C e^{-\alpha x} + D e^{\alpha x}$$

Five unknown variables: A, B, C, D and α

(3) Using boundary conditions to determine unknown variables:

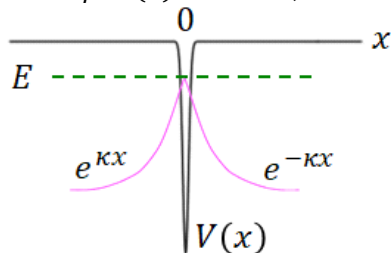
$$\psi^{(-)}(-\infty) = 0 \Rightarrow A = 0 \Rightarrow \psi^{(-)}(x) = B e^{\alpha x}, \quad x < 0$$

$$\psi^{(+)}(\infty) = 0 \Rightarrow D = 0 \Rightarrow \psi^{(+)}(x) = C e^{-\alpha x}, \quad x > 0$$

Continuity of $\psi(x)$

$$\psi^{(-)}(0) = \psi^{(+)}(0) \Rightarrow B = C$$

$$\left. \begin{aligned} \psi^{(-)}(x) &= B e^{\alpha x}, & x < 0 \\ \psi^{(+)}(x) &= B e^{-\alpha x}, & x > 0 \end{aligned} \right\} \Rightarrow \psi(x) = B e^{\alpha|x|}$$



Continuity of $d\psi/dx$

$$\left(\left. \frac{d\psi^{(-)}}{dx} \right|_{x=0} = B\alpha \right) \stackrel{?}{=} \left(\left. \frac{d\psi^{(+)}}{dx} \right|_{x=0} = -B\alpha \right)$$

The derivative at $x = 0$ is discontinuous.

$\Rightarrow \psi(x)$ does not satisfy the Schrödinger equation.

$\Rightarrow$ Sturm-Liouville theory can not be applied.

Integration just below a boundary to just above a boundary

$$\begin{aligned} \int_{-\epsilon}^{\epsilon} \frac{d^2\psi}{dx^2} dx &= -\frac{2m}{\hbar^2} \int_{-\epsilon}^{\epsilon} (V_0\delta(x) + E)\psi dx \\ \Rightarrow \left. \frac{d\psi^{(+)}}{dx} \right|_{\epsilon} - \left. \frac{d\psi^{(-)}}{dx} \right|_{-\epsilon} &= -\frac{2m}{\hbar^2} \int_{-\epsilon}^{\epsilon} V_0\delta(x)\psi dx - \frac{2m}{\hbar^2} E \int_{-\epsilon}^{\epsilon} \psi dx \\ \Rightarrow -B\alpha - B\alpha &= -\frac{2m}{\hbar^2} V_0\psi(0) \\ \Rightarrow -2B\alpha &= -\frac{2m}{\hbar^2} V_0B \\ \Rightarrow \alpha &= \frac{mV_0}{\hbar^2} \end{aligned}$$

There is only one α value satisfying the boundary conditions.

Using the normalization condition:

$$\begin{aligned} \int_{-\infty}^{\infty} |\psi|^2 dx &= \frac{|B|^2}{\alpha} = 1 \Rightarrow B = \sqrt{\alpha} \\ \psi(x) &= \sqrt{\alpha} e^{-\alpha|x|} \end{aligned}$$

(4) Eigenenergy and eigenfunction

$$\begin{aligned} E &= -\frac{\hbar^2\alpha^2}{2m} = -\frac{mV_0^2}{2\hbar^2} \\ \psi(x) &= \sqrt{\alpha} e^{-\alpha|x|} = \frac{\sqrt{mV_0}}{\hbar} e^{-mV_0|x|/\hbar^2} \end{aligned}$$

C. GROUND STATE PROPERTIES

(1) The ground state

$$\begin{aligned} E &= -\frac{mV_0^2}{2\hbar^2} > -\infty, \text{ there is only one bound state.} \\ \psi(x) &= \frac{\sqrt{mV_0}}{\hbar} e^{-mV_0|x|/\hbar^2} \end{aligned}$$

(2) The average position of the particle

$$\langle \hat{x} \rangle = \alpha \int_{-\infty}^{\infty} x e^{-2\alpha|x|} dx = \alpha \left[\int_{-\infty}^0 x e^{2\alpha x} dx + \int_0^{\infty} x e^{-2\alpha x} dx \right]$$

Let $x = -u$, $dx = -du$

$$\int_{-\infty}^0 x e^{2\alpha x} dx = \int_{\infty}^0 -u e^{-2\alpha u} (-du) = - \int_0^{\infty} u e^{-2\alpha u} du$$

$$\langle \hat{x} \rangle = \alpha \left[- \int_0^{\infty} x e^{-2\alpha x} dx + \int_0^{\infty} x e^{-2\alpha x} dx \right] = 0$$

The average momentum of the particle

Since the derivative at $x = 0$ is discontinuous

$$\langle \hat{p} \rangle \neq -i\hbar\alpha \left[\int_{-\infty}^0 e^{\alpha x} \frac{d}{dx} e^{\alpha x} dx + \int_0^{\infty} e^{-\alpha x} \frac{d}{dx} e^{-\alpha x} dx \right]$$

Fourier transformation

$$\begin{aligned} \phi(k) &= \sqrt{\frac{\alpha}{2\pi}} \int_{-\infty}^{\infty} e^{-\alpha|x|} e^{-ikx} dx \\ &= \sqrt{\frac{\alpha}{2\pi}} \int_{-\infty}^{\infty} e^{-\alpha|x|} (\cos kx - i \sin kx) dx \end{aligned}$$

Since the sine is an odd function, the latter term vanishes.

$$\phi(k) = \sqrt{\frac{\alpha}{2\pi}} \int_0^{\infty} e^{-\alpha|x|} \cos kx dx$$

$$\text{Since } \int_{-\infty}^0 e^{\alpha x} \cos kx dx = \int_{-\infty}^0 e^{-\alpha u} \cos ku (-du) = \int_0^{\infty} e^{-\alpha u} \cos ku du$$

$$\begin{aligned} \phi(k) &= 2 \sqrt{\frac{\alpha}{2\pi}} \int_0^{\infty} e^{-\alpha x} \cos kx dx \\ &= \sqrt{\frac{\alpha}{2\pi}} \int_0^{\infty} e^{-\alpha x} (e^{ikx} + e^{-ikx}) dx \\ &= \sqrt{\frac{\alpha}{2\pi}} \left[\frac{e^{(ik-\alpha)x}}{ik-\alpha} - \frac{e^{-(ik+\alpha)x}}{ik+\alpha} \right]_0^{\infty} \\ &= \sqrt{\frac{\alpha}{2\pi}} \left[\frac{-1}{ik-\alpha} + \frac{1}{ik+\alpha} \right] \\ &= \sqrt{\frac{\alpha}{2\pi}} \frac{2\alpha}{k^2 + \alpha^2} \\ &= \sqrt{\frac{1}{2\pi\alpha}} \frac{1}{1 + k^2/\alpha^2} \\ \langle \hat{p} \rangle &= \hbar \langle \hat{k} \rangle = \frac{2\hbar}{\pi\alpha} \int_{-\infty}^{\infty} \frac{k}{(1 + k^2/\alpha^2)^2} dk = 0 \end{aligned}$$

For a bound state, the expectation value of momentum is zero.

(3) The Heisenberg uncertainty relation

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

Calculate the uncertainty for position:

$$\langle \hat{x}^2 \rangle = \alpha \left[\int_{-\infty}^0 x^2 e^{2\alpha x} dx + \int_0^{\infty} x^2 e^{-2\alpha x} dx \right]$$

$$\text{Since } \int_{-\infty}^0 x^2 e^{2\alpha x} dx = \int_{-\infty}^0 u^2 e^{-2\alpha u} (-du) = \int_0^{\infty} u^2 e^{-2\alpha u} du$$

$$\Rightarrow \langle \hat{x}^2 \rangle = 2\alpha \int_0^{\infty} x^2 e^{-2\alpha x} dx$$

Using integration by parts

$$= 2\alpha \times \frac{1}{\alpha} \int_0^{\infty} x e^{-2\alpha x} dx$$

Using integration by parts

$$= 2 \times \frac{1}{2\alpha} \int_0^{\infty} e^{-2\alpha x} dx$$

$$= \frac{1}{2\alpha^2}$$

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = \frac{1}{2\alpha^2} - 0 = \frac{1}{2\alpha^2}$$

Calculate the uncertainty for momentum:

$$\langle \hat{p}^2 \rangle = \hbar^2 \langle \hat{k}^2 \rangle = \frac{2\hbar^2}{\pi\alpha} \int_{-\infty}^{\infty} \frac{k^2 dk}{(1 + k^2/\alpha^2)^2} = \frac{2\alpha^2 \hbar^2}{\pi} \int_{-\infty}^{\infty} \frac{u^2 du}{(1 + u^2)^2}$$

$$\text{Since } \int_{-\infty}^{\infty} \frac{u^2}{(1 + u^2)^2} du = \frac{1}{2} \int_{-\infty}^{\infty} \frac{1}{1 + u^2} du = \frac{\tan^{-1} u}{2} \Big|_{-\infty}^{\infty} = \frac{\pi}{2}$$

$$\Rightarrow \langle \hat{p}^2 \rangle = \frac{2\alpha^2 \hbar^2}{\pi} \times \frac{\pi}{2} = \alpha^2 \hbar^2$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 = \alpha^2 \hbar^2 - 0 = \alpha^2 \hbar^2$$

$$(\Delta x)^2 (\Delta p)^2 = \frac{1}{2\alpha^2} \times \alpha^2 \hbar^2 = \frac{\hbar^2}{2}$$

$$\Delta x \Delta p = \frac{\hbar}{\sqrt{2}} > \frac{\hbar}{2}$$

As a particle is confined in a δ -potential, so

$$\Delta x = \frac{1}{\sqrt{2}\alpha}$$

The uncertainty for momentum is,

$$\Delta p = \frac{\hbar}{\sqrt{2}\Delta x} = \alpha \hbar$$

The minimum energy of the system is

$$E_{\min} = -\frac{(\Delta p)^2}{2m} = -\frac{\alpha^2 \hbar^2}{2m} = E > V(0) = -\infty$$

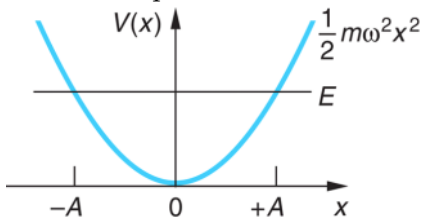
The lowest energy $E_{\min}$ lying above $-\infty$ is equal to the ground state energy E and called the zero-point energy. The zero point energy is the smallest possible energy a physical system can possess, which is a direct consequence of the uncertainty principle.

$\underbrace{\Delta x \Delta p > \frac{\hbar}{2}}_{\text{uncertainty principle}} \quad \Leftrightarrow \quad \underbrace{E_{\min} = -\frac{\alpha^2 \hbar^2}{2m} = E > V(0)}_{\text{zero point energy}}$
--

3-4 The Parabolic Potential Well

A. POWER SERIES SOLUTIONS

- (1) Consider a potential of the form



The Hamiltonian operator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2$$

The Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \frac{1}{2} m \omega^2 x^2 \psi(x) = E \psi(x)$$

- (2) Solutions:

$$\frac{d^2}{dx^2} \psi(x) + \left(\frac{2mE}{\hbar^2} - \frac{m^2 \omega^2 x^2}{\hbar^2} \right) \psi(x) = 0$$

Let $\lambda = \frac{2mE}{\hbar^2}$ and $\alpha = \frac{m\omega}{\hbar}$

$$\frac{d^2}{dx^2} \psi(x) + (\lambda - \alpha^2 x^2) \psi(x) = 0$$

Consider asymptotic solution: $x \rightarrow \pm\infty$

$$\frac{d^2}{dx^2} \psi(x) - \alpha^2 x^2 \psi(x) = 0$$

$$\Rightarrow \psi(x) = A e^{-\alpha x^2/2}$$

Suppose the general solutions $\psi(x) = A e^{-\alpha x^2/2} H(x)$ and substitute into the Schrödinger equation

$$\frac{d^2}{dx^2} H(x) - 2\alpha x \frac{d}{dx} H(x) + (\lambda - \alpha) H(x) = 0$$

Let $u = \sqrt{\alpha} x$

$$\frac{dH}{du} = \frac{dH}{dx} \frac{dx}{du} = \frac{1}{\sqrt{\alpha}} \frac{dH}{dx}$$

$$\frac{d^2 H}{du^2} = \frac{1}{\alpha} \frac{d^2 H}{dx^2}$$

$$\Rightarrow \frac{d^2}{du^2} H(u) - 2u \frac{d}{du} H(u) + \left(\frac{\lambda}{\alpha} - 1 \right) H(u) = 0$$

Hermite [her-'meet] differential equation

The power series method give a solution called Hermite polynomials

$$H(u) \rightarrow H_n(u) = (-1)^n e^{u^2} \frac{d^n}{du^n} e^{-u^2} \text{ is Hermite polynomial}$$

where $\frac{\lambda}{\alpha} - 1 = 2n, \quad n = 0, 1, 2, 3, \dots, \infty$

$$\psi(u) \rightarrow \psi_n(u) = A e^{-u^2/2} H_n(u) = A e^{-\alpha x^2/2} H_n(\sqrt{\alpha} x)$$

OS:

Recursion relation of Hermite polynomials

$$H_{n+1}(u) = 2u H_n(u) - 2n H_{n-1}(u)$$

Hermite polynomials

$$H_0(u) = 1$$

$$H_1(u) = 2u$$

$$H_2(u) = 2u H_1(u) - 2H_0(u) = 4u^2 - 2$$

$$H_3(u) = 2u H_2(u) - 2 \times 2H_1(u) = 8u^3 - 12u$$

Orthogonality

$$\int_{-\infty}^{\infty} H_m(u) H_n(u) e^{-u^2} du = \sqrt{\pi} 2^n n! \delta_{nm}$$

Using the normalization condition:

$$A^2 \int_{-\infty}^{\infty} H_n(\sqrt{\alpha} x) H_n(\sqrt{\alpha} x) e^{-\alpha x^2} dx = \frac{A^2}{\sqrt{\alpha}} \int_{-\infty}^{\infty} H_n(u) H_n(u) e^{-u^2} du$$

$$= \frac{A^2 \sqrt{\pi} 2^n n!}{\sqrt{\alpha}}$$

$$= 1$$

$$\Rightarrow A = \frac{1}{\sqrt{2^n n!}} \left(\frac{\alpha}{\pi} \right)^{1/4}$$

$$\Rightarrow \psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{\alpha}{\pi} \right)^{1/4} e^{-\alpha x^2/2} H_n(\sqrt{\alpha} x)$$

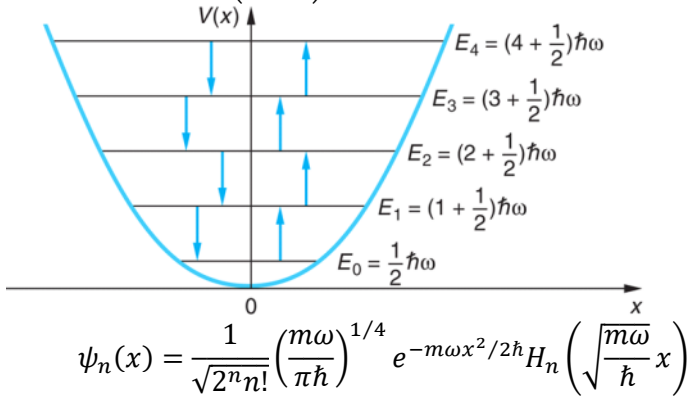
(3) Eigenenergy and eigenfunction:

Since $\frac{\lambda}{\alpha} - 1 = 2n$ and $\alpha = \frac{m\omega}{\hbar}$

$$\lambda_n = \alpha(2n + 1)$$

$$\Rightarrow \frac{2mE_n}{\hbar^2} = \frac{m\omega}{\hbar} (2n+1)$$

$$\Rightarrow E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, 3, \dots, \infty$$



B. VARIATIONAL METHOD (MINIMIZATION)

- (1) Assuming a trial wavefunction

$$\psi_0 = Ae^{-x^2/2a^2}$$

Using the normalization condition:

$$A^2 \int_{-\infty}^{\infty} e^{-x^2/2a^2} e^{-x^2/2a^2} dx = A^2 \underbrace{\int_{-\infty}^{\infty} e^{-x^2/a^2} dx}_{\text{Gauss integral}} = A^2 \sqrt{\pi} a = 1$$

$$\Rightarrow A = \left(\frac{1}{\pi a^2} \right)^{1/4}$$

$$\Rightarrow \psi_0 = \left(\frac{1}{\pi a^2} \right)^{1/4} e^{-x^2/2a^2}$$

- (2) Find a by looking for the extreme value of $\langle \hat{\mathcal{H}} \rangle(a)$

$$\langle \hat{\mathcal{H}} \rangle = \frac{\langle \hat{p}^2 \rangle}{2m} + \langle V(x) \rangle$$

$$\begin{aligned} \frac{\langle \hat{p}^2 \rangle}{2m} &= -\frac{\hbar^2}{2m\sqrt{\pi}a} \int_{-\infty}^{\infty} e^{-x^2/2a^2} \frac{d^2}{dx^2} e^{-x^2/2a^2} dx \\ &= -\frac{\hbar^2}{2\sqrt{\pi}ma} \int_{-\infty}^{\infty} e^{-x^2/2a^2} \left(-\frac{1}{a^2} + \frac{x^2}{a^4} \right) e^{-x^2/2a^2} dx \end{aligned}$$

$$\begin{aligned}
\text{Since } \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} dx &= \frac{a^2}{2} \sqrt{\pi} a \\
\frac{\langle \hat{p}^2 \rangle}{2m} &= -\frac{\hbar^2}{2\sqrt{\pi}ma} \left(-\frac{1}{a^2} \sqrt{\pi} a + \frac{1}{a^4} \frac{a^2}{2} \sqrt{\pi} a \right) = \frac{\hbar^2}{4ma^2} \\
\langle V(x) \rangle &= \frac{1}{2} m\omega^2 \left(\frac{1}{\sqrt{\pi}a} \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} dx \right) = \frac{1}{4} m\omega^2 a^2 \\
\langle \hat{H} \rangle(a) &= \frac{\hbar^2}{4ma^2} + \frac{m\omega^2 a^2}{4}
\end{aligned}$$

The extreme value of $\langle \hat{H} \rangle(a)$

$$\begin{aligned}
\frac{\partial \langle \hat{H} \rangle}{\partial a} = 0 &\Rightarrow -\frac{\hbar^2}{2ma^3} + \frac{m\omega^2 a}{2} = 0 \Rightarrow a^2 = \frac{\hbar}{m\omega} \\
\psi_0(x) &= \left(\frac{1}{\pi a^2} \right)^{1/4} e^{-x^2/2a^2} = \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} e^{-m\omega x^2/2\hbar}
\end{aligned}$$

C. GROUND STATE PROPERTIES

- (1) The ground state $n = 0$

$$E_0 = \frac{1}{2} \hbar \omega$$

$$\text{Let } a^2 = \frac{\hbar}{m\omega}$$

$$\psi_0 = \left(\frac{1}{a\sqrt{\pi}} \right)^{1/2} e^{-x^2/2a^2} \dots \text{Gauss function}$$

- (2) The average position and momentum of the particle

$$\langle \hat{x} \rangle = \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/2a^2} x e^{-x^2/2a^2} dx = \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} x e^{-x^2/a^2} dx = 0$$

$$\begin{aligned}
\langle \hat{p} \rangle &= \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/2a^2} \left(-i\hbar \frac{\partial}{\partial x} \right) e^{-x^2/2a^2} dx \\
&= \frac{i\hbar}{a^2} \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} x e^{-x^2/a^2} dx \\
&= 0
\end{aligned}$$

The particle is moving back and forth, and so its average position and average momentum are zero.

- (3) The Heisenberg uncertainty relation

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

Calculate the uncertainty for position:

$$\begin{aligned} \langle \hat{x}^2 \rangle &= \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/2a^2} x^2 e^{-x^2/2a^2} dx \\ &= \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} x^2 e^{-x^2/a^2} dx \\ &= \frac{1}{a\sqrt{\pi}} \times \frac{a^2}{2} \times \sqrt{\pi} a \\ &= \frac{a^2}{2} \end{aligned}$$

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = \frac{a^2}{2} - 0 = \frac{a^2}{2}$$

Calculate the uncertainty for momentum:

$$\begin{aligned} \langle \hat{p}^2 \rangle &= \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/2a^2} \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) e^{-x^2/2a^2} dx \\ &= -\hbar^2 \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/2a^2} \left(-\frac{1}{a^2} + \frac{x^2}{a^4} \right) e^{-x^2/2a^2} dx \\ &= -\hbar^2 \frac{1}{a\sqrt{\pi}} \left(-\frac{1}{a^2} \sqrt{\pi} a + \frac{1}{a^4} \frac{a^2}{2} \sqrt{\pi} a \right) \\ &= \frac{\hbar^2}{2a^2} \end{aligned}$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 = \frac{\hbar^2}{2a^2} - 0 = \frac{\hbar^2}{2a^2}$$

$$(\Delta x)^2 (\Delta p)^2 = \frac{a^2}{2} \times \frac{\hbar^2}{2a^2} = \frac{\hbar^2}{4}$$

$$\Delta x \Delta p = \frac{\hbar}{2}$$

As a particle is confined in a parabolic potential, so

$$\Delta x = \frac{a}{\sqrt{2}} = \sqrt{\frac{\hbar}{2m\omega}}$$

The uncertainty for momentum is,

$$\Delta p = \frac{\hbar}{2\Delta x} = \frac{\hbar}{\sqrt{2}a} = \sqrt{\frac{\hbar m\omega}{2}}$$

The minimum energy of the system is

$$E_{\min} = \frac{(\Delta p)^2}{2m} + \frac{m\omega^2(\Delta x)^2}{2} = \frac{\hbar m\omega}{4m} + \frac{m\omega^2 \hbar}{4m\omega} = \frac{\hbar\omega}{2} = E_0 > 0$$

Alternative approach:

Using the uncertainty relation

$$\Delta x \Delta p = \frac{\hbar}{2} \Rightarrow \Delta p = \frac{\hbar}{2\Delta x}$$

The minimum energy of the system is

$$E_{\min} = \frac{(\Delta p)^2}{2m} + \frac{m\omega^2(\Delta x)^2}{2} = \frac{\hbar^2}{8m(\Delta x)^2} + \frac{m\omega^2(\Delta x)^2}{2}$$

Since $\frac{dE_{\min}}{d(\Delta x)} = 0$

$$-\frac{\hbar^2}{4m(\Delta x)^3} + m\omega^2(\Delta x) = 0$$

$$\Rightarrow (\Delta x)^2 = \frac{\hbar}{2m\omega}$$

$$\Rightarrow E_{\min} = \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2} = E_0 > V(0) = 0$$

The lowest energy $E_{\min}$ lying above zero is equal to the ground state energy E_0 and called the zero-point energy. The zero point energy is the smallest possible energy a physical system can possess, which is a direct consequence of the uncertainty principle.

$\underbrace{\Delta x \Delta p = \frac{\hbar}{2}}_{\text{uncertainty principle}} \Leftrightarrow \underbrace{E_{\min} = \frac{\hbar\omega}{2} = E_0 > V(0)}_{\text{zero point energy}}$

D. EXCITED STATE PROPERTIES

- (1) The first excited state $n = 1$

$$E_1 = \frac{3}{2}\hbar\omega$$

Let $a^2 = \frac{\hbar}{m\omega}$

$$\psi_1 = \left(\frac{1}{2a\sqrt{\pi}}\right)^{1/2} 2\left(\frac{x}{a}\right) e^{-x^2/2a^2}$$

- (2) The second excited state $n = 2$

$$E_2 = \frac{5}{2}\hbar\omega$$

Let $a^2 = \frac{\hbar}{m\omega}$

$$\psi_2 = \left(\frac{1}{8a\sqrt{\pi}} \right)^{1/2} \left[2 - 4 \left(\frac{x}{a} \right)^2 \right] e^{-x^2/2a^2}$$

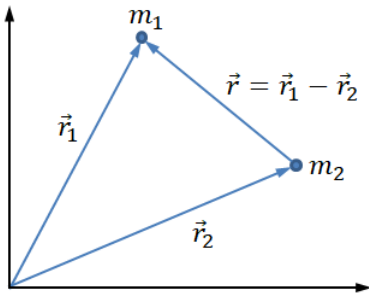
- (3) The energy difference between adjacent energy levels is independent of the quantum number and is the same.

$$\Delta E = E_2 - E_1 = E_1 - E_0 = \hbar\omega$$

3-5 The Hydrogen Atom

A. CENTRAL FORCE MOTION

- (1) Consider the general 3 dimensional two-body problem:



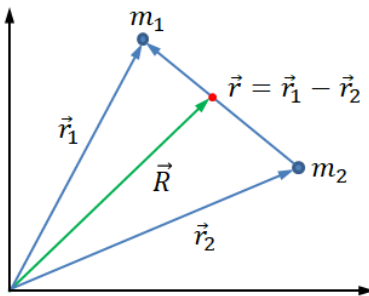
2 masses m_1 and m_2 need 6 coordinates.

Assume that the 2 bodies interact with a potential $V(\vec{r})$, which depends only on the distance $\vec{r} = |\vec{r}_1 - \vec{r}_2|$ between m_1 and m_2 (central force).

The Lagrangian is

$$L = \frac{1}{2}m_1\dot{\vec{r}}_1^2 + \frac{1}{2}m_2\dot{\vec{r}}_2^2 - V(\vec{r})$$

- (2) Center of mass coordinate:



$$(m_1 + m_2)\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 \Rightarrow \vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

Relative coordinate $\vec{r} = \vec{r}_1 - \vec{r}_2$:

- (3) The inverse coordinate relations:

$$\vec{R} = \frac{m_1\vec{r}_1 + m_2(\vec{r}_1 - \vec{r})}{m_1 + m_2} = \vec{r}_1 - \frac{m_2}{m_1 + m_2}\vec{r} \Rightarrow \vec{r}_1 = \vec{R} + \frac{m_2}{m_1 + m_2}\vec{r}$$

$$\vec{R} = \frac{m_1(\vec{r}_2 + \vec{r}) + m_2\vec{r}_2}{m_1 + m_2} = \vec{r}_2 + \frac{m_1}{m_1 + m_2}\vec{r} \Rightarrow \vec{r}_2 = \vec{R} - \frac{m_1}{m_1 + m_2}\vec{r}$$

The Lagrangian becomes

$$\begin{aligned} L &= \frac{1}{2}m_1 \left(\dot{\vec{R}}^2 + \left(\frac{m_2}{m_1 + m_2} \right)^2 \dot{\vec{r}}^2 + 2 \frac{m_2}{m_1 + m_2} \left| \dot{\vec{R}} \cdot \dot{\vec{r}} \right| \right) \\ &\quad + \frac{1}{2}m_2 \left(\dot{\vec{R}}^2 + \left(\frac{m_1}{m_1 + m_2} \right)^2 \dot{\vec{r}}^2 - 2 \frac{m_1}{m_1 + m_2} \left| \dot{\vec{R}} \cdot \dot{\vec{r}} \right| \right) - V(\vec{r}) \\ &= \frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2 + \frac{1}{2} \frac{m_1 m_2^2 + m_2 m_1^2}{(m_1 + m_2)^2} \dot{\vec{r}}^2 - V(\vec{r}) \\ &= \frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2 + \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \dot{\vec{r}}^2 - V(\vec{r}) \end{aligned}$$

Define the reduced mass μ

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad \text{alternative } \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$$

The Lagrangian becomes

$$L = \frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2 + \frac{1}{2}\mu\dot{\vec{r}}^2 - V(\vec{r})$$

- (4) For the 2-body (central force problem), the motion separates into 2 distinct parts:

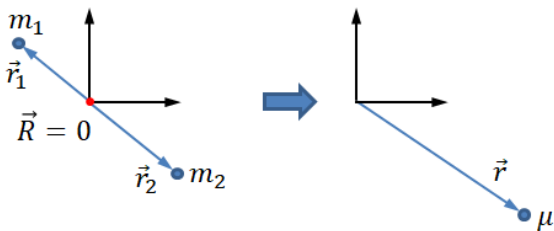
$$\text{The center of mass motion: } L_{\text{CM}} = \frac{1}{2}(m_1 + m_2)\dot{\vec{R}}^2$$

$$\text{The relative motion: } L_{\text{rel}} = \frac{1}{2}\mu\dot{\vec{r}}^2 - V(\vec{r})$$

By transforming coordinates from $(\vec{r}_1, \vec{r}_2)$ to $(\vec{R}, \vec{r})$, the 2-body problem has been separated into 2 independent one-body problem.

- (5) The motion of the center of mass is identical to the trivial motion of a free particle. It corresponds to a uniform translation of the center of mass through space, i.e., trivial and uninteresting. Clearly, all of the interesting physics is in the relative motion part.

For convenience, take the origin of coordinates at the CM: $\vec{R} = 0$



$$\vec{r}_1 = \frac{m_2}{m_1 + m_2} \vec{r}, \quad \vec{r}_2 = -\frac{m_1}{m_1 + m_2} \vec{r} \Rightarrow m_1 \vec{r}_1 + m_2 \vec{r}_2 = 0$$

$$L = L_{\text{rel}} = \frac{1}{2} \mu \dot{\vec{r}}^2 - V(\vec{r}) = T(\dot{\vec{r}}) - V(\vec{r})$$

$$\Rightarrow H = T(\dot{\vec{r}}) + V(\vec{r}) = \frac{1}{2} \mu \dot{\vec{r}}^2 + V(\vec{r}) = \frac{\vec{p}^2}{2\mu} + V(\vec{r})$$

B. THE HYDROGEN ATOM: ONE-ELECTRON ATOM

(1) Consider a potential of the form

$$V(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r} \dots \text{Coulomb potential}$$

Since $V(r)$ depends only on r of the electron from the force center, there is no orientation dependence.

$\Rightarrow$ The system has spherical symmetry.

The Hamiltonian operator in spherical coordinates is

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + V(x) = -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

where

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

$$\text{Let } \hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) \right] + \frac{\hbar^2 \hat{L}^2}{2\mu r^2} - \frac{e^2}{4\pi\epsilon_0 r}$$

The Schrödinger equation is

$$-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) \psi + \frac{\hat{L}^2}{2\mu r^2} \psi - \frac{e^2}{4\pi\epsilon_0 r} \psi = E\psi$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) \psi - \frac{\hat{L}^2}{\hbar^2} \psi + \frac{2\mu r^2}{\hbar^2} \frac{e^2}{4\pi\epsilon_0 r} \psi + \frac{2\mu r^2}{\hbar^2} E\psi = 0$$

$$\left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \right] \psi - \frac{\hat{L}^2}{\hbar^2} \psi = 0$$

Using separation of variables: $\psi \equiv \psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$

$$\underbrace{\frac{1}{R} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \right]}_{=\lambda} R - \underbrace{\frac{1}{Y} \frac{\hat{L}^2}{\hbar^2} Y}_{=\lambda} = 0$$

We obtain two ordinary differential equations:

$$\begin{aligned} \frac{1}{R} \left[\frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{2\mu r^2}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \right] R &= \lambda \dots \text{radial equation} \\ -\frac{1}{Y} \frac{\hat{L}^2}{\hbar^2} Y &= \lambda \dots \text{angular equation} \end{aligned}$$

From angular equation, we have

$$\begin{aligned} &\left(-\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \right) Y(\theta, \phi) = \lambda \hbar^2 Y(\theta, \phi) \\ \Rightarrow &\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) + \lambda Y(\theta, \phi) = 0 \\ \Rightarrow &\left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \lambda \sin^2 \theta \right] Y(\theta, \phi) + \frac{\partial^2}{\partial \phi^2} Y(\theta, \phi) = 0 \end{aligned}$$

Using separation of variables: $Y(\theta, \phi) = P(\theta)\Phi(\phi)$

$$\underbrace{\frac{1}{P} \left[\sin \theta \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \lambda \sin^2 \theta \right] P}_{=m^2} - \underbrace{\frac{1}{\Phi} \frac{\partial^2}{\partial \phi^2} \Phi}_{=m^2} = 0$$

We obtain two ordinary differential equations:

$$\begin{aligned} -\frac{1}{\Phi} \frac{d^2}{d\phi^2} \Phi &= m^2 \dots \text{azimuthal equation} \\ \frac{1}{P} \left[\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) + \lambda \sin^2 \theta \right] P &= m^2 \\ &\dots \theta\text{-dependent differential equation} \end{aligned}$$

(2) After using separation of variables, we have three differential equations

$$\begin{aligned} \frac{1}{R} \left[\frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{2\mu r^2}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \right] R &= \lambda \dots \text{radial equation} \\ \frac{1}{P} \left[\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) + \lambda \sin^2 \theta \right] P &= m^2 \\ &\dots \theta\text{-dependent differential equation} \end{aligned}$$

$$\frac{1}{\Phi} \frac{d^2}{d\phi^2} \Phi = -m^2 \dots \text{azimuthal equation}$$

- Solution of azimuthal equation

$$\frac{d^2 \Phi}{d\phi^2} = -m^2 \Phi \Rightarrow \Phi = A e^{im\phi}$$

Using the periodic boundary condition:

$$\begin{aligned} \Phi(\phi + 2\pi) &= \Phi(\phi) \\ e^{im(\phi+2\pi)} &= e^{im\phi} \Rightarrow e^{im2\pi} = 1, \quad m = 0, \pm 1, \pm 2, \pm 3, \dots, \pm \infty \end{aligned}$$

Using the normalization condition:

$$\int_0^{2\pi} \Phi^* \Phi d\phi = A^2 2\pi = 1 \Rightarrow A = \sqrt{\frac{1}{2\pi}}$$

$$\Phi(\phi) = \sqrt{\frac{1}{2\pi}} e^{im\phi}$$

- Solution of θ -dependent differential equation

$$\begin{aligned} \sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) P(\theta) + \lambda \sin^2 \theta P(\theta) &= m^2 P(\theta) \\ \Rightarrow \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) P(\theta) + \left[\lambda - \frac{m^2}{\sin^2 \theta} \right] P(\theta) &= 0 \end{aligned}$$

Let $z = \cos \theta$. We have

$$\begin{aligned} \frac{dz}{d\theta} &= -\sin \theta \\ \frac{d}{dz} &= \frac{d}{d\theta} \frac{d\theta}{dz} = -\frac{1}{\sin \theta} \frac{d}{d\theta} \\ \sin \theta \frac{d}{d\theta} &= -\sin^2 \theta \frac{d}{dz} = -(1-z^2) \frac{d}{dz} \end{aligned}$$

We obtain the associated Legendre differential equation

$$\frac{d}{dz} \left((1-z^2) \frac{dP(z)}{dz} \right) + \left[\lambda - \frac{m^2}{1-z^2} \right] P(z) = 0$$

The power series method give a solution called the associated Legendre polynomials

$$P_{\ell, m}(z) = B \frac{(1-z^2)^{\frac{|m|}{2}}}{2^\ell \ell!} \frac{d^{\ell+|m|}}{dz^{\ell+|m|}} (z^2-1)^\ell$$

where

$$\begin{aligned} \lambda &= \ell(\ell+1) \\ \ell &= 0, 1, 2, \dots, \infty \end{aligned}$$

$$m = 0, \pm 1, \pm 2, \dots, \pm \ell$$

We obtain the spherical harmonic function

$$Y_{\ell,m}(\theta, \phi) = P_{\ell,m}(\cos \theta) \Phi_m(\phi) = P_{\ell,m}(\cos \theta) e^{im\phi}$$

Using the normalization condition:

$$\int_0^\pi \int_0^{2\pi} Y_{\ell,m}^*(\theta, \phi) Y_{\ell',m'}(\theta, \phi) \sin \theta \, d\theta \, d\phi = \delta_{\ell,\ell'} \delta_{m,m'}$$

$$\Rightarrow 2B^2 = 1 \Rightarrow B = \sqrt{\frac{1}{2}}$$

$$\Rightarrow Y_{\ell,m}(\theta, \phi) = \sqrt{\frac{1}{4\pi} \frac{(1-z^2)^{\frac{|m|}{2}}}{2^\ell \ell!}} \frac{d^{\ell+|m|}}{dz^{\ell+|m|}} (z^2 - 1)^\ell e^{im\phi}$$

$\ell = 0$	$m = 0$	$Y_{00} = \sqrt{\frac{1}{4\pi}}$
$\ell = 1$	$m = 1$	$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta \, e^{i\phi}$
	$m = 0$	$Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta$
	$m = -1$	$Y_{1-1} = \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{-i\phi}$

- Solution of radial equation

$$\left[\frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{2\mu r^2}{\hbar^2} \left(E + \frac{e^2}{4\pi\epsilon_0 r} \right) \right] R(r) = \ell(\ell+1)R(r)$$

$$\left[r^2 \frac{d^2}{dr^2} + 2r \frac{d}{dr} + \frac{2\mu E}{\hbar^2} r^2 + \frac{2\mu}{\hbar^2} \frac{e^2}{4\pi\epsilon_0} r - \ell(\ell+1) \right] R(r) = 0$$

Let $u(r) = rR(r)$

$$\frac{d^2 u(r)}{dr^2} = r \frac{d^2 R(r)}{dr^2} + 2 \frac{dR(r)}{dr}$$

$$\frac{d^2 u(r)}{dr^2} + \left(\frac{2\mu E}{\hbar^2} + \frac{2\mu}{\hbar^2} \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} - \frac{\ell(\ell+1)}{r^2} \right) u(r) = 0$$

Set $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{e^2 \mu}$ Bohr atomic radius and $k^2 = \frac{2\mu E}{\hbar^2}$

We obtain the associated Laguerre differential equation

$$\frac{d^2 u(r)}{dr^2} + \left(k^2 + \frac{2}{a_0 r} - \frac{\ell(\ell+1)}{r^2} \right) u(r) = 0$$

The power series method give a solution called the associated Laguerre polynomials

$$R_{n,\ell}(r) = \frac{u_{n,\ell}(r)}{r} = C e^{-r/na_0} \left(\frac{r}{a_0} \right)^{|\ell|/2} L_n^{|\ell|} \left(\frac{r}{a_0} \right)$$

where

$$n = 1, 2, 3, \dots, \infty$$

$$\ell = 0, 1, 2, \dots, n-1$$

Using the normalization condition:

$$\int_0^\infty R_{n,\ell}^*(r) R_{n',\ell'}(r) r^2 dr = \delta_{n,n'} \delta_{\ell,\ell'} \Rightarrow \frac{a_0^3}{4} C^2 = 1 \Rightarrow C = \sqrt{\frac{4}{a_0^3}}$$

$$R_{n,\ell}(r) = \sqrt{\frac{4}{a_0^3}} e^{-r/na_0} \left(\frac{r}{a_0} \right)^{|\ell|/2} L_n^{|\ell|} \left(\frac{r}{a_0} \right)$$

$$n = 1 \quad \ell = 0 \quad R_{10} = \frac{2}{\sqrt{a_0^3}} e^{-r/a_0}$$

$$n = 2 \quad \ell = 0 \quad R_{20} = \frac{1}{\sqrt{2a_0^3}} \left(1 - \frac{r}{2a_0} \right) e^{-r/2a_0}$$

$$\ell = 1 \quad R_{21} = \frac{1}{2\sqrt{6a_0^3}} \frac{r}{a_0} e^{-r/2a_0}$$

(3) Eigenenergy and eigenfunction

$$E_n = -\frac{13.6}{n^2}$$

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

$$= \sqrt{\frac{1}{\pi a_0^3}} e^{-r/na_0} \left(\frac{r}{a_0} \right)^{|\ell|/2} L_n^{|\ell|} \left(\frac{r}{a_0} \right) P_{\ell m}(\cos \theta) e^{im\phi}$$

where

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{e^2\mu}$$

$$n = 1, 2, 3, \dots, \infty$$

$$\ell = 0, 1, 2, \dots, n-1$$

$$m = 0, \pm 1, \pm 2, \pm 3, \dots, \pm \ell$$

Spectroscopic notation:

For historical reasons, each possible ℓ value is denoted by a letter of the alphabet according to the following sequence:

ℓ	0	1	2	3	4
Notation	<i>s</i>	<i>p</i>	<i>d</i>	<i>f</i>	<i>g</i>

OS:

The reason for the non-alphabetic order of the first four letters is that they are taken from the names of four series of spectral lines, called sharp, principal, diffuse and fine, observed in the spectra of alkali atoms such as Sodium.

C. GROUND STATE PROPERTIES

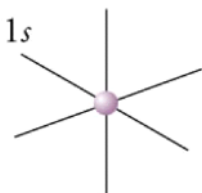
- (1) The ground state $n = 1, \ell = 0, m = 0$

$$E_{100} = -\frac{13.6}{n^2} = -13.6$$

$$\psi_{100}(r, \theta, \phi) = R_{10}(r)Y_{00}(\theta, \phi) = \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0}$$

where

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{e^2\mu}$$

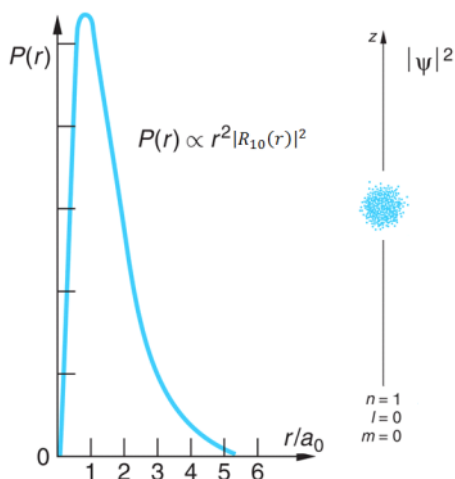


- (2) The average position and momentum of the particle

$$\begin{aligned}\langle \hat{r} \rangle &= \frac{4}{a_0^3} \int_0^\infty r \cdot |R_{10}(r)|^2 r^2 dr \\ &= \frac{4}{a_0^3} \int_0^\infty r^3 e^{-2r/a_0} dr \\ &= \frac{4}{a_0^3} 3! \left(\frac{a_0}{2}\right)^4 \\ &= \frac{3}{2} a_0\end{aligned}$$

The average position is not the atomic radius.

The probability density of the electron found between r and $r + dr$ shells is $|R_{10}(r)|^2 r^2$,



The most likely distance of r

$$\frac{d}{dr} |R_{10}(r)|^2 r^2 = 0$$

$$\Rightarrow \frac{d}{dr} (r^2 e^{-2r/a_0}) = 2r e^{-2r/a_0} - \frac{2}{a_0} r^2 e^{-2r/a_0} = 0$$

$$\Rightarrow r_{\max} = a_0$$

a_0 is the most possible position of finding the particle. In other words, a_0 implies the radius, not the average radius, of the Hydrogen atom.

(3) The Heisenberg uncertainty relation

$$(\Delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

$$(\Delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

Calculate the uncertainty for position:

Since $x = r \sin \theta \cos \phi$

$$\begin{aligned} \langle \hat{x} \rangle &= \int \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} \cdot x \cdot \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\ &= \frac{1}{\pi a_0^3} \int e^{-r/a_0} \cdot r \sin \theta \cos \phi \cdot e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\ &= \frac{1}{\pi a_0^3} \int_0^\infty r^3 e^{-2r/a_0} dr \int_0^\pi \sin^2 \theta d\theta \underbrace{\int_0^{2\pi} \cos \phi d\phi}_{=0} \\ &= 0 \end{aligned}$$

$$\begin{aligned}
\langle \hat{x}^2 \rangle &= \int \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} \cdot x^2 \cdot \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\
&= \frac{1}{\pi a_0^3} \int_0^\infty e^{-r/a_0} \cdot r^2 \sin^2 \theta \cos^2 \phi \cdot e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\
&= \frac{1}{\pi a_0^3} \int_0^\infty r^4 e^{-2r/a_0} dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} \cos^2 \phi d\phi
\end{aligned}$$

Since

$$\begin{aligned}
\int_0^{2\pi} \cos^2 \phi d\phi &= \frac{1}{2} \int_0^{2\pi} (1 + \cos 2\phi) d\phi = \pi \\
\int_0^\pi \sin^3 \theta d\theta &= \int_0^\pi \left(\frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta \right) d\theta = \frac{3}{4} \times 2 - \frac{1}{4} \times \frac{2}{3} = \frac{4}{3} \\
\int_0^\infty r^4 e^{-2r/a_0} dr &= 4! \left(\frac{a_0}{2} \right)^5 = \frac{3a_0^5}{4} \\
\Rightarrow \langle \hat{x}^2 \rangle &= \frac{1}{\pi a_0^3} \times \frac{3a_0^5}{4} \times \frac{4}{3} \times \pi = a_0^2 \\
\Rightarrow (\Delta x)^2 &= \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = a_0^2 - 0 = a_0^2
\end{aligned}$$

Calculate the uncertainty for momentum:

In spherical coordinates,

$$\begin{aligned}
\frac{\partial}{\partial x} &= \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r} \frac{1}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi} \\
\langle \hat{p}_x \rangle &= \int \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} \cdot \left(-i\hbar \frac{\partial}{\partial x} \right) \cdot \sqrt{\frac{1}{\pi a_0^3}} e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\
&= \frac{1}{\pi a_0^3} \int e^{-r/a_0} \left(-i\hbar \sin \theta \cos \phi \frac{\partial}{\partial r} \right) e^{-r/a_0} r^2 dr \sin \theta d\theta d\phi \\
&= \frac{-i\hbar}{\pi a_0^3} \int_0^\infty e^{-r/a_0} \left(-\frac{1}{a_0} \right) e^{-r/a_0} r^2 dr \int_0^\pi \sin^2 \theta d\theta \underbrace{\int_0^{2\pi} \cos \phi d\phi}_{=0} \\
&= 0
\end{aligned}$$

Since ψ_{100} is spherical symmetry,

$$\langle \hat{p}_x^2 \rangle = \langle \hat{p}_y^2 \rangle = \langle \hat{p}_z^2 \rangle = \frac{1}{3} \langle \hat{p}^2 \rangle$$

$$\begin{aligned}
\langle \hat{p}^2 \rangle &= -\frac{\hbar^2}{\pi a_0^3} \int e^{-r/a_0} \nabla^2 e^{-r/a_0} d^3r \\
&= -\frac{\hbar^2}{\pi a_0^3} \int e^{-r/a_0} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) e^{-r/a_0} d^3r \\
&= -\frac{\hbar^2}{\pi a_0^3} \int e^{-r/a_0} \left(-\frac{2}{a_0 r} + \frac{1}{a_0^2} \right) e^{-r/a_0} r^2 dr \int \sin \theta d\theta \int d\phi \\
&= -\frac{\hbar^2}{\pi a_0^3} \left(-\frac{2}{a_0} 1! \left(\frac{a_0}{2} \right)^2 + \frac{1}{a_0^2} 2! \left(\frac{a_0}{2} \right)^3 \right) \times 4\pi \\
&= -\frac{\hbar^2}{\pi a_0^3} \times -\frac{a_0}{4} \times 4\pi \\
&= \frac{\hbar^2}{a_0^2}
\end{aligned}$$

$$(\Delta p_x)^2 = \langle \hat{p}_x^2 \rangle - \langle \hat{p}_x \rangle^2 = \frac{\hbar^2}{3a_0^2} - 0 = \frac{\hbar^2}{3a_0^2}$$

$$(\Delta x)^2 (\Delta p_x)^2 = a_0^2 \times \frac{\hbar^2}{3a_0^2} = \frac{\hbar^2}{3}$$

$$\Delta x \Delta p_x = \frac{\hbar}{\sqrt{3}} \approx 0.57\hbar > \frac{\hbar}{2}$$

OS:

Fourier transform of $R_{10}(x)$

$$\begin{aligned}
\phi(k) &= \frac{1}{(2\pi)^{3/2}} \sqrt{\frac{1}{\pi a_0^3}} \int e^{-r/a_0} e^{-i\vec{k} \cdot \vec{r}} r^2 \sin \theta dr d\theta d\phi \\
&= \frac{2\pi}{(2\pi a_0)^{3/2}} \sqrt{\frac{1}{\pi}} \int e^{-r/a_0} e^{-ikr \cos \theta} r^2 \sin \theta dr d\theta
\end{aligned}$$

Since

$$\int_0^\pi e^{-ikr \cos \theta} \sin \theta d\theta = \frac{1}{ikr} e^{-ikr \cos \theta} \Big|_0^\pi = \frac{1}{ikr} (e^{ikr} - e^{-ikr})$$

thus, we obtain

$$\begin{aligned}
\phi(k) &= \frac{1}{i\pi\sqrt{2}a_0^{3/2}k} \int_0^\infty e^{-r/a_0} \frac{1}{r} (e^{ikr} - e^{-ikr}) r^2 dr \\
&= \frac{1}{i\pi\sqrt{2}a_0^{3/2}k} \int_0^\infty (re^{(ik-1/a_0)r} - re^{-(ik+1/a_0)r}) dr \\
&= \frac{1}{i\pi\sqrt{2}a_0^{3/2}k} \left[\frac{1}{(1/a_0 - ik)^2} - \frac{1}{(1/a_0 + ik)^2} \right] \\
&= \frac{1}{i\pi\sqrt{2}a_0^{3/2}k} \frac{2(2ik/a_0)}{\left[(1/a_0)^2 + k^2\right]^2} \\
&= \frac{2\sqrt{2}a_0^{3/2}}{\pi} \frac{1}{(1 + k^2a_0^2)^2} \\
\langle \hat{p}_x \rangle &= \frac{8a_0^3}{\pi^2} \int \frac{1}{(1 + k^2a_0^2)^2} \cdot \hbar k \cdot \frac{1}{(1 + k^2a_0^2)^2} k^2 dk \sin\theta d\theta d\phi \\
&= \frac{8a_0^3\hbar}{\pi^2} \int_0^\infty \frac{1}{(1 + k^2a_0^2)^4} k^2 dk \int_0^\pi \sin\theta \sin\theta d\theta \int_0^{2\pi} \cos\phi d\phi \\
&= \frac{8a_0^3\hbar}{\pi^2} \int_0^\infty \frac{k^3}{(1 + k^2a_0^2)^4} dr \int_0^\pi \sin^2\theta d\theta \underbrace{\int_0^{2\pi} d(\sin\phi)}_{=0} \\
&= 0 \\
\langle \hat{p}^2 \rangle &= \frac{32a_0^3}{\pi} \int_0^\infty \frac{\hbar^2 k^2}{(1 + k^2a_0^2)^4} k^2 dk = \frac{32\hbar^2}{\pi a_0^2} \int_0^\infty \frac{u^4 du}{(1 + u^2)^4}
\end{aligned}$$

Since

$$\begin{aligned}
\int_0^\infty \frac{u^4}{(1 + u^2)^4} du &= \frac{1}{32} \int_0^\infty \frac{1}{1 + u^2} du = \frac{\tan^{-1} u}{32} \Big|_0^\infty = \frac{\pi}{32} \\
\Rightarrow \langle \hat{p}^2 \rangle &= \frac{32\hbar^2}{\pi a_0^2} \times \frac{\pi}{32} = \frac{\hbar^2}{a_0^2}
\end{aligned}$$

As a particle is confined in a Coulomb potential, so

$$\Delta r = \Delta x = a_0 = \frac{4\pi\epsilon_0\hbar^2}{e^2\mu}$$

The uncertainty for momentum is,

$$\Delta p = \frac{\hbar}{a_0} = \frac{e^2\mu}{4\pi\epsilon_0\hbar}$$

The minimum energy of the system is

$$\begin{aligned}
E_{\min} &= \frac{(\Delta p)^2}{2\mu} - \frac{e^2}{4\pi\epsilon_0\Delta r} \\
&= \left(\frac{e^2}{4\pi\epsilon_0\hbar}\right)^2 \frac{\mu^2}{2\mu} - \frac{e^2}{4\pi\epsilon_0} \frac{e^2\mu}{4\pi\epsilon_0\hbar^2} \\
&= \left(\frac{1}{2} - 1\right) \left(\frac{e^2}{4\pi\epsilon_0\hbar}\right)^2 \mu \\
&= -\frac{1}{2} \left(\frac{e^2}{4\pi\epsilon_0\hbar}\right)^2 \mu = E_{100}
\end{aligned}$$

Alternative approach:

For a hydrogen atom, the uncertainty in momentum is

$$(\Delta p)^2 = \frac{\hbar^2}{(\Delta r)^2}$$

The minimum energy of the system is

$$E_{\min} = \frac{(\Delta p)^2}{2\mu} - \frac{e^2}{4\pi\epsilon_0\Delta r} = \frac{\hbar^2}{2\mu(\Delta r)^2} - \frac{e^2}{4\pi\epsilon_0\Delta r}$$

Since $\frac{dE_{\min}}{d(\Delta r)} = 0$

$$-\frac{\hbar^2}{\mu(\Delta r)^3} + \frac{e^2}{4\pi\epsilon_0(\Delta r)^2} = 0$$

$$\Rightarrow \Delta r = \frac{4\pi\epsilon_0\hbar^2}{e^2\mu}$$

$$\Rightarrow E_{\min} = -\frac{e^4\mu}{32\pi^2\epsilon_0^2\hbar^2} = -\frac{1}{2} \left(\frac{e^2}{4\pi\epsilon_0\hbar}\right)^2 \mu = E_{100} > V(0) = -\infty$$

The lowest energy $E_{\min}$ is larger than $-\infty$ and called the zero-point energy. The zero point energy is the smallest possible energy a physical system can possess, which is in accordance with the uncertainty principle. The zero-point energy gives a basic explanation of the stability of the ground state of the hydrogen atom.

$\underbrace{\Delta x \Delta p_x > \frac{\hbar}{2}}_{\text{uncertainty principle}} \Leftrightarrow \underbrace{E_{\min} = E_{100} = -13.6 \text{ eV} > V(0)}_{\text{zero point energy}}$
--

EXAMPLES:

1. Using Heisenberg uncertainty principle to prove that there are no

electrons in the nucleus.

ANSWER:

$$\Delta p = \frac{\hbar}{\Delta x}$$

$$\Delta E = \frac{(\Delta p)^2}{2\mu} = \frac{\hbar^2}{2\mu(\Delta x)^2} = \frac{(\hbar c)^2}{2\mu c^2(\Delta x)^2}$$

If electrons are in the nucleus, $\Delta x \approx 10^{-14}$ (m)

$$\Delta E = \frac{(197.327)^2}{2 \times 0.511 \times 10^2} \approx 381 \text{ (MeV)} \gg E_{\text{exp}} \approx 1 \text{ (MeV)}$$

If electrons are in the atom, $\Delta x \approx 10^{-10}$ (m)

$$\Delta E = \frac{(197.327)^2}{2 \times 0.511 \times (10^5)^2} \approx 3.81 \text{ (eV)} \approx E_{200}$$

OS:

In 1899, scientist found that β -particles are electrons with energy around 1 MeV. In 1911, Rutherford proposed his atomic nuclear model where the size of the nucleus is 10^{-14} m. Many physicist thought that electrons and protons exist inside the nucleus to explain the β -particle. When N. Bohr proposed his atomic model, not all electrons stay inside the nucleus, but part of electrons reside outside the nucleus, i.e., Bohr's orbits.

(4) $\Rightarrow$ All the binding potentials have $\langle \hat{p} \rangle = 0$.

$\Rightarrow$ All the binding potentials have a zero-point energy, i.e., the lowest allowed values of total energy lies above the minimum value of the potential energy.

D. EXCITED STATES PROPERTIES

(1) The first excited state $n = 2, \ell = 0, 1$

$$\begin{cases} \ell = 0, & m = 0 \\ \ell = 1, & m = -1, 0, 1 \end{cases}$$

$$E_{200} = -\frac{13.6}{2^2} = -3.4 = E_{211} = E_{210} = E_{21-1}$$

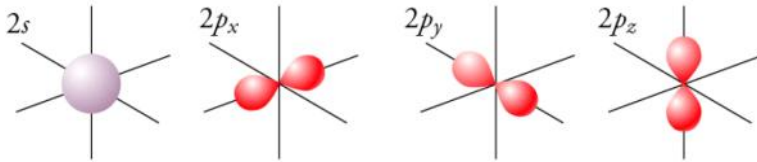
Since

$$\hat{H}\psi_{200} = E_{200}\psi_{200}, \quad \psi_{200} = \frac{1}{2\sqrt{2\pi a_0^3}} \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0}$$

$$\hat{\mathcal{H}}\psi_{211} = E_{211}\psi_{211}, \quad \psi_{211} = \frac{1}{8\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta e^{i\phi}$$

$$\hat{\mathcal{H}}\psi_{210} = E_{210}\psi_{210}, \quad \psi_{210} = \frac{1}{4\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta$$

$$\hat{\mathcal{H}}\psi_{21-1} = E_{21-1}\psi_{21-1}, \quad \psi_{21-1} = \frac{1}{8\sqrt{2\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos\theta e^{-i\phi}$$



$$\psi_{200} \neq \psi_{211} \neq \psi_{210} \neq \psi_{21-1}$$

$\Rightarrow$ There are 4 states with the same energy and these states are called degenerate states.

(2) The general expression of the expectation value of $\mathcal{O}(r)$

$$\langle r \rangle = \frac{a_0}{2} (3n^2 - \ell(\ell + 1))$$

$$\langle r^2 \rangle = \frac{a_0^2 n^2}{2} (5n^2 + 1 - 3\ell(\ell + 1))$$

—Chapter 4—

Heisenberg Matrix Mechanics

4-1 The Matrix

A. OBSERVABLE AND MATRIX

- (1) Heisenberg: all observable quantities are always related to a pair of Bohr orbits, but never just a single one.

n\m	1	2	3	4
1	0	1	0	0
2	1	0	$\sqrt{2}$	0
3	0	$\sqrt{2}$	0	$\sqrt{3}$
4	0	0	$\sqrt{3}$	0

- (2) Heisenberg conjectured that each observable, such as x and p in classical mechanics, corresponds to an infinite matrix.

$$x \rightarrow \mathbf{X} = \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$p \rightarrow \mathbf{P} = i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & -1 & 0 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The elements of these matrices are the transition probability between two Bohr orbits.

B. MATRIX ALGEBRA

- (1) Vector

1. column vector

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix}$$

2. row vector

$$\mathbf{c}^\dagger = (c_1^* \quad c_2^* \quad c_3^* \quad \cdots)$$

3. Inner product (expectation)

$$\mathbf{c}^\dagger \cdot \mathbf{c} = (c_1^* \quad c_2^* \quad c_3^* \quad \cdots) \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \sum_n c_n^* c_n = c\text{-number}$$

(2) Matrix and eigenvalue equation

A product of a column vector $\mathbf{c}$ and a row vector $\mathbf{c}^\dagger$ produces a matrix, which is called outer product:

$$\mathbf{c}\mathbf{c}^\dagger = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} (c_1^* \quad c_2^* \quad c_3^* \quad \cdots) = \begin{pmatrix} c_1 c_1^* & c_1 c_2^* & c_1 c_3^* & \cdots \\ c_2 c_1^* & c_2 c_2^* & c_2 c_3^* & \cdots \\ c_3 c_1^* & c_3 c_2^* & c_3 c_3^* & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \mathbf{A}$$

The eigenvalue equation of a matrix:

$$\mathbf{A} \cdot \mathbf{c} = \lambda \cdot \mathbf{c}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ a_{31} & a_{32} & a_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \lambda \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix}$$

where $\mathbf{c}$ is called the eigenvector and λ is called the eigenvalue.

(3) Commutator

$$[\mathbf{X}, \mathbf{P}] = \mathbf{X}\mathbf{P} - \mathbf{P}\mathbf{X}$$

$$\begin{aligned} &= \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & -1 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &\quad - i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & -1 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & \cdots \\ 0 & \sqrt{2} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= i \frac{\hbar}{2} \left[\begin{pmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} - \begin{pmatrix} -1 & 0 & 0 & \cdots \\ 0 & -1 & 0 & \cdots \\ 0 & 0 & -1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \right] \\ &= i \frac{\hbar}{2} \begin{pmatrix} 2 & 0 & 0 & \cdots \\ 0 & 2 & 0 & \cdots \\ 0 & 0 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= i\hbar \mathbf{1} \end{aligned}$$

C. EIGENVALUE EQUATION OF A HERMITIAN MATRIX

- (1) Hermitian matrix (the matrix is its own complex conjugate transpose),
 $\mathbf{A}^\dagger = \mathbf{A}$

EXAMPLES:

1. The position matrix $\mathbf{X}$

$$\mathbf{X}^\dagger = \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots \\ 1 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & \sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} = \mathbf{X}$$

2. The momentum matrix $\mathbf{P}$

$$\begin{aligned} \mathbf{P}^\dagger &= -i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots \\ -1 & 0 & \sqrt{2} & 0 & \cdots \\ 0 & -\sqrt{2} & 0 & \sqrt{3} & \cdots \\ 0 & 0 & -\sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= i \sqrt{\frac{m\hbar\omega}{2}} \begin{pmatrix} 0 & -1 & 0 & 0 & \cdots \\ 1 & 0 & -\sqrt{2} & 0 & \cdots \\ 0 & \sqrt{2} & 0 & -\sqrt{3} & \cdots \\ 0 & 0 & \sqrt{3} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \\ &= \mathbf{P} \end{aligned}$$

- (2) The eigenvalue equation of a Hermitian matrix

$$\mathbf{A} \cdot \mathbf{u}_n = \lambda_n \mathbf{u}_n$$

1. Take expectation on both sides

$$\mathbf{u}_n^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n = \lambda_n \mathbf{u}_n^\dagger \cdot \mathbf{u}_n = \lambda_n$$

Take complex conjugate transpose on both sides

$$(\mathbf{u}_n^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n)^\dagger = \lambda_n^*$$

Since

$$(\mathbf{u}_n^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n)^\dagger = \mathbf{u}_n^\dagger \cdot \mathbf{A}^\dagger \cdot \mathbf{u}_n = \mathbf{u}_n^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n = \lambda_n$$

thus **the eigenvalues are real**:

$$\lambda_n^* = \lambda_n$$

2. Take expectation on both sides for $k \neq n$

$$\mathbf{u}_k^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n = \lambda_n \mathbf{u}_k^\dagger \cdot \mathbf{u}_n$$

Take complex conjugate transpose on both sides

$$\left(\mathbf{u}_k^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n\right)^\dagger = \lambda_n^* \left(\mathbf{u}_k^\dagger \cdot \mathbf{u}_n\right)^\dagger = \lambda_n \mathbf{u}_n^\dagger \cdot \mathbf{u}_k \cdots \cdots \quad (\text{a})$$

Since

$$\left(\mathbf{u}_k^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_n\right)^\dagger = \mathbf{u}_n^\dagger \cdot \mathbf{A}^\dagger \cdot \mathbf{u}_k = \mathbf{u}_n^\dagger \cdot \mathbf{A} \cdot \mathbf{u}_k = \lambda_k \mathbf{u}_n^\dagger \cdot \mathbf{u}_k \cdots \cdots \quad (\text{b})$$

(a) - (b) gives

$$(\lambda_n - \lambda_k) \mathbf{u}_n^\dagger \cdot \mathbf{u}_k = 0$$

thus **the eigenvectors are orthonormal**:

$$\mathbf{u}_n^\dagger \cdot \mathbf{u}_k = \delta_{nk}, \quad \delta_{nk} = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$

- (3) Since $\mathbf{u}_n$ are orthonormal, i.e., $\mathbf{u}_n$ are linear independent, any vector $\mathbf{c}$ in the system can be written as a linear combination of eigenvectors $\mathbf{u}_n$, i.e.,

$$\mathbf{c} = \sum_n c_n \mathbf{u}_n, \text{ where } c_n = \mathbf{u}_n^\dagger \cdot \mathbf{c} = \sum_m c_m \underbrace{\mathbf{u}_n^\dagger \cdot \mathbf{u}_m}_{=\delta_{nm}}$$

Since

$$\begin{aligned} \mathbf{c}^\dagger \cdot \mathbf{c} &= \sum_m c_m^* \mathbf{u}_m^\dagger \cdot \sum_n c_n \mathbf{u}_n \\ &= \sum_{mn} c_m^* c_n \mathbf{u}_m^\dagger \cdot \mathbf{u}_n \\ &= \sum_{mn} c_m^* c_n \delta_{mn} \\ &= \sum_n |c_n|^2 \\ &= 1 \end{aligned}$$

$|c_n|^2$ is the probability of finding a particle associated with the eigenvalue λ_n .

EXAMPLES:

1. Consider a 2×2 Hermitian matrix

$$\mathbf{A} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The secular equation is

$$|\mathbf{A} - \lambda_n \mathbf{1}| = 0 \Rightarrow \begin{vmatrix} -\lambda_n & -i \\ i & -\lambda_n \end{vmatrix} = 0 \Rightarrow \lambda_n^2 - 1 = 0$$

$$\lambda_n = \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \dots \text{eigenvalues}$$

From the eigenvalue equation, we have

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \lambda_n \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\text{For } \lambda_1 = 1, \quad \begin{cases} -ic_2 = c_1 \\ ic_1 = c_2 \end{cases}$$

Using the normalization condition:

$$|c_1|^2 + |c_2|^2 = |c_1|^2 [1 + 1] = 1$$

$$\Rightarrow c_1 = \frac{1}{\sqrt{2}}, \quad c_2 = \frac{i}{\sqrt{2}}$$

$$\Rightarrow \mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \dots \text{eigenvector}$$

Similarly,

$$\text{For } \lambda_2 = -1, \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \dots \text{eigenvector}$$

2. Consider a 3×3 Hermitian matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

The secular equation is

$$|\mathbf{A} - \lambda_n \mathbf{1}| = 0 \Rightarrow \begin{vmatrix} -\lambda_n & 1 & 0 \\ 1 & -\lambda_n & 1 \\ 0 & 1 & -\lambda_n \end{vmatrix} = 0$$

$$\Rightarrow -\lambda_n(\lambda_n^2 - 1) - (-\lambda_n) = \lambda_n(2 - \lambda_n^2) = 0$$

$$\Rightarrow \lambda_n = \begin{Bmatrix} \sqrt{2} \\ 0 \\ -\sqrt{2} \end{Bmatrix} \dots \text{eigenvalues}$$

From the eigenvalue equation, we have

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \lambda_n \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\text{For } \lambda_1 = \sqrt{2} \Rightarrow \begin{cases} c_2 = \sqrt{2}c_1 \\ c_1 + c_3 = \sqrt{2}c_2 \\ c_2 = \sqrt{2}c_3 \end{cases}$$

Using the normalization condition:

$$\begin{aligned} |c_1|^2 + |c_2|^2 + |c_3|^2 &= |c_1|^2 [1 + 2 + 1] = 1 \\ \Rightarrow c_1 &= \frac{1}{2}, \quad c_2 = \frac{1}{\sqrt{2}}, \quad c_3 = \frac{1}{2} \\ \Rightarrow \mathbf{u}_1 &= \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} \cdots \text{eigenvector} \end{aligned}$$

Similarly,

$$\text{For } \lambda_2 = 0, \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdots \text{eigenvector}$$

$$\text{For } \lambda_3 = -\sqrt{2}, \quad \mathbf{u}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} \cdots \text{eigenvector}$$

4-2 Linear Algebra & Dirac Notation

A. LINEAR ALGEBRA OF SCHRÖDINGER WAVE MECHANICS

(1) Time-independent Schrödinger equation

$$\underbrace{\left[\frac{\hat{p}^2}{2m} + V(x) \right]}_{=\hat{\mathcal{H}}} \psi_n(x) = E_n \psi_n(x)$$

where $\hat{\mathcal{H}}$ is a Hermitian differential operator,

$$\int \left(\hat{\mathcal{H}} \psi_n(x) \right)^* \psi'_n(x) dx = \int \psi_n^*(x) \left(\hat{\mathcal{H}} \psi'_n(x) \right) dx$$

and ψ_n (eigenfunctions) are orthonormal,

$$\int \psi_m^*(x) \psi_n(x) dx = \delta_{mn}$$

Any wavefunction can be written as a linear combination of eigenfunctions $\psi_n(x)$, i.e.,

$$\psi(x) = \sum_n c_n \psi_n(x), \text{ where } c_n = \int \psi_n^*(x) \psi(x) dx$$

Born rule and Heisenberg's uncertainty principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

(2) Abstract of Schrödinger equations

$$\hat{A} \psi_n(x) = \lambda_n \psi_n(x) \cdots \cdots \text{eigenvalue equation}$$

1. Since ψ_n are orthonormal, ψ_n are linear independent, i.e., $\psi_n(x)$ form a basis of a vector space $\mathbb{V}$.

$$\boxed{\begin{matrix} \mathbb{V} \\ \{\psi_n(x)\} \end{matrix}}$$

2. The complex conjugate of ψ_n also form a basis of a vector space $\tilde{\mathbb{V}}$, called dual space. (The dual space is defined automatically for any vector space.)

$$\boxed{\begin{matrix} \tilde{\mathbb{V}} \\ \{\psi_n^*(x)\} \end{matrix}}$$

$$\boxed{\{\psi_n^*(x)\} \quad \tilde{V}}$$

Note: $\psi_n^*(x)$ and $\psi_n(x)$ are independent.

Abstract of uncertainty principle

The Hermitian operators $\hat{x}$ and $\hat{p}$ satisfy the commutation relation,

$$[\hat{x}, \hat{p}] = i\hbar$$

(3) Hilbert space of quantum system

1. Inner product (expectation)

(A particular mapping between vector space and dual space and is not automatic.)

$$\int_{-\infty}^{\infty} \psi^*(x) \psi'(x) dx = c\text{-number}$$

$$\left(\int_{-\infty}^{\infty} \psi^*(x) \psi'(x) dx \right)^* = \int_{-\infty}^{\infty} \psi'^*(x) \psi(x) dx = c\text{-number}$$

2. Completeness relation

Since

$$\begin{aligned} \psi(x) &= \sum_n c_n \psi_n(x) \\ &= \sum_n \int \psi_n^*(x') \psi(x') dx' \psi_n(x) \\ &= \int \underbrace{\sum_n \psi_n(x) \psi_n^*(x')}_{=\delta(x-x')} \psi(x') dx' \end{aligned}$$

we obtain

$$\sum_n \psi_n(x) \psi_n^*(x') = \delta(x - x') \dots \text{completeness relation}$$

A completeness relation for a set of eigenfunctions is the identity assuring that there is no basis function "missing". Thus the basis spans the entire Hilbert space such that every wavefunction in the Hilbert space can be expressed as a linear combination of this basis.

$\Rightarrow$ If the vector space is equipped with an inner product which associates a complex number, called Hilbert space $\mathbb{H}_S$.

$$\boxed{\begin{matrix} \mathbb{H}_S \\ \{\psi_n(x)\} \end{matrix}}$$

$$[\hat{x}, \hat{p}] = i\hbar$$

B. LINEAR ALGEBRA OF HEISENBERG MATRIX MECHANICS

- (1) Eigenvalue equation for an observable

$$\mathbf{H} \cdot \mathbf{u}_n = E_n \mathbf{u}_n$$

where $\mathbf{H}$ is a Hermitian matrix,

$$\mathbf{H}^\dagger = \mathbf{H}$$

and $\mathbf{u}_n$ (eigenvectors) are orthonormal,

$$\mathbf{u}_m^\dagger \cdot \mathbf{u}_n = \delta_{mn}$$

Any vector can be written as a linear combination of eigenvectors $\mathbf{u}_n$, i.e.,

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \sum_n c_n \mathbf{u}_n, \text{ where } c_n = \mathbf{u}_n^\dagger(x) \cdot \mathbf{c}$$

Born rule and Heisenberg's uncertainty principle

$$[\mathbf{X}, \mathbf{P}] = \mathbf{XP} - \mathbf{PX} = i\hbar \mathbf{1}$$

- (2) Abstract of eigenvalue equations

$$\mathbf{A} \cdot \mathbf{u}_n = \lambda_n \mathbf{u}_n$$

1. Since $\mathbf{u}_n$ are orthonormal, $\mathbf{u}_n$ are linear independent, i.e., $\mathbf{u}_n$ form a basis of a vector space $\mathbb{V}$.

$$\boxed{\begin{matrix} \mathbb{V} \\ \{\mathbf{u}_n\} \end{matrix}}$$

2. The complex conjugate of $\mathbf{u}_n$ also form a basis of a vector space $\widehat{\mathbb{V}}$, called dual space. (The dual space is defined automatically for any vector space.)

$$\boxed{\begin{matrix} \widehat{\mathbb{V}} \\ \{\mathbf{u}_n^\dagger\} \end{matrix}}$$

Note: $\mathbf{u}_n^\dagger$ and $\mathbf{u}_n$ are independent.

Abstract of uncertainty principle

The Hermitian matrices $\mathbf{X}$ and $\mathbf{P}$ satisfy the commutation relation,

$$[\mathbf{X}, \mathbf{P}] = i\hbar \mathbf{1}$$

(3) Hilbert space of quantum system

1. Inner product (expectation)

(A particular mapping between vector space and dual space and is not automatic.)

$$\mathbf{c}^\dagger \cdot \mathbf{c} = \begin{pmatrix} c_1^* & c_2^* & c_3^* & \cdots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \sum_n c_n^* c_n = c\text{-number}$$

2. Completeness relation

Since

$$\mathbf{c} = \sum_n c_n \mathbf{u}_n = \sum_n \mathbf{u}_n^\dagger \cdot \mathbf{c} \mathbf{u}_n = \sum_n \mathbf{u}_n \mathbf{u}_n^\dagger \cdot \mathbf{c}$$

we obtain

$$\sum_n \mathbf{u}_n \mathbf{u}_n^\dagger = \mathbf{1} \cdots \cdots \text{completeness relation}$$

The eigenfunctions satisfy the completeness relation.

$\Rightarrow$ If the vector space is equipped with an inner product which associates a complex number, called Hilbert space $\mathbb{H}_{\mathcal{H}}$.

$\mathbb{H}_{\mathcal{H}}$
 $\{\mathbf{u}_n\}$

 $[\mathbf{X}, \mathbf{P}] = i\hbar \mathbf{1}$

C. ALGEBRAIC ISOMORPHIC HILBERT SPACE IN DIRAC NOTATION

(1) Algebraic isomorphic Hilbert space

Hilbert space of Schrödinger's mechanics:

uncertainty
principle
 $\Delta x \Delta p \geq \hbar/2$

abstract

$$[\hat{x}, \hat{p}] = i\hbar$$

$$\hat{\mathcal{H}}\psi_n(x) = E_n\psi_n(x) \longrightarrow \psi(x) = \sum_n c_n \psi_n(x) \xrightarrow{\text{abstract}} \boxed{\mathbb{H}_{\mathcal{S}} \{ \psi_n(x) \}}$$

Hilbert space of Heisenberg's mechanics:

$$[\mathbf{X}, \mathbf{P}] = i\hbar \mathbf{1}$$

$$\mathbf{H} \cdot \mathbf{u}_n = \lambda_n \cdot \mathbf{u}_n \longrightarrow \mathbf{c} = \sum_n c_n \mathbf{u}_n \xrightarrow{\text{abstract}} \boxed{\mathbb{H}_{\mathcal{H}} \{ \mathbf{u}_n \}}$$

Mathematically, von Neumann showed that Schrödinger's wave mechanics and Heisenberg's matrix mechanics are algebraic isomorphic and isometric realizations of the Hilbert space, i.e.,

$$[\hat{x}, \hat{p}] = i\hbar$$

$$[\mathbf{X}, \mathbf{P}] = i\hbar \mathbf{1}$$

$$\boxed{\mathbb{H}_{\mathcal{S}} \{ \psi_n(x) \}} \cong \boxed{\mathbb{H}_{\mathcal{H}} \{ \mathbf{u}_n \}}$$

both mechanics are mathematical equivalence, but in different forms.

- (2) Dirac invented a useful alternative notation for the Hilbert space which is now referred to as Dirac notation (or called bra-ket notation).

$$\boxed{\mathbb{H}_{\mathcal{D}} \{ |\psi_n\rangle \}} \quad [\hat{x}, \hat{p}] = i\hbar$$

Dirac notation enables us to do calculations in a general Hilbert space without needing to worry about the space's representation.

OS:

In general, we use it to set up a problem and then choose the most expeditious way, such as matrix algebra or differential calculus, to execute the calculation.

(3) Hilbert space in Dirac notation

1. Ket space (vector space)

$$\boxed{\{\psi_n\}}^{\mathbb{V}}$$

$|\psi_n\rangle$ are the basis of the ket space.

2. Bra space (dual space)

$$\boxed{\langle\psi_n|}^{\tilde{\mathbb{V}}}$$

Note: $\langle\psi_n| \equiv |\psi_n\rangle^\dagger$ and $|\psi_n\rangle$ are independent.

3. Inner product (expectation)

$$\langle\psi_n|\psi\rangle = \text{c-number}$$

4. Completeness relation

Since

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle \dots\dots (a)$$

where c_n is the coefficient and given by

$$\langle\psi_n|\psi\rangle = \sum_m c_m \underbrace{\langle\psi_n|\psi_m\rangle}_{=\delta_{nm}} = c_n \dots\dots (b)$$

Insert equation (b) into equation (a) to obtain

$$|\psi\rangle = \sum_n \langle\psi_n|\psi\rangle |\psi_n\rangle = \sum_n |\psi_n\rangle \langle\psi_n||\psi\rangle \dots\dots (c)$$

Thus, we have

$$\sum_n |\psi_n\rangle \langle\psi_n| = \hat{1}$$

which is also called **the resolution of identity**.

D. OPERATORS IN DIRAC NOTATION

(1) Hermitian operator (the operator is its own complex conjugate)

$$\hat{A}^\dagger = \hat{A}$$

Its eigenvalue equation is given by:

$$\hat{A}|\psi_n\rangle = \lambda_n|\psi_n\rangle$$

1. Take expectation on both sides

$$\langle\psi_n|\hat{A}|\psi_n\rangle = \lambda_n\langle\psi_n|\psi_n\rangle = \lambda_n$$

Take complex conjugate on both sides

$$\langle\psi_n|\hat{A}|\psi_n\rangle^* = \lambda_n^*$$

Since

$$\langle\psi_n|\hat{A}|\psi_n\rangle^* = \langle\psi_n|\hat{A}^\dagger|\psi_n\rangle = \langle\psi_n|\hat{A}|\psi_n\rangle = \lambda_n$$

thus **the eigenvalues are real**:

$$\lambda_n = \lambda_n^*$$

2. Take expectation on both sides for $k \neq n$

$$\langle\psi_k|\hat{A}|\psi_n\rangle = \lambda_n\langle\psi_k|\psi_n\rangle$$

Take complex conjugate on both sides

$$\langle\psi_k|\hat{A}|\psi_n\rangle^* = \lambda_n^*\langle\psi_k|\psi_n\rangle^* = \lambda_n\langle\psi_n|\psi_k\rangle \cdots \cdots \text{ (a)}$$

Since

$$\langle\psi_k|\hat{A}|\psi_n\rangle^* = \langle\psi_n|\hat{A}^\dagger|\psi_k\rangle = \langle\psi_n|\hat{A}|\psi_k\rangle = \lambda_k\langle\psi_n|\psi_k\rangle \cdots \cdots \text{ (b)}$$

(a) - (b) gives

$$(\lambda_n - \lambda_k)\langle\psi_n|\psi_k\rangle = 0$$

thus **the eigenkets are orthonormal**:

$$\langle\psi_n|\psi_k\rangle = \delta_{nk}, \quad \delta_{nk} = \begin{cases} 1, & n = k \\ 0, & n \neq k \end{cases}$$

Since $|\psi_n\rangle$ are orthonormal, i.e., $|\psi_n\rangle$ are linear independent, thus $|\psi_n\rangle$ form a basis of a Hilbert space $\mathbb{H}$.

(2) Projection operator

Define

$$\hat{P}_n = \underbrace{|\psi_n\rangle\langle\psi_n|}_{\substack{\text{outer product} \\ \text{of eigenket}}}$$

and $\hat{P}_n$ satisfies the following property:

$$\hat{P}_n^2 = |\psi_n\rangle\langle\psi_n|\underbrace{\cdot|\psi_n\rangle\langle\psi_n|}_{=1} = |\psi_n\rangle\langle\psi_n| = \hat{P}_n$$

Thus, $\hat{P}_n$ is called the projection operator. Using the resolution of identity, we obtain

$$|\psi\rangle = \underbrace{\sum_n |\psi_n\rangle\langle\psi_n|}_{\hat{1}} |\psi\rangle = \sum_n \hat{P}_n |\psi\rangle$$

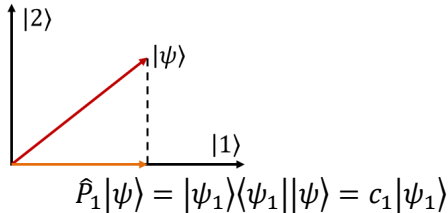
$\hat{P}_n$ acting on a ket $|\psi\rangle$ will project the ket to a eigenket $|\psi_n\rangle$ with the

coefficient $\langle \psi_n | \psi \rangle$, i.e.,

$$\hat{P}_n |\psi\rangle = |\psi_n\rangle \underbrace{\langle \psi_n | \psi \rangle}_{=c_n} = c_n |\psi_n\rangle$$

EXAMPLES:

1. The projection of a ket $|\psi\rangle$ onto the eigenket $|1\rangle$



- (3) Commutator

The Hermitian operators $\hat{x}$ and $\hat{p}$ satisfy the commutation relation,

$$[\hat{x}, \hat{p}] = i\hbar$$

E. REPRESENTATION OF KET SPACE

- (1) Dirac notation represented in **matrix**

1. We shall then express the ket $|\psi\rangle$ as a column vector and bra $\langle \psi|$ as a row vector.

$$|\psi\rangle \rightarrow \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix}$$

$$\langle \psi| \rightarrow \mathbf{c}^\dagger = (c_1^* \quad c_2^* \quad c_3^* \quad \dots)$$

Inner product: a product of a bra $\langle \psi|$ and a ket $|\psi\rangle$ produces a value.

$$\langle \psi | \psi \rangle = \mathbf{c}^\dagger \cdot \mathbf{c} = (c_1^* \quad c_2^* \quad c_3^* \quad \dots) \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{pmatrix} = \sum_n c_n^* c_n$$

2. Expand the operator $\hat{A}$ in terms of *countable* eigenkets

$$\begin{aligned}
\hat{A} &= \underbrace{\sum_m |\psi_m\rangle\langle\psi_m|}_{\hat{1}} \cdot \hat{A} \cdot \underbrace{\sum_n |\psi_n\rangle\langle\psi_n|}_{\hat{1}} \\
&= \sum_{mn} \underbrace{\langle\psi_m|\hat{A}|\psi_n\rangle}_{=a_{mn}} |\psi_m\rangle\langle\psi_n| \\
&= \sum_{mn} a_{mn} |\psi_m\rangle\langle\psi_n|
\end{aligned}$$

Since $|\psi_m\rangle\langle\psi_n|$ is an outer product (a product of a ket $|\psi\rangle$ and a bra $\langle\psi|$ produces a matrix), for example,

$$\begin{aligned}
|\psi_2\rangle\langle\psi_3| &\rightarrow \mathbf{u}_2 \mathbf{u}_3^\dagger = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 & \dots \end{pmatrix} \\
&= \begin{pmatrix} 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}
\end{aligned}$$

thus, we obtain

$$\hat{A} \rightarrow \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{1n} & \dots \\ a_{21} & a_{22} & a_{2n} & \dots \\ a_{m1} & a_{m2} & a_{33} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

EXAMPLES:

1. Consider a ket as

$$|\psi\rangle = -i\frac{4}{5}|0\rangle + \frac{3}{5}|1\rangle$$

Find the inner product $\langle\psi|\psi\rangle$ and outer product $|\psi\rangle\langle\psi|$

ANSWER:

In matrix representation, we choose unit vectors as a basis:

$$\begin{aligned}
|0\rangle &\rightarrow \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
|1\rangle &\rightarrow \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}
\end{aligned}$$

Thus, we have

$$|\psi\rangle \rightarrow \boldsymbol{\Psi} = \begin{pmatrix} -i\frac{4}{5} \\ \frac{3}{5} \end{pmatrix}$$

Inner Product (a product of a bra and ket):

$$\langle\psi|\psi\rangle \rightarrow \Psi^\dagger \cdot \Psi = \begin{pmatrix} i\frac{4}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} -i\frac{4}{5} \\ \frac{3}{5} \end{pmatrix} = \frac{16}{25} + \frac{9}{25} = 1$$

Outer Product (a product of a ket and bra):

$$|\psi\rangle\langle\psi| \rightarrow \Psi\Psi^\dagger = \begin{pmatrix} -i\frac{4}{5} \\ \frac{3}{5} \end{pmatrix} \begin{pmatrix} i\frac{4}{5} & \frac{3}{5} \end{pmatrix} = \begin{pmatrix} \frac{16}{25} & -i\frac{12}{25} \\ i\frac{12}{25} & \frac{9}{25} \end{pmatrix}$$

2. An operator $\hat{A}$ acting on a system with a basis including two independent kets $\{|\psi_1\rangle, |\psi_2\rangle\}$

$$\hat{A}|\psi_1\rangle = i|\psi_2\rangle$$

$$\hat{A}|\psi_2\rangle = -i|\psi_1\rangle$$

Expand $\hat{A}$ in terms of eigenkets and express $\hat{A}$ as a matrix

ANSWER:

- Expand $\hat{A}$ in terms of eigenkets:

$$\hat{A} = \sum_{mn} a_{mn} |\psi_m\rangle\langle\psi_n|$$

where

$$a_{11} = \langle\psi_1|\hat{A}|\psi_1\rangle = i\langle\psi_1|\psi_2\rangle = 0$$

$$a_{22} = \langle\psi_2|\hat{A}|\psi_2\rangle = -i\langle\psi_2|\psi_1\rangle = 0$$

$$a_{12} = \langle\psi_1|\hat{A}|\psi_2\rangle = -i\langle\psi_1|\psi_1\rangle = -i$$

$$a_{21} = \langle\psi_2|\hat{A}|\psi_1\rangle = i\langle\psi_2|\psi_2\rangle = i$$

Thus, we obtain

$$\hat{A} = -i|\psi_1\rangle\langle\psi_2| + i|\psi_2\rangle\langle\psi_1|$$

- Express $\hat{A}$ as a matrix:

$$\mathbf{A} = \begin{matrix} & \begin{matrix} |\psi_1\rangle & |\psi_2\rangle \end{matrix} \\ \begin{matrix} \langle\psi_1| \\ \langle\psi_2| \end{matrix} & \begin{pmatrix} \langle\psi_1|\hat{A}|\psi_1\rangle & \langle\psi_1|\hat{A}|\psi_2\rangle \\ \langle\psi_2|\hat{A}|\psi_1\rangle & \langle\psi_2|\hat{A}|\psi_2\rangle \end{pmatrix} \end{matrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

(2) Dirac notation represented in **wavefunction**

- The resolution of identity

Linear combination of a uncountable basis $|\mathbf{x}\rangle$

$$|\psi\rangle = \int_{-\infty}^{\infty} \psi(x)|x\rangle dx \cdots (a)$$

where $\psi(x)$ is the coefficient and given by

$$\langle x|\psi\rangle = \int_{-\infty}^{\infty} \psi(x') \underbrace{\langle x|x'\rangle}_{=\delta(x-x')} dx' = \psi(x) \cdots (b)$$

Insert equation (b) into equation (a) to obtain

$$|\psi\rangle = \int_{-\infty}^{\infty} \langle x|\psi\rangle |x\rangle dx = \int_{-\infty}^{\infty} |x\rangle \langle x| |\psi\rangle dx \cdots (c)$$

Thus, we have

$$\int_{-\infty}^{\infty} |x\rangle \langle x| dx = \hat{1}$$

which is called the resolution of identity.

2. Projection operator

Define

$$\hat{P}(x) = |x\rangle \langle x|$$

From equation (c), we have

$$|\psi\rangle = \int \hat{P}(x) |\psi\rangle dx$$

$\hat{P}(x)$ acting on a ket $|\psi\rangle$ will project the ket to a eigenket $|x\rangle$ with the coefficient $\langle x|\psi\rangle$, i.e.,

$$\hat{P}(x) |\psi\rangle = |x\rangle \underbrace{\langle x|\psi\rangle}_{=\psi(x)} = \psi(x) |x\rangle$$

3. Express the operator as a differential operator

Position operator:

$$\begin{aligned} \langle x|\hat{x}|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|\hat{x}|x'\rangle \langle x'|\psi\rangle dx' \\ &= \int_{-\infty}^{\infty} x' \underbrace{\langle x|x'\rangle}_{=\delta(x-x')} \psi(x') dx' \\ &= x\psi(x) \\ \Rightarrow \langle x|\hat{x}|\psi\rangle &= \hat{x}\langle x|\psi\rangle = x\psi(x) \end{aligned}$$

Momentum operator:

$$\begin{aligned}
\langle x|\hat{p}|\psi\rangle &= \langle x|\hat{p} \underbrace{\int_{-\infty}^{\infty} |p'\rangle\langle p'| dp'}_{\hat{1}} |\psi\rangle \\
&= \int_{-\infty}^{\infty} \langle x|\hat{p}|p'\rangle\langle p'|\psi\rangle dp' \\
&= \int_{-\infty}^{\infty} p' \langle x|p'\rangle\langle p'|\psi\rangle dp'
\end{aligned}$$

Since

$$\langle x|x'\rangle = \int_{-\infty}^{\infty} \langle x|p'\rangle\langle p'|x'\rangle dp'$$

and

$$\begin{aligned}
\langle x|x'\rangle &= \delta(x' - x) \\
&= \int_{-\infty}^{\infty} e^{-ip'(x'-x)/\hbar} \frac{dp'}{2\pi\hbar} \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\hbar}} e^{ip'x/\hbar} \frac{1}{\sqrt{2\pi\hbar}} e^{-ip'x'/\hbar} dp' \\
\Rightarrow \langle x|p'\rangle &= \frac{1}{\sqrt{2\pi\hbar}} e^{ip'x/\hbar} \\
\langle x|\hat{p}|\psi\rangle &= \int_{-\infty}^{\infty} p' \frac{1}{\sqrt{2\pi\hbar}} e^{ip'x/\hbar} \langle p'|\psi\rangle dp' \\
&= \int_{-\infty}^{\infty} -i\hbar \frac{\partial}{\partial x} \frac{1}{\sqrt{2\pi\hbar}} e^{ip'x/\hbar} \langle p'|\psi\rangle dp' \\
&= \int_{-\infty}^{\infty} -i\hbar \frac{\partial}{\partial x} \langle x|p'\rangle\langle p'|\psi\rangle dp' \\
&= -i\hbar \frac{\partial}{\partial x} \langle x|\psi\rangle \\
&= \hat{p}\psi(x) \\
\Rightarrow \langle x|\hat{p}|\psi\rangle &= \hat{p}\langle x|\psi\rangle = -i\hbar \frac{\partial}{\partial x} \psi(x)
\end{aligned}$$

EXAMPLES:

1. The commutation relation of $\hat{x}$ and $\hat{p}$

$$\begin{aligned}
\langle x | [\hat{x}, \hat{p}] | \psi \rangle &= \langle x | \hat{x} \hat{p} - \hat{p} \hat{x} | \psi \rangle \\
&= \langle x | \hat{x} \hat{p} | \psi \rangle - \langle x | \hat{p} \hat{x} | \psi \rangle \\
&= \int_{-\infty}^{\infty} \langle x | \hat{x} | x' \rangle \langle x' | \hat{p} | \psi \rangle dx' \\
&\quad - \int_{-\infty}^{\infty} \langle x | \hat{p} | x' \rangle \langle x' | \hat{x} | \psi \rangle dx' \\
&= \int_{-\infty}^{\infty} \hat{x} \langle x | x' \rangle \hat{p} \langle x' | \psi \rangle dx' \\
&\quad - \int_{-\infty}^{\infty} \hat{p} \langle x | x' \rangle \hat{x} \langle x' | \psi \rangle dx' \\
&= \int_{-\infty}^{\infty} \hat{x} \delta(x - x') \hat{p} \psi(x') dx' \\
&\quad - \int_{-\infty}^{\infty} \hat{p} \delta(x - x') \hat{x} \psi(x') dx' \\
&= [\hat{x} \hat{p} - \hat{p} \hat{x}] \psi(x) \\
&= i\hbar \langle x | \psi \rangle \\
\Rightarrow [\hat{x}, \hat{p}] &= \hat{x} \hat{p} - \hat{p} \hat{x} = i\hbar
\end{aligned}$$

F. UNITARY TRANSFORMATION

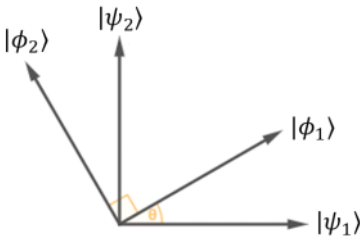
- (1) For a Hilbert space $\mathbb{H}$, there are many possible sets of basis as

$$\boxed{\begin{array}{c} \mathbb{H} \\ \{|\psi_n\rangle\} \end{array}} = \boxed{\begin{array}{c} \mathbb{H} \\ \{|\psi'_n\rangle\} \end{array}} = \boxed{\begin{array}{c} \mathbb{H} \\ \{|\phi_n\rangle\} \end{array}}$$

Between any two such sets of basis, there exists a linear transformation that preserves inner product (and hence length), known as a **unitary transformation**.

- (2) A unitary transformation is equivalent to a change of basis

Consider two sets of basis $\{|\psi_n\rangle\}$ and $\{|\phi_n\rangle\}$,



where the relationships between two bases are

$$\begin{aligned} |\phi_1\rangle &= a_{11}|\psi_1\rangle + a_{21}|\psi_2\rangle + \cdots + a_{n1}|\psi_n\rangle \\ |\phi_2\rangle &= a_{12}|\psi_1\rangle + a_{22}|\psi_2\rangle + \cdots + a_{n2}|\psi_n\rangle \\ &\vdots \\ |\phi_n\rangle &= a_{1n}|\psi_1\rangle + a_{2n}|\psi_2\rangle + \cdots + a_{nn}|\psi_n\rangle \end{aligned} \quad \text{or} \quad |\phi_n\rangle = \sum_m a_{mn} |\psi_m\rangle$$

We have, in the matrix form,

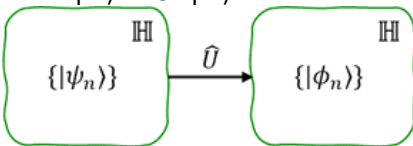
$$\begin{pmatrix} |\phi_1\rangle \\ |\phi_2\rangle \\ \vdots \\ |\phi_n\rangle \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \\ \vdots \\ |\psi_n\rangle \end{pmatrix}$$

We then write the relationships in the Dirac notation:

$$\begin{pmatrix} |\phi_1\rangle \\ |\phi_2\rangle \\ \vdots \\ |\phi_n\rangle \end{pmatrix} \rightarrow |\Phi\rangle, \quad \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \\ \vdots \\ |\psi_n\rangle \end{pmatrix} \rightarrow |\Psi\rangle, \quad \text{and} \quad \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix} \rightarrow \hat{U}^\dagger$$

and obtain

$$|\Phi\rangle = \hat{U}^\dagger |\Psi\rangle$$



Since

$$\langle \Phi | \Phi \rangle = \langle \Psi | \hat{U} \hat{U}^\dagger | \Psi \rangle = \langle \Psi | \Psi \rangle$$

we obtain

$$\hat{U} \hat{U}^\dagger = \hat{1} \Rightarrow \hat{U}^\dagger = \hat{U}^{-1}$$

Thus, $\hat{U}$ is a unitary operator that maps one basis set to the other.

EXAMPLES:

1. There are two sets of orthonormal basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ and

$\{|\phi_1\rangle, |\phi_2\rangle\}$ belonging to the Hilbert space. The relationships between these two basis are given by:

$$|\phi_1\rangle = \frac{1}{\sqrt{2}}|\psi_1\rangle + \frac{i}{\sqrt{2}}|\psi_2\rangle$$

$$|\phi_2\rangle = \frac{1}{\sqrt{2}}|\psi_1\rangle - \frac{i}{\sqrt{2}}|\psi_2\rangle$$

Write a ket $|\varphi\rangle = 2|\psi_1\rangle + 3|\psi_2\rangle$ in terms of the new basis $\{|\phi_1\rangle, |\phi_2\rangle\}$.

ANSWER:

Write the change of basis in the matrix form:

$$\begin{pmatrix} |\phi_1\rangle \\ |\phi_2\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \end{pmatrix}$$

Thus, we obtain

$$\mathbf{U}^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$$

and

$$\mathbf{b} = \mathbf{U}^\dagger \cdot \mathbf{c} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 + 3i \\ 2 - 3i \end{pmatrix}$$

In Dirac notation:

$$|\varphi\rangle = \frac{2 + 3i}{\sqrt{2}}|\phi_1\rangle + \frac{2 - 3i}{\sqrt{2}}|\phi_2\rangle$$

- (3) A similarity transformation is equivalent to a change of operator
Consider an operator $\hat{A}$ acting on the ket $|\Psi\rangle$ in the basis $\{|\psi_n\rangle\}$,

$$|\Psi'\rangle = \hat{A}|\Psi\rangle$$

Since

$$|\Phi'\rangle = \hat{U}^\dagger |\Psi'\rangle = \hat{U}^\dagger \hat{A} |\Psi\rangle = \hat{U}^\dagger \hat{A} \hat{U} |\Phi\rangle$$

The operator $\hat{A}$ in the basis $\{|\psi_n\rangle\}$ is equal to the operator $\hat{U}^\dagger \hat{A} \hat{U}$ in the basis $\{|\phi_n\rangle\}$.

Let

$$\hat{B} \equiv \hat{U}^\dagger \hat{A} \hat{U} \dots \text{similarity transformation}$$

- (4) If $\hat{A}$ is diagonalizable, then $\hat{B}$ is diagonalized, i.e., $\hat{B} = \hat{D}$, and then we obtain

$$\hat{D} = \hat{U}^\dagger \hat{A} \hat{U}$$

Recall the basic linear algebra, the matrix $\mathbf{U}$ that diagonalizes $\mathbf{A}$ has

the eigenvectors of $\mathbf{v}_n$ as its columns:

$$\mathbf{U} = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n)$$

Then we obtain

$$\mathbf{A} = \mathbf{U} \cdot \mathbf{D} \cdot \mathbf{U}^\dagger$$

$$\begin{aligned} &= (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \cdots) \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots \\ 0 & \lambda_2 & 0 & \cdots \\ 0 & 0 & \lambda_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \mathbf{v}_1^\dagger \\ \mathbf{v}_2^\dagger \\ \mathbf{v}_3^\dagger \\ \vdots \end{pmatrix} \\ &= (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \cdots) \begin{pmatrix} \lambda_1 \mathbf{v}_1^\dagger \\ \lambda_2 \mathbf{v}_2^\dagger \\ \lambda_3 \mathbf{v}_3^\dagger \\ \vdots \end{pmatrix} \\ &= \sum_n \lambda_n \mathbf{v}_n \mathbf{v}_n^\dagger \end{aligned}$$

In Dirac notation:

$$\hat{A} = \sum_n \lambda_n |\phi_n\rangle \langle \phi_n|$$

where $|\phi_n\rangle$ are the eigenkets of the Hermitian operator $\hat{A}$. Thus, $\hat{A}$ is expanded in its spectrum λ_n by projection operators $|\phi_n\rangle \langle \phi_n|$, called *spectral representation*.

EXAMPLES:

1. An operator $\hat{A}$ acting on a system with a basis including two independent kets $\{|\psi_1\rangle, |\psi_2\rangle\}$

$$\hat{A}|\psi_1\rangle = i|\psi_2\rangle$$

$$\hat{A}|\psi_2\rangle = -i|\psi_1\rangle$$

Diagonalize $\hat{A}$.

ANSWER:

- Since

$$\hat{A} = \sum_{mn} a_{mn} |\psi_m\rangle \langle \psi_n|$$

where

$$a_{mn} = \langle \psi_m | \hat{A} | \psi_n \rangle$$

we obtain

$$\mathbf{A} = \begin{pmatrix} \langle \psi_1 | \hat{A} | \psi_1 \rangle & \langle \psi_1 | \hat{A} | \psi_2 \rangle \\ \langle \psi_2 | \hat{A} | \psi_1 \rangle & \langle \psi_2 | \hat{A} | \psi_2 \rangle \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The eigenvalues and eigenvectors are

$$\begin{vmatrix} 0 & -i \\ i & 0 \end{vmatrix} - \lambda \mathbf{1} = 0 \Rightarrow \begin{vmatrix} -\lambda & -i \\ i & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$$

$$\lambda_1 = 1, \quad \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\lambda_2 = -1, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Thus, we can obtain

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Alternative, we can construct a unitary matrix by writing $\mathbf{U}$ in terms of its columns as follows:

$$\mathbf{U} = (\mathbf{v}_1 \quad \mathbf{v}_2) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$$

$$\mathbf{U}^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$$

The similarity transformation gives

$$\begin{aligned} \mathbf{D} &= \mathbf{U}^\dagger \cdot \mathbf{A} \cdot \mathbf{U} \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} 1 & -1 \\ i & i \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

Thus, $\mathbf{D}$ is a diagonal matrix with respect to the basis $\{\mathbf{v}_n\}$.

- Spectrum expression of $\hat{A}$

$$\hat{A} = |\phi_1\rangle\langle\phi_1| - |\phi_2\rangle\langle\phi_2|$$

or

$$\mathbf{A} = \mathbf{U} \cdot \mathbf{D} \cdot \mathbf{U}^\dagger = (\mathbf{v}_1 \quad \mathbf{v}_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathbf{v}_1^\dagger \\ \mathbf{v}_2^\dagger \end{pmatrix} = \mathbf{v}_1 \mathbf{v}_1^\dagger - \mathbf{v}_2 \mathbf{v}_2^\dagger$$

2. Consider a two-dimensional Hilber space with an operator $\hat{\sigma}_x$

$$\hat{\sigma}_x | +x \rangle = +1 | +x \rangle$$

$$\hat{\sigma}_x | -x \rangle = -1 | -x \rangle$$

The spectral decomposition of $\hat{\sigma}_x$.

ANSWER:

Since $\hat{\sigma}_x$ is a diagonalized operator with respect to the basis $\{|+x\rangle, |-x\rangle\}$, the spectral decomposition of $\hat{\sigma}_x$ in Dirac notation is given by

$$\hat{\sigma}_x = |+x\rangle\langle+x| - |-x\rangle\langle-x|$$

4-3 Postulates of Quantum Mechanics

A. POSTULATES OF QUANTUM MECHANICS

The postulates of quantum mechanics connect the mathematical formalism, i.e., the used objects of Hilbert space with a physical interpretation, i.e., with "meaning".

- I. At each instant the state of a physical system is represented by a ket $|\psi\rangle$ in the space of states.

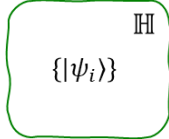
COMMENTS:

1. A ket describes the state a single system at a single instant not an ensemble of systems.
2. The space of states is a vector space, which is formed through a set of the basis $\{|\psi_i\rangle\}$. Any state of the system can be written in terms of the basis,

$$|\psi\rangle = \sum_i c_i |\psi_i\rangle$$

3. If the space of states is equipped with an inner product which associates a complex number to a ket and bra, called Hilbert space.

$$\langle\psi'|\psi\rangle = c\text{-number}$$



Hilbert space
 $\{|\psi_i\rangle\}$

An inner product of a ket and bra is also called an overlap, the overlap between quantum states.

- II. Every observable attribute of a physical system is described by an operator that acts on the states that describe the system.

COMMENTS:

1. Anything that can be measured in an experiment is called an observable, i.e., the measuring apparatus of a particular physical quantity. The observable is associated with a linear operator $\hat{A}$.
2. A measurement is the interaction of measuring apparatus with

quantum mechanical system, i.e., an operator acting on a state $\hat{A}|\psi\rangle$.

3. After the measurement, the state of the system may be different than it was before the measurement, i.e.,

$$\hat{A}|\psi\rangle = |\psi'\rangle$$

- III. The only possible result of the measurement of an observable A is one of the eigenvalues of the corresponding operator $\hat{A}$.

COMMENTS:

1. The measurement operation yields the result λ , i.e., the operator acts on the states and it does not change the state,

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle$$

2. The results of measurements must be real, i.e., the eigenvalues are real. Operators associated with real eigenvalues are *Hermitian* operators.

$$\hat{A}^\dagger = \hat{A} \Rightarrow \hat{A}|\psi\rangle = \lambda|\psi\rangle, \quad \lambda \text{ is real}$$

3. The eigenstates of a Hermitian operator are orthogonal.

$\langle\psi_m|\psi_n\rangle = \delta_{mn} \Rightarrow$ *The eigenstates can span the space of states, so they form a basis $\{|\psi_n\rangle\}$ for a Hilbert space.*

Any state of the system can be written in terms of the eigenstates,

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle$$

- IV. When a measurement of an observable A is made on a generic state $|\psi\rangle$, the probability of obtaining an eigenvalue λ_n is given by the square of the inner product of $|\psi\rangle$ with the eigenstate $|\psi_n\rangle$, i.e.,

$$|\langle\psi_n|\psi\rangle|^2.$$

COMMENTS:

1. The concept of probability implies that we must consider a large number of measurements performed on identical systems, all of them prepared in the same state $|\psi\rangle$.

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle$$

2. The measurement operation yields the result λ_n with a

probability proportional to c_n^2 .

$$\langle \psi_n | \psi \rangle = \sum_m c_m \underbrace{\langle \psi_n | \psi_m \rangle}_{=\delta_{nm}} = c_n$$

c_n is known as the "probability amplitude" to measure λ_n .

3. If the states are normalized.

$$\langle \psi | \psi \rangle = 1 \Rightarrow \sum_{mn} c_m^* c_n \underbrace{\langle \psi_m | \psi_n \rangle}_{=\delta_{mn}} = \sum_n c_n^2 = 1$$

4. The expected value of the observable A is

$$\begin{aligned} \langle \hat{A} \rangle &= \langle \psi | \hat{A} | \psi \rangle \\ &= \sum_{mn} c_m^* c_n \langle \psi_m | \hat{A} | \psi_n \rangle \\ &= \sum_{mn} c_m^* c_n \lambda_n \underbrace{\langle \psi_m | \psi_n \rangle}_{=\delta_{mn}} \\ &= \sum_n \lambda_n |c_n|^2 \end{aligned}$$

- V. Immediately after the measurement of an observable A has yielded a value λ_n , the state of the system is the normalized eigenstate $|\psi_n\rangle$

COMMENTS:

1. Before the measurement, the system is in the state $|\psi\rangle$.
2. Once we measure the state $|\psi\rangle$, we obtain a value λ_n associated with the probability $|\langle \psi_n | \psi \rangle|^2$.
3. If we immediately remeasure the system, the results of the second measurement are not statistically distributed, the result is always λ_n again.

$\Rightarrow$ Once we measure the state, $|\psi\rangle$ **instantaneously collapses** into the eigenstate $|\psi_n\rangle$.

$\Rightarrow$ The description of measurement in terms of the projection operator $\hat{P}_n = |\psi_n\rangle\langle\psi_n|$, i.e.,

$$\hat{P}_n |\psi\rangle = \langle \psi_n | \psi \rangle |\psi_n\rangle \Rightarrow |\psi_n\rangle = \frac{\hat{P}_n}{\langle \psi_n | \psi \rangle} |\psi\rangle$$

Since

$$|\langle \psi_n | \psi \rangle|^2 = \langle \psi_n | \psi \rangle \langle \psi | \psi_n \rangle = \langle \psi | \underbrace{|\psi_n\rangle\langle\psi_n|}_{\hat{P}_n} | \psi \rangle = \langle \psi | \hat{P}_n | \psi \rangle$$

$$\Rightarrow \langle \psi_n | \psi \rangle = \sqrt{\langle \psi | \hat{P}_n | \psi \rangle}$$

Thus, we obtain

$$|\psi_n\rangle = \frac{\hat{P}_n}{\sqrt{\langle \psi | \hat{P}_n | \psi \rangle}} |\psi\rangle$$

VI. The time evolution of a quantum system preserves the normalization of the associated state. The time evolution of the state of a quantum system is described by $|\Psi(t)\rangle = \hat{U}(t, t')|\Psi(t')\rangle$, for some unitary operator $\hat{U}$.

COMMENTS:

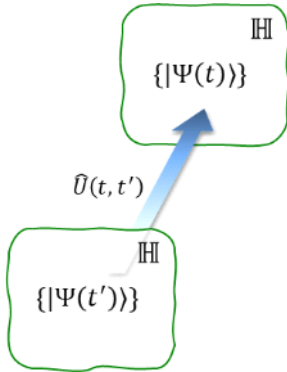
$$\langle \Psi(t) | \Psi(t) \rangle = \langle \Psi(t') | \hat{U}^\dagger(t, t') \hat{U}(t, t') | \Psi(t') \rangle$$

Since inner products between a ket and bra remain constant over time (conservation of probability), we obtain

$$\langle \Psi(t) | \Psi(t) \rangle = \langle \Psi(t') | \Psi(t') \rangle$$

$$\Rightarrow \hat{U}^\dagger(t, t') \hat{U}(t, t') = 1$$

$$\Rightarrow \hat{U}(t, t') \text{ is a unitary operator}$$



B. SUPERPOSITION PRINCIPLES

(1) Pure state

A quantum system can be described by a single state $|\psi\rangle$. The state $|\psi\rangle$ can be represented as a linear combination of a basis $|\psi_n\rangle$:

$$|\psi\rangle = \sum_n c_n |\psi_n\rangle, \text{ where } \sum_n c_n^2 = 1$$

The expectation value of an arbitrary operator $\hat{A}$

$$\begin{aligned}
\langle \hat{A} \rangle &= \sum_n \langle \psi_n | c_n^* \cdot \hat{A} \cdot \sum_m c_m | \psi_m \rangle \\
&= \sum_{n=m} c_n^2 \langle \psi_n | \hat{A} | \psi_n \rangle + \sum_{\substack{nm \\ n \neq m}} c_n^* c_m \langle \psi_n | \hat{A} | \psi_m \rangle
\end{aligned}$$

The relative phases of c_n and c_m are responsible for the quantum interference effects.

EXAMPLES:

1. Consider a system is described by a state $|\psi\rangle$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle + |\psi_2\rangle) = \frac{1}{\sqrt{2}}(|\phi_1\rangle + |\phi_2\rangle)$$

There are two observables A and B such that

$$\hat{A}|\psi_1\rangle = a_1|\psi_1\rangle \text{ and } \hat{A}|\psi_2\rangle = a_2|\psi_2\rangle$$

$$\hat{B}|\phi_1\rangle = b_1|\phi_1\rangle \text{ and } \hat{B}|\phi_2\rangle = b_2|\phi_2\rangle$$

These eigenstates are related by

$$|\psi_1\rangle = \frac{3}{5}|\phi_1\rangle + \frac{4}{5}|\phi_2\rangle, \quad |\psi_2\rangle = \frac{4}{5}|\phi_1\rangle - \frac{3}{5}|\phi_2\rangle$$

- (a) After A is measured, and a_1 is obtained, B is measured immediately. What is the probability to obtain b_1 or b_2 ?
- (b) Right after the measurement of B , the system is at either $|\phi_1\rangle$ or $|\phi_2\rangle$ state. Then A is measured again. What is the probability to obtain a_1 ?

ANSWER:

- (a) After A is measured, and a_1 is obtained

$$|\psi\rangle \xrightarrow{\text{collapse}} |\psi_1\rangle = \frac{3}{5}|\phi_1\rangle + \frac{4}{5}|\phi_2\rangle$$

B is measured immediately and obtain

$$\mathcal{P}(b_1) = \frac{9}{25}$$

$$\mathcal{P}(b_2) = \frac{16}{25}$$

- (b) If the system is in the state $|\phi_1\rangle = \frac{3}{5}|\psi_1\rangle + \frac{4}{5}|\psi_2\rangle$

$$\mathcal{P}(a_1|b_1) = \frac{9}{25} \times \mathcal{P}(b_1) = \frac{9}{25} \times \frac{9}{25}$$

If the system is in the state $|\phi_2\rangle = \frac{4}{5}|\psi_1\rangle - \frac{3}{5}|\psi_2\rangle$

$$\mathcal{P}(a_1|b_2) = \frac{16}{25} \times \mathcal{P}(b_2) = \frac{16}{25} \times \frac{16}{25}$$

The total probability

$$\begin{aligned}\mathcal{P}(a_1) &= \mathcal{P}(a_1|b_1) + \mathcal{P}(a_1|b_2) \\ &= \frac{9}{25} \times \frac{9}{25} + \frac{16}{25} \times \frac{16}{25} \\ &= \frac{337}{625}\end{aligned}$$

(2) Mixed state

A statistical ensemble of pure states: $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_i\rangle\}$

p_1 is the probability of finding an individual system of the ensemble described by the state $|\psi_1\rangle$.

p_2 is the probability of finding an individual system of the ensemble described by the state $|\psi_2\rangle$.

$$\begin{aligned}&\vdots \\ &\sum_i p_i = 1\end{aligned}$$

The expectation value of an arbitrary operator $\hat{A}$

$$\langle \hat{A} \rangle = \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle$$

There are no contributions to the ensemble averages coming from interferences between different states.

EXAMPLES:

1. Imaging two systems, system A contains all particles in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle + |\psi_2\rangle)$$

and system B contains half of the particles in the state $|\psi_1\rangle$ and half in the state $|\psi_2\rangle$.

What is the probability of finding a state?

ANSWER:

- (a) For A:

Before the measurement:

There is a 0% chance that the particle is in either of those states, and a 100% chance that the particle is in the superposition state $|\psi\rangle$.

After the measurement:

We obtain λ_1 and 50% chance in the state $|\psi_1\rangle$. We obtain λ_2 and 50% chance in the state $|\psi_2\rangle$.

○ For B:

Before the measurement:

The particle is definitely in the state $|\psi_1\rangle$ or the state $|\psi_2\rangle$.

The particle has 50% chance in the state $|\psi_1\rangle$ and 50% chance in the state $|\psi_2\rangle$.

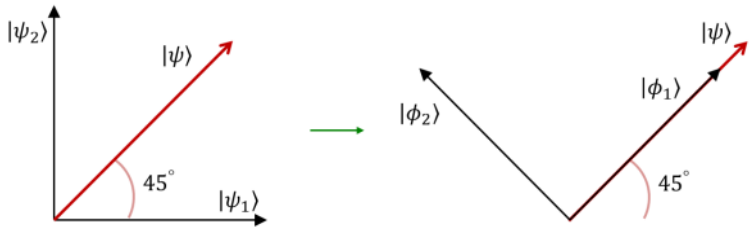
After the measurement:

We obtain λ_1 and 50% chance in the state $|\psi_1\rangle$. We obtain λ_2 and 50% chance in the state $|\psi_2\rangle$

$\Rightarrow$ Measuring the state $|\psi_1\rangle$ or $|\psi_2\rangle$, cannot distinguish A and B.

(b) Change of basis (unitary transformation)

Rotate the instrument 45° . We have two new axes $|\phi_1\rangle$ and $|\phi_2\rangle$:



$$\begin{pmatrix} |\phi_1\rangle \\ |\phi_2\rangle \end{pmatrix} = \begin{pmatrix} \cos \frac{\pi}{4} & \sin \frac{\pi}{4} \\ -\sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix} \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \end{pmatrix} \\ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} |\psi_1\rangle \\ |\psi_2\rangle \end{pmatrix}$$

Thus, we obtain

$$|\phi_1\rangle = \frac{1}{\sqrt{2}} (|\psi_1\rangle + |\psi_2\rangle)$$

$$|\phi_2\rangle = \frac{1}{\sqrt{2}} (-|\psi_1\rangle + |\psi_2\rangle)$$

Inverse the relationship, we have

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} (|\phi_1\rangle - |\phi_2\rangle)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} (|\phi_1\rangle + |\phi_2\rangle)$$

After the measurement:

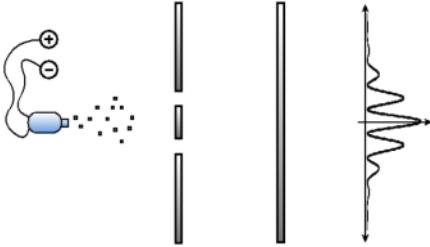
For A, we obtain λ_1 and there is a 100% chance in the state $|\phi_1\rangle$.

For B, we obtain λ_1 and 50% in the state $|\phi_1\rangle$, and λ_2 and 50% in the state $|\phi_2\rangle$.

$\Rightarrow$ The probabilities in A of different quantum states result from probability amplitudes. The probabilities in B of different quantum states do not derive from probability amplitudes: the probabilities of the different quantum states in B are exactly like classical probabilities.

C. THE DOUBLE-SLIT EXPERIMENT OF ELECTRONS

(1) A schematic diagram of the double-slit experiment



Electrons have 50% chance going through the slit 1 and 50% chance going through slit 2. Each electron is described by a single state $|\psi\rangle$,

$$|\psi\rangle = \sqrt{\frac{a}{2}}(|y_1\rangle + |y_2\rangle), \text{ where } a = |y_1 - y_2|$$

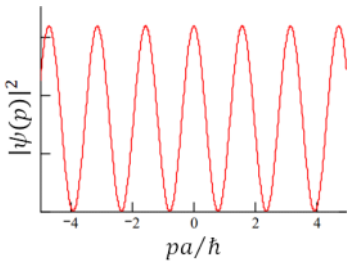
where the slit performs a position measurement on electrons.

Since slits localize the electron in the y -direction, the uncertainty principle demands a compensating delocalization in the y -component of the momentum. The wavefunction in momentum is

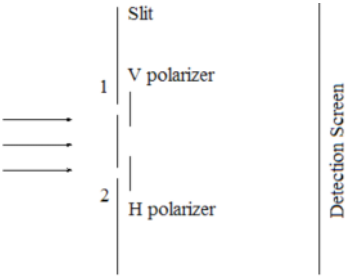
$$\psi(p) = \langle p|\psi\rangle = \sqrt{\frac{a}{2}}(\langle p|y_1\rangle + \langle p|y_2\rangle) = \frac{\sqrt{a}}{2\sqrt{\pi\hbar}}(e^{-ipy_1/\hbar} + e^{-ipy_2/\hbar})$$

The probability density is

$$\begin{aligned} |\psi(p)|^2 &= \frac{a}{4\pi\hbar}(2 + e^{ip(y_1-y_2)/\hbar} + e^{-ip(y_1-y_2)/\hbar}) \\ &= \frac{a}{2\pi\hbar}\left(1 + \cos\frac{pa}{\hbar}\right) \end{aligned}$$



- (2) A schematic diagram of the double-slit experiment with polarizers behind the slits, where V and H stand for vertical and horizontal, respectively.



$$|\psi\rangle = \sqrt{\frac{a}{2}} (|y_1\rangle|V\rangle + |y_2\rangle|H\rangle)$$

where $|V\rangle$ and $|H\rangle$ represent the vertical and horizontal polarization states.

$$\begin{aligned}\psi(p) &= \langle p|\psi\rangle \\ &= \sqrt{\frac{a}{2}} (\langle p|y_1\rangle|V\rangle + \langle p|y_2\rangle|H\rangle) \\ &= \frac{\sqrt{a}}{2\sqrt{\pi\hbar}} (e^{-ipy_1/\hbar}|V\rangle + e^{-ipy_2/\hbar}|H\rangle) \\ |\psi(p)|^2 &= \frac{a}{4\pi\hbar} (1 + 1) = \frac{a}{2\pi\hbar}\end{aligned}$$

⇒ Due to the presence of the polarizing films the interference fringes disappear.

D. SYMMETRY AND CONSERVATION

- (1) In classical mechanics, the symmetry implies that [c.f.1-5-B]

$$[f, H]_{q,p} = 0$$

where f is called the constant of motion, i.e., a conserved quantity.

In quantum mechanics, the variables are replaced by operators. Thus,

we have

$$\begin{aligned} [\hat{O}, \hat{H}] &= \hat{O}\hat{H} - \hat{H}\hat{O} = 0 \\ \Rightarrow \hat{H} &= \hat{O}^{-1}\hat{H}\hat{O} \text{ or } \hat{H} = \hat{O}\hat{H}\hat{O}^{-1} \end{aligned}$$

If $\hat{O}$ is a unitary operator, $\hat{O}^{-1} = \hat{O}^\dagger$, then we obtain

$$\hat{H} = \hat{O}^\dagger \hat{H} \hat{O} \text{ or } \hat{H} = \hat{O} \hat{H} \hat{O}^\dagger$$

This is a similarity transformation to itself.

NOTE:

The Hamiltonian is left invariant by the transformation does not mean that its eigenstates are invariant.

(2) Suppose that

$$[\hat{O}, \hat{H}] = 0$$

1. The eigenvalues of $\hat{O}$ are conserved quantities, namely, a good quantum number.
2. The eigenstate of $\hat{O}$ is also a eigenstate of $\hat{H}$.

PROOF:

$$\text{Let } \hat{O}|\psi\rangle = \Lambda|\psi\rangle$$

Since $[\hat{O}, \hat{H}] = 0 \Rightarrow \hat{O}\hat{H} = \hat{H}\hat{O}$, we obtain

$$\begin{aligned} \hat{O}(\hat{H}|\psi\rangle) &= \hat{H}\hat{O}|\psi\rangle = \hat{H}\Lambda|\psi\rangle = \Lambda(\hat{H}|\psi\rangle) \\ \Rightarrow \hat{H}|\psi\rangle &\text{ is the eigenstate of } \hat{O}. \end{aligned}$$

$\Rightarrow \hat{H}|\psi\rangle$ must be the same eigenstate as $|\psi\rangle$, only multiplied by a scalar, i.e., $\hat{H}|\psi\rangle = \lambda|\psi\rangle$.

$\Rightarrow |\psi\rangle$ is the common eigenstate of $\hat{O}$ and $\hat{H}$.

■

EXAMPLES:

1. Consider reflection symmetry in 1D: $\hat{P}^\dagger x \hat{P} = -x$

Thus, we have

$$\hat{P}^2 = \hat{1} \Rightarrow \hat{P}\hat{P}^\dagger = \hat{1} \Rightarrow \hat{P}^\dagger = \hat{P}^{-1}$$

If the potential has even parity, $\hat{P}^\dagger V(x) \hat{P} = \hat{P}V(x) \hat{P} = V(x)$, then

$$[\hat{P}, \hat{H}] = 0$$

$\Rightarrow \hat{H}$ and $\hat{P}$ can be simultaneously diagonalized.

Since

$$\hat{P}^2 = \hat{1}$$

the simultaneous eigenstates must obey

$$\hat{P}|\psi\rangle = \pm|\psi\rangle$$

As the Hamiltonian is invariant under reflection, but its eigenstates are not. The transformed eigenstates get multiplied by a phase in the factor can only be 0 or π .

4-4 The Ladder Operator Method for Harmonic Oscillator

A. LADDER OPERATORS

- (1) The time-independent Schrödinger equation for a harmonic oscillator

$$\hat{H}\psi_n(x) = \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right)\psi_n(x) = E_n\psi_n(x)$$

where the eigenenergy and eigenfunction are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, \dots, \infty$$

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-m\omega x^2/2\hbar} H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right)$$

- (2) States and operators in Dirac notation:

$$\psi_n(x) \rightarrow |\psi_n\rangle$$

$$\hat{x} \rightarrow \hat{x}$$

$$\hat{p} \rightarrow \hat{p}$$

The Schrödinger equation becomes

$$\hat{H}|\psi_n\rangle = \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right)|\psi_n\rangle = \hbar\omega \left(n + \frac{1}{2} \right)|\psi_n\rangle$$

$$\Rightarrow \frac{1}{\hbar\omega} \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right)|\psi_n\rangle = \left(n + \frac{1}{2} \right)|\psi_n\rangle$$

$$\Rightarrow \left[\frac{1}{\hbar\omega} \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right) - \frac{1}{2} \right] |\psi_n\rangle = n|\psi_n\rangle$$

Define an operator $\hat{n}$ such that $\hat{n}|\psi_n\rangle = n|\psi_n\rangle$,

$$\hat{n} = \frac{1}{\hbar\omega} \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right) - \frac{1}{2}$$

$\Rightarrow$ The eigenvalues of the operator $\hat{n}$ are the integers n and $\hat{n}$ is called the number operator.

Thus, we can let $|\psi_n\rangle \rightarrow |n\rangle$

$$\hat{H}|n\rangle = \hbar\omega \left(\hat{n} + \frac{1}{2} \right)|n\rangle = \hbar\omega \left(n + \frac{1}{2} \right)|n\rangle$$

- (3) Let the position and momentum operators as

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger)$$

$$\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^\dagger - \hat{a})$$

The inverse relations are

$$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p})$$

Since $\hat{a} \neq \hat{a}^\dagger$, $\hat{a}$ and $\hat{a}^\dagger$ are nonhermitian operators, we consider the product of $\hat{a}$ and $\hat{a}^\dagger$, such that $(\hat{a}^\dagger\hat{a})^\dagger = \hat{a}^\dagger\hat{a}$ is a hermitian operator,

$$\begin{aligned}\hat{a}^\dagger\hat{a} &= \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p})\frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p}) \\ &= \frac{1}{2m\hbar\omega}(\hat{p}^2 + m^2\omega^2\hat{x}^2 + im\omega\hat{x}\hat{p} - im\omega\hat{p}\hat{x}) \\ &= \frac{1}{\hbar\omega}\left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - \frac{1}{2}\right) \\ &= \hat{n} \\ \Rightarrow \hat{H}|n\rangle &= \hbar\omega\left(\hat{n} + \frac{1}{2}\right)|n\rangle = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)|n\rangle\end{aligned}$$

(4) The commutation relation between $\hat{a}$ and $\hat{a}^\dagger$

$$\begin{aligned}[\hat{a}, \hat{a}^\dagger] &= \frac{1}{2m\hbar\omega}[(m\omega\hat{x} + i\hat{p}), (m\omega\hat{x} - i\hat{p})] \\ &= \frac{1}{2m\hbar\omega}([m\omega\hat{x}, m\omega\hat{x}] + [i\hat{p}, m\omega\hat{x}] + [m\omega\hat{x}, -i\hat{p}] + [i\hat{p}, -i\hat{p}]) \\ &= \frac{i}{2\hbar}[\hat{p}, \hat{x}] - \frac{i}{2\hbar}[\hat{x}, \hat{p}] \\ &= \frac{i}{2\hbar}(-i\hbar) - \frac{i}{2\hbar}(i\hbar) \\ &= 1 \\ \Rightarrow \hat{a}\hat{a}^\dagger &= \hat{a}^\dagger\hat{a} + 1\end{aligned}$$

Thus, we can show that

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

where $\hat{a}^\dagger$ is a *raising* operator that increases the value of n by 1.

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$$

where $\hat{a}$ is a *lowering* operator that decreases the value of n by 1.

PROOF:

- Consider a new state that is created by the action of $\hat{a}^\dagger$, i.e.,

$$\hat{a}^\dagger|n\rangle = |\tilde{n}'\rangle$$

Thus

$$\hat{n}\hat{a}^\dagger|n\rangle = \hat{n}|n'\rangle = n'|n'\rangle$$

On the other hand,

$$\begin{aligned}\hat{n}\hat{a}^\dagger|n\rangle &= (\hat{a}^\dagger\hat{a})\hat{a}^\dagger|n\rangle = \hat{a}^\dagger(\hat{a}^\dagger\hat{a} + 1)|n\rangle = \hat{a}^\dagger(n+1)|n\rangle \\ &\Rightarrow n' = n+1\end{aligned}$$

Assume $\hat{a}^\dagger|n\rangle = A|n+1\rangle$ where A is a constant.

$$\langle\hat{a}^\dagger n|\hat{a}^\dagger n\rangle = A^2\langle n+1|n+1\rangle = A^2$$

On the other hand,

$$\begin{aligned}\langle\hat{a}^\dagger n|\hat{a}^\dagger n\rangle &= \langle n|\hat{a}\hat{a}^\dagger|n\rangle = \langle n|\hat{a}^\dagger\hat{a} + 1|n\rangle = n+1 \\ &\Rightarrow A = \sqrt{n+1} \\ &\Rightarrow \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle\end{aligned}$$

- Consider a new state that is created by the action of $\hat{a}$, i.e.,

$$\hat{a}|n\rangle = |n'\rangle$$

Thus

$$\hat{n}\hat{a}|n\rangle = \hat{n}|n'\rangle = n'|n'\rangle$$

On the other hand,

$$\begin{aligned}\hat{n}\hat{a}|n\rangle &= (\hat{a}^\dagger\hat{a})\hat{a}|n\rangle = (\hat{a}\hat{a}^\dagger - 1)\hat{a}|n\rangle = (\hat{a}\hat{n} - \hat{a})|n\rangle = \hat{a}(n-1)|n\rangle \\ &\Rightarrow n' = n-1\end{aligned}$$

Assume $\hat{a}|n\rangle = A|n-1\rangle$ where A is a constant.

$$\langle\hat{a}n|\hat{a}n\rangle = A^2\langle n-1|n-1\rangle = A^2$$

On the other hand,

$$\begin{aligned}\langle\hat{a}n|\hat{a}n\rangle &= \langle n|\hat{a}^\dagger\hat{a}|n\rangle = \langle n|\hat{n}|n\rangle = n \\ &\Rightarrow A^2 = \sqrt{n} \\ &\Rightarrow \hat{a}|n\rangle = \sqrt{n}|n-1\rangle\end{aligned}$$

■

EXAMPLES:

- Using

$$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

to find the ground state wavefunction.

ANSWER:

Using the lowering operator $\hat{a}$ acts on the ground state:

$$\hat{a}|0\rangle = \hat{a}|\psi_0\rangle = 0$$

$$\Rightarrow \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})|\psi_0\rangle = 0$$

We can ignore the prefactor, i.e.,

$$(m\omega\hat{x} + i\hat{p})|\psi_0\rangle = 0$$

Multiplying $\langle x|$ from the left side,

$$\langle x|(m\omega\hat{x} + i\hat{p})|\psi_0\rangle = 0$$

$$\Rightarrow m\omega x\langle x|\psi_0\rangle + i\hat{p}\langle x|\psi_0\rangle = 0$$

$$\Rightarrow m\omega x\psi_0(x) + i\left(-i\hbar\frac{d}{dx}\right)\psi_0(x) = 0$$

$$\Rightarrow \frac{d\psi_0(x)}{\psi_0(x)} = -\frac{m\omega}{\hbar}x dx$$

Integral on both sides

$$\ln \psi_0(x) = -\frac{m\omega}{2\hbar}x^2 + C$$

$$\Rightarrow \psi_0(x) = e^{-m\omega x^2/2\hbar + C} = Ae^{-m\omega x^2/2\hbar}$$

Using the normalization condition:

$$\int_{-\infty}^{\infty} |\psi_0(x)|^2 dx = A^2 \int_{-\infty}^{\infty} e^{-m\omega x^2/\hbar} dx = A^2 \sqrt{\frac{\pi\hbar}{m\omega}} = 1$$

$$\Rightarrow A^2 = \sqrt{\frac{m\omega}{\pi\hbar}}$$

We obtain

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}$$

B. CREATION AND ANNIHILATION OPERATORS

(1) The ground state

$$\langle \hat{x} \rangle = \langle n|\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle n|\hat{a}^\dagger + \hat{a}|n\rangle = 0$$

$$\langle \hat{p} \rangle = \langle n|\hat{p}|n\rangle = i\sqrt{\frac{m\hbar\omega}{2}} \langle n|\hat{a}^\dagger - \hat{a}|n\rangle = 0$$

$$\begin{aligned}
\langle \hat{x}^2 \rangle &= \langle n | \hat{x}^2 | n \rangle \\
&= \frac{\hbar}{2m\omega} \langle n | \hat{a}^\dagger \hat{a}^\dagger + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a} \hat{a} | n \rangle \\
&= \frac{\hbar}{2m\omega} \langle n | 2\hat{a}^\dagger \hat{a} + 1 | n \rangle \\
&= \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right) \\
\langle \hat{p}^2 \rangle &= \langle n | \hat{p}^2 | n \rangle \\
&= -\frac{m\hbar\omega}{2} \langle n | \hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a} \hat{a} | n \rangle \\
&= -\frac{m\hbar\omega}{2} \langle n | -2\hat{a}^\dagger \hat{a} - 1 | n \rangle \\
&= m\hbar\omega \left(n + \frac{1}{2} \right) \\
(\Delta x)^2 &= \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2 = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right) \\
(\Delta p)^2 &= \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2 = m\hbar\omega \left(n + \frac{1}{2} \right) \\
(\Delta x)^2 (\Delta p)^2 &= \frac{\hbar}{m\omega} m\hbar\omega \left(n + \frac{1}{2} \right)^2 = \hbar^2 \left(n + \frac{1}{2} \right)^2 \\
\Delta x \Delta p &= \hbar \left(n + \frac{1}{2} \right)
\end{aligned}$$

For the ground state $n = 0$, we have

$$\Delta x \Delta p = \frac{\hbar}{2}$$

Since

$$E_0 = \langle 0 | \hat{H} | 0 \rangle = \hbar\omega \langle 0 | \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) | 0 \rangle = \frac{\hbar\omega}{2}$$

which implies $\langle 0 | \hat{a}^\dagger \hat{a} | 0 \rangle = \langle \hat{a} 0 | \hat{a} 0 \rangle = 0 \Rightarrow \hat{a} | 0 \rangle = 0$

$\Rightarrow$ The ground state $|0\rangle$ of the harmonic oscillator defined by $\hat{a}|0\rangle = 0$ is a minimum uncertainty state with energy $E_0 = \hbar\omega/2$.

(2) The excited states

Since $\hat{a}|0\rangle = 0$ it follows that,

$$|1\rangle = \frac{\hat{a}^\dagger}{\sqrt{1}} |0\rangle$$

$$\begin{aligned}
 |2\rangle &= \frac{\hat{a}^\dagger}{\sqrt{2}} |1\rangle = \frac{\hat{a}^\dagger}{\sqrt{2}} \frac{\hat{a}^\dagger}{\sqrt{1}} |0\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2!}} |0\rangle \\
 &\vdots \\
 |n\rangle &= \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle
 \end{aligned}$$

Thus $|n\rangle$ is obtained by applying $\hat{a}^\dagger$ to the ground state n times and the associated energy is

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right)$$

Define a quantum (photon) for the transition between adjacent energy levels:

$$\Delta E = E_n - E_{n-1} = \hbar\omega \left(n + \frac{1}{2} \right) - \hbar\omega \left((n-1) + \frac{1}{2} \right) = \hbar\omega$$

Since

$$\frac{E_n - E_0}{\Delta E} = \left(n + \frac{1}{2} \right) - \frac{1}{2} = n$$

the excited states are referred to as virtual particles (quasi-particles).

EXAMPLES:

1. Considering the excited state $n = 2$

$$\frac{(\hat{a}^\dagger)^2}{\sqrt{2!}} |0\rangle = |2\rangle$$

The excited state is referred to the number of photons

$$n = \frac{E_2 - E_0}{\Delta E} = \frac{\hbar\omega \left(2 + \frac{1}{2} \right) - \frac{1}{2} \hbar\omega}{\hbar\omega} = 2$$

Therefore, $|0\rangle$ is referred to the vacuum state. $\hat{a}^\dagger$ is called "**creation**" operator that creates a particle and $\hat{a}$ is called "**annihilation**" operator that annihilates a particle.

C. MINIMUM UNCERTAINTY STATES

- (1) Consider an arbitrary annihilation operator $\hat{a}' = \hat{a} - \phi$, where ϕ is a constant. Thus, we obtain

$$\hat{a}'|0\rangle = (\hat{a} - \phi)|0\rangle = \phi|0\rangle \neq 0$$

Assume the states $|\phi\rangle$ with $\langle\phi|\phi\rangle = 1$, such that $(\hat{a} - \phi)|\phi\rangle = 0$. Thus, we obtain,

$$\hat{a}|\phi\rangle - \phi|\phi\rangle = 0 \Rightarrow \hat{a}|\phi\rangle = \phi|\phi\rangle$$

where $|\phi\rangle$ are called coherent states.

- (2) From $\hat{a}|\phi\rangle = \phi|\phi\rangle$ it follows that $\langle\phi|\hat{a}^\dagger = \langle\phi|\phi^*$. So,

$$\langle\phi|\hat{a}^\dagger\hat{a}|\phi\rangle = \phi^*\phi = |\phi|^2$$

$$\langle\phi|\hat{a} + \hat{a}^\dagger|\phi\rangle = \phi + \phi^*$$

$$\langle\phi|\hat{a} - \hat{a}^\dagger|\phi\rangle = \phi - \phi^*$$

$$\begin{aligned}\langle\phi|(\hat{a} + \hat{a}^\dagger)^2|\phi\rangle &= \langle\phi|(\hat{a}\hat{a} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^\dagger)|\phi\rangle \\ &= \phi^2 + \phi^*\phi + 1 + \phi^*\phi + \phi^{*2} \\ &= (\phi + \phi^*)^2 + 1\end{aligned}$$

In a similar way it follows that:

$$\begin{aligned}\langle\phi|(\hat{a} - \hat{a}^\dagger)^2|\phi\rangle &= \langle\phi|(\hat{a}\hat{a} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^\dagger)|\phi\rangle \\ &= \phi^2 - \phi^*\phi - 1 - \phi^*\phi + \phi^{*2} \\ &= (\phi - \phi^*)^2 - 1\end{aligned}$$

Now we can calculate the uncertainty for position and momentum:

$$\begin{aligned}(\Delta x)^2 &= \langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2 \\ &= \frac{\hbar}{2m\omega}\langle\phi|(\hat{a} + \hat{a}^\dagger)^2|\phi\rangle - \frac{\hbar}{2m\omega}(\langle\phi|\hat{a} + \hat{a}^\dagger|\phi\rangle)^2 \\ &= \frac{\hbar}{2m\omega}\left((\phi + \phi^*)^2 + 1 - (\phi + \phi^*)^2\right) \\ &= \frac{\hbar}{2m\omega} \\ (\Delta p)^2 &= \langle\hat{p}^2\rangle - \langle\hat{p}\rangle^2 \\ &= -\frac{\hbar m\omega}{2}\langle\phi|(\hat{a} - \hat{a}^\dagger)^2|\phi\rangle - \left(-\frac{\hbar m\omega}{2}\right)(\langle\phi|\hat{a} - \hat{a}^\dagger|\phi\rangle)^2 \\ &= -\frac{\hbar m\omega}{2}\left((\phi - \phi^*)^2 - 1 - (\phi - \phi^*)^2\right) \\ &= \frac{\hbar m\omega}{2}\end{aligned}$$

$$(\Delta x)^2(\Delta p)^2 = \frac{\hbar}{2m\omega} \times \frac{\hbar m\omega}{2} = \frac{\hbar^2}{4} \Rightarrow \Delta x \Delta p = \frac{\hbar}{2}$$

The coherent states satisfy the minimum uncertainty relation.

(3) Since the coherent states satisfy

$$1. \quad \Delta x \Delta p = \frac{\hbar}{2}$$

$$2. \quad (\hat{a} - \phi)|\phi\rangle = 0$$

the coherent states are the minimum uncertainty states.

4-5 Time-Evolution Operators

A. UNITARY OPERATOR $\hat{U}$

- (1) If a system is in the state $|\Psi(t')\rangle$, defined at some initial time t' , then the state will evolve in time to the state

$$|\Psi(t)\rangle = \hat{U}(t, t')|\Psi(t')\rangle$$

Assume that time is a continuous parameter

$$\lim_{t \rightarrow t'} |\Psi(t)\rangle = \lim_{t \rightarrow t'} \hat{U}(t, t') |\Psi(t')\rangle = |\Psi(t')\rangle \Rightarrow \hat{U}(t', t') = \hat{1}$$

Since

$$|\Psi(t)\rangle = \hat{U}(t, t')|\Psi(t')\rangle$$

$$|\Psi(t)\rangle = \hat{U}(t, t_1)|\Psi(t_1)\rangle = \hat{U}(t, t_1)\hat{U}(t_1, t')|\Psi(t')\rangle$$

we obtain

$$\hat{U}(t, t') = \hat{U}(t, t_1)\hat{U}(t_1, t'), \text{ for } t > t_1 > t' \dots \text{decomposition}$$

- (2) Solve $\hat{U}(t, t')$ from the equation of motion:

$$\begin{aligned} \frac{\partial}{\partial t} \hat{U}(t, t') &= \lim_{\epsilon \rightarrow 0} \frac{\hat{U}(t + \epsilon, t') - \hat{U}(t, t')}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\hat{U}(t + \epsilon, t)\hat{U}(t, t') - \hat{U}(t, t')}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{[\hat{U}(t + \epsilon, t) - \hat{1}]\hat{U}(t, t')}{\epsilon} \end{aligned}$$

The infinitesimal time evolution of the operator $\hat{U}(t + \epsilon, t)$ describes dynamical evolution under the influence of the Hamiltonian $\hat{H}(t)$ as,

$$\begin{aligned} \hat{U}(t + \epsilon, t) &= \hat{U}(t, t) - \frac{i}{\hbar} \hat{H}(t) \hat{U}(t, t) \epsilon = \hat{1} - \frac{i}{\hbar} \hat{H}(t) \epsilon \\ \Rightarrow \frac{\partial}{\partial t} \hat{U}(t, t') &= \lim_{\epsilon \rightarrow 0} \frac{-\frac{i}{\hbar} \hat{H}(t) \epsilon \hat{U}(t, t')}{\epsilon} = -\frac{i}{\hbar} \hat{H}(t) \hat{U}(t, t') \end{aligned}$$

Integrate with respect to time on both sides

$$\begin{aligned} \int_{t'}^t \frac{\partial}{\partial t_1} \hat{U}(t_1, t') dt_1 &= -\frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) \hat{U}(t_1, t') dt_1 \\ \hat{U}(t, t') - \hat{1} &= -\frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) \hat{U}(t_1, t') dt_1 \\ \hat{U}(t, t') &= \hat{1} - \frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) \hat{U}(t_1, t') dt_1 \end{aligned}$$

Iterate:

$$\begin{aligned}
\hat{U}(t, t') &= \hat{1} - \frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) dt_1 \left[\hat{1} - \frac{i}{\hbar} \int_{t'}^{t_1} \hat{H}(t_2) \hat{U}(t_2, t') dt_2 \right] \\
&= \hat{1} - \frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) dt_1 + \left(-\frac{i}{\hbar} \right)^2 \int_{t'}^t \hat{H}(t_1) dt_1 \int_{t'}^{t_1} \hat{H}(t_2) \hat{U}(t_2, t') dt_2
\end{aligned}$$

Re-iterate:

$$\hat{U}(t, t') = \hat{1} - \frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) dt_1 + \left(-\frac{i}{\hbar} \right)^2 \int_{t'}^t \hat{H}(t_1) dt_1 \int_{t'}^{t_1} \hat{H}(t_2) dt_2 + \dots$$

where $t > t_1 > t_2 > \dots > t'$.

Considering the dummy variables t_1 and t_2 , the integral can be rewritten into a compact form

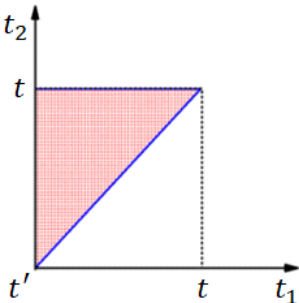
$$\begin{aligned}
&\int_{t'}^t \hat{H}(t_1) dt_1 \int_{t'}^{t_1} \hat{H}(t_2) dt_2 \\
&= \frac{1}{2} \int_{t'}^t \hat{H}(t_1) dt_1 \int_{t'}^{t_1} \hat{H}(t_2) dt_2 + \frac{1}{2} \int_{t'}^t \hat{H}(t_2) dt_2 \int_{t'}^{t_2} \hat{H}(t_1) dt_1 \\
&= \frac{1}{2} \int_{t'}^t \int_{t'}^t \hat{H}(t_1) \hat{H}(t_2) dt_1 dt_2 \Theta(t_1 - t_2) \\
&\quad + \frac{1}{2} \int_{t'}^t \int_{t'}^t \hat{H}(t_2) \hat{H}(t_1) dt_2 dt_1 \Theta(t_2 - t_1)
\end{aligned}$$

$$\text{where } \Theta(t_1 - t_2) = \begin{cases} 1, & t_1 \geq t_2 \\ 0, & t_1 < t_2 \end{cases}$$

Let $\hat{T}$ be the time ordering operator,

$$\begin{aligned}
\hat{T}[\hat{H}(t_1) \hat{H}(t_2)] &= \begin{cases} \hat{H}(t_1) \hat{H}(t_2), & \text{if } t_1 > t_2 \\ \hat{H}(t_2) \hat{H}(t_1), & \text{if } t_2 > t_1 \end{cases} \\
\Rightarrow \int_{t'}^t \hat{H}(t_1) dt_1 \int_{t'}^{t_1} \hat{H}(t_2) dt_2 &= \frac{1}{2} \int_{t'}^t \int_{t'}^t \hat{T}[\hat{H}(t_1) \hat{H}(t_2)] dt_1 dt_2
\end{aligned}$$

Pictorially, the area of integration is



Thus, we obtain

$$\begin{aligned}
\int_{t'}^t \int_{t'}^{t_1} dt_1 dt_2 &= \frac{1}{2} \int_{t'}^t \int_{t'}^t \hat{T}[\dots] dt_1 dt_2 = \frac{1}{2!} \int_{t'}^t \int_{t'}^t \hat{T}[\dots] dt_1 dt_2 \\
\int_{t'}^t \int_{t'}^{t_1} \int_{t'}^{t_2} dt_1 dt_2 dt_3 &= \frac{1}{6} \int_{t'}^t \int_{t'}^t \int_{t'}^t \hat{T}[\dots] dt_1 dt_2 dt_3 \\
&= \frac{1}{3!} \int_{t'}^t \int_{t'}^t \int_{t'}^t \hat{T}[\dots] dt_1 dt_2 dt_3 \\
&\vdots \\
\int_{t'}^t \int_{t'}^{t_1} \dots \int_{t'}^{t_N} dt_1 \dots dt_N &= \frac{1}{N!} \int_{t'}^t \int_{t'}^t \dots \int_{t'}^t \hat{T}[\dots] dt_1 \dots dt_N \\
\hat{U}(t, t') &= \hat{1} - \frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) dt_1 + \frac{1}{2!} \left(-\frac{i}{\hbar} \right)^2 \int_{t'}^t \int_{t'}^t \hat{T}[\hat{H}(t_1) \hat{H}(t_2)] dt_1 dt_2 + \dots \\
&= \hat{T} \left[\exp \left(-\frac{i}{\hbar} \int_{t'}^t \hat{H}(t_1) dt_1 \right) \right]
\end{aligned}$$

B. SCHRÖDINGER PICTURE AND HEISENBERG PICTURE

- (1) Properties of the time-evolution operator $\hat{U}(t) \equiv \hat{U}(t, 0)$

Since

$$\hat{U}(t, 0) \hat{U}(0, t) = \hat{U}(t, t) = \hat{1}$$

we obtain

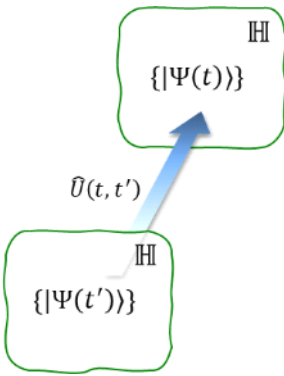
$$\hat{U}^{-1}(t, 0) = \hat{U}(0, t) \dots \text{time-reversal}$$

Since

$$\begin{aligned}
\langle \Psi(t) | \Psi(t) \rangle &= \langle \Psi(0) | \hat{U}^\dagger(t, 0) \hat{U}(t, 0) | \Psi(0) \rangle \\
\langle \Psi(t) | \Psi(t) \rangle &= \langle \Psi(0) | \Psi(0) \rangle
\end{aligned}$$

we obtain

$$\begin{aligned}
\hat{U}^\dagger(t, 0) \hat{U}(t, 0) &= \hat{1} \\
\Rightarrow \hat{U}^\dagger(t, 0) &= \hat{U}^{-1}(t, 0) \dots \text{unitary} \\
\Rightarrow \hat{U}^\dagger(t, 0) &= \hat{U}^{-1}(t, 0) = \hat{U}(0, t)
\end{aligned}$$



- (2) The time dependence of the expectation value is preserved by similarity transformation when a change of basis is made (unitary transformation).

$$\underbrace{\langle \Psi(t) | \hat{A} | \Psi(t) \rangle}_{\text{Schrödinger picture}} = \langle \Psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \Psi(0) \rangle = \underbrace{\langle \Psi(0) | \hat{A}(t) | \Psi(0) \rangle}_{\text{Heisenberg picture}}$$

COMMENTS:

The time evolution of the expectation value

$$\underbrace{\langle \Psi(t) | \hat{A} | \Psi(t) \rangle}_{\text{Schrödinger picture}} = \underbrace{\langle \Psi(0) | \hat{A}(t) | \Psi(0) \rangle}_{\text{Heisenberg picture}}$$

Schrödinger picture:

The time-dependent state is given by the unitary transformation.

$$|\Psi(t)\rangle = \hat{U}(t) |\Psi(0)\rangle$$

Heisenberg picture:

The time-dependent operator is given by the similarity transformation.

$$\hat{A}(t) = \hat{U}^\dagger(t) \hat{A} \hat{U}(t)$$

EXAMPLES:

1. Harmonic oscillator

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar}$$

$$\frac{d}{dt} \hat{a}(t) = -\frac{i}{\hbar} [\hat{a}(t), \hat{H}] = -\frac{i}{\hbar} e^{i\hat{H}t/\hbar} [\hat{a}, \hat{H}] e^{-i\hat{H}t/\hbar}$$

$$\frac{d}{dt} \hat{a}^\dagger(t) = -\frac{i}{\hbar} [\hat{a}^\dagger(t), \hat{H}] = -\frac{i}{\hbar} e^{i\hat{H}t/\hbar} [\hat{a}^\dagger, \hat{H}] e^{-i\hat{H}t/\hbar}$$

Since $[\hat{a}, \hat{H}] = \hbar\omega \hat{a}$ and $[\hat{a}^\dagger, \hat{H}] = -\hbar\omega \hat{a}^\dagger$

$$\begin{aligned}\frac{d}{dt}\hat{a}(t) &= -i\omega e^{i\hat{H}t/\hbar}\hat{a}e^{-i\hat{H}t/\hbar} = -i\omega\hat{a}(t) \Rightarrow \hat{a}(t) = e^{-i\omega t}\hat{a} \\ \frac{d}{dt}\hat{a}^\dagger(t) &= i\omega e^{i\hat{H}t/\hbar}\hat{a}^\dagger e^{-i\hat{H}t/\hbar} = i\omega\hat{a}^\dagger(t) \Rightarrow \hat{a}^\dagger(t) = e^{i\omega t}\hat{a}^\dagger\end{aligned}$$

$$\begin{aligned}\hat{x}(t) &= \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger(t) + \hat{a}(t)) \\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{e^{i\omega t}}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p}) + \frac{e^{-i\omega t}}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})\right) \\ &= \left(\frac{e^{i\omega t} + e^{-i\omega t}}{2}\hat{x} + \frac{e^{i\omega t} - e^{-i\omega t}}{2i}\frac{\hat{p}}{m\omega}\right) \\ &= \hat{x}\cos\omega t + \frac{\hat{p}}{m\omega}\sin\omega t \\ \hat{p}(t) &= i\sqrt{\frac{m\omega\hbar}{2}}(\hat{a}^\dagger(t) - \hat{a}(t)) = \hat{p}\cos\omega t - m\omega\hat{x}\sin\omega t\end{aligned}$$

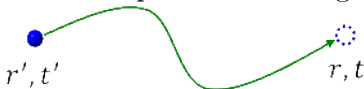
—Chapter 5—

Feynman Path Integral

5-1 Path Integral

A. PROPAGATOR

- (1) Consider a particle that is going to move from (r', t') to (r, t)



The probability of finding a particle at position r at time t is

$$|\Psi(r, t)|^2 = |\langle r | \Psi(t) \rangle|^2,$$

where $|\Psi(t)\rangle = \hat{U}(t, t')|\Psi(t')\rangle$ is a state evolving in time.

Thus, we obtain

$$\begin{aligned} \Psi(r, t) &= \langle r | \Psi(t) \rangle \\ &= \langle r | \hat{U}(t, t') | \Psi(t') \rangle \\ &= \langle r | \hat{U}(t, t') \underbrace{\int |r'\rangle \langle r' | dr'}_{\hat{1}} | \Psi(t') \rangle \\ &= \int \langle r | \hat{U}(t, t') | r' \rangle \langle r' | \Psi(t') \rangle dr' \\ &= \int K(r, t; r', t') \Psi(r', t') dr' \dots (a) \end{aligned}$$

where $K(r, t; r', t')$ is referred to as the propagator

$$K(r, t; r', t') = \langle r | \hat{U}(t, t') | r' \rangle$$

OS:

Huygens' principle:

Equation (a) illustrates that the wavefunction at r and t is determined by the wavefunction at the original time t' , receiving contributions from all r' which are weighted by the amplitude K

EXAMPLES:

1. The propagation of a free particle from (x', t') to (x, t)

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

$$\hat{U}(t, t') = e^{-\frac{i}{\hbar} \int_{t'}^t \hat{H} dt_1} = e^{-\frac{i}{\hbar} \hat{H} \cdot (t-t')} = e^{-i \frac{\hat{p}^2 \cdot (t-t')}{2m\hbar}}$$

$$\begin{aligned}
K(x, t; x', t') &= \langle x | e^{-i\frac{\hat{p}^2 \cdot (t-t')}{2m\hbar}} | x' \rangle \\
&= \int \langle x | e^{-i\frac{\hat{p}^2 \cdot (t-t')}{2m\hbar}} | p \rangle \langle p | x' \rangle dp \\
&= \int e^{-i\frac{p^2 \cdot (t-t')}{2m\hbar}} \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}} \frac{e^{-ipx'/\hbar}}{\sqrt{2\pi\hbar}} dp \\
&= \frac{1}{2\pi\hbar} \int e^{-\frac{i(t-t')}{2m\hbar} p^2 + \frac{i(x-x')}{\hbar} p} dp
\end{aligned}$$

Using the Gauss integral

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} e^{b^2/4a}$$

We obtain

$$\begin{aligned}
K(x, t; x', t') &= \frac{1}{2\pi\hbar} \sqrt{\frac{\pi}{i(t-t')/2m\hbar}} e^{\left(\frac{i(x-x')}{\hbar}\right)^2 / 4 \left(\frac{i(t-t')}{2m\hbar}\right)} \\
&= \sqrt{\frac{m}{2\pi i\hbar(t-t')}} e^{\frac{im(x-x')^2}{2\hbar(t-t')}}
\end{aligned}$$

2. Using the normalization condition:

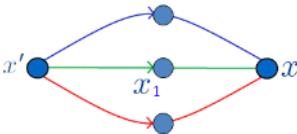
$$\int K(x, t'; x', t') dx' = 1$$

We obtain

$$\begin{aligned}
K(x, t'; x', t') &= \lim_{t \rightarrow t'} \langle x | e^{-i\frac{\hat{p}^2 \cdot (t-t')}{2m\hbar}} | x' \rangle = \langle x | x' \rangle \\
&= \lim_{t \rightarrow t'} \sqrt{\frac{m}{2\pi i\hbar(t-t')}} e^{-\frac{m(x-x')^2}{2i\hbar(t-t')}} = \delta(x-x')
\end{aligned}$$

From equations above, we get $\langle x | x' \rangle = \delta(x-x')$

(2) Split the time interval into two pieces



$$\begin{aligned}
K(x, t; x', t') &= \langle x | \hat{U}(t, t_1) \hat{U}(t_1, t') | x' \rangle \\
&= \int \langle x | \hat{U}(t, t_1) | x_1 \rangle \langle x_1 | \hat{U}(t_1, t') | x' \rangle dx_1 \\
&= \int K(x, t; x_1, t_1) K(x_1, t_1; x', t') dx_1
\end{aligned}$$

The propagation from (x', t') to (x, t) can be viewed as the sum of all paths that go through intermediate states (x_1, t_1) .

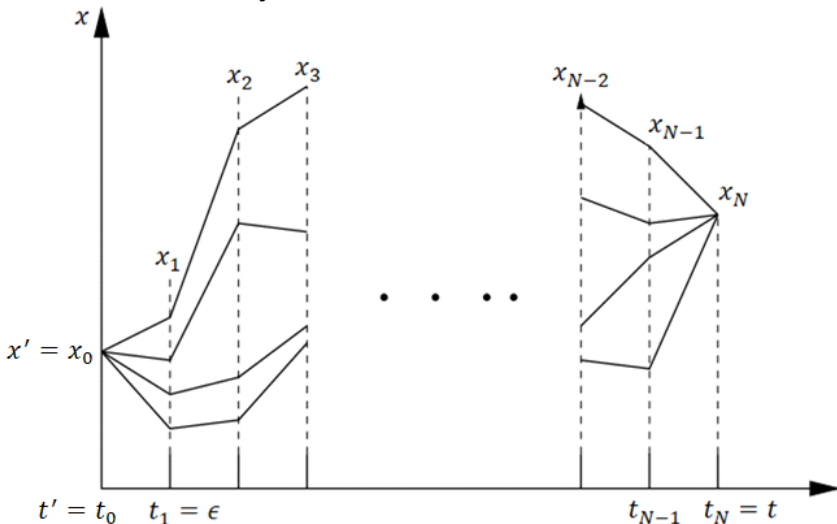
(3) Split the time interval into N infinitesimal pieces

The propagator of a single particle can be decomposed by dividing the time interval $t - t'$ into N infinitesimal pieces, i.e.,

$$K(x, t; x', t') = \langle x | \hat{U}(t, t_{N-1}) \hat{U}(t_{N-1}, t_{N-2}) \cdots \hat{U}(t_1, t') | x' \rangle$$

Insert $N - 1$ resolutions of identity

$$\begin{aligned}
K(x, t; x', t') &= \langle x | \hat{U}(t, t_{N-1}) \underbrace{\int |x_{N-1}\rangle \langle x_{N-1}| dx_{N-1}}_{\hat{1}} \hat{U}(t_{N-1}, t_{N-2}) \cdots \\
&\quad \times \hat{U}(t_2, t_1) \underbrace{\int |x_1\rangle \langle x_1| dx_1}_{\hat{1}} \hat{U}(t_1, t') | x' \rangle \\
&= \int dx_{N-1} \cdots dx_1 \langle x | \hat{U}(t, t_{N-1}) | x_{N-1} \rangle \cdots \langle x_1 | \hat{U}(t_1, t') | x' \rangle \\
&= \int dx_{N-1} \cdots dx_1 K(x, t; x_{N-1}, t_{N-1}) \cdots K(x_1, t_1; x', t')
\end{aligned}$$



The propagation from (x', t') to (x, t) can be viewed as the sum of all

paths that go through intermediate states on each time slice.

(4) The propagator on each time slice

Let $t' = t_0$ and $t = t_N$

In the infinitesimal time interval,

$$\frac{t_N - t_0}{N} = \epsilon, \quad t_i - t_{i-1} = \epsilon$$

The Hamiltonian can be considered as time-independent

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$$

$$\hat{U}(t_i, t_{i-1}) = e^{-\frac{i}{\hbar} \hat{H} \cdot (t_i - t_{i-1})} = e^{-\frac{i \hat{H} \epsilon}{\hbar}}$$

$$\begin{aligned} K(x_i, t_i; x_{i-1}, t_{i-1}) &= \langle x_i | e^{-\frac{i \hat{H} \epsilon}{\hbar}} | x_{i-1} \rangle \\ &= \langle x_i | \left(1 - \frac{i \epsilon}{\hbar} \hat{H} + O(\epsilon^2) \right) | x_{i-1} \rangle \\ &= \langle x_i | x_{i-1} \rangle - \frac{i \epsilon}{\hbar} \langle x_i | \hat{H} | x_{i-1} \rangle + O(\epsilon^2) \end{aligned}$$

Here

$$\begin{aligned} \langle x_i | x_{i-1} \rangle &= \delta(x_i - x_{i-1}) = \int e^{\frac{i p_i (x_i - x_{i-1})}{\hbar}} \frac{dp_i}{2\pi\hbar} \\ -\frac{i \epsilon}{\hbar} \langle x_i | \hat{H} | x_{i-1} \rangle &= -\frac{i \epsilon}{\hbar} \int \langle x_i | \left(\frac{\hat{p}^2}{2m} + V(\hat{x}) \right) | p_i \rangle \langle p_i | x_{i-1} \rangle dp_i \\ &= -\frac{i \epsilon}{\hbar} \int \left(\frac{p_i^2}{2m} + V(x_i) \right) \langle x_i | p_i \rangle \langle p_i | x_{i-1} \rangle dp_i \\ &= -\frac{i \epsilon}{\hbar} \int \left(\frac{p_i^2}{2m} + V(x_i) \right) \frac{e^{\frac{i p_i x_i}{\hbar}}}{\sqrt{2\pi\hbar}} \frac{e^{-\frac{i p_i x_{i-1}}{\hbar}}}{\sqrt{2\pi\hbar}} dp_i \\ &= -\frac{i \epsilon}{\hbar} \int \left(\frac{p_i^2}{2m} + V(x_i) \right) e^{\frac{i p_i (x_i - x_{i-1})}{\hbar}} \frac{dp_i}{2\pi\hbar} \\ K(x_i, t_i; x_{i-1}, t_{i-1}) &= \int \left[1 - \frac{i \epsilon}{\hbar} \left(\frac{p_i^2}{2m} + V(x_i) \right) + O(\epsilon^2) \right] e^{\frac{i p_i (x_i - x_{i-1})}{\hbar}} \frac{dp_i}{2\pi\hbar} \end{aligned}$$

$$\text{Since } 1 - \frac{i \epsilon}{\hbar} \left(\frac{p_i^2}{2m} + V(x_i) \right) + O(\epsilon^2) = e^{-\frac{i \epsilon}{\hbar} \left(\frac{p_i^2}{2m} + V(x_i) \right)}$$

we obtain

$$K(x_i, t_i; x_{i-1}, t_{i-1}) = \int e^{-\frac{i\epsilon}{\hbar} \left(\frac{p_i^2}{2m} + V(x_i) \right)} e^{\frac{ip_i(x_i - x_{i-1})}{\hbar}} \frac{dp_i}{2\pi\hbar}$$

Since $e^{\frac{ip_i(x_i - x_{i-1})}{\hbar}} = e^{\frac{i\epsilon p_i (x_i - x_{i-1})}{\hbar \epsilon}} = e^{\frac{i\epsilon p_i \dot{x}_i}{\hbar}}$
we obtain

$$\begin{aligned} K(x_i, t_i; x_{i-1}, t_{i-1}) &= \int e^{-\frac{i\epsilon}{\hbar} \left(\frac{p_i^2}{2m} + V(x_i) - p_i \dot{x}_i \right)} \frac{dp_i}{2\pi\hbar} \\ &= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} e^{\frac{i\epsilon m \dot{x}_i^2}{\hbar \cdot 2}} e^{-\frac{i\epsilon}{\hbar} V(x_i)} \\ &= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} e^{\frac{i\epsilon}{\hbar} \left(\frac{m \dot{x}_i^2}{2} - V(x_i) \right)} \end{aligned}$$

Take $L(x_i, \dot{x}_i) = \frac{m \dot{x}_i^2}{2} - V(x_i) \dots\dots$ Lagrangian

$$K(x_i, t_i; x_{i-1}, t_{i-1}) = \sqrt{\frac{m}{2\pi i \hbar \epsilon}} e^{\frac{i\epsilon}{\hbar} L(x_i, \dot{x}_i, t)}$$

(5) The propagator for one "path"

$$K_{\text{path}} = \prod_{i=1}^N K(x_i, t_i; x_{i-1}, t_{i-1}) = \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{i}{\hbar} \sum_{i=1}^N L(x_i, \dot{x}_i) \epsilon}$$

In the limit $N \rightarrow \infty, \epsilon \rightarrow 0$, the summation goes over to an integral

$$\begin{aligned} \sum_{i=1}^N L(x_i, \dot{x}_i) \epsilon &\rightarrow \int_{t_0}^t L(x, \dot{x}) dt' \\ K_{\text{path}} &= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{i}{\hbar} \int_{t_0}^t L(x, \dot{x}) dt'} \end{aligned}$$

(6) Summing over all " N -legged" paths

$$\begin{aligned} K(x, t; x_0, t_0) &= \int K_{\text{path}} \prod_{k=1}^{N-1} dx_k \\ &= \int \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{i}{\hbar} \int_{t_0}^t L(x, \dot{x}) dt'} \prod_{k=1}^{N-1} dx_k \end{aligned}$$

Let $Dx(t) = \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \prod_{k=1}^{N-1} dx_k \dots\dots$ called measure

$$K(x, t; x_0, t_0) = \int e^{\frac{i}{\hbar} \int_{t_0}^t L(x, \dot{x}) dt'} Dx(t) = \int e^{\frac{iS[x(t)]}{\hbar}} Dx(t)$$

where $S[x(t)] = \int_{t_0}^t L(x, \dot{x}) dt'$

Feynman's conjecture: The propagator is the sum of all paths connecting x_0 and x . Each path is assigned a weight $e^{\frac{iS[x(t)]}{\hbar}}$.

EXAMPLES:

1. Harmonic oscillator

$$m\ddot{x} = -kx = -\frac{d}{dx}V$$

$$V = \int kx dx = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2, \quad \omega^2 = \frac{k}{m}$$

$$T = \frac{1}{2}m\dot{x}^2$$

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2$$

$$S = \int_a^b L(x, \dot{x}) dt = \int_a^b \left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2 \right) dt$$

Considering

$$\frac{d}{dt}(x\dot{x}) = \dot{x}^2 + x\ddot{x} = \dot{x}^2 + x\left(-\frac{k}{m}x\right) = \dot{x}^2 - \omega^2 x^2$$

We obtain

$$\begin{aligned} S &= \int_a^b \left[\frac{1}{2}m \left[\left(\frac{d}{dt}(x\dot{x}) \right) + \omega^2 x^2 \right] - \frac{1}{2}m\omega^2 x^2 \right] dt \\ &= \int_a^b \left[\frac{1}{2}m \left(\frac{d}{dt}(x\dot{x}) \right) + \frac{1}{2}m\omega^2 x^2 - \frac{1}{2}m\omega^2 x^2 \right] dt \\ &= \int_a^b \frac{1}{2}m \left(\frac{d}{dt}(x\dot{x}) \right) dt \\ &= \frac{1}{2}m(x_b\dot{x}_b - x_a\dot{x}_a) \end{aligned}$$

Let

$$\begin{aligned} x &= A \cos \omega t + B \sin \omega t, & t_a &= t_0, & t_b &= t \\ x(t_0) &= x_0 = A \\ \dot{x} &= -A\omega \sin \omega t + B\omega \cos \omega t \end{aligned}$$

We obtain

$$S = \frac{m\omega}{2} \frac{(x^2 + x_0^2) \cos \omega(t - t_0) - 2xx_0}{\sin \omega(t - t_0)}$$

B. DERIVE SCHRÖDINGER EQUATION

(1) Method I: time evolution of wavefunction

In the infinitesimal time interval $\epsilon \rightarrow 0$, the Hamiltonian is time-independent

$$\begin{aligned}
 \Psi(x, t + \epsilon) &= \langle x | \Psi(t + \epsilon) \rangle \\
 &= \langle x | e^{-i\hat{H}\epsilon/\hbar} | \Psi(t) \rangle \\
 &= \int \langle x | e^{-i\hat{H}\epsilon/\hbar} | x_0 \rangle \langle x_0 | \Psi(t) \rangle dx_0 \\
 &= \int K(x, t + \epsilon; x_0, t) \Psi(x_0, t) dx_0 \\
 &= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} \int e^{\frac{i}{\hbar} \left[\frac{m(x-x_0)^2}{2\epsilon} - \epsilon V(x) \right]} \Psi(x_0, t) dx_0
 \end{aligned}$$

Let $\eta = x_0 - x$

$$\Psi(x, t + \epsilon) = \sqrt{\frac{m}{2\pi i \hbar \epsilon}} \int e^{\frac{im\eta^2}{2\hbar\epsilon}} e^{-\frac{i}{\hbar}\epsilon V(x)} \Psi(x + \eta, t) d\eta$$

Since $\epsilon \rightarrow 0$ and $\eta \rightarrow 0$, we expand

$$\begin{aligned}
 &e^{-\frac{i}{\hbar}\epsilon V(x)} \Psi(x + \eta, t) \\
 &= \left(1 - \frac{i\epsilon}{\hbar} V(x) + \dots \right) \left(\Psi(x, t) + \eta \frac{d\Psi}{dx} + \frac{\eta^2}{2} \frac{d^2\Psi}{dx^2} + \dots \right) \\
 &= \Psi(x, t) + \eta \frac{\partial \Psi(x, t)}{\partial x} + \frac{\eta^2}{2} \frac{\partial^2 \Psi(x, t)}{\partial x^2} - \frac{i\epsilon}{\hbar} V(x) \Psi(x, t) + \dots \\
 \Rightarrow \Psi(x, t + \epsilon) &= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} \int e^{\frac{im\eta^2}{2\hbar\epsilon}} \left[\Psi(x, t) + \eta \frac{\partial \Psi(x, t)}{\partial x} + \frac{\eta^2}{2} \frac{\partial^2 \Psi(x, t)}{\partial x^2} \right. \\
 &\quad \left. - \frac{i\epsilon}{\hbar} V(x) \Psi(x, t) \right] d\eta
 \end{aligned}$$

Gauss integral

$$\begin{aligned}
 \int e^{\frac{im\eta^2}{2\hbar\epsilon}} d\eta &= \int e^{-\frac{m\eta^2}{2i\hbar\epsilon}} d\eta = \sqrt{\frac{2\pi i \hbar \epsilon}{m}} \\
 \int e^{\frac{im\eta^2}{2\hbar\epsilon}} \eta^2 d\eta &= \int e^{-\frac{m\eta^2}{2i\hbar\epsilon}} \eta^2 d\eta = \frac{1}{2} \frac{2i\hbar\epsilon}{m} \sqrt{\frac{2\pi i \hbar \epsilon}{m}} = -\frac{\hbar\epsilon}{im} \sqrt{\frac{2\pi i \hbar \epsilon}{m}}
 \end{aligned}$$

$$\begin{aligned}
\Rightarrow \Psi(x, t + \epsilon) &= \sqrt{\frac{m}{2\pi i \hbar \epsilon}} \left[\sqrt{\frac{2\pi i \hbar \epsilon}{m}} \Psi(x, t) - \frac{\hbar \epsilon}{im} \sqrt{\frac{2\pi i \hbar \epsilon}{m}} \frac{1}{2} \frac{\partial^2 \Psi(x, t)}{\partial x^2} \right. \\
&\quad \left. - \frac{i\epsilon}{\hbar} V(x) \sqrt{\frac{2\pi i \hbar \epsilon}{m}} \Psi(x, t) \right] \\
&= \Psi(x, t) - \frac{\hbar \epsilon}{i2m} \frac{\partial^2}{\partial x^2} \Psi(x, t) - \frac{i\epsilon}{\hbar} V(x) \Psi(x, t) \\
\Rightarrow \frac{i\hbar [\Psi(x, t + \epsilon) - \Psi(x, t)]}{\epsilon} &= \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x, t)
\end{aligned}$$

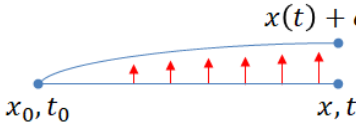
As $\epsilon \rightarrow 0$

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi(x, t)$$

(2) Method II: path integral

$$K(x, t) = \int e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \dots \dots (a)$$

We shift the trajectory $x(t)$ by a small amount $\delta x(t)$ with the boundary condition that x_0 is held fixed, i.e., $\delta x(t_0) = 0$.



L.H.S. of equation (a):

$$K(x + \delta x(t), t) - K(x) = \frac{\partial}{\partial x} K(x, t) \delta x(t)$$

R.H.S. of equation (a):

$$\begin{aligned}
&\int e^{\frac{i}{\hbar} S[x(t) + \delta x(t)]} Dx(t) - \int e^{\frac{iS[x(t)]}{\hbar}} Dx(t) = \int \frac{i\delta S}{\hbar} e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \\
&\Rightarrow \frac{\partial}{\partial x} K(x, t) \delta x(t) = \int \frac{i\delta S}{\hbar} e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \\
&\Rightarrow -i\hbar \frac{\partial}{\partial x} K(x, t) \delta x(t) = \int \delta S e^{\frac{iS[x(t)]}{\hbar}} Dx(t)
\end{aligned}$$

Recall that the action changes by

$$\begin{aligned}
\delta S &= S[x(t) + \delta x(t)] - S[x(t)] \\
&= \int_{t_0}^t \left(\frac{\partial L}{\partial x} \delta x + \frac{\partial L}{\partial \dot{x}} \delta \dot{x} \right) dt \\
&= \int_{t_0}^t \frac{\partial L}{\partial x} \delta x dt + \frac{\partial L}{\partial \dot{x}} \delta x \Big|_{t_0}^t - \int_{t_0}^t \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) \delta x dt \\
&= \frac{\partial L}{\partial \dot{x}} \delta x \Big|_{t_0}^t + \int_{t_0}^t \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt \\
&= \frac{\partial L}{\partial \dot{x}} \delta x(t) \\
&= p(t) \delta x(t) \\
&\Rightarrow -i\hbar \frac{\partial}{\partial x} K(x, t) \delta x(t) = \int p(t) \delta x(t) e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \\
&\Rightarrow -i\hbar \frac{\partial}{\partial x} K(x, t) = \int p(t) e^{\frac{iS[x(t)]}{\hbar}} Dx(t)
\end{aligned}$$

The equation of motion of $K(x, t)$

$$\frac{\partial}{\partial t} K(x, t) = \int \frac{i}{\hbar} \frac{\partial S}{\partial t} e^{\frac{iS[x(t)]}{\hbar}} Dx(t)$$

Recall that $\frac{\partial S}{\partial t} = -H(t)$, [c.f.1-5]

$$\begin{aligned}
\frac{\partial}{\partial t} K(x, t) &= \int \frac{i}{\hbar} (-H(t)) e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \\
&\Rightarrow i\hbar \frac{\partial}{\partial t} K(x, t) = \int H(t) e^{\frac{iS[x(t)]}{\hbar}} Dx(t)
\end{aligned}$$

If $H = \frac{p^2}{2m} + V(x)$

$$\begin{aligned}
i\hbar \frac{\partial}{\partial t} K(x, t) &= \int \left(\frac{p^2}{2m} + V(x) \right) e^{\frac{iS[x(t)]}{\hbar}} Dx(t) \\
&= \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] K(x, t)
\end{aligned}$$

5-2 Propagator of a Free Particle and a Harmonic Oscillator

A. FREE PARTICLE IN GAUSS INTEGRAL APPROACH

(1) Path integral in discretized time slices

$$K(x, t; x_0, t_0) = \int e^{\frac{i}{\hbar} \int L(x, \dot{x}) dt'} Dx(t) = \int e^{\frac{iS[x(t)]}{\hbar}} Dx(t)$$

The Lagrangian is

$$L(x, \dot{x}) = \frac{m\dot{x}^2}{2}$$

As N is large and ϵ is finite, the integral can be written as a summation

$$S = \epsilon \sum_{i=1}^N \frac{m\dot{x}_i^2}{2} = \frac{m}{2} \epsilon \sum_{i=1}^N \left(\frac{x_i - x_{i-1}}{\epsilon} \right)^2 = \frac{m}{2} \sum_{i=1}^N \frac{(x_i - x_{i-1})^2}{\epsilon}$$

$$\text{Let } y_i = \sqrt{\frac{m}{2\hbar\epsilon}} x_i \Rightarrow x_i = \sqrt{\frac{2\hbar\epsilon}{m}} y_i, \text{ and } dx_i = \sqrt{\frac{2\hbar\epsilon}{m}} dy_i$$

The action becomes

$$S = \frac{m}{2} \sum_{i=1}^N \frac{(x_i - x_{i-1})^2}{\epsilon} = \hbar \sum_{i=1}^N (y_i - y_{i-1})^2$$

The measure becomes

$$\begin{aligned} Dx(t) &= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \prod_{k=1}^{N-1} dx_k \\ &= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \left(\frac{2\hbar\epsilon}{m} \right)^{(N-1)/2} \prod_{k=1}^{N-1} dy_k \\ &= \sqrt{\frac{m}{2\hbar\epsilon}} (i\pi)^{-N/2} dy_1 dy_2 \cdots dy_{N-1} \\ K(x, t; x_0, t_0) &= \sqrt{\frac{m}{2\hbar\epsilon}} (i\pi)^{-N/2} \int e^{i \sum_{i=1}^N (y_i - y_{i-1})^2} dy_1 dy_2 \cdots dy_{N-1} \end{aligned}$$

(2) The integration over dy_1 , i.e., $i = 1$ yields

$$\begin{aligned}\int e^{i(y_1-y_0)^2+i(y_2-y_1)^2} dy_1 &= \int e^{-\frac{2}{i}\left(y_1-\frac{y_2+y_0}{2}\right)^2+i\frac{(y_2-y_0)^2}{2}} dy_1 \\ &= \sqrt{\frac{\pi i}{2}} e^{i\frac{(y_2-y_0)^2}{2}}\end{aligned}$$

The integration over dy_2 , i.e., $i = 2$ yields

$$\begin{aligned}\sqrt{\frac{\pi i}{2}} \int e^{\frac{i}{2}(y_2-y_0)^2+i(y_3-y_2)^2} dy_2 &= \sqrt{\frac{\pi i}{2}} \int e^{-\frac{3}{i2}\left(y_2-\frac{y_0+2y_3}{3}\right)^2+i\frac{(y_3-y_0)^2}{3}} dy_2 \\ &= \sqrt{\frac{\pi i}{2}} \sqrt{\frac{2\pi i}{3}} e^{i\frac{(y_3-y_0)^2}{3}} \\ &= \sqrt{\frac{(\pi i)^2}{3}} e^{\frac{i}{3}(y_3-y_0)^2} \\ \int e^{i\sum_{i=1}^N (y_i-y_{i-1})^2} dy_1 dy_2 \cdots dy_{N-1} &= \frac{(\pi i)^{(N-1)/2}}{\sqrt{N}} e^{\frac{i}{N}(y_N-y_0)^2} \\ K(x, t; x_0, t_0) &= \sqrt{\frac{m}{2\hbar\epsilon}} (i\pi)^{-N/2} \frac{(\pi i)^{(N-1)/2}}{\sqrt{N}} e^{\frac{i}{N}(y_N-y_0)^2} \\ &= \sqrt{\frac{m}{2\pi i\hbar N\epsilon}} e^{\frac{im}{2\hbar N\epsilon}(x_N-x_0)^2} \\ &= \sqrt{\frac{m}{2\pi i\hbar(t-t_0)}} e^{\frac{im(x-x_0)^2}{2\hbar(t-t_0)}}\end{aligned}$$

B. FREE PARTICLE IN MULTI-DIMENSIONAL GAUSS INTEGRAL APPROACH

- (1) Write the action in the matrix form:

The action in discretized time slices is

$$\begin{aligned}
S &= \epsilon \sum_{i=1}^N \frac{m}{2} \left(\frac{x_i - x_{i-1}}{\epsilon} \right)^2 \\
&= \frac{m}{2\epsilon} \left[(x_N - x_{N-1})^2 + \cdots + (x_2 - x_1)^2 + (x_1 - x_0)^2 \right] \\
&= \frac{m}{2\epsilon} [x_N^2 - 2x_N x_{N-1} + x_{N-1}^2 \cdots + x_2^2 - 2x_2 x_1 + x_1^2 + x_1^2 - 2x_1 x_0 + x_0^2] \\
&= \frac{m}{2\epsilon} [(x_N^2 + x_0^2) - (2x_N x_{N-1} + 2x_1 x_0) + (2x_{N-1}^2 \cdots 2x_2^2 - 2x_2 x_1 + 2x_1^2)]
\end{aligned}$$

The 2nd term can be rewritten as

$$\frac{m}{2\epsilon} [2x_N x_{N-1} + 2x_1 x_0] = \frac{m}{\epsilon} (x_N, 0, \dots, 0, x_0) \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix}$$

$$\text{Let } \mathbf{x} = \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix} \text{ and } \mathbf{b} = -\frac{im}{\hbar\epsilon} \begin{pmatrix} x_N \\ 0 \\ \vdots \\ 0 \\ x_0 \end{pmatrix}$$

$$\frac{m}{2\epsilon} [2x_N x_{N-1} + 2x_1 x_0] = \frac{\hbar}{i} \mathbf{b}^T \cdot \mathbf{x}$$

The 3rd term can be rewritten as

$$\begin{aligned}
&\frac{m}{2\epsilon} [2x_{N-1}^2 + \cdots + 2x_2^2 - 2x_2 x_1 + 2x_1^2] \\
&= \frac{m}{2\epsilon} (x_{N-1}, x_{N-2}, \dots, x_2, x_1) \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & \ddots & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix}
\end{aligned}$$

$$\text{Let } \mathbf{x} = \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix} \text{ and } \mathbf{A} = -\frac{im}{\hbar\epsilon} \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & \ddots & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

$$\frac{m}{2\epsilon} [2x_{N-1}^2 + \cdots + 2x_2^2 - 2x_2 x_1 + 2x_1^2] = -\frac{\hbar}{i} \frac{1}{2} \mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x}$$

$$\Rightarrow S = \frac{m}{2\epsilon} (x_N^2 + x_0^2) + \frac{\hbar}{i} \left(-\frac{1}{2} \mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x} + \mathbf{b}^T \cdot \mathbf{x} \right)$$

$$\begin{aligned}
K(x, t; x_0, t_0) &= \int e^{iS/\hbar} D\mathbf{x}(t) \\
&= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{im}{2\hbar\epsilon}(x_N^2 + x_0^2)} \int \exp\left(-\frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) \prod_{k=1}^{N-1} dx_k
\end{aligned}$$

The multi-dimensional Gauss integral

$$\int \exp\left(-\frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) \prod_{k=1}^{N-1} dx_k = \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} \exp\left(\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right)$$

Thus, we obtain

$$\begin{aligned}
K(x, t; x_0, t_0) &= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{im}{2\hbar\epsilon}(x_N^2 + x_0^2)} \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} \exp\left(\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right) \\
&= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} \exp\left(\frac{im}{2\hbar\epsilon}(x_N^2 + x_0^2) + \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right)
\end{aligned}$$

(2) Calculate $\det \mathbf{A}$ and $\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$

$$\text{Let } \mathbf{K}_N = \begin{pmatrix} 1 & -a & 0 & 0 & 0 \\ -a & 1 & -a & 0 & 0 \\ 0 & -a & \ddots & -a & 0 \\ 0 & 0 & -a & 1 & -a \\ 0 & 0 & 0 & -a & 1 \end{pmatrix}, \quad a = \frac{1}{2}$$

$$\begin{aligned}
\det \mathbf{A} &= \left(\frac{2m}{i\hbar\epsilon} \right)^{N-1} \det \mathbf{K}_N \\
&= \left(\frac{2m}{i\hbar\epsilon} \right)^{N-1} \frac{\lambda_+^N - \lambda_-^N}{\lambda_+ - \lambda_-}, \quad \text{where } \lambda_{\pm} = \frac{1}{2} (1 \pm \sqrt{1 - 4a^2}) = \frac{1}{2} \\
&= \left(\frac{2m}{i\hbar\epsilon} \right)^{N-1} (\lambda_+^{N-1} + \lambda_+^{N-2} \lambda_- + \dots + \lambda_+ \lambda_-^{N-2} + \lambda_-^{N-1}) \\
&= \left(\frac{2m}{i\hbar\epsilon} \right)^{N-1} N \left(\frac{1}{2} \right)^{N-1} \\
&= \left(\frac{m}{i\hbar\epsilon} \right)^{N-1} N
\end{aligned}$$

$$\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} = \frac{1}{2} \left(\frac{im}{\hbar\epsilon} \right)^2 (x_N, 0, \dots, 0, x_0) \cdot \mathbf{A}^{-1} \cdot \begin{pmatrix} x_N \\ 0 \\ \vdots \\ 0 \\ x_0 \end{pmatrix}$$

Here, we need only $(1,1)$, $(N-1, N-1)$, $(1, N-1)$, and $(N-1, 1)$ components of $\mathbf{A}^{-1}$.

$$\begin{aligned}
 (\mathbf{A}^{-1})_{1,1} &= (\mathbf{A}^{-1})_{N-1,N-1} = \frac{i\hbar\epsilon \det \mathbf{K}_{N-2}}{2m \det \mathbf{K}_{N-1}} = \frac{i\hbar\epsilon}{2m} \frac{N-1}{N} \\
 (\mathbf{A}^{-1})_{1,N-1} &= (\mathbf{A}^{-1})_{N-1,1} \\
 &= \frac{i\hbar\epsilon (-1)^{N-2}}{2m \det \mathbf{K}_{N-1}} \det \begin{pmatrix} -1/2 & 1 & -1/2 & 0 \\ 0 & -1/2 & \ddots & -1/2 \\ 0 & 0 & -1/2 & 1 \\ 0 & 0 & 0 & -1/2 \end{pmatrix} \\
 &= \frac{i\hbar\epsilon}{2m} \frac{2^{N-1}}{N} \frac{1}{2^{N-2}} \\
 &= \frac{i\hbar\epsilon}{mN} \\
 \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} &= \frac{1}{2} \left(\frac{im}{\hbar\epsilon} \right)^2 \frac{i\hbar\epsilon}{m} \frac{N-1}{N} (x_N^2 + x_0^2) + \frac{1}{2} \left(\frac{im}{\hbar\epsilon} \right)^2 \frac{i\hbar\epsilon}{mN} 2x_N x_0
 \end{aligned}$$

(3) The prefactor

$$\begin{aligned}
 \left(\frac{m}{2\pi i\hbar\epsilon} \right)^{N/2} \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} &= \left(\frac{m}{2\pi i\hbar\epsilon} \right)^{N/2} \sqrt{\frac{(2\pi)^{N-1}}{\left(\frac{m}{i\hbar\epsilon} \right)^{N-1} N}} \\
 &= \sqrt{\frac{m}{2\pi i\hbar N\epsilon}} \\
 &= \sqrt{\frac{m}{2\pi i\hbar(t-t_0)}}
 \end{aligned}$$

The exponent

$$\begin{aligned}
& \frac{im}{\hbar 2\epsilon} (x_N^2 + x_0^2) + \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \\
&= \frac{im}{\hbar 2\epsilon} (x_N^2 + x_0^2) + \frac{1}{2} \left(\frac{im}{\hbar \epsilon} \right)^2 \frac{i\hbar \epsilon}{m} \frac{N-1}{N} (x_N^2 + x_0^2) + \frac{1}{2} \left(\frac{im}{\hbar \epsilon} \right)^2 \frac{i\hbar \epsilon}{mN} 2x_N x_0 \\
&= \frac{im}{2\hbar \epsilon} \frac{1}{N} (x_N^2 + x_0^2) - \frac{im}{2\hbar N \epsilon} 2x_N x_0 \\
&= \frac{im}{2\hbar} \frac{(x_N - x_0)^2}{N \epsilon} \\
&= \frac{im(x - x_0)^2}{2\hbar(t - t_0)}
\end{aligned}$$

The propagator of the free particle is

$$K(x, t; x_0, t_0) = \sqrt{\frac{m}{2\pi i \hbar (t - t_0)}} e^{\frac{im(x - x_0)^2}{2\hbar(t - t_0)}}$$

C. HARMONIC OSCILLATOR IN MULTI-DIMENSIONAL GAUSS INTEGRAL APPROACH

- (1) Write the action in the matrix form:

The action in discretized time slices is

$$\begin{aligned}
S &= \epsilon \left[\sum_{i=1}^N \frac{m}{2} \left(\frac{x_i - x_{i-1}}{\epsilon} \right)^2 - \frac{1}{2} m \omega^2 \sum_{i=1}^{N-1} x_i^2 - \frac{1}{2} m \omega^2 \frac{(x_N^2 + x_0^2)}{2} \right] \\
&= \frac{m}{2\epsilon} \sum_{i=1}^N (x_i - x_{i-1})^2 - \frac{\epsilon m \omega^2}{2} \sum_{i=1}^{N-1} x_i^2 - \frac{\epsilon m \omega^2}{2} \frac{(x_N^2 + x_0^2)}{2}
\end{aligned}$$

The last term is there to correctly account for all $N - 1$ time slices for the potential energy term $-\int V dt'$, but to retain the symmetry between the initial and final positions, we took their average.

$$\begin{aligned}
S = & \frac{m}{2\epsilon}(x_N^2 + x_0^2) - \frac{m}{\epsilon}(x_N, 0, \dots, 0, x_0) \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix} \\
& + \frac{m}{2\epsilon}(x_{N-1}, x_{N-2}, \dots, x_2, x_1) \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & \ddots & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix} \\
& - \frac{\epsilon m \omega^2}{2}(x_{N-1}, x_{N-2}, \dots, x_2, x_1) \cdot \mathbf{1} \cdot \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix}
\end{aligned}$$

$$- \frac{\epsilon m \omega^2 (x_N^2 + x_0^2)}{2}$$

Let $\mathbf{x} = \begin{pmatrix} x_{N-1} \\ x_{N-2} \\ \vdots \\ x_2 \\ x_1 \end{pmatrix}$, $\mathbf{b} = -\frac{im}{\hbar\epsilon} \begin{pmatrix} x_N \\ 0 \\ \vdots \\ 0 \\ x_0 \end{pmatrix}$, and

$$\begin{aligned}
\mathbf{A} = & -\frac{i2m}{\hbar\epsilon} \begin{pmatrix} 1 & -1/2 & 0 & 0 & 0 \\ -1/2 & 1 & -1/2 & 0 & 0 \\ 0 & -1/2 & \ddots & -1/2 & 0 \\ 0 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 0 & -1/2 & 1 \end{pmatrix} + \frac{i\epsilon m \omega^2}{\hbar} \mathbf{1} \\
= & \frac{m}{i\hbar\epsilon} (2 - \omega^2 \epsilon^2) \begin{pmatrix} 1 & -a & 0 & 0 & 0 \\ -a & 1 & -a & 0 & 0 \\ 0 & -a & \ddots & -a & 0 \\ 0 & 0 & -a & 1 & -a \\ 0 & 0 & 0 & -a & 1 \end{pmatrix}
\end{aligned}$$

where $a = \frac{\frac{1}{2} \frac{2m}{i\hbar\epsilon}}{\frac{m}{i\hbar\epsilon} (2 - \omega^2 \epsilon^2)} = \frac{1}{2 - \omega^2 \epsilon^2}$

The action becomes

$$S = \frac{m}{2\epsilon}(x_N^2 + x_0^2) - \frac{\epsilon m \omega^2}{4}(x_N^2 + x_0^2) + \frac{\hbar}{i} \left(-\frac{1}{2} \mathbf{x}^T \cdot \mathbf{A} \cdot \mathbf{x} + \mathbf{b}^T \mathbf{x} \right)$$

The propagator is

$$\begin{aligned}
K(x, t; x_0, t_0) &= \int \exp \left(\frac{im}{2\hbar\epsilon} (x_N^2 + x_0^2) - \frac{i\epsilon m \omega^2}{4\hbar} (x_N^2 + x_0^2) - \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} \right) D\mathbf{x}(t) \\
&= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} \exp \left(\frac{im}{2\hbar\epsilon} (x_N^2 + x_0^2) - \frac{i\epsilon m \omega^2}{4\hbar} (x_N^2 + x_0^2) \right) \\
&\quad \times \int \exp \left(-\frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} \right) \prod_{k=1}^{N-1} dx_k
\end{aligned}$$

The multi-dimensional Gauss integral

$$\int \exp \left(-\frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} \right) \prod_{k=1}^{N-1} dx_k = \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} \exp \left(\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \right)$$

We obtain

$$\begin{aligned}
K(x, t; x_0, t_0) &= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} e^{\frac{im}{2\hbar\epsilon}(x_N^2 + x_0^2) - \frac{i\epsilon m \omega^2}{4\hbar}(x_N^2 + x_0^2)} \sqrt{\frac{(2\pi)^{N-1}}{\det \mathbf{A}}} \exp \left(\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \right) \\
&= \left(\frac{m}{2\pi i \hbar \epsilon} \right)^{N/2} (2\pi)^{(N-1)/2} \frac{1}{\sqrt{\det \mathbf{A}}} \\
&\quad \times \exp \left(\frac{im}{2\hbar\epsilon} (x_N^2 + x_0^2) - \frac{i\epsilon m \omega^2}{4\hbar} (x_N^2 + x_0^2) + \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \right)
\end{aligned}$$

(2) Calculate $\det \mathbf{A}$ and $\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$

$$\text{Let } \mathbf{K}_N = \begin{pmatrix} 1 & -a & 0 & 0 & 0 \\ -a & 1 & -a & 0 & 0 \\ 0 & -a & \ddots & -a & 0 \\ 0 & 0 & -a & 1 & -a \\ 0 & 0 & 0 & -a & 1 \end{pmatrix}, \quad \text{where } a = \frac{1}{2 - \omega^2 \epsilon^2}$$

$$\det \mathbf{A} = \left(\frac{m}{i\hbar\epsilon} \right)^{N-1} \det \mathbf{K}_N = \left(\frac{m}{i\hbar\epsilon} \right)^{N-1} (2 - \omega^2 \epsilon^2)^{N-1} \frac{\lambda_+^N - \lambda_-^N}{\lambda_+ - \lambda_-}$$

$$\text{where } \lambda_{\pm} = \frac{1}{2} \left(1 \pm \sqrt{1 - \frac{4}{(2 - \omega^2 \epsilon^2)^2}} \right)$$

In the limit $\epsilon = \frac{t - t_0}{N}$ and $N \rightarrow \infty$

$$\lambda_{\pm} \approx \frac{1}{2} (1 \pm i\omega\epsilon)$$

$$\frac{\lambda_+^N - \lambda_-^N}{\lambda_+ - \lambda_-} \approx \frac{1}{2^N} \frac{(1 + i\omega\epsilon)^N - (1 - i\omega\epsilon)^N}{(1 + i\omega\epsilon) - (1 - i\omega\epsilon)}$$

Since $\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$

$$\frac{\lambda_+^N - \lambda_-^N}{\lambda_+ - \lambda_-} = \frac{1}{2^N} \frac{e^{i\omega(t-t_0)} - e^{-i\omega(t-t_0)}}{i\omega\epsilon} = \frac{N}{2^{N-1}} \frac{\sin \omega(t-t_0)}{\omega(t-t_0)}$$

$$\det \mathbf{A} = \left(\frac{m}{i\hbar\epsilon}\right)^{N-1} (2 - \omega^2\epsilon^2)^{N-1} \frac{N}{2^{N-1}} \frac{\sin \omega(t-t_0)}{\omega(t-t_0)}$$

$$= \left(\frac{m}{i\hbar\epsilon}\right)^{N-1} N \left(1 - \frac{1}{2}\omega^2\epsilon^2\right)^{N-1} \frac{\sin \omega(t-t_0)}{\omega(t-t_0)}$$

$$\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} = \frac{1}{2} \left(\frac{im}{\hbar\epsilon}\right)^2 (x_N, 0, \dots, 0, x_0) \cdot \mathbf{A}^{-1} \cdot \begin{pmatrix} x_N \\ 0 \\ \vdots \\ 0 \\ x_0 \end{pmatrix}$$

Here, we need only (1,1), (N-1, N-1), (1, N-1), and (N-1, 1) components of $\mathbf{A}^{-1}$.

$$\begin{aligned} (\mathbf{A}^{-1})_{1,1} &= (\mathbf{A}^{-1})_{N-1,N-1} \\ &= \frac{i\hbar\epsilon}{m} (2 - \omega^2\epsilon^2)^{-1} \frac{\det \mathbf{K}_{N-2}}{\det \mathbf{K}_{N-1}} \\ &= \frac{i\hbar\epsilon}{m} (2 - \omega^2\epsilon^2)^{-1} \frac{\lambda_+^{N-1} - \lambda_-^{N-1}}{\lambda_+^N - \lambda_-^N} \end{aligned}$$

$$\begin{aligned} (\mathbf{A}^{-1})_{1,N-1} &= (\mathbf{A}^{-1})_{N-1,1} \\ &= \frac{i\hbar\epsilon}{m} (2 - \omega^2\epsilon^2)^{-1} \frac{(-1)^{N-2}}{\det K_{N-1}} \det \begin{pmatrix} -a & 1 & -a & 0 \\ 0 & -a & \ddots & -a \\ 0 & 0 & -a & 1 \\ 0 & 0 & 0 & -a \end{pmatrix} \\ &= \frac{i\hbar\epsilon}{m} (2 - \omega^2\epsilon^2)^{-1} \frac{\lambda_+ - \lambda_-}{\lambda_+^N - \lambda_-^N} a^{N-2} \\ &= \frac{i\hbar\epsilon}{m} (2 - \omega^2\epsilon^2)^{-1} \frac{2^{N-1}}{N} \frac{\omega(t-t_0)}{\sin \omega(t-t_0)} \left(\frac{1}{2 - \omega^2\epsilon^2}\right)^{N-2} \\ &= \frac{i\hbar\epsilon}{Nm} \frac{\omega(t-t_0)}{\sin \omega(t-t_0)} \left(\frac{1}{1 - \omega^2\epsilon^2/2}\right)^{N-1} \end{aligned}$$

Since $\lim_{\substack{N \rightarrow \infty \\ \epsilon \rightarrow 0}} \left(1 - \frac{1}{2}\omega^2\epsilon^2\right)^{-(N-1)} = 1$

$$\begin{aligned}
(\mathbf{A}^{-1})_{1,N-1} &= (\mathbf{A}^{-1})_{N-1,1} = \frac{i\hbar\epsilon}{Nm} \frac{\omega(t-t_0)}{\sin \omega(t-t_0)} \\
\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} &= \frac{1}{2} \left(\frac{im}{\hbar\epsilon} \right)^2 \frac{i\hbar\epsilon}{m} \left((2 - \omega^2 \epsilon^2)^{-1} \frac{\lambda_+^{N-1} - \lambda_-^{N-1}}{\lambda_+^N - \lambda_-^N} (x_N^2 + x_0^2) \right. \\
&\quad \left. + \frac{2}{N} \frac{\omega(t-t_0)}{\sin \omega(t-t_0)} x_N x_0 \right) \\
&= \frac{-im}{2\hbar\epsilon} \left(\frac{1}{2 - \omega^2 \epsilon^2} \frac{\lambda_+^{N-1} - \lambda_-^{N-1}}{\lambda_+^N - \lambda_-^N} (x_N^2 + x_0^2) + \frac{2\epsilon\omega}{\sin \omega(t-t_0)} x_N x_0 \right)
\end{aligned}$$

(3) The prefactor

$$\begin{aligned}
&\left(\frac{m}{2\pi i\hbar\epsilon} \right)^{N/2} (2\pi)^{N/2-1/2} \frac{1}{\sqrt{\det \mathbf{A}}} \\
\frac{1}{\sqrt{\det \mathbf{A}}} &= \left(\frac{m}{i\hbar\epsilon} \right)^{-(N-1)/2} N^{-1/2} \left(1 - \frac{1}{2} \omega^2 \epsilon^2 \right)^{-(N-1)/2} \sqrt{\frac{\omega(t-t_0)}{\sin \omega(t-t_0)}}
\end{aligned}$$

Since $\lim_{N \rightarrow \infty} \left(1 - \frac{1}{2} \omega^2 \epsilon^2 \right)^{-(N-1)} = 1$

$$\frac{1}{\sqrt{\det \mathbf{A}}} = \left(\frac{m}{i\hbar\epsilon} \right)^{-(N-1)/2} N^{-1/2} \sqrt{\frac{\omega(t-t_0)}{\sin \omega(t-t_0)}}$$

The prefactor is

$$\begin{aligned}
&\left(\frac{m}{2\pi i\hbar\epsilon} \right)^{N/2} (2\pi)^{(N-1)/2} \left(\frac{m}{i\hbar\epsilon} \right)^{-(N-1)/2} N^{-1/2} \sqrt{\frac{\omega(t-t_0)}{\sin \omega(t-t_0)}} \\
&= (2\pi)^{-1/2} \left(\frac{1}{\epsilon} \right)^{1/2} N^{-1/2} \sqrt{\frac{\omega(t-t_0)}{\sin \omega(t-t_0)}} \\
&= \sqrt{\frac{m\omega}{2\pi i\hbar \sin \omega(t-t_0)}}
\end{aligned}$$

The exponent

$$\begin{aligned}
&\frac{im}{2\hbar\epsilon} (x_N^2 + x_0^2) + \frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \\
&= \frac{im}{2\hbar\epsilon} \left[\left(1 - \frac{1}{2 - \omega^2 \epsilon^2} \frac{\lambda_+^{N-1} - \lambda_-^{N-1}}{\lambda_+^N - \lambda_-^N} \right) (x_N^2 + x_0^2) - \frac{2\epsilon\omega}{\sin \omega(t-t_0)} x_N x_0 \right]
\end{aligned}$$

Here

$$\begin{aligned}
1 - \frac{1}{2 - \omega^2 \epsilon^2} \frac{\lambda_+^{N-1} - \lambda_-^{N-1}}{\lambda_+^N - \lambda_-^N} &= 1 - \frac{1}{2} \frac{2^N}{2^{N-1}} \frac{(1 + i\omega\epsilon)^{N-1} - (1 - i\omega\epsilon)^{N-1}}{(1 + i\omega\epsilon)^N - (1 - i\omega\epsilon)^N} \\
&= i\omega\epsilon \frac{(1 + i\omega\epsilon)^{N-1} + (1 - i\omega\epsilon)^{N-1}}{(1 + i\omega\epsilon)^N - (1 - i\omega\epsilon)^N} \\
&= \omega\epsilon \frac{\cos \omega(t - t_0)}{\sin \omega(t - t_0)}
\end{aligned}$$

The exponent is

$$\begin{aligned}
&= \frac{im}{2\hbar} \left[\omega \frac{\cos \omega(t - t_0)}{\sin \omega(t - t_0)} (x_N^2 + x_0^2) - \frac{2\omega}{\sin \omega(t - t_0)} x_N x_0 \right] \\
&= \frac{i}{\hbar} \frac{m\omega}{2} \frac{(x_N^2 + x_0^2) \cos \omega(t - t_0) - 2x_N x_0}{\sin \omega(t - t_0)}
\end{aligned}$$

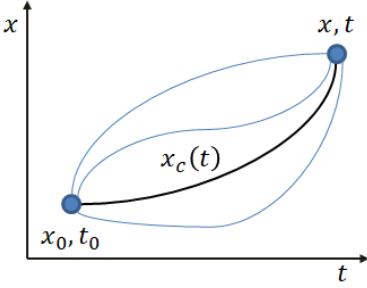
We obtain

$$\begin{aligned}
K(x, t; x_0, t_0) &= \sqrt{\frac{m\omega}{2\pi i \hbar \sin \omega(t - t_0)}} \\
&\quad \times \exp \left[\frac{i}{\hbar} \frac{m\omega}{2} \frac{(x^2 + x_0^2) \cos \omega(t - t_0) - 2xx_0}{\sin \omega(t - t_0)} \right]
\end{aligned}$$

5-3 Stationary Phase Approximation

A. STATIONARY PHASE APPROXIMATION

- (1) Fluctuations around the classical path



$$x(t) = x_c(t) + \eta(t)$$

$$\dot{x}(t) = \dot{x}_c(t) + \dot{\eta}(t)$$

$$S[x(t)] = S[x_c(t) + \eta(t)]$$

$$= \int_{t_0}^t L(x_c + \eta, \dot{x}_c + \dot{\eta}) dt'$$

$$= \underbrace{\int_{t_0}^t L(x_c, \dot{x}_c) dt'}_{S_c} + \underbrace{\int_{t_0}^t \left(\left. \frac{\partial L}{\partial x} \right|_{x_c} \eta + \left. \frac{\partial L}{\partial \dot{x}} \right|_{x_c} \dot{\eta} \right) dt'}_{S_1}$$

$$+ \underbrace{\int_{t_0}^t \left(\left. \frac{1}{2} \frac{\partial^2 L}{\partial x^2} \right|_{x_c} \eta^2 + \left. \frac{\partial^2 L}{\partial x \partial \dot{x}} \right|_{x_c} \eta \dot{\eta} + \left. \frac{1}{2} \frac{\partial^2 L}{\partial \dot{x}^2} \right|_{x_c} \dot{\eta}^2 \right) dt'}_{S_2} + \dots$$

- (2) The zeroth order term:

$$S_c = \int_{t_0}^t L(x_c, \dot{x}_c) dt' = \text{constant}$$

The first order term:

Since the classical path causes the action to be stationary, i.e., the first-order variations in the action about a classical path vanish.

$$\begin{aligned}
S_1 &= \int_{t_0}^t \left(\frac{\partial L}{\partial x} \bigg|_{x_c} \eta + \frac{\partial L}{\partial \dot{x}} \bigg|_{x_c} \dot{\eta} \right) dt' \\
&= \int_{t_0}^t \frac{\partial L}{\partial x} \bigg|_{x_c} \eta dt' + \frac{\partial L}{\partial \dot{x}_c} \delta x_c \bigg|_{t_0}^t - \int_{t_0}^t \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \bigg|_{x_c} \eta dt' \\
&= \int_{t_0}^t \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right)_{x_c} \eta dt' \\
&= 0
\end{aligned}$$

The second order term:

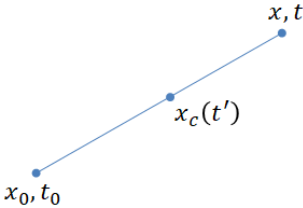
$$\begin{aligned}
S_2 &= \int_{t_0}^t \left(\frac{1}{2} \frac{\partial^2 L}{\partial x^2} \bigg|_{x_c} \eta^2 + \frac{\partial^2 L}{\partial x \partial \dot{x}} \bigg|_{x_c} \eta \dot{\eta} + \frac{1}{2} \frac{\partial^2 L}{\partial \dot{x}^2} \bigg|_{x_c} \dot{\eta}^2 \right) dt' \\
&\Rightarrow S[x(t)] = S_c + S_2 + \dots
\end{aligned}$$

Feynman path integral

$$K(x, t; x_0, t_0) = \int e^{\frac{iS[x(t)]}{\hbar}} Dx(t) = e^{\frac{iS_c}{\hbar}} \int e^{\frac{iS_2}{\hbar}} Dx(t) = C e^{\frac{iS_c}{\hbar}}$$

- (3) The propagation of a free particle from (x_0, t_0) to (x, t)

The classical path for a free particle is a straight line in the $x-t$ plane



$$\frac{x' - x_0}{t' - t_0} = \frac{x - x_0}{t - t_0} \Rightarrow x_c(t') = x' = x_0 + \frac{x - x_0}{t - t_0} (t' - t_0)$$

Since $\frac{x - x_0}{t - t_0} = v$ is the constant velocity

$$x_c(t') = x_0 + v(t' - t_0)$$

$$L_c = T - V = \frac{1}{2} m \dot{x}_c^2(t') = \frac{1}{2} m v^2 = \frac{m}{2} \left(\frac{x - x_0}{t - t_0} \right)^2 = \text{constant}$$

$$S_c = \int_{t_0}^t L_c dt' = \frac{m}{2} \left(\frac{x - x_0}{t - t_0} \right)^2 \int_{t_0}^t dt' = \frac{m}{2} \frac{(x - x_0)^2}{(t - t_0)}$$

$$K(x, t; x_0, t_0) = C e^{\frac{iS_c}{\hbar}} = C e^{i \frac{m(x-x_0)^2}{2\hbar(t-t_0)}}$$

Using the normalization condition:

$$\begin{aligned}
K(x, t_0; x_0, t_0) &= \delta(x - x_0) = \lim_{\alpha \rightarrow 0} \frac{1}{\sqrt{\pi\alpha}} e^{-\frac{(x-x_0)^2}{\alpha}} \\
\Rightarrow C &= \frac{1}{\sqrt{\pi}} \sqrt{-\frac{im}{2\hbar(t-t_0)}} = \sqrt{\frac{m}{2\pi i\hbar(t-t_0)}} \\
\Rightarrow K(x, t; x_0, t_0) &= \sqrt{\frac{m}{2\pi i\hbar(t-t_0)}} e^{i\frac{m(x-x_0)^2}{2\hbar(t-t_0)}}
\end{aligned}$$

(4) The propagation of a harmonic oscillator from (x_0, t_0) to (x, t)

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2 x^2$$

The Lagrange's equation

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = m\omega^2 x - m\ddot{x} = 0$$

$$x_c(t') = A \cos \omega(t' - t_0) + B \sin \omega(t' - t_0)$$

Using boundary conditions:

$$x(t_0) = x_0 = A$$

$$x(t) = x = x_0 \cos \omega(t - t_0) + B \sin \omega(t - t_0)$$

$$\Rightarrow B = \frac{x - x_0 \cos \omega(t - t_0)}{\sin \omega(t - t_0)}$$

$$\begin{aligned}
x_c(t') &= x_0 \cos \omega(t' - t_0) + \frac{x - x_0 \cos \omega(t - t_0)}{\sin \omega(t - t_0)} \sin \omega(t' - t_0) \\
&= x_0 \frac{\cos \omega(t' - t_0) \sin \omega(t - t_0) - \cos \omega(t - t_0) \sin \omega(t' - t_0)}{\sin \omega(t - t_0)}
\end{aligned}$$

$$+ x \frac{\sin \omega(t' - t_0)}{\sin \omega(t - t_0)}$$

$$= x_0 \frac{\sin \omega(t - t')}{\sin \omega(t - t_0)} + x \frac{\sin \omega(t' - t_0)}{\sin \omega(t - t_0)}$$

$$\dot{x}_c(t') = -x_0 \omega \frac{\cos \omega(t - t')}{\sin \omega(t - t_0)} + x \omega \frac{\cos \omega(t' - t_0)}{\sin \omega(t - t_0)}$$

$$\begin{aligned}
L_c &= \frac{1}{2} m \dot{x}_c^2(t') - \frac{1}{2} m \omega^2 x_c^2(t') \\
&= \frac{1}{2} m \left(-x_0 \omega \frac{\cos \omega(t-t')}{\sin \omega(t-t_0)} + x \omega \frac{\cos \omega(t'-t_0)}{\sin \omega(t-t_0)} \right)^2 \\
&\quad - \frac{1}{2} m \omega^2 \left(x_0 \frac{\sin \omega(t-t')}{\sin \omega(t-t_0)} + x \frac{\sin \omega(t'-t_0)}{\sin \omega(t-t_0)} \right)^2 \\
&= \frac{m \omega^2}{2 \sin^2 \omega(t-t_0)} \left[x_0^2 \cos 2\omega(t-t') + x^2 \cos 2\omega(t'-t_0) \right. \\
&\quad \left. - 2xx_0 \cos \omega(t+t_0-2t') \right] \\
S_c &= \int_{t_0}^t L_c dt' \\
&= \frac{m \omega^2}{2 \sin^2 \omega(t-t_0)} \left[\frac{1}{2\omega} (x_0^2 \sin 2\omega(t-t_0) + x^2 \sin 2\omega(t-t_0)) \right. \\
&\quad \left. - 2xx_0 \left(-\frac{1}{2\omega} \right) (\sin \omega(t+t_0-2t) - \sin \omega(t+t_0-2t_0)) \right] \\
&= \frac{m \omega (x^2 + x_0^2) \cos \omega(t-t_0) - 2xx_0}{2 \sin \omega(t-t_0)}
\end{aligned}$$

$$K(x, t; x_0, t_0) = C e^{\frac{iS_c}{\hbar}}$$

Using the normalization condition:

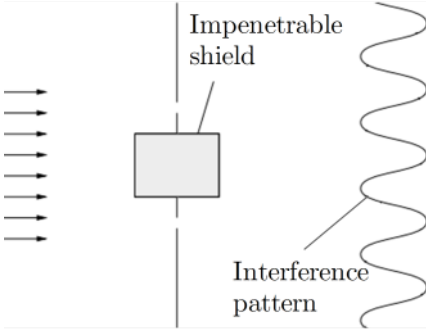
$$K(x, t_0; x_0, t_0) = \delta(x - x_0) = \lim_{\alpha \rightarrow 0} \frac{1}{\sqrt{\pi\alpha}} e^{-\frac{(x-x_0)^2}{\alpha}}$$

we obtain

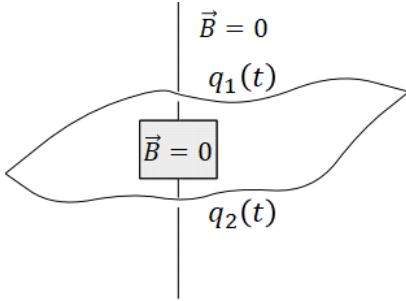
$$\begin{aligned}
\lim_{t \rightarrow t_0} \frac{iS_c}{\hbar} &= \lim_{t \rightarrow t_0} \frac{im\omega}{2\hbar} \frac{x^2 + x_0^2 - 2xx_0}{\sin \omega(t-t_0)} = \lim_{t \rightarrow t_0} \frac{im\omega(x-x_0)^2}{2\hbar \sin \omega(t-t_0)} \\
\Rightarrow C &= \frac{1}{\sqrt{\pi}} \sqrt{-\frac{im\omega}{2\hbar \sin \omega(t-t_0)}} = \sqrt{\frac{m\omega}{2\pi i \hbar \sin \omega(t-t_0)}} \\
\Rightarrow K(x, t; x_0, t_0) &= \sqrt{\frac{m\omega}{2\pi i \hbar \sin \omega(t-t_0)}} \\
&\quad \times \exp \left[\frac{i}{\hbar} \frac{m\omega}{2} \frac{(x^2 + x_0^2) \cos \omega(t-t_0) - 2xx_0}{\sin \omega(t-t_0)} \right]
\end{aligned}$$

B. THE AHARONOV-BOHM EFFECT (1959)

- (1) Aharonov and Bohm proposed a refinement of electron double-slit experiment: magnetic flux is confined within the shaded area and electrons are excluded from this area by a perfect shield.



- (2) No magnetic flux ($\vec{B}_{\text{in}} = 0$ and $\vec{B}_{\text{out}} = 0$):



Consider two paths $q_1(t)$ and $q_2(t)$ passing through slits-1 and -2, respectively, and which arrive at the same spot on the screen.

The free particle propagator for one slit is

$$K_1 = \int_{\text{slit-1}} e^{\frac{iS}{\hbar}} Dq = C e^{\frac{i\theta_1}{\hbar}}$$

The propagator for a particle to reach a given point on the screen is

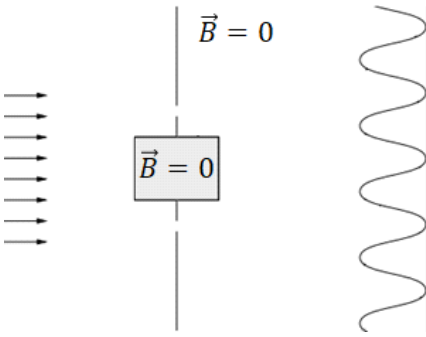
$$K = K_1 + K_2 = C e^{\frac{i\theta_1}{\hbar}} + C e^{\frac{i\theta_2}{\hbar}} = C e^{\frac{i\theta_1}{\hbar}} \left(1 + e^{\frac{i(\theta_2 - \theta_1)}{\hbar}} \right)$$

Let the relative phase $\phi_{21} = \theta_2 - \theta_1$

$$K = C e^{\frac{i\theta_1}{\hbar}} \left(1 + e^{\frac{i\phi_{21}}{\hbar}} \right)$$

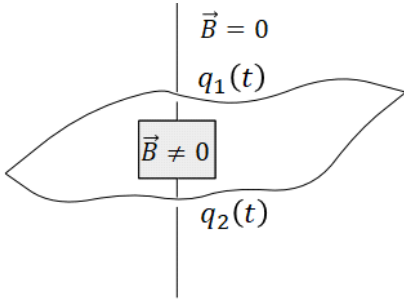
The probability is the modulus squared of the propagator

$$|K|^2 = C^2 \left| 1 + e^{\frac{i\phi_{21}}{\hbar}} \right|^2 = C^2 2 \left(1 + \cos \frac{\phi_{21}}{\hbar} \right) = C^2 4 \cos^2 \frac{\phi_{21}}{2\hbar}$$



The interference pattern appears on the screen.

- (3) With a nonzero and arbitrary flux passing through the shielded region:



The free particle propagator for one slit becomes

$$\begin{aligned}
 K'_1 &= \int_{\text{slit-1}} e^{\frac{iS'}{\hbar}} Dq = \int_{\text{slit-1}} e^{\frac{i}{\hbar} \int_{t_0}^t L'(\dot{q}, q) dt'} Dq \\
 L'(\dot{q}, q) &= \frac{1}{2} m v^2 - \frac{e}{c} \vec{v} \cdot \vec{A}(q) = L(\dot{q}, q) - \frac{e}{c} \vec{v} \cdot \vec{A}(q) \\
 S'[q] &= \int_{t_0}^t L(\dot{q}, q) dt' - \frac{e}{c} \int \vec{v} \cdot \vec{A} dt' \\
 &= S - \frac{e}{c} \int \frac{d\vec{q}(t')}{dt'} \cdot \vec{A}(q(t')) dt' \\
 &= S - \frac{e}{c} \int d\vec{q} \cdot \vec{A}(q) \\
 K'_1 &= \int_{\text{slit-1}} e^{\frac{iS}{\hbar} - \frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}} Dq \\
 &= e^{-\frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}} \int_{\text{slit-1}} e^{\frac{iS}{\hbar}} Dq \\
 &= C e^{\frac{i\theta_1}{\hbar}} e^{-\frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}}
 \end{aligned}$$

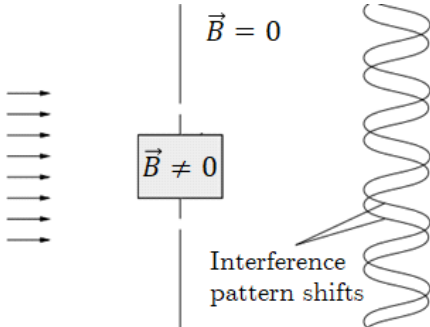
$$\begin{aligned}
K' &= K'_1 + K'_2 \\
&= C e^{\frac{i\theta_1}{\hbar}} e^{-\frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}} + C e^{\frac{i\theta_2}{\hbar}} e^{-\frac{ie}{\hbar c} \int_2 d\vec{q} \cdot \vec{A}} \\
&= C e^{\frac{i\theta_1}{\hbar}} e^{-\frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}} \left(1 + e^{\frac{i(\theta_2 - \theta_1)}{\hbar}} e^{-\frac{ie}{\hbar c} (\int_2 d\vec{q} \cdot \vec{A} - \int_1 d\vec{q} \cdot \vec{A})} \right)
\end{aligned}$$

Since $\frac{e}{c} \left(\int_2 d\vec{q} \cdot \vec{A} - \int_1 d\vec{q} \cdot \vec{A} \right) = \frac{e}{c} \oint d\vec{q} \cdot \vec{A} = \phi$

$$K' = C e^{\frac{i\theta_1}{\hbar}} e^{-\frac{ie}{\hbar c} \int_1 d\vec{q} \cdot \vec{A}} \left(1 + e^{\frac{i\phi_{21}}{\hbar}} e^{-\frac{i\phi}{\hbar}} \right)$$

$$\begin{aligned}
|K'|^2 &= C^2 \left| 1 + e^{\frac{i(\phi_{21} - \phi)}{\hbar}} \right|^2 \\
&= C^2 2 \left(1 + \cos \left(\frac{\phi_{21} - \phi}{\hbar} \right) \right) \\
&= C^2 4 \cos^2 \left(\frac{\phi_{21} - \phi}{2\hbar} \right)
\end{aligned}$$

The change of the phase due to the magnetic field is only dependent of the flux inside the closed loop bounded by the two paths.



The interference pattern is shifted in spite of the fact that the electrons are perfectly shielded from the magnetic field and feel no magnetic force whatsoever.

⇒ The probability amplitude of each path is changed by the vector potential $\vec{A}(\mathbf{q})$.

⇒ Exist many paths connecting $\mathbf{x}_0$ and $\mathbf{x}$. Each path is assigned a weight $e^{iS[\mathbf{x}(t)]/\hbar}$.

5-4 Gauge Transformation

A. GAUGE TRANSFORMATION IN ELECTROMAGNETISM

- (1) Considering an electron in the electromagnetic field described by the scalar potential φ and the vector potential $\vec{A}$.

The Lagrangian is

$$L = \frac{1}{2}mv^2 + e\varphi - \frac{e}{c}\vec{v} \cdot \vec{A}, \quad [\text{c.f.1-2}]$$

Define the conjugate momentum

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} - \frac{e}{c}\vec{A}$$

The Hamiltonian is

$$H = \vec{p} \cdot \vec{v} - L = \frac{1}{2m} \left(\vec{p} + \frac{e}{c}\vec{A} \right)^2 - e\varphi, \quad [\text{c.f.1-3}]$$

- (2) The Hamilton's equations under the gauge transformation
Let

$$\varphi'(\vec{r}, t) = \varphi - \frac{1}{c} \frac{\partial f(\vec{r}, t)}{\partial t}, \quad \vec{A}'(\vec{r}, t) = \vec{A} + \nabla f(\vec{r}, t)$$

where $f(\vec{r}, t)$ is an arbitrary scalar function in space-time.

The Hamiltonian becomes

$$\begin{aligned} H' &= \frac{1}{2m} \left(\vec{p} + \frac{e}{c}\vec{A}' \right)^2 - e\varphi' \\ &= \frac{1}{2m} \left(\vec{p} + \frac{e}{c}\vec{A} + \frac{e}{c}\nabla f \right)^2 - e\varphi + \frac{e}{c} \frac{\partial f}{\partial t} \\ &= \underbrace{\frac{1}{2m} \left(\vec{p} + \frac{e}{c}\vec{A} \right)^2 - e\varphi}_{=H} + \frac{1}{2m} \left(2 \left(\vec{p} + \frac{e}{c}\vec{A} \right) \frac{e}{c} \nabla f + \frac{e^2}{c^2} (\nabla f)^2 \right) + \frac{e}{c} \frac{\partial f}{\partial t} \\ &= H + \frac{1}{2m} \left(2 \left(\vec{p} + \frac{e}{c}\vec{A} \right) \frac{e}{c} \nabla f + \frac{e^2}{c^2} (\nabla f)^2 \right) + \frac{e}{c} \frac{\partial f}{\partial t} \end{aligned}$$

The Hamilton's equations become

$$\dot{q}' = \frac{\partial H'}{\partial p} = \dot{q} + \frac{1}{m} \left(\frac{e}{c} \nabla f \right) = \dot{q} + \frac{e}{mc} \nabla f$$

$\Rightarrow$ The Hamilton's equations do not retain the same form under the gauge transformation.

- (3) The Schrödinger equation is gauge invariant only if the wavefunction

itself transforms covariantly with a position-dependent phase.

PROOF:

The Schrödinger equation in the presence of an electromagnetic field:

$$\begin{aligned}\hat{\mathcal{H}} &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A} \right)^2 - e\varphi \\ i\hbar \frac{\partial}{\partial t} \Psi(x, t) &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A} \right)^2 \Psi(x, t) - e\varphi \Psi(x, t) \\ \left(i\hbar \frac{\partial}{\partial t} + e\varphi \right) \Psi(x, t) &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A} \right)^2 \Psi(x, t) \dots\dots (a)\end{aligned}$$

The Schrödinger equation under the gauge transformation

$$\begin{aligned}\hat{\mathcal{H}}' &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A}' \right)^2 - e\varphi' \\ i\hbar \frac{\partial}{\partial t} \Psi'(x, t) &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A}' \right)^2 \Psi'(x, t) - e\varphi' \Psi'(x, t) \\ \left(i\hbar \frac{\partial}{\partial t} + e\varphi' \right) \Psi'(x, t) &= \frac{1}{2m} \left(\hat{p} + \frac{e}{c} \vec{A}' \right)^2 \Psi'(x, t) \dots\dots (b)\end{aligned}$$

Assume $\Psi'(x, t) = e^{-i\alpha(x, t)} \Psi(x, t)$ to make equations (a) and (b) the same.

R.H.S.:

$$\begin{aligned}\hat{p} \Psi'(x, t) &= -i\hbar \nabla \left(e^{-i\alpha(x, t)} \Psi(x, t) \right) \\ &= -e^{-i\alpha(x, t)} i\hbar \nabla \Psi(x, t) - e^{-i\alpha(x, t)} \hbar \nabla \alpha(x, t) \Psi(x, t) \\ &= e^{-i\alpha(x, t)} (-i\hbar \nabla - \hbar \nabla \alpha(x, t)) \Psi(x, t) \\ &= e^{-i\alpha(x, t)} \left(\hat{p} - \hbar \nabla \alpha(x, t) \right) \Psi(x, t) \\ \left(\hat{p} + \frac{e}{c} \vec{A}' \right)^2 e^{-i\alpha(x, t)} \Psi(x, t) &= \left(\hat{p} + \frac{e}{c} \vec{A} + \frac{e}{c} \nabla f \right) e^{-i\alpha(x, t)} \left(\hat{p} - \hbar \nabla \alpha(x, t) + \frac{e}{c} \vec{A} + \frac{e}{c} \nabla f \right) \Psi(x, t) \\ &= e^{-i\alpha(x, t)} \left(\hat{p} - \hbar \nabla \alpha(x, t) + \frac{e}{c} \vec{A} + \frac{e}{c} \nabla f \right)^2 \Psi(x, t)\end{aligned}$$

In order to obtain

$$\left(\hat{p} + \frac{e}{c} \vec{A}' \right)^2 e^{-i\alpha(x, t)} \Psi(x, t) = e^{-i\alpha(x, t)} \left(\hat{p} + \frac{e}{c} \vec{A} \right)^2 \Psi(x, t)$$

Let $\frac{e}{c} \nabla f - \hbar \nabla \alpha(x, t) = 0$

$$\alpha(x, t) = \frac{ef(x, t)}{\hbar c}$$

L.H.S.:

$$\begin{aligned} & \left(i\hbar \frac{\partial}{\partial t} + e\varphi' \right) e^{-i\alpha(x,t)} \Psi(x,t) \\ &= e^{-i\alpha(x,t)} \left(i\hbar \frac{\partial}{\partial t} + \hbar \frac{\partial \alpha}{\partial t} + e\varphi - \frac{e}{c} \frac{\partial f}{\partial t} \right) \Psi(x,t) \end{aligned}$$

In order to obtain

$$\left(i\hbar \frac{\partial}{\partial t} + e\varphi' \right) e^{-i\alpha(x,t)} \Psi(x,t) = e^{-i\alpha(x,t)} \left(i\hbar \frac{\partial}{\partial t} + e\varphi \right) \Psi(x,t)$$

$$\text{Let } \hbar \frac{\partial \alpha}{\partial t} - \frac{e}{c} \frac{\partial f}{\partial t} = 0$$

$$\alpha(x,t) = \frac{ef(x,t)}{\hbar c}$$

$$\Rightarrow \Psi'(x,t) = e^{-i\frac{e}{\hbar c}f(x,t)} \Psi(x,t)$$

■

(4) The propagator under the gauge transformation

$$K = \int e^{iS[x]/\hbar} Dx, \quad S[x] = \int_{t_0}^t L(\dot{x}, x) dt'$$

$$L' = L - \frac{e}{c} \frac{d}{dt} f(x,t)$$

$$S'[x] = S[x] - \frac{e}{c} (f(x,t) - f(x_0, t_0))$$

The propagator becomes

$$\begin{aligned} K' &= \int e^{iS'[x]/\hbar} Dx \\ &= \int e^{i\left(S[x] - \frac{e}{c}(f(x,t) - f(x_0, t_0))\right)/\hbar} Dx \\ &= e^{-\frac{ie}{\hbar c}(f(x,t) - f(x_0, t_0))} K \end{aligned}$$

B. GAUGE TRANSFORMATION AS UNITARY TRANSFORMATION

(1) Unitary transformation

$$\Psi'(x,t) = e^{-\frac{ie}{\hbar c}f(x,t)} \Psi(x,t)$$

$$\langle x | \Psi'(x,t) \rangle = e^{-\frac{ie}{\hbar c}f(x,t)} \langle x | \Psi(x,t) \rangle = \langle x | e^{-\frac{ie}{\hbar c}f(x,t)} | \Psi(x,t) \rangle$$

$$\text{Let } \hat{U} = e^{-\frac{ie}{\hbar c}f(x,t)}$$

$$| \Psi'(x,t) \rangle = e^{-\frac{ie}{\hbar c}f(x,t)} | \Psi(x,t) \rangle = \hat{U} | \Psi(x,t) \rangle$$

Since $\hat{U}^\dagger \hat{U} = e^{\frac{ie}{\hbar c} f(x,t)} e^{-\frac{ie}{\hbar c} f(x,t)} = \hat{1}$, $\hat{U}$ is a unitary operator.

The probability density is invariant under the gauge transformation

$$\langle \Psi' | \Psi' \rangle = \langle \Psi | \hat{U}^\dagger \hat{U} | \Psi \rangle = \langle \Psi | \Psi \rangle$$

(2) Similarity transformation

$$\hat{U}^\dagger \hat{A}' \hat{U} = \hat{U}^\dagger \hat{A}' \hat{U} - \hat{U}^\dagger \hat{U} \hat{A}' + \hat{A}' = \hat{U}^\dagger [\hat{A}', \hat{U}] + \hat{A}'$$

EXAMPLES:

1. Canonical momentum $\hat{p}$

$$\begin{aligned} [\hat{p}, \hat{U}] \psi &= (\hat{p} \hat{U} - \hat{U} \hat{p}) \psi \\ &= -i\hbar \nabla \left(e^{-\frac{ie}{\hbar c} f(x,t)} \psi \right) - e^{-\frac{ie}{\hbar c} f(x,t)} (-i\hbar \nabla) \psi \\ &= -\frac{e}{c} e^{-\frac{ie}{\hbar c} f(x,t)} \psi \nabla f - i\hbar e^{-\frac{ie}{\hbar c} f(x,t)} \nabla \psi + i\hbar e^{-\frac{ie}{\hbar c} f(x,t)} \nabla \psi \\ &= -\frac{e}{c} e^{-\frac{ie}{\hbar c} f(x,t)} \psi \nabla f \\ [\hat{p}, \hat{U}] &= -\frac{e}{c} \hat{U} \nabla f \\ \hat{U}^\dagger \hat{p} \hat{U} &= \hat{U}^\dagger \left(-\frac{e}{c} \hat{U} \nabla f \right) + \hat{p} = \hat{p} - \frac{e}{c} \nabla f \end{aligned}$$

The canonical momentum $\hat{p}$ is gauge dependent and thus is not a true physical quantity.

2. Mechanical momentum $\hat{p} + e\vec{A}/c$

$$\begin{aligned} \left[\hat{p} + \frac{e\vec{A}}{c} + \frac{e}{c} \nabla f, \hat{U} \right] &= -\frac{e}{c} \hat{U} \nabla f \\ \hat{U}^\dagger \left(\hat{p} + \frac{e}{c} \vec{A} + \frac{e}{c} \nabla f \right) \hat{U} &= -\frac{e}{c} \nabla f + \left(\hat{p} + \frac{e}{c} \vec{A} + \frac{e}{c} \nabla f \right) = \hat{p} + \frac{e}{c} \vec{A} \end{aligned}$$

The mechanical momentum is gauge invariant.

Appendix – Gauss Integral

A. GAUSS INTEGRAL

(1) Gauss integral

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-ax^2} dx &= \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} e^{-a(x-x_0)^2} dx &= \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} e^{-ax^2+bx} dx &= \int_{-\infty}^{\infty} e^{-a\left(x-\frac{b}{2a}\right)^2 + \frac{b^2}{4a}} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}} \\ \int_{-\infty}^{\infty} e^{-ax^2} x^2 dx &= \frac{1}{2a} \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} e^{-ax^2} x^4 dx &= \frac{3}{4a^2} \sqrt{\frac{\pi}{a}}\end{aligned}$$

(2) Multi-dimensional Gauss integral

$$\begin{aligned}1. \int \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x}\right) \prod_{k=1}^N dx_k &= \frac{(2\pi)^{N/2}}{\sqrt{\det \mathbf{A}}} \\ 2. \int \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}\right) \prod_{k=1}^N dx_k &= \frac{(2\pi)^{N/2}}{\sqrt{\det \mathbf{A}}} \exp\left(\frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\right)\end{aligned}$$

where $\mathbf{A}$ is symmetric and positive definite.

PROOF 2:

The matrix $\mathbf{A}$ can be diagonalized by the similarity transformation:

$$\mathbf{D} = \mathbf{U}^T \mathbf{A} \mathbf{U}, \quad \mathbf{U}^T = \mathbf{U}^{-1}$$

Change of variables

$$\mathbf{y} = \mathbf{U}^T (\mathbf{x} - \mathbf{A}^{-1} \mathbf{b})$$

Since

$$\begin{aligned}-\frac{1}{2}\mathbf{y}^T \mathbf{D} \mathbf{y} &= -\frac{1}{2}(\mathbf{x} - \mathbf{A}^{-1} \mathbf{b})^T \mathbf{U} \cdot \mathbf{U}^T \mathbf{A} \mathbf{U} \cdot \mathbf{U}^T (\mathbf{x} - \mathbf{A}^{-1} \mathbf{b}) \\ &= -\frac{1}{2}(\mathbf{x}^T - \mathbf{b}^T \mathbf{A}^{-1}) \mathbf{A} (\mathbf{x} - \mathbf{A}^{-1} \mathbf{b}) \\ &= -\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \frac{1}{2}\mathbf{x}^T \mathbf{b} + \frac{1}{2}\mathbf{b}^T \mathbf{x} - \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b} \\ &= -\frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x} - \frac{1}{2}\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}\end{aligned}$$

$$-\frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{x} = -\frac{1}{2}\mathbf{y}^T\mathbf{D}\mathbf{y} + \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}$$

we obtain

$$\begin{aligned} I &= \exp\left(\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}\right) \prod_{i=1} \int_{-\infty}^{\infty} e^{-\frac{1}{2}y_i^2 D_{ii}} dy_i \\ &= \exp\left(\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}\right) \prod_{i=1} \sqrt{\frac{2\pi}{D_{ii}}} \\ &= \frac{(2\pi)^{N/2}}{\sqrt{\det\mathbf{A}}} \exp\left(\frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}\right) \end{aligned}$$

where $\det\mathbf{D} = \det\mathbf{A}$

■

$$3. \int \exp(-\mathbf{z}^\dagger \mathbf{A} \mathbf{z} + \mathbf{b}^\dagger \mathbf{z} + \mathbf{z}^\dagger \mathbf{b}) \prod_{k=1}^N \frac{dz_k^* dz_k}{2\pi i} = \frac{1}{\det \mathbf{A}} \exp(\mathbf{b}^\dagger \mathbf{A}^{-1} \mathbf{b})$$

where $\mathbf{A}$ is Hermitian and positive definite.

PROOF:

$$\mathbf{D} = \mathbf{U}^\dagger \mathbf{A} \mathbf{U}, \quad \mathbf{U}^\dagger = \mathbf{U}^{-1}$$

Change of variables

$$\mathbf{z}' = \mathbf{U}^\dagger (\mathbf{z} - \mathbf{A}^{-1} \mathbf{b})$$

$$\mathbf{z}'^\dagger = (\mathbf{z}^\dagger - \mathbf{b}^\dagger \mathbf{A}^{-1}) \mathbf{U}$$

$$-\mathbf{z}'^\dagger \mathbf{D} \mathbf{z}' = -(\mathbf{z}'^\dagger - \mathbf{b}^\dagger \mathbf{A}^{-1}) \mathbf{U} \cdot \mathbf{U}^\dagger \mathbf{A} \mathbf{U} \cdot \mathbf{U}^\dagger (\mathbf{z}' - \mathbf{A}^{-1} \mathbf{b})$$

$$= -(\mathbf{z}'^\dagger - \mathbf{b}^\dagger \mathbf{A}^{-1}) \mathbf{A} (\mathbf{z}' - \mathbf{A}^{-1} \mathbf{b})$$

$$= -\mathbf{z}'^\dagger \mathbf{A} \mathbf{z}' + \mathbf{z}'^\dagger \mathbf{b} + \mathbf{b}^\dagger \mathbf{z}' - \mathbf{b}^\dagger \mathbf{A}^{-1} \mathbf{b}$$

The Jacobian is $|\det \mathbf{U}| = 1$

$$Z(J^*, J) = e^{\mathbf{b}^\dagger \mathbf{A}^{-1} \mathbf{b}} \int_{-\infty}^{\infty} \prod_{i=1}^n \frac{dz_i'^* dz_i'}{2\pi i} e^{-\mathbf{z}'^\dagger \mathbf{D} \mathbf{z}'} = e^{\mathbf{b}^\dagger \mathbf{A}^{-1} \mathbf{b}} (\det \mathbf{A})^{-1}$$

■

B. DETERMINANT

(1) For a $N \times N$ matrix of the form

$$\mathbf{K}_N = \begin{pmatrix} 1 & -a & 0 & 0 & 0 \\ -a & 1 & -a & 0 & 0 \\ 0 & -a & \ddots & -a & 0 \\ 0 & 0 & -a & 1 & -a \\ 0 & 0 & 0 & -a & 1 \end{pmatrix}$$

$$\det \mathbf{K}_N = \frac{\lambda_+^{N+1} - \lambda_-^{N+1}}{\lambda_+ - \lambda_-}, \quad \lambda_{\pm} = \frac{1}{2} \left(1 \pm \sqrt{1 - 4a^2} \right)$$

If $\det \mathbf{K}_N$ is singular, $\det \mathbf{K}_N = \lambda_+^N + \lambda_+^{N-1} \lambda_- + \cdots + \lambda_+ \lambda_-^{N-1} + \lambda_-^N$

(2) The recursion relation

$$\det \mathbf{K}_N = \det \mathbf{K}_{N-1} - a^2 \det \mathbf{K}_{N-2}$$

(3) Components of $(\mathbf{K}_N^{-1})$

$$(\mathbf{K}_N^{-1})_{1,1} = \frac{\det \mathbf{K}_{N-2}}{\det \mathbf{K}_{N-1}}$$

$$(\mathbf{K}_N^{-1})_{1,N-1} = \frac{(-1)^{N-2}}{\det \mathbf{K}_{N-1}} \det \begin{pmatrix} -a & 1 & -a & 0 \\ 0 & -a & 1 & -a \\ 0 & 0 & \ddots & 1 \\ 0 & 0 & 0 & -a \end{pmatrix}$$

—Chapter 6—

Orbital and Spin Operators

6-1 Quantization of Orbital Angular Momentum

A. ANGULAR MOMENTUM OPERATORS

(1) In Cartesian coordinate

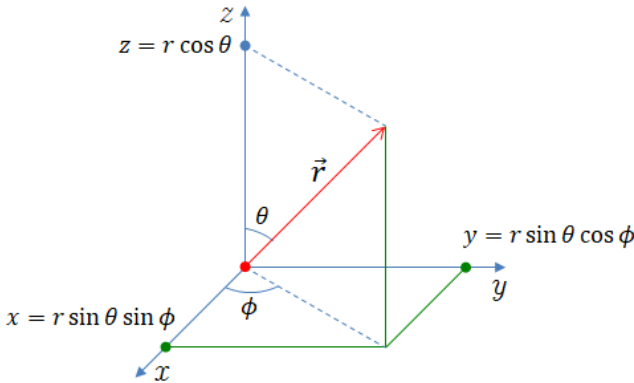
$$\vec{L} = \vec{r} \times \vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$$

$$\hat{L}_y = z\hat{p}_x - x\hat{p}_z = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right)$$

$$\hat{L}_z = x\hat{p}_y - y\hat{p}_x = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

(2) In spherical coordinate



$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\frac{\partial}{\partial r} = \frac{\partial x}{\partial r} \frac{\partial}{\partial x} + \frac{\partial y}{\partial r} \frac{\partial}{\partial y} + \frac{\partial z}{\partial r} \frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \theta} \frac{\partial}{\partial z}$$

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}$$

$$\begin{cases} \frac{\partial x}{\partial r} = \sin \theta \cos \phi \\ \frac{\partial y}{\partial r} = \sin \theta \sin \phi \\ \frac{\partial z}{\partial r} = \cos \theta \end{cases} \quad \begin{cases} \frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi \\ \frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi \\ \frac{\partial z}{\partial \theta} = -r \sin \theta \end{cases} \quad \begin{cases} \frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi \\ \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi \\ \frac{\partial z}{\partial \phi} = 0 \end{cases}$$

$$\frac{\partial}{\partial r} = \sin \theta \cos \phi \frac{\partial}{\partial x} + \sin \theta \sin \phi \frac{\partial}{\partial y} + \cos \theta \frac{\partial}{\partial z} \dots (a)$$

$$\frac{\partial}{\partial \theta} = r \cos \theta \cos \phi \frac{\partial}{\partial x} + r \cos \theta \sin \phi \frac{\partial}{\partial y} - r \sin \theta \frac{\partial}{\partial z} \dots (b)$$

$$\frac{\partial}{\partial \phi} = -r \sin \theta \sin \phi \frac{\partial}{\partial x} + r \sin \theta \cos \phi \frac{\partial}{\partial y} \dots (c)$$

$$\frac{\partial}{\partial x} = \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r} \frac{1}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi}$$

$$\frac{\partial}{\partial y} = \sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \sin \phi \frac{\partial}{\partial \theta} + \frac{1}{r} \frac{1}{\sin \theta} \cos \phi \frac{\partial}{\partial \phi}$$

$$\frac{\partial}{\partial z} = \cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta}$$

$$\hat{\mathcal{L}}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right)$$

$$= -i\hbar r \sin \theta \sin \phi \left(\cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta} \right)$$

$$+ i\hbar r \cos \theta \left(\sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \sin \phi \frac{\partial}{\partial \theta} + \frac{1}{r} \frac{1}{\sin \theta} \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$= i\hbar \left(\sin^2 \theta \sin \phi \frac{\partial}{\partial \theta} - \cos^2 \theta \sin \phi \frac{\partial}{\partial \theta} - \frac{\cos \theta}{\sin \theta} \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$= i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\begin{aligned}
\hat{L}_y &= -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \\
&= -i\hbar r \cos \theta \left(\sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r} \frac{1}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi} \right) \\
&\quad + i\hbar r \sin \theta \cos \phi \left(\cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta} \right) \\
&= i\hbar \left(-\cos^2 \theta \cos \phi \frac{\partial}{\partial \theta} - \sin^2 \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{\cos \theta}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi} \right) \\
&= i\hbar \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \\
\hat{L}_z &= -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \\
&= -i\hbar r \sin \theta \cos \phi \left(\sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \sin \phi \frac{\partial}{\partial \theta} + \frac{1}{r} \frac{1}{\sin \theta} \cos \phi \frac{\partial}{\partial \phi} \right) \\
&\quad + i\hbar r \sin \theta \sin \phi \left(\sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r} \frac{1}{\sin \theta} \sin \phi \frac{\partial}{\partial \phi} \right) \\
&= -i\hbar \left(\cos^2 \phi \frac{\partial}{\partial \phi} + \sin^2 \phi \frac{\partial}{\partial \phi} \right) \\
&= -i\hbar \frac{\partial}{\partial \phi}
\end{aligned}$$

(3) Commutation relations

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= \hat{L}_x \hat{L}_y - \hat{L}_y \hat{L}_x = i\hbar \hat{L}_z \\
[\hat{L}_y, \hat{L}_z] &= \hat{L}_y \hat{L}_z - \hat{L}_z \hat{L}_y = i\hbar \hat{L}_x \\
[\hat{L}_z, \hat{L}_x] &= \hat{L}_z \hat{L}_x - \hat{L}_x \hat{L}_z = i\hbar \hat{L}_y
\end{aligned}$$

PROOF:

$$\begin{aligned}
\hat{L}_x \hat{L}_y &= -\hbar^2 \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \\
&= -\hbar^2 \left(-\sin \phi \cos \phi \frac{\partial^2}{\partial \theta^2} + \sin^2 \phi \frac{\partial}{\partial \theta} \cot \theta \frac{\partial}{\partial \phi} \right. \\
&\quad \left. + \cot \theta \cos \phi \sin \phi \frac{\partial}{\partial \theta} + \cot^2 \theta \cos^2 \phi \frac{\partial}{\partial \phi} \right)
\end{aligned}$$

$$\begin{aligned}
\hat{L}_y \hat{L}_x &= -\hbar^2 \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \\
&= -\hbar^2 \left(-\sin \phi \cos \phi \frac{\partial^2}{\partial \theta^2} - \cos^2 \phi \frac{\partial}{\partial \theta} \cot \theta \frac{\partial}{\partial \phi} \right. \\
&\quad \left. + \cot \theta \cos \phi \sin \phi \frac{\partial}{\partial \theta} - \cot^2 \theta \sin^2 \phi \frac{\partial}{\partial \phi} \right) \\
\hat{L}_x \hat{L}_y - \hat{L}_y \hat{L}_x &= -\hbar^2 \left(\frac{\partial}{\partial \theta} \cot \theta \frac{\partial}{\partial \phi} + \cot^2 \theta \frac{\partial}{\partial \phi} \right) \\
&= -\hbar^2 (-\csc^2 \theta + \cot^2 \theta) \frac{\partial}{\partial \phi} \\
&= \hbar^2 \frac{\partial}{\partial \phi} \\
&= i\hbar \hat{L}_z
\end{aligned}$$

■

(4) Square of angular momentum operator

$$\begin{aligned}
\hat{L}^2 &= \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \\
&= -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \cot \theta \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial \phi^2} \right) \\
&= -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \\
&= -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]
\end{aligned}$$

Commutators

$$[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = [\hat{L}^2, \hat{L}_z] = 0$$

PROOF:

- Method I:

$$\begin{aligned}
\hat{L}^2 \hat{L}_z &= i\hbar^3 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \frac{\partial}{\partial \phi} \\
\hat{L}_z \hat{L}^2 &= i\hbar^3 \frac{\partial}{\partial \phi} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\
\hat{L}^2 \hat{L}_z - \hat{L}_z \hat{L}^2 &= 0
\end{aligned}$$

- Method II:

$$\begin{aligned}
[\hat{L}^2, \hat{L}_z] &= [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] + [\hat{L}_z^2, \hat{L}_z] \\
&= \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\
&= i\hbar(-\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x + \hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y) \\
&= 0
\end{aligned}$$

■

B. EIGENVALUES AND EIGENFUNCTIONS

- (1) $\psi(\theta, \phi)$ is a simultaneous eigenfunction of $\hat{L}^2$ and $\hat{L}_z$ if and only if $[\hat{L}^2, \hat{L}_z] = 0$.

PROOF:

" $\Rightarrow$ "

$$\begin{aligned}
&\begin{cases} \hat{L}_z\psi = \lambda\psi \\ \hat{L}^2\psi = \Lambda\psi \end{cases} \\
&\hat{L}^2\hat{L}_z\psi = \lambda\hat{L}^2\psi = \lambda\Lambda\psi \dots\dots (a) \\
&\hat{L}_z\hat{L}^2\psi = \Lambda\hat{L}_z\psi = \Lambda\lambda\psi \dots\dots (b) \\
&(a) - (b) \\
&(\hat{L}^2\hat{L}_z - \hat{L}_z\hat{L}^2)\psi = [\hat{L}^2, \hat{L}_z]\psi = (\lambda\Lambda - \Lambda\lambda)\psi = 0
\end{aligned}$$

" $\Leftarrow$ ": Two cases to consider

1. Nondegenerate case:

Suppose $\hat{L}_z\psi = \lambda\psi$

Since $[\hat{L}^2, \hat{L}_z] = 0 \Rightarrow \hat{L}^2\hat{L}_z = \hat{L}_z\hat{L}^2$, we obtain

$$\begin{aligned}
&\hat{L}_z(\hat{L}^2\psi) = \hat{L}^2\hat{L}_z\psi = \hat{L}^2\lambda\psi = \lambda(\hat{L}^2\psi) \\
&\Rightarrow \hat{L}^2\psi \text{ is the eigenfunction of } \hat{L}_z. \\
&\Rightarrow \hat{L}^2\psi \text{ must be the same eigenfunction as } \psi, \text{ only multiplied by a} \\
&\text{scalar, i.e., } \hat{L}^2\psi = \Lambda\psi.
\end{aligned}$$

2. Degenerate case:

Suppose $\begin{cases} \hat{L}_z\psi_1 = \lambda\psi_1 \\ \hat{L}_z\psi_2 = \lambda\psi_2 \end{cases}$, and $c_1\psi_1 + c_2\psi_2 = 0$ for $c_1 = c_2 = 0$

Let the linear combination $\psi = c_1\psi_1 + c_2\psi_2$

$$\hat{L}_z\psi = \hat{L}_z(c_1\psi_1 + c_2\psi_2) = \lambda(c_1\psi_1 + c_2\psi_2) = \lambda\psi$$

Since $[\hat{L}^2, \hat{L}_z] = 0 \Rightarrow \hat{L}^2\hat{L}_z = \hat{L}_z\hat{L}^2$, we obtain

$$\hat{L}_z(\hat{L}^2\psi) = \hat{L}^2(\hat{L}_z\psi) = \lambda(c_1\hat{L}^2\psi_1 + c_2\hat{L}^2\psi_2)$$

We need to prove the existence of

$$\begin{aligned}
& c_1 \hat{\mathcal{L}}^2 \psi_1 + c_2 \hat{\mathcal{L}}^2 \psi_2 = \Lambda (c_1 \psi_1 + c_2 \psi_2) \\
\text{Assume } & \begin{cases} \hat{\mathcal{L}}^2 \psi_1 = b_{11} \psi_1 + b_{21} \psi_2 \\ \hat{\mathcal{L}}^2 \psi_2 = b_{12} \psi_1 + b_{22} \psi_2 \end{cases} \\
& c_1 (b_{11} \psi_1 + b_{21} \psi_2) + c_2 (b_{12} \psi_1 + b_{22} \psi_2) = \Lambda (c_1 \psi_1 + c_2 \psi_2) \\
& \Rightarrow \begin{cases} c_1 b_{11} + c_2 b_{12} = \Lambda c_1 \\ c_1 b_{21} + c_2 b_{22} = \Lambda c_2 \end{cases} \\
& \Rightarrow \begin{vmatrix} b_{11} - \Lambda & b_{12} \\ b_{21} & b_{22} - \Lambda \end{vmatrix} = 0
\end{aligned}$$

There are two roots Λ_1 and Λ_2 .

For $\Lambda = \Lambda_1$

$$\begin{aligned}
& (\Lambda_1 - b_{11})c_1 = b_{12}c_2 \text{ and } |c_1|^2 + |c_2|^2 = 1 \\
& c_1 = \frac{b_{12}}{\sqrt{(\Lambda_1 - b_{11})^2 + b_{12}^2}}, \quad c_2 = \frac{(\Lambda_1 - b_{11})}{\sqrt{(\Lambda_1 - b_{11})^2 + b_{12}^2}}
\end{aligned}$$

For $\Lambda = \Lambda_2$

$$\begin{aligned}
& (\Lambda_2 - b_{11})c_1 = b_{12}c_2 \text{ and } |c_1|^2 + |c_2|^2 = 1 \\
& c_1 = \frac{b_{12}}{\sqrt{(\Lambda_2 - b_{11})^2 + b_{12}^2}}, \quad c_2 = \frac{(\Lambda_2 - b_{11})}{\sqrt{(\Lambda_2 - b_{11})^2 + b_{12}^2}} \\
& \Rightarrow \hat{\mathcal{L}}^2 \psi = c_1 \hat{\mathcal{L}}^2 \psi_1 + c_2 \hat{\mathcal{L}}^2 \psi_2 = \Lambda (c_1 \psi_1 + c_2 \psi_2) = \Lambda \psi
\end{aligned}$$

■

- (2) For two conserved quantities, e.g., $[\hat{\mathcal{L}}_z, \hat{\mathcal{H}}] = 0$ and $[\hat{\mathcal{L}}^2, \hat{\mathcal{H}}] = 0$, if $[\hat{\mathcal{L}}^2, \hat{\mathcal{L}}_z] = 0$, the common eigenfunctions are nondegenerate.

PROOF:

Suppose ψ_1 and ψ_2 are degenerate eigenfunctions of $\hat{\mathcal{H}}$, and $c_1 \psi_1 + c_2 \psi_2 = 0$ for $c_1 = c_2 = 0$

$$\text{Since } [\hat{\mathcal{L}}_z, \hat{\mathcal{H}}] = 0 \Rightarrow \begin{cases} \hat{\mathcal{H}} \psi_1 = E \psi_1 \\ \hat{\mathcal{L}}_z \psi_1 = \lambda \psi_1 \end{cases}$$

$$\text{Since } [\hat{\mathcal{L}}^2, \hat{\mathcal{H}}] = 0 \Rightarrow \begin{cases} \hat{\mathcal{H}} \psi_2 = E \psi_2 \\ \hat{\mathcal{L}}^2 \psi_2 = \Lambda \psi_2 \end{cases}$$

$$\text{If } [\hat{\mathcal{L}}^2, \hat{\mathcal{L}}_z] = 0$$

$\Rightarrow \hat{\mathcal{L}}^2$ and $\hat{\mathcal{L}}_z$ have the common eigenfunction.

$\Rightarrow \psi_1 = c \psi_2$, ψ_1 and ψ_2 are nondegenerate.

$\Rightarrow$ The common eigenfunctions are nondegenerate.

OS:

If the symmetry group of a system is an abelian group, then all irreducible representations are 1D, and hence yield no degeneracy. ■

(3) Find the common eigenfunctions $\psi(\theta, \phi)$ of $\hat{L}_z$ and $\hat{L}^2$

1. Suppose $\Phi(\phi)$ is the eigenfunction of $\hat{L}_z$, i.e.,

$$\begin{aligned}\hat{L}_z \Phi(\phi) &= \lambda \Phi(\phi) \\ \Rightarrow -i\hbar \frac{\partial}{\partial \phi} \Phi(\phi) &= \lambda \Phi(\phi) \\ \Rightarrow \Phi(\phi) &= e^{i\lambda\phi/\hbar}\end{aligned}$$

Since $\Phi(\phi + 2\pi) = \Phi(\phi)$, we obtain

$$\begin{aligned}e^{i\lambda(\phi+2\pi)/\hbar} &= e^{i\lambda\phi/\hbar} \\ \Rightarrow e^{i\lambda 2\pi/\hbar} &= 1, \quad \frac{\lambda}{\hbar} = 0, \pm 1, \pm 2, \dots, \pm \infty \\ \Rightarrow \lambda &= m_\ell \hbar, \quad m_\ell = 0, \pm 1, \pm 2, \dots, \pm \infty \\ \Rightarrow \Phi(\phi) &\rightarrow \Phi_{m_\ell}(\phi) = e^{im_\ell\phi}\end{aligned}$$

2. Since $[\hat{L}^2, \hat{L}_z] = 0$, and let $\psi(\theta, \phi) = P(\theta)\Phi(\phi)$, we obtain

$$\begin{aligned}\hat{L}_z \psi(\theta, \phi) &= \lambda \psi(\theta, \phi) \\ \hat{L}^2 \psi(\theta, \phi) &= \Lambda \psi(\theta, \phi) \\ \Rightarrow \hat{L}^2 P(\theta) \Phi(\phi) &= \Lambda P(\theta) \Phi(\phi) \\ \Rightarrow -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] P(\theta) \Phi(\phi) &= \Lambda P(\theta) \Phi(\phi) \\ \Rightarrow -\hbar^2 \left[\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) - \frac{m_\ell^2}{\sin^2 \theta} \right] P(\theta) &= \Lambda P(\theta) \\ \Rightarrow \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) P(\theta) + \left[\frac{\Lambda}{\hbar^2} - \frac{m_\ell^2}{\sin^2 \theta} \right] P(\theta) &= 0\end{aligned}$$

$$\text{Let } z = \cos \theta \Rightarrow \frac{dz}{d\theta} = -\sin \theta \Rightarrow \frac{d}{dz} = \frac{d\theta}{dz} \frac{d}{d\theta} = -\frac{1}{\sin \theta} \frac{d}{d\theta}$$

We obtain the associated Legendre differential equations

$$\frac{d}{dz} (1 - z^2) \frac{dP(z)}{dz} + \left[\frac{\Lambda}{\hbar^2} - \frac{m_\ell^2}{1 - z^2} \right] P(z) = 0$$

The power series method give a solution called the associated Legendre polynomials

$$P_{\ell, m_\ell}(z) = \frac{(1-z^2)^{\frac{|m_\ell|}{2}}}{2^\ell \ell!} \frac{d^{\ell+|m_\ell|}}{dz^{\ell+|m_\ell|}} (z^2-1)^\ell$$

where

$$\Lambda = \ell(\ell+1)\hbar^2$$

$$\ell = 0, 1, 2, 3, \dots, \infty$$

$$m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$$

The common eigenfunctions are $\psi_{\ell, m_\ell}(\theta, \phi) = P_{\ell, m_\ell}(\theta) \Phi_{m_\ell}(\phi)$

$$\hat{L}^2 \psi_{\ell, m_\ell}(\theta, \phi) = \ell(\ell+1)\hbar^2 \psi_{\ell, m_\ell}(\theta, \phi)$$

$$\hat{L}_z \psi_{\ell, m_\ell}(\theta, \phi) = m_\ell \hbar \psi_{\ell, m_\ell}(\theta, \phi)$$

3. Since $[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x \neq 0$ and $[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y \neq 0$,

$\psi_{\ell, m_\ell}(\theta, \phi)$ are not the eigenfunctions of $\hat{L}_x$ and $\hat{L}_y$, i.e.,

$$\hat{L}_x \psi_{\ell, m_\ell}(\theta, \phi) = \psi_{\ell', m'_\ell}(\theta, \phi)$$

$$\hat{L}_y \psi_{\ell, m_\ell}(\theta, \phi) = \psi_{\ell', m'_\ell}(\theta, \phi)$$

(4) Physical interpretation of eigenvalues

1. The magnitude of orbital angular momentum is quantized for electron orbits.

$$\ell(\ell+1)\hbar^2, \quad \ell = 0, 1, 2, 3, \dots, \infty$$

2. The direction of orbital angular momentum is also quantized, i.e., the orbital angular momentum can have only certain directions is called space quantization.

$$m_\ell \hbar, \quad m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$$

3. The expectation values are

$$\langle \hat{L}^2 \rangle = \int \psi_{\ell, m_\ell}^*(\theta, \phi) \hat{L}^2 \psi_{\ell, m_\ell}(\theta, \phi) d\Omega = \ell(\ell+1)\hbar^2$$

$$\langle \hat{L}_z \rangle = \int \psi_{\ell, m_\ell}^*(\theta, \phi) \hat{L}_z \psi_{\ell, m_\ell}(\theta, \phi) d\Omega = m_\ell \hbar$$

$$\langle \hat{L}_x \rangle = \int \psi_{\ell, m_\ell}^*(\theta, \phi) \hat{L}_x \psi_{\ell, m_\ell}(\theta, \phi) d\Omega = 0$$

$$\langle \hat{L}_y \rangle = \int \psi_{\ell, m_\ell}^*(\theta, \phi) \hat{L}_y \psi_{\ell, m_\ell}(\theta, \phi) d\Omega = 0$$

4. Spectroscopic notation

For historical reasons, each possible ℓ value is denoted by a letter of the alphabet according to the following sequence:

ℓ	0	1	2	3	4
Notation	<i>s</i>	<i>p</i>	<i>d</i>	<i>f</i>	<i>g</i>

OS:

The reason for the non-alphabetic order of the first four letters is that they are taken from the names of four series of spectral lines, called sharp, principal, diffuse and fine, observed in the spectra of alkali atoms such as Sodium.

6-2 Magnetic Moment

A. ZEEMAN EFFECT (1896)

- (1) Zeeman effect (Pieter Zeeman): Spectral lines split into two or more separate emission lines by an external magnetic field.

Hydrogen atom



Sodium doublet (D line)



No Field



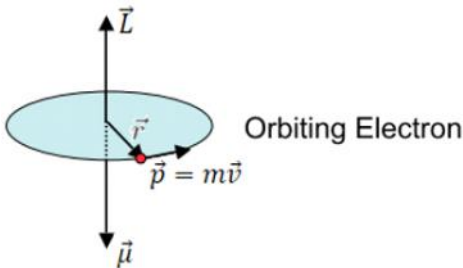
Weak Field

The number of lines formed varies, but the spread is proportional to the strength of the applied field.

⇒ Splitting proportional to the magnetic field indicates an interaction with a moving charge.

B. LARMOR THEOREM (1897)

- (1) Orbital magnetism:



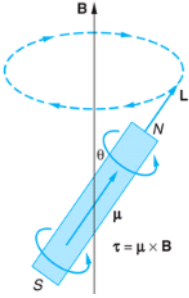
$$\vec{\mu} = i\vec{A} = -\frac{ev}{2\pi r} \pi r^2 \vec{n} = -\frac{e}{2} r^2 \omega \vec{n}$$

$$\vec{L} = \vec{r} \times m\vec{v} = mr^2 \omega \vec{n}$$

$$\Rightarrow \vec{\mu} = -\frac{e}{2m} \vec{L}$$

Larmor theorem (Sir Hoseph Larmor): If a system of charged particles is rotating, the magnetic moment is proportional to its angular momentum.

- (2) Electron magnetic moment in the presence of a uniform magnetic field



The external magnetic field exerts a torque on the magnetic moment and cause the magnetic moment to precess around the direction of the magnetic field.

$$\text{Torque } \vec{\tau} = \vec{\mu} \times \vec{B} = -\frac{e}{2m} \vec{L} \times \vec{B} = -\frac{e}{2m} LB \sin \theta \vec{\phi}$$

$$\text{Since } \vec{\tau} = \frac{d\vec{L}}{dt} = mr^2 \omega \frac{d\vec{n}}{dt} = mr^2 \omega \left(-\frac{\sin \theta d\phi}{dt} \vec{\phi} \right) = -mr^2 \omega^2 \sin \theta \vec{\phi}$$

we obtain

$$-\frac{e}{2m} LB \sin \theta = -mr^2 \omega^2 \sin \theta$$

$$\Rightarrow \frac{e}{2m} mr^2 \omega B = mr^2 \omega^2$$

$$\Rightarrow \boxed{\omega = \frac{eB}{2m}} \dots \text{Larmor frequency (precession angular velocity)}$$

C. SPACE QUANTIZATION

- (1) Bohr (1913) quantized the angular momentum and obtained the discretized energy levels.
- (2) Arnold Sommerfeld (1919) quantized the z -component of the angular momentum.

COMMENTS:

1. The Hamiltonian in the presence of a electric and magnetic field

$$\begin{aligned}
\hat{\mathcal{H}} &= \frac{1}{2m} \left(\hat{\mathbf{p}} + \frac{e}{c} \vec{A} \right)^2 - e\Phi \\
&= \frac{1}{2m} \left(\hat{\mathbf{p}} + \frac{e}{c} \vec{A} \right) \cdot \left(\hat{\mathbf{p}} + \frac{e}{c} \vec{A} \right) - e\Phi \\
&= \frac{\hat{\mathbf{p}}^2}{2m} + \frac{e}{2mc} \left(\hat{\mathbf{p}} \cdot \vec{A} + \vec{A} \cdot \hat{\mathbf{p}} \right) + \frac{e^2}{2mc^2} A^2 - e\Phi
\end{aligned}$$

Since the commutator of $\vec{A}$ and $\hat{\mathbf{p}}$

$$\begin{aligned}
[\vec{A}, \hat{\mathbf{p}}] &= \vec{A} \cdot \hat{\mathbf{p}} - \hat{\mathbf{p}} \cdot \vec{A} \\
&= \vec{A} \cdot (-i\hbar \nabla) - [(-i\hbar \nabla) \cdot \vec{A} + (-i\hbar \vec{A} \cdot \nabla)] \\
&= i\hbar \nabla \cdot \vec{A}
\end{aligned}$$

Adopting the Coulomb gauge condition, i.e., $\nabla \cdot \vec{A} = 0$

$$[\vec{A}, \hat{\mathbf{p}}] = 0 \Rightarrow \hat{\mathbf{p}} \cdot \vec{A} = \vec{A} \cdot \hat{\mathbf{p}}$$

The Hamiltonian is simplified

$$\hat{\mathcal{H}} = \frac{\hat{\mathbf{p}}^2}{2m} + \frac{e}{mc} \vec{A} \cdot \hat{\mathbf{p}} + \frac{e^2}{2mc^2} A^2 - e\Phi$$

Let $\hat{\mathcal{H}}_0 = \frac{\hat{\mathbf{p}}^2}{2m} - e\Phi$ hydrogen atom model

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \frac{e}{mc} \vec{A} \cdot \hat{\mathbf{p}} + \frac{e^2}{2mc^2} A^2$$

- For a uniform magnetic field, in the gauge invariance take the symmetric gauge

$$\vec{A} = \frac{1}{2} \vec{B} \times \vec{r}$$

The second term becomes

$$\frac{e}{mc} \vec{A} \cdot \hat{\mathbf{p}} = \frac{e}{2mc} (\vec{B} \times \vec{r}) \cdot \hat{\mathbf{p}} = \frac{e}{2mc} (\vec{r} \times \hat{\mathbf{p}}) \cdot \vec{B} = \frac{e}{2mc} \hat{\mathcal{L}} \cdot \vec{B}$$

The third term becomes

$$\begin{aligned}
\frac{e^2}{2mc^2} A^2 &= \frac{e^2}{8mc^2} (\vec{B} \times \vec{r})^2 = \frac{e^2}{8mc^2} \left(B^2 r^2 - (\vec{r} \cdot \vec{B})^2 \right) \\
\Rightarrow \hat{\mathcal{H}} &= \hat{\mathcal{H}}_0 + \frac{e}{2mc} \hat{\mathcal{L}} \cdot \vec{B} + \frac{e^2}{8mc^2} \left(B^2 r^2 - (\vec{r} \cdot \vec{B})^2 \right)
\end{aligned}$$

- If the field lies along the z-axis $\vec{B} = B\hat{z}$,

$$\frac{e}{2mc} \hat{\mathcal{L}} \cdot \vec{B} = \frac{e}{2mc} \hat{\mathcal{L}}_z B$$

$$\begin{aligned}
\frac{e^2}{8mc^2} \left(B^2 r^2 - (\vec{r} \cdot \vec{B})^2 \right) &= \frac{e^2}{8mc^2} (B^2(x^2 + y^2 + z^2) - (zB)^2) \\
&= \frac{e^2 B^2}{8mc^2} (x^2 + y^2) \\
\Rightarrow \hat{\mathcal{H}} &= \hat{\mathcal{H}}_0 + \frac{e}{2mc} \hat{\mathcal{L}}_z B + \frac{e^2 B^2}{8mc^2} (x^2 + y^2)
\end{aligned}$$

Since $\omega = \frac{eB}{2mc}$

we obtain

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \omega \hat{\mathcal{L}}_z + \frac{1}{2} m \omega^2 (x^2 + y^2) = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}' + \hat{\mathcal{H}}''$$

where

$$\hat{\mathcal{H}}' = \omega \hat{\mathcal{L}}_z \dots \text{paramagnetic term}$$

$$\hat{\mathcal{H}}'' = \frac{1}{2} m \omega^2 (x^2 + y^2) \dots \text{diamagnetic term}$$

4. Estimate energies of $\hat{\mathcal{H}}_0$, $\hat{\mathcal{H}}'$, and $\hat{\mathcal{H}}''$:

Coulomb term:

$$\Phi = \frac{e}{4\pi\epsilon_0 \langle r \rangle} \approx \frac{e}{4\pi\epsilon_0 a_0}, \quad E_c = \frac{\hbar^2}{2ma_0^2} \sim 13.6 \text{ eV}$$

Paramagnetic term:

$$\langle \hat{\mathcal{L}}_z \rangle \approx \hbar, \quad E_p = \hbar \omega = \hbar \frac{eB}{2mc} = \mu_B B \sim 5.8 \times 10^{-5} \text{ eV}$$

where μ_B is the Bohr magneton and

$$\mu_B = \frac{e\hbar}{2mc} = 9.27 \times 10^{-24} \text{ joule/tesla} = 5.8 \times 10^{-5} \text{ eV/tesla}$$

Diamagnetic term:

$$\begin{aligned}
(x^2 + y^2) &\approx a_0^2 \text{ (Bohr radius)} \\
E_d &= \frac{1}{2} m \omega^2 a_0^2 = \frac{\hbar^2 \omega^2}{4} \frac{2ma_0^2}{\hbar^2} = \frac{\mu_B^2}{4E_c} \sim 6.2 \times 10^{-11} \text{ eV} \\
&\Rightarrow E_c \gg E_p \gg E_d
\end{aligned}$$

$\Rightarrow$ The diamagnetic term is negligible and the paramagnetic term effects as a small perturbation.

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}' = \hat{\mathcal{H}}_0 + \omega \hat{\mathcal{L}}_z$$

5. The eigenfunctions of $\hat{\mathcal{H}}$

Since $[\hat{\mathcal{H}}_0, \hat{\mathcal{L}}_z] = 0 \Rightarrow [\hat{\mathcal{H}}_0, \hat{\mathcal{H}}'] = 0$, $\hat{\mathcal{H}}_0$ and $\hat{\mathcal{H}}'$ have the common eigenfunctions.

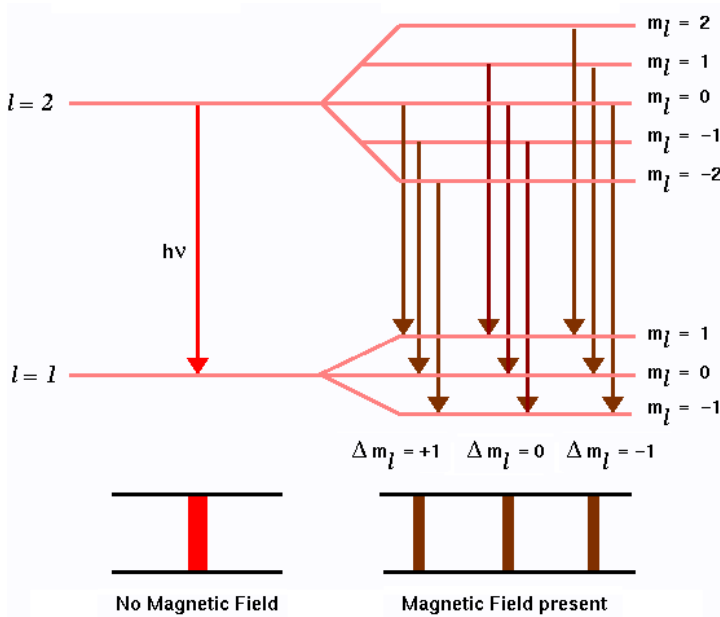
Suppose that $\psi_{n,\ell,m_\ell}(r, \theta, \phi)$ is the eigenfunction of $\hat{\mathcal{H}}_0$, i.e.,

$$\hat{\mathcal{H}}_0 \psi_{n,\ell,m_\ell} = E_{n,\ell,m_\ell}^0 \psi_{n,\ell,m_\ell}, \quad E_{n,\ell,m_\ell}^0 = -\frac{13.6}{n^2}$$

ψ_{n,ℓ,m_ℓ} is also the eigenfunction of $\hat{\mathcal{H}}'$, i.e.,

$$\begin{aligned}\hat{\mathcal{H}}'\psi_{n,\ell,m_\ell} &= E'_{n,\ell,m_\ell}\psi_{n,\ell,m_\ell}, & E'_{n,\ell,m_\ell} &= m_\ell\hbar\omega \\ \Rightarrow E_{n,\ell,m_\ell} &= -\frac{13.6}{n^2} + m_\ell\hbar\omega, & m_\ell &= 0, \pm 1, \pm 2, \dots, \pm\ell\end{aligned}$$

Theoretically, for a single atom:



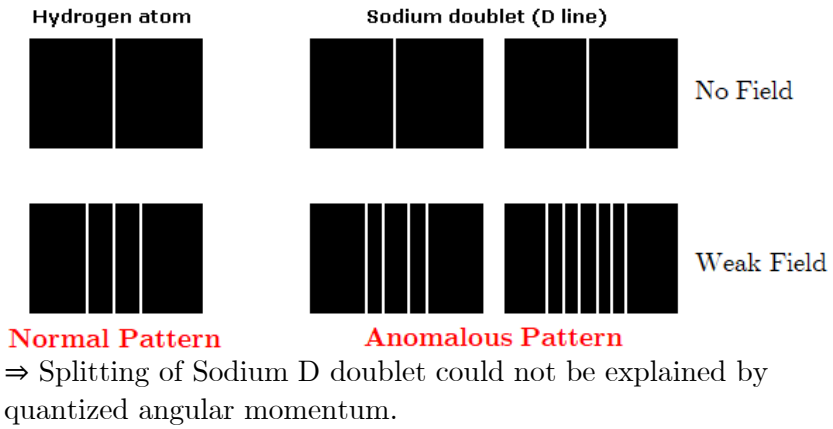
$\Rightarrow$ The energy depends only on the magnitude of the z -component of the angular momentum.

$\Rightarrow$ The energy depends on the particular orientations of the angular momentum of a particle, called *space quantization*.

$\Rightarrow$ Spectral line of $(2\ell + 1)$ -fold denegeracy with multiplets separated by

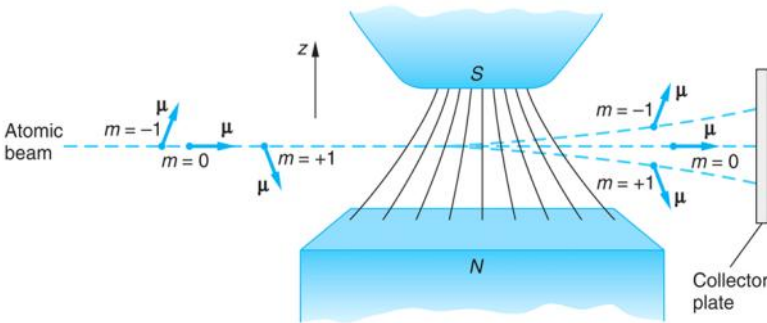
$$\hbar\omega = \hbar \frac{eB}{2mc} = \mu_B B \propto B$$

Experimentally, for the Hydrogen and Sodium atom:



D. STERN-GERLACH EXPERIMENT (1922)

- (1) Goal: determine the component of μ , i.e., the quantization of the direction of the angular momentum.
- (2) Setup: a magnet with a strong field gradient along the z -axis $\vec{B} = B\hat{z}$



A torque rotating the magnetic moment through some angle does work so that there is an energy associated with this interaction.

$$U = -\vec{\mu} \cdot \vec{B} = -\mu_z B = \frac{e}{2m} L_z B = \omega L_z$$

The neutral silver atoms feel a net external force

$$\langle \vec{F}_B \rangle = -\nabla \langle \hat{U} \rangle \text{ and } \hat{U} = -\hat{\mu}_z B(z) = \omega(z) \hat{L}_z$$

L.H.S.:

$$\langle \vec{F}_B \rangle = -\nabla \langle \hat{U} \rangle = \langle \hat{\mu}_z \rangle \nabla B(z) = \mu_z \frac{\partial B(z)}{\partial z} \hat{k}$$

R.H.S.:

$$\begin{aligned}
\langle \vec{F}_B \rangle &= -\nabla \langle \hat{U} \rangle \\
&= -\langle \hat{\mathcal{L}}_z \rangle \nabla \omega(z) \\
&= -m_\ell \hbar \frac{e}{2mc} \frac{\partial B(z)}{\partial z} \hat{k} \\
&= -m_\ell \mu_B \frac{\partial B(z)}{\partial z} \hat{k}
\end{aligned}$$

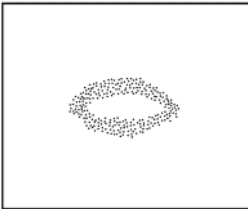
The magnetic dipole moment is

$$\mu_z = -m_\ell \mu_B$$

Stern and Gerlach had assumed that the silver atoms were in a state with $\ell = 1$, and Sommerfeld's theory predicted spatial separation into 3 peaks, i.e., $m_\ell = -1, 0, 1$.

(3) Results:

Actual results



1. The atom has a magnetic dipole moment.
2. The electrons are deflected either up or down by a constant amount, in roughly equal numbers.

⇒ Stern and Gerlach interpreted the results that the directional quantization existed.

OS:

As it turned out, while they were right that something directional was quantized in the experiment, they were mistaken as to what it was.

E. THE BIRTH OF THE CONCEPT OF SPIN

(1) Pauli exclusion principle (early 1925):

"No two electrons could occupy exactly the same quantum state."

It only worked if the electrons have four quantum numbers rather than three.

EXAMPLES:

1. For a Helium, two electrons in the ground state have identical set of three quantum numbers, i.e.,

$$n = 1, \ell = 0, m_\ell = 0$$

The exclusion principle cannot be maintained.

- (2) Uhlenbeck and Goudsmit (mid 1925):

Propose that the fourth quantum number of the electron can take on only one of two possible values $\frac{1}{2}$ or $-\frac{1}{2}$, which is called "spin" and successfully explains the anomalous Zeeman effect.

- (3) Fraser (1927) confirmed that the ground state of the hydrogen atom is at $\ell = 0$, i.e., ψ_{100} .

Phipps and Taylor (1927) did a Stern-Gerlach experiment with hydrogen atoms.

Since $\ell = 0$ and $m_\ell = 0$, according to Sommerfeld's space-quantization model, one can predict only one atomic beam.

⇒ Experimental results: observed two beams.

⇒ The separation of the beam into two components had nothing to do with the so-called "space-quantization" in the atom.

- (4) Pauli (1927) pointed out that the observed two beams is entirely due to the spin of the single electron, and the spin has no classical analog. Comparison between the orbital magnetic moment and the spin magnetic moment:

$$\hat{\mu}_\ell = -\mu_B \frac{\hat{L}}{\hbar}, \quad (\text{has a classical analog})$$

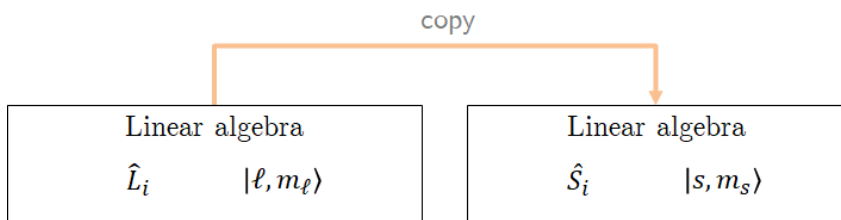
$$\hat{\mu}_s = -2\mu_B \frac{\hat{S}}{\hbar}, \quad (\text{has no classical analog})$$

The z-component of the magnetic moment:

$$\mu_{\ell z} = -m_\ell \mu_B, \quad m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$$

$$\mu_{sz} = -2m_s \mu_B, \quad m_s = \pm \frac{1}{2}$$

The algebra of spin operators is a carbon copy of the algebra of orbital angular momentum operators:



- (5) Dirac (1928) studied the relativistic quantum mechanics and give the explanation of the electron's spin magnetic moment $\hat{\mu}_s = -g_s\mu_B\hat{S}/\hbar$ with $g_s = 2$ (the experimental result is $g_s = 2.002319$).

6-3 Algebra of Orbital Operators

A. ALGEBRA OF ORBITAL OPERATORS

(1) Dirac notation of orbital operators

1. Canonical commutation relations

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$$

2. The square of the angular momentum operator

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

3. The common eigenstates of $\hat{L}^2$ and $\hat{L}_z$

$$\hat{L}^2|\ell, m_\ell\rangle = \ell(\ell+1)\hbar^2|\ell, m_\ell\rangle, \quad \ell = 0, 1, 2, 3, \dots, \infty$$

$$\hat{L}_z|\ell, m_\ell\rangle = m_\ell\hbar|\ell, m_\ell\rangle, \quad m_\ell = 0, \pm 1, \pm 2, \dots, \pm \ell$$

(2) Define the raising and lowering operators

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y$$

$$\hat{L}_- = \hat{L}_x - i\hat{L}_y$$

Both $\hat{L}_+$ and $\hat{L}_-$ are nonhermitian operators.

The commutation relations:

$$1. [\hat{L}^2, \hat{L}_+] = [\hat{L}^2, \hat{L}_x] + i[\hat{L}^2, \hat{L}_y] = 0$$

$$2. [\hat{L}^2, \hat{L}_-] = [\hat{L}^2, \hat{L}_x] - i[\hat{L}^2, \hat{L}_y] = 0$$

$$3. [\hat{L}_z, \hat{L}_+] = [\hat{L}_z, \hat{L}_x] + i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y + i(-i\hbar\hat{L}_x) = \hbar\hat{L}_+ \\ \Rightarrow \hat{L}_z\hat{L}_+ = \hat{L}_+\hat{L}_z + \hbar\hat{L}_+$$

$$4. [\hat{L}_z, \hat{L}_-] = [\hat{L}_z, \hat{L}_x] - i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y - i(-i\hbar\hat{L}_x) = -\hbar\hat{L}_- \\ \Rightarrow \hat{L}_z\hat{L}_- = \hat{L}_-\hat{L}_z - \hbar\hat{L}_-$$

Since $[\hat{L}^2, \hat{L}_-] = [\hat{L}^2, \hat{L}_+] = 0$, the operation of $\hat{L}_-$ and $\hat{L}_+$ have no effect on the eigenvalue ℓ . Thus, we can show that

$$1. \hat{L}_-|\ell, m_\ell\rangle = \sqrt{\ell(\ell+1) - m_\ell(m_\ell-1)}\hbar|\ell, m_\ell-1\rangle$$

$$2. \hat{L}_+|\ell, m_\ell\rangle = \sqrt{\ell(\ell+1) - m_\ell(m_\ell+1)}\hbar|\ell, m_\ell+1\rangle$$

PROOF 1 & 2:

1. Consider a new state that is created by the action of $\hat{L}_-$, i.e.,

$$\hat{L}_-|\ell, m_\ell\rangle = |\ell, m'_\ell\rangle$$

Thus

$$\hat{L}_z\hat{L}_-|\ell, m_\ell\rangle = \hat{L}_z|\ell, m'_\ell\rangle = m'_\ell\hbar|\ell, m'_\ell\rangle$$

On the other hand,

$$\begin{aligned}\hat{L}_z\hat{L}_-|\ell, m_\ell\rangle &= (\hat{L}_-\hat{L}_z - \hbar\hat{L}_-)|\ell, m_\ell\rangle \\ &= (m_\ell\hbar\hat{L}_- - \hbar\hat{L}_-)|\ell, m_\ell\rangle \\ &= (m_\ell - 1)\hbar|\ell, m'_\ell\rangle \\ \Rightarrow m'_\ell &= m_\ell - 1\end{aligned}$$

Assume $\hat{L}_-|\ell, m_\ell\rangle = A|\ell, m'_\ell\rangle$ where A is a constant.

$$\langle\hat{L}_-\ell, m_\ell|\hat{L}_-\ell, m_\ell\rangle = A^2\langle\ell, m'_\ell|\ell, m'_\ell\rangle = A^2$$

On the other hand,

$$\langle\hat{L}_-\ell, m_\ell|\hat{L}_-\ell, m_\ell\rangle = \langle\ell, m_\ell|\hat{L}_+\hat{L}_-|\ell, m_\ell\rangle$$

Since

$$\begin{aligned}\hat{L}_+\hat{L}_- &= (\hat{L}_x + i\hat{L}_y)(\hat{L}_x - i\hat{L}_y) \\ &= \hat{L}_x^2 + \hat{L}_y^2 - i(\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x) \\ &= \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z\end{aligned}$$

we obtain

$$\begin{aligned}\langle\hat{L}_-\ell, m_\ell|\hat{L}_-\ell, m_\ell\rangle &= \langle\ell, m_\ell|\hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z|\ell, m_\ell\rangle \\ &= \ell(\ell + 1)\hbar^2 - m_\ell^2\hbar^2 + m_\ell\hbar^2 \\ &= [\ell(\ell + 1) - m_\ell(m_\ell - 1)]\hbar^2 \\ \Rightarrow A &= \sqrt{\ell(\ell + 1) - m_\ell(m_\ell - 1)}\hbar \\ \Rightarrow \hat{L}_-|\ell, m_\ell\rangle &= \sqrt{\ell(\ell + 1) - m_\ell(m_\ell - 1)}\hbar|\ell, m_\ell - 1\rangle\end{aligned}$$

■

2. Consider a new state that is created by the action of $\hat{L}_+$, i.e.,

$$\hat{L}_+|\ell, m_\ell\rangle = |\ell, m'_\ell\rangle$$

Thus

$$\hat{L}_z\hat{L}_+|\ell, m_\ell\rangle = \hat{L}_z|\ell, m'_\ell\rangle = m'_\ell\hbar|\ell, m'_\ell\rangle$$

On the other hand,

$$\begin{aligned}\hat{L}_z\hat{L}_+|\ell, m_\ell\rangle &= (\hat{L}_+\hat{L}_z + \hbar\hat{L}_+)|\ell, m_\ell\rangle \\ &= (m_\ell\hbar\hat{L}_+ + \hbar\hat{L}_+)|\ell, m_\ell\rangle \\ &= (m_\ell + 1)\hbar|\ell, m'_\ell\rangle \\ \Rightarrow m'_\ell &= m_\ell + 1\end{aligned}$$

Assume $\hat{L}_+|\ell, m_\ell\rangle = A|\ell, m'_\ell\rangle$ where A is a constant.

$$\langle \hat{L}_+ \ell, m_\ell | \hat{L}_+ \ell, m_\ell \rangle = A^2 \langle \ell, m'_\ell | \ell, m'_\ell \rangle = A^2$$

On the other hand,

$$\langle \hat{L}_+ \ell, m_\ell | \hat{L}_+ \ell, m_\ell \rangle = \langle \ell, m_\ell | \hat{L}_- \hat{L}_+ | \ell, m_\ell \rangle$$

Since

$$\begin{aligned} \hat{L}_- \hat{L}_+ &= (\hat{L}_x - i\hat{L}_y)(\hat{L}_x + i\hat{L}_y) \\ &= \hat{L}_x^2 + \hat{L}_y^2 + i(\hat{L}_x \hat{L}_y - \hat{L}_y \hat{L}_x) \\ &= \hat{L}^2 - \hat{L}_z^2 - \hbar \hat{L}_z \end{aligned}$$

we obtain

$$\begin{aligned} \langle \hat{L}_+ \ell, m_\ell | \hat{L}_+ \ell, m_\ell \rangle &= \langle \ell, m_\ell | \hat{L}^2 - \hat{L}_z^2 - \hbar \hat{L}_z | \ell, m_\ell \rangle \\ &= \ell(\ell + 1)\hbar^2 - m_\ell^2 \hbar^2 - m_\ell \hbar^2 \\ &= [\ell(\ell + 1) - m_\ell(m_\ell + 1)]\hbar^2 \\ \Rightarrow A &= \sqrt{\ell(\ell + 1) - m_\ell(m_\ell + 1)}\hbar \\ \Rightarrow \hat{L}_+ | \ell, m_\ell \rangle &= \sqrt{\ell(\ell + 1) - m_\ell(m_\ell + 1)}\hbar | \ell, m_\ell + 1 \rangle \end{aligned}$$

■

(3) The possible values of ℓ and m_ℓ

1. $\hat{L}_+$ raises m_ℓ by 1:

Since $\hat{L}_+ | \ell, \ell \rangle = \sqrt{\ell(\ell + 1) - \ell(\ell + 1)}\hbar | \ell, \ell + 1 \rangle = 0$, and

$\sqrt{\ell(\ell + 1) - m_\ell(m_\ell + 1)} \geq 0$, we obtain

$$m_{\ell, \max} = \ell$$

2. $\hat{L}_-$ lowers m_ℓ by 1:

Since $\hat{L}_- | \ell, -\ell \rangle = \sqrt{\ell(\ell + 1) - (-\ell)(-\ell - 1)}\hbar | \ell, -\ell - 1 \rangle = 0$, and

$\sqrt{\ell(\ell + 1) - m_\ell(m_\ell - 1)} \geq 0$, we obtain

$$m_{\ell, \min} = -\ell$$

$$\Rightarrow m_\ell = \underbrace{\ell, \ell - 1, \dots, 1, 0, 1, -\ell + 1, -\ell}_{N \text{ integer steps}}$$

The values of m_ℓ go from $m_{\ell, \max} = \ell$ to $m_{\ell, \min} = -\ell$ in N integer steps

$$m_{\ell, \max} - m_{\ell, \min} = 2\ell = N$$

$$\Rightarrow \ell = \frac{N}{2}$$

ℓ , mathematically, could be an integer or a half-integer, i.e., $\ell =$

$0, \frac{1}{2}, 1, \frac{3}{2}, \dots, \infty$, BUT the angular momentum cannot be a half-integer.

Thus, we have $\ell = 0, 1, \dots, \infty$.

B. REPRESENTATION FOR $\ell = 1$

- (1) Choose the eigenstate of $\hat{L}_z$ as basis states

There are three basis states $\{|\ell, m_\ell\rangle\} = \{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$ for $\ell = 1$ and the basis states form a *vector space* $\mathbb{U}_{n,\ell} = \mathbb{U}_{2,1}$.

- (2) Wavefunction representation

$$|1, 1\rangle \rightarrow \psi_{1,1}(\theta, \phi) = \frac{1}{8\sqrt{\pi}a_0^{3/2}} \left(\frac{r}{a_0}\right) e^{-r/2a_0} \sin \theta e^{i\phi}$$

$$|1, 0\rangle \rightarrow \psi_{1,0}(\theta, \phi) = \frac{1}{4\sqrt{\pi}a_0^{3/2}} \left(\frac{r}{a_0}\right) e^{-r/2a_0} \cos \theta$$

$$|1, -1\rangle \rightarrow \psi_{1,-1}(\theta, \phi) = \frac{1}{8\sqrt{\pi}a_0^{3/2}} \left(\frac{r}{a_0}\right) e^{-r/2a_0} \sin \theta e^{-i\phi}$$

The angular momentum operators are

$$\hat{L}_x \rightarrow \hat{\mathcal{L}}_x = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y \rightarrow \hat{\mathcal{L}}_y = i\hbar \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_z \rightarrow \hat{\mathcal{L}}_z = -i\hbar \frac{\partial}{\partial \phi}$$

- (3) Matrix representation

Choose the eigenstate of $\hat{L}_z$ as basis states

$$|1, 1\rangle \rightarrow \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|1, 0\rangle \rightarrow \mathbf{u}_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$|1, -1\rangle \rightarrow \mathbf{u}_{-1} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Since

$$\hat{L}_+ |1, 0\rangle = \sqrt{1(1+1) - 0(0+1)}\hbar |1, 1\rangle = \sqrt{2}\hbar |1, 1\rangle$$

$$\hat{L}_+|1, -1\rangle = \sqrt{1(1+1) - (-1)(-1+1)}\hbar|1, 0\rangle = \sqrt{2}\hbar|1, 0\rangle$$

$$\hat{L}_-|1, 1\rangle = \sqrt{1(1+1) - 1(1-1)}\hbar|1, 0\rangle = \sqrt{2}\hbar|1, 0\rangle$$

$$\hat{L}_-|1, 0\rangle = \sqrt{1(1+1) - 0(0-1)}\hbar|1, -1\rangle = \sqrt{2}\hbar|1, -1\rangle$$

we obtain

$$\hat{L}_+ \rightarrow \mathbf{L}_+ = \sqrt{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\hat{L}_- \rightarrow \mathbf{L}_- = \sqrt{2}\hbar \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{L}_x \rightarrow \mathbf{L}_x = \frac{1}{2}(\mathbf{L}_+ + \mathbf{L}_-) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{L}_y \rightarrow \mathbf{L}_y = \frac{1}{2i}(\mathbf{L}_+ - \mathbf{L}_-) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

$$\hat{L}_z \rightarrow \mathbf{L}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

EXAMPLES:

1. Suppose $|\psi\rangle = \frac{1}{\sqrt{5}}|1, 1\rangle + \frac{\sqrt{3}}{\sqrt{5}}|1, 0\rangle + \frac{1}{\sqrt{5}}|1, -1\rangle$

The expectation value of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$

ANSWER:

$$\begin{aligned} \langle\psi|\hat{L}_z|\psi\rangle &= \frac{1}{5}\langle 1,1|\hat{L}_z|1,1\rangle + \frac{3}{5}\langle 1,0|\hat{L}_z|1,0\rangle + \frac{1}{5}\langle 1,-1|\hat{L}_z|1,-1\rangle \\ &= 0 \\ \langle\psi|\hat{L}_x|\psi\rangle &= \frac{1}{2}\left(\frac{\sqrt{3}}{5}\langle 1,1|\hat{L}_+|1,0\rangle + \frac{\sqrt{3}}{5}\langle 1,0|\hat{L}_+|1,-1\rangle \right. \\ &\quad \left. + \frac{\sqrt{3}}{5}\langle 1,0|\hat{L}_-|1,1\rangle + \frac{\sqrt{3}}{5}\langle 1,-1|\hat{L}_-|1,0\rangle\right) \\ &= \frac{\sqrt{3}}{10}(\sqrt{2}\hbar + \sqrt{2}\hbar + \sqrt{2}\hbar + \sqrt{2}\hbar) \\ &= \frac{2\sqrt{6}}{5}\hbar \end{aligned}$$

$$\begin{aligned}
\langle \psi | \hat{L}_y | \psi \rangle &= \frac{1}{2i} \left(\frac{\sqrt{3}}{5} \langle 1,1 | \hat{L}_+ | 1,0 \rangle + \frac{\sqrt{3}}{5} \langle 1,0 | \hat{L}_+ | 1,-1 \rangle \right. \\
&\quad \left. - \frac{\sqrt{3}}{5} \langle 1,0 | \hat{L}_- | 1,1 \rangle - \frac{\sqrt{3}}{5} \langle 1,-1 | \hat{L}_- | 1,0 \rangle \right) \\
&= \frac{\sqrt{3}}{10i} (\sqrt{2}\hbar + \sqrt{2}\hbar - \sqrt{2}\hbar - \sqrt{2}\hbar) = 0
\end{aligned}$$

2. Suppose $\Psi = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix}$

The expectation value of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$

ANSWER:

$$\begin{aligned}
\Psi^\dagger \cdot \mathbf{L}_z \cdot \Psi &= \frac{1}{\sqrt{5}} (1 \quad \sqrt{3} \quad 1) \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix} \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\Psi^\dagger \cdot \mathbf{L}_x \cdot \Psi &= \frac{1}{\sqrt{5}} (1 \quad \sqrt{3} \quad 1) \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix} \\
&= \frac{\hbar}{5\sqrt{2}} (1 \quad \sqrt{3} \quad 1) \begin{pmatrix} \sqrt{3} \\ 2 \\ \sqrt{3} \end{pmatrix} \\
&= \frac{\hbar}{5\sqrt{2}} (\sqrt{3} + 2\sqrt{3} + \sqrt{3}) \\
&= \frac{2\sqrt{6}}{5} \hbar
\end{aligned}$$

$$\begin{aligned}
\Psi^\dagger \cdot \mathbf{L}_y \cdot \Psi &= \frac{1}{\sqrt{5}} (1 \quad \sqrt{3} \quad 1) \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix} \\
&= \frac{\hbar}{5\sqrt{2}} (1 \quad \sqrt{3} \quad 1) \begin{pmatrix} -i\sqrt{3} \\ 0 \\ i\sqrt{3} \end{pmatrix} \\
&= \frac{\hbar}{5\sqrt{2}} (-i\sqrt{3} + 0 + i\sqrt{3}) \\
&= 0
\end{aligned}$$

3. Suppose $\Psi = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix}$

Using unitary transformation to find the expectation value of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$

ANSWER:

The matrix form of $\mathbf{L}_x$:

$$\mathbf{L}_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Diagonalize $\mathbf{L}_x$ and obtain the eigenvalues and eigenvectors:

$$|\mathbf{L}_x - \lambda \mathbf{1}| = 0 \Rightarrow -\lambda(\lambda^2 - 1) + \lambda = 0 \Rightarrow \lambda = \pm\sqrt{2}, 0$$

$$\lambda = \sqrt{2} \Rightarrow \mathbf{v}_1 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

$$\lambda = -\sqrt{2} \Rightarrow \mathbf{v}_{-1} = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}$$

$$\lambda = 0 \Rightarrow \mathbf{v}_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

The unitary matrix has the eigenvectors of $\mathbf{v}_n$ as its columns,

$$\mathbf{U}_x = (\mathbf{v}_1 \quad \mathbf{v}_0 \quad \mathbf{v}_{-1}) = \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}$$

Change of vector:

$$\begin{aligned} \Phi &= \mathbf{U}_x^\dagger \cdot \Psi \\ &= \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix}_z \\ &= \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 + \sqrt{6} \\ 0 \\ 2 - \sqrt{6} \end{pmatrix}_x \end{aligned}$$

The similarity transformation (change of operator):

$$\mathbf{B} = \mathbf{U}_x^\dagger \mathbf{L}_x \mathbf{U}_x = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{aligned}
\Rightarrow \boldsymbol{\Phi}^\dagger \cdot \mathbf{B} \cdot \boldsymbol{\Phi} &= \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 + \sqrt{6} & 0 & 2 - \sqrt{6} \end{pmatrix} \\
&\quad \times \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \frac{1}{2\sqrt{5}} \begin{pmatrix} 2 + \sqrt{6} \\ 0 \\ 2 - \sqrt{6} \end{pmatrix} \\
&= \frac{2\sqrt{6}}{5} \hbar
\end{aligned}$$

The matrix form of $\mathbf{L}_y$:

$$\mathbf{L}_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

Diagonalize $\mathbf{L}_y$ and obtain the eigenvalues and eigenvectors:

$$|\mathbf{L}_y - \lambda \mathbf{1}| = 0 \Rightarrow -\lambda(\lambda^2 - 1) + \lambda = 0 \Rightarrow \lambda = \pm\sqrt{2}, 0$$

$$\lambda = \sqrt{2} \Rightarrow \mathbf{v}_1 = \frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2} \\ -1 \end{pmatrix}$$

$$\lambda = 0 \Rightarrow \mathbf{v}_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\lambda = -\sqrt{2} \Rightarrow \mathbf{v}_{-1} = \frac{1}{2} \begin{pmatrix} -1 \\ i\sqrt{2} \\ 1 \end{pmatrix}$$

The unitary matrix has the eigenvectors of $\mathbf{v}_n$ as its columns,

$$\mathbf{U}_y = (\mathbf{v}_1 \quad \mathbf{v}_0 \quad \mathbf{v}_{-1}) = \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & -1 \\ i\sqrt{2} & 0 & i\sqrt{2} \\ -1 & \sqrt{2} & 1 \end{pmatrix}$$

Change of vector:

$$\begin{aligned}
\boldsymbol{\Phi} &= \mathbf{U}_y^\dagger \cdot \boldsymbol{\Psi} \\
&= \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & -1 \\ i\sqrt{2} & 0 & i\sqrt{2} \\ -1 & \sqrt{2} & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ \sqrt{3} \\ 1 \end{pmatrix}_z \\
&= \frac{1}{2\sqrt{5}} \begin{pmatrix} \sqrt{6} \\ 0 \\ \sqrt{6} \end{pmatrix}_y
\end{aligned}$$

The similarity transformation:

$$\begin{aligned}
\mathbf{B} &= \mathbf{U}_y^\dagger \mathbf{L}_y \mathbf{U}_y = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \\
\Rightarrow \boldsymbol{\Phi}^\dagger \cdot \mathbf{B} \cdot \boldsymbol{\Phi} &= \frac{1}{2\sqrt{5}} (\sqrt{6} \quad 0 \quad \sqrt{6}) \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \frac{1}{2\sqrt{5}} \begin{pmatrix} \sqrt{6} \\ 0 \\ \sqrt{6} \end{pmatrix} \\
&= 0
\end{aligned}$$

6-4 Algebra of Spin Operators

A. ALGEBRA OF SPIN OPERATORS

(1) Dirac notation of spin operators

1. Canonical commutation relations

$$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z$$

$$[\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x$$

$$[\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y$$

2. The square of the spin operator

$$\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2$$

3. The common eigenstates of $\hat{S}^2$ and $\hat{S}_z$

$$\hat{S}^2|s, m_s\rangle = s(s+1)\hbar^2|s, m_s\rangle$$

$$\hat{S}_z|s, m_s\rangle = m_s\hbar|s, m_s\rangle$$

(2) Define the raising and lowering operators

$$\hat{S}_+ = \hat{S}_x + i\hat{S}_y$$

$$\hat{S}_- = \hat{S}_x - i\hat{S}_y$$

Both $\hat{S}_+$ and $\hat{S}_-$ are nonhermitian operators.

The commutation relations:

$$1. [\hat{S}^2, \hat{S}_+] = [\hat{S}^2, \hat{S}_x] + i[\hat{S}^2, \hat{S}_y] = 0$$

$$2. [\hat{S}^2, \hat{S}_-] = [\hat{S}^2, \hat{S}_x] - i[\hat{S}^2, \hat{S}_y] = 0$$

$$3. [\hat{S}_z, \hat{S}_+] = [\hat{S}_z, \hat{S}_x] + i[\hat{S}_z, \hat{S}_y] = i\hbar\hat{S}_y + i(-i\hbar\hat{S}_x) = \hbar\hat{S}_+ \\ \Rightarrow \hat{S}_z\hat{S}_+ = \hat{S}_+\hat{S}_z + \hbar\hat{S}_+$$

$$4. [\hat{S}_z, \hat{S}_-] = [\hat{S}_z, \hat{S}_x] - i[\hat{S}_z, \hat{S}_y] = i\hbar\hat{S}_y - i(-i\hbar\hat{S}_x) = -\hbar\hat{S}_- \\ \Rightarrow \hat{S}_z\hat{S}_- = \hat{S}_-\hat{S}_z - \hbar\hat{S}_-$$

Since $[\hat{S}^2, \hat{S}_-] = [\hat{S}^2, \hat{S}_+] = 0$, the operation of $\hat{S}_-$ and $\hat{S}_+$ have no effect on the eigenvalue s . Thus, we can show that

$$\hat{S}_-|s, m_s\rangle = \sqrt{s(s+1) - m_s(m_s-1)}\hbar|s, m_s-1\rangle$$

$$\hat{S}_+|s, m_s\rangle = \sqrt{s(s+1) - m_s(m_s+1)}\hbar|s, m_s+1\rangle$$

(3) The possible values of s and m_s

1. $\hat{S}_+$ raises m_s by 1:

Since $\hat{S}_+|s, s\rangle = \sqrt{s(s+1) - s(s+1)}\hbar|s, s+1\rangle = 0$, and

$\sqrt{s(s+1) - m_s(m_s+1)} \geq 0$, we obtain

$$m_{s,\max} = s$$

2. $\hat{S}_-$ lowers m_s by 1:

Since $\hat{S}_-|s, -s\rangle = \sqrt{s(s+1) - (-s)(-s-1)}\hbar|s, -s-1\rangle = 0$, and

$\sqrt{s(s+1) - m_s(m_s-1)} \geq 0$, we obtain

$$m_{s,\min} = -\ell$$

$$\Rightarrow m_s = \underbrace{s, s-1, \dots, 1, 0, 1, -s+1, -s}_{N \text{ integer steps}}$$

The values of m_s go from $m_{s,\max} = s$ to $m_{s,\min} = -s$ in N integer steps

$$m_{s,\max} - m_{s,\min} = 2s = N$$

$$\Rightarrow s = \frac{N}{2}$$

s , mathematically, could be an integer or a half-integer, i.e., $s =$

$0, \frac{1}{2}, 1, \frac{3}{2}, \dots, \infty$. There is no a priori reason to exclude the half-integer values of s . The half-integer values of s not only empirically fit experimental data, but also are verified through the spin algebra.

B. SPIN BASIS STATES

- (1) Single spin- $\frac{1}{2}$ particle

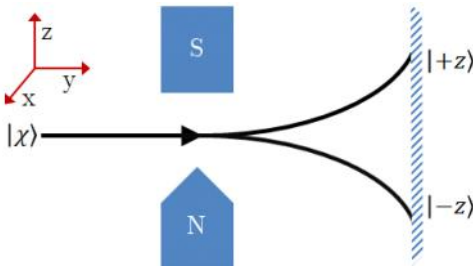
$$s = \frac{1}{2} \Rightarrow m_s = \frac{1}{2}, -\frac{1}{2}$$

There are two spin basis states $\{|s, m_s\rangle\} = \left\{\left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \left|\frac{1}{2}, -\frac{1}{2}\right\rangle\right\}$ and the basis states form a *vector space* $\mathbb{V}_{1/2}$.

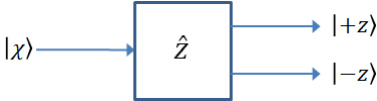
Any spin state of the system can be written as a linear combination of spin basis states $\left\{\left|\frac{1}{2}, +\frac{1}{2}\right\rangle, \left|\frac{1}{2}, -\frac{1}{2}\right\rangle\right\}$, i.e.,

$$|\chi\rangle = a \left|\frac{1}{2}, +\frac{1}{2}\right\rangle + b \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

- (2) Experimentally, spin- $\frac{1}{2}$ particles pass through a Stern-Gerlach (SG) apparatus and the SG device is oriented in the z -direction:



Schematic representation



Conventionally, we choose the spin states are the eigenstates of $\hat{S}_z$, i.e.,

$$\hat{S}_z|+z\rangle = +\frac{\hbar}{2}|+z\rangle, \quad \text{spin up}$$

$$\hat{S}_z|-z\rangle = -\frac{\hbar}{2}|-z\rangle, \quad \text{spin down}$$

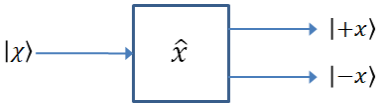
where

$$|+z\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \text{ and } |-z\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

Thus, any spin state of the system can be written as a linear combination of eigenstates $\{|+z\rangle, |-z\rangle\}$, i.e.,

$$|\chi\rangle = a|+z\rangle + b|-z\rangle$$

(3) If the SG device is oriented in the x -direction, i.e.,



Thus, we have

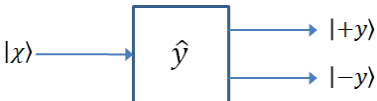
$$\hat{S}_x|+x\rangle = +\frac{\hbar}{2}|+x\rangle, \quad \text{spin up}$$

$$\hat{S}_x|-x\rangle = -\frac{\hbar}{2}|-x\rangle, \quad \text{spin down}$$

where

$$|+x\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \text{ and } |-x\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

(4) If the SG device is oriented in the y -direction, i.e.,



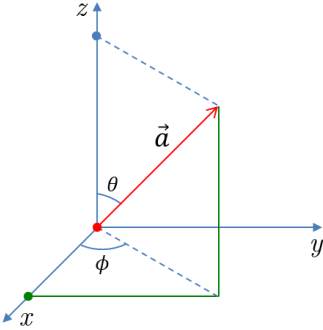
Thus, we have

$$\begin{aligned}\hat{S}_y|+y\rangle &= +\frac{\hbar}{2}|+y\rangle, & \text{spin up} \\ \hat{S}_y|-y\rangle &= -\frac{\hbar}{2}|-y\rangle, & \text{spin down}\end{aligned}$$

where

$$|+y\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \text{ and } |-y\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

- (5) If the SG device is oriented along $\vec{a} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ where $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$.



$$\begin{aligned}\hat{S}_a &= \vec{a} \cdot \hat{S} \\ &= \sin \theta \cos \phi \hat{S}_x + \sin \theta \sin \phi \hat{S}_y + \cos \theta \hat{S}_z \\ &= \frac{1}{2} \sin \theta \cos \phi (\hat{S}_+ + \hat{S}_-) + \frac{1}{2i} \sin \theta \sin \phi (\hat{S}_+ - \hat{S}_-) + \cos \theta \hat{S}_z \\ &= \frac{1}{2} \sin \theta e^{-i\phi} \hat{S}_+ + \frac{1}{2} \sin \theta e^{i\phi} \hat{S}_- + \cos \theta \hat{S}_z\end{aligned}$$

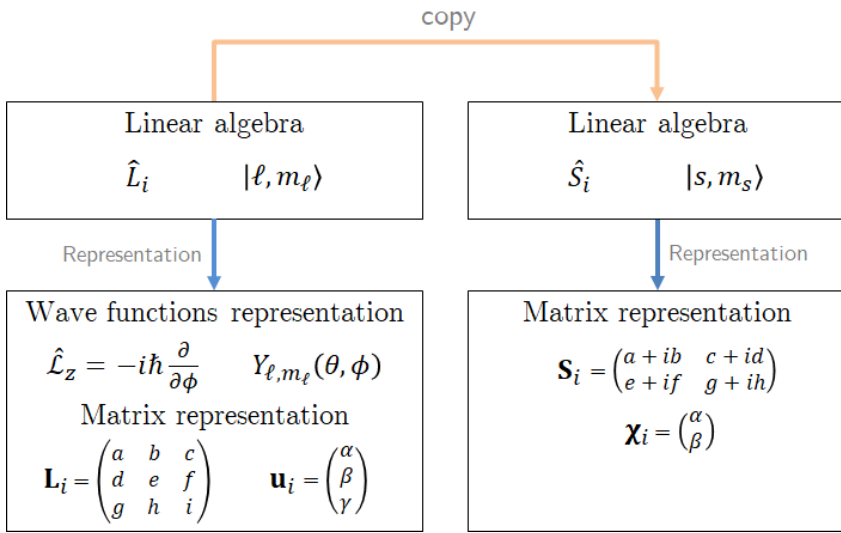
Thus, we have

$$\begin{aligned}\hat{S}_a|+a\rangle &= +\frac{\hbar}{2}|+a\rangle, & \text{spin up} \\ \hat{S}_a|-a\rangle &= -\frac{\hbar}{2}|-a\rangle, & \text{spin down}\end{aligned}$$

where

$$|+a\rangle = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \text{ and } |-a\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

C. MATRIX REPRESENTATION AND PAULI MATRICES



(1) Choose the eigenstate of $\hat{S}_z$ as basis states

$$|+z\rangle \rightarrow \mathbf{u}_{+z} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|-z\rangle \rightarrow \mathbf{u}_{-z} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(2) Since

$$\hat{S}_+|-z\rangle = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right) - \left(-\frac{1}{2}\right)\left(-\frac{1}{2} + 1\right)}\hbar|+z\rangle = \hbar|+z\rangle$$

$$\hat{S}_-|+z\rangle = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right) - \frac{1}{2}\left(\frac{1}{2} - 1\right)}\hbar|-z\rangle = \hbar|-z\rangle$$

we obtain

$$\hat{S}_+ \rightarrow \mathbf{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\hat{S}_- \rightarrow \mathbf{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

and

$$\hat{S}_x \rightarrow \mathbf{S}_x = \frac{1}{2}(\mathbf{S}_+ + \mathbf{S}_-) = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{S}_y \rightarrow \mathbf{S}_y = \frac{1}{2i}(\mathbf{S}_+ - \mathbf{S}_-) = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{S}_z \rightarrow \mathbf{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{aligned}
\hat{S}_a \rightarrow \mathbf{S}_a &= \frac{1}{2} \sin \theta e^{-i\phi} \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \sin \theta e^{i\phi} \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\
&\quad + \cos \theta \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\
&= \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}
\end{aligned}$$

(3) Define Pauli matrix

$$\boldsymbol{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Thus, we have

$$\begin{aligned}
\mathbf{S}_x &= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2} \boldsymbol{\sigma}_x \\
\mathbf{S}_y &= \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar}{2} \boldsymbol{\sigma}_y \\
\mathbf{S}_z &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{\hbar}{2} \boldsymbol{\sigma}_z
\end{aligned}$$

D. CHANGE OF SPIN BASIS

(1) Change $\hat{S}_x$ -basis states $\{|+x\rangle, |-x\rangle\}$ to $\hat{S}_z$ -basis states $\{|+z\rangle, |-z\rangle\}$

$$|+x\rangle = \frac{1}{\sqrt{2}}|+z\rangle + \frac{1}{\sqrt{2}}|-z\rangle$$

$$|-x\rangle = \frac{1}{\sqrt{2}}|+z\rangle - \frac{1}{\sqrt{2}}|-z\rangle$$

PROOF:

Let

$$|+x\rangle \equiv a|+z\rangle + b|-z\rangle$$

$$|-x\rangle \equiv c|+z\rangle + d|-z\rangle$$

Since $\langle \pm z | \pm x \rangle = \text{unknown}$, the unitary operator $\hat{U}$ cannot be defined.

Thus, we use raising and lowering operators as follows:

$$\begin{aligned}
\hat{S}_x|+x\rangle &= \frac{1}{2}(\hat{S}_+ + \hat{S}_-)(a|+z\rangle + b|-z\rangle) \\
&= a\frac{\hat{S}_+}{2}|+z\rangle + b\frac{\hat{S}_+}{2}|-z\rangle + a\frac{\hat{S}_-}{2}|+z\rangle + b\frac{\hat{S}_-}{2}|-z\rangle \\
&= b\frac{\hat{S}_+}{2}|-z\rangle + a\frac{\hat{S}_-}{2}|+z\rangle \\
&= b\frac{\hbar}{2}|+z\rangle + a\frac{\hbar}{2}|-z\rangle
\end{aligned}$$

Since

$$\hat{S}_x|+x\rangle = \frac{\hbar}{2}(b|+z\rangle + a|-z\rangle) = \frac{\hbar}{2}|+x\rangle$$

we obtain

$$\begin{aligned} |+x\rangle &= b|+z\rangle + a|-z\rangle \equiv a|+z\rangle + b|-z\rangle \\ \Rightarrow a &= b \end{aligned}$$

Using the normalization condition:

$$\begin{aligned} |a|^2 + |b|^2 &= 1 \Rightarrow |a| = \frac{1}{\sqrt{2}} \\ \Rightarrow |+x\rangle &= \frac{1}{\sqrt{2}}|+z\rangle + \frac{1}{\sqrt{2}}|-z\rangle \end{aligned}$$

Similarly,

$$\begin{aligned} \hat{S}_x|-x\rangle &= \frac{1}{2}(\hat{S}_+ + \hat{S}_-)(c|+z\rangle + d|-z\rangle) \\ &= d\frac{\hbar}{2}|+z\rangle + c\frac{\hbar}{2}|-z\rangle \\ &= \left(-\frac{\hbar}{2}\right)(-d|+z\rangle - c|-z\rangle) \\ \Rightarrow |-x\rangle &= -d|+z\rangle - c|-z\rangle \equiv c|+z\rangle + d|-z\rangle \\ \Rightarrow d &= -c \end{aligned}$$

Using the normalization condition:

$$\begin{aligned} |c|^2 + |d|^2 &= 1 \Rightarrow |c| = \frac{1}{\sqrt{2}} \\ \Rightarrow |-x\rangle &= \frac{1}{\sqrt{2}}|+z\rangle - \frac{1}{\sqrt{2}}|-z\rangle \end{aligned}$$

■

- (2) Change $\hat{S}_y$ -basis states $\{|+y\rangle, |-y\rangle\}$ states to $\hat{S}_z$ -basis states $\{|+z\rangle, |-z\rangle\}$

$$\begin{aligned} |+y\rangle &= \frac{1}{\sqrt{2}}|+z\rangle + \frac{i}{\sqrt{2}}|-z\rangle \\ |-y\rangle &= \frac{1}{\sqrt{2}}|+z\rangle - \frac{i}{\sqrt{2}}|-z\rangle \end{aligned}$$

PROOF:

Let

$$\begin{aligned} |+y\rangle &\equiv a|+z\rangle + b|-z\rangle \\ |-y\rangle &\equiv c|+z\rangle + d|-z\rangle \end{aligned}$$

We obtain

$$\begin{aligned}
\hat{S}_y|+y\rangle &= \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)(a|+z\rangle + b|-z\rangle) \\
&= -ib\frac{\hbar}{2}|+z\rangle + ia\frac{\hbar}{2}|-z\rangle \\
&= \frac{\hbar}{2}(-ib|+z\rangle + ia|-z\rangle) \\
\Rightarrow |y\rangle &= -ib|+z\rangle + ia|-z\rangle \equiv a|+z\rangle + b|-z\rangle \\
\Rightarrow b &= ia
\end{aligned}$$

Using the normalization condition:

$$\begin{aligned}
|a|^2 + |b|^2 &= 1 \Rightarrow |a| = \frac{1}{\sqrt{2}} \\
\Rightarrow |y\rangle &= \frac{1}{\sqrt{2}}|+z\rangle + \frac{i}{\sqrt{2}}|-z\rangle
\end{aligned}$$

Similarly,

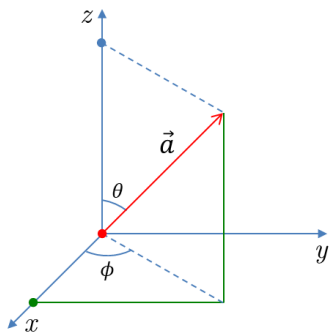
$$\begin{aligned}
\hat{S}_y|-y\rangle &= \frac{1}{2i}(\hat{S}_+ - \hat{S}_-)(c|+z\rangle + d|-z\rangle) \\
&= -id\frac{\hbar}{2}|+z\rangle + ic\frac{\hbar}{2}|-z\rangle \\
&= \left(-\frac{\hbar}{2}\right)(id|+z\rangle - ic|-z\rangle) \\
\Rightarrow |-y\rangle &= id|+z\rangle - ic|-z\rangle \equiv c|+z\rangle + d|-z\rangle \\
\Rightarrow d &= -ic
\end{aligned}$$

Using the normalization condition:

$$\begin{aligned}
|c|^2 + |d|^2 &= 1 \Rightarrow |c| = \frac{1}{\sqrt{2}} \\
\Rightarrow |-y\rangle &= \frac{1}{\sqrt{2}}|+z\rangle - \frac{i}{\sqrt{2}}|-z\rangle
\end{aligned}$$

■

- (3) Change $\hat{S}_a$ -basis states $\{|+a\rangle, |-a\rangle\}$ along $\vec{a} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$ to $\hat{S}_z$ -basis states $\{|+z\rangle, |-z\rangle\}$



$$|+a\rangle = \cos\frac{\theta}{2}|+z\rangle + \sin\frac{\theta}{2}e^{i\phi}|-z\rangle$$

$$|-a\rangle = -\sin\frac{\theta}{2}e^{-i\phi}|+z\rangle + \cos\frac{\theta}{2}|-z\rangle$$

where $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$.

PROOF:

Let

$$|+a\rangle \equiv a|+z\rangle + b|-z\rangle$$

$$|-a\rangle \equiv c|+z\rangle + d|-z\rangle$$

We obtain

$$\begin{aligned}\hat{S}_a|+a\rangle &= \left(\frac{1}{2}\sin\theta e^{-i\phi}\hat{S}_+ + \frac{1}{2}\sin\theta e^{i\phi}\hat{S}_- + \cos\theta\hat{S}_z\right)|+a\rangle \\ &= \frac{\hbar}{2}\left[(b\sin\theta e^{-i\phi} + a\cos\theta)|+z\rangle + (a\sin\theta e^{i\phi} - b\cos\theta)|-z\rangle\right] \\ \Rightarrow |a\rangle &= (b\sin\theta e^{-i\phi} + a\cos\theta)|+z\rangle + (a\sin\theta e^{i\phi} - b\cos\theta)|-z\rangle \\ &\equiv a|+z\rangle + b|-z\rangle \\ \Rightarrow b\sin\theta e^{-i\phi} + a\cos\theta &= a \text{ or } a\sin\theta e^{i\phi} - b\cos\theta = b \\ \Rightarrow b &= e^{i\phi}\frac{(1 - \cos\theta)}{\sin\theta}a = e^{i\phi}\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}a\end{aligned}$$

Using the normalization condition:

$$\begin{aligned}|a|^2 + |b|^2 &= |a|^2\left[1 + \frac{\sin^2\frac{\theta}{2}}{\cos^2\frac{\theta}{2}}\right] = |a|^2\frac{1}{\cos^2\frac{\theta}{2}} = 1 \Rightarrow |a| = \cos\frac{\theta}{2} \\ \Rightarrow |a\rangle &= \cos\frac{\theta}{2}|+z\rangle + \sin\frac{\theta}{2}e^{i\phi}|-z\rangle\end{aligned}$$

Similarly,

$$\begin{aligned}
\hat{S}_a|-a\rangle &= \left(\frac{1}{2} \sin \theta e^{-i\phi} \hat{S}_+ + \frac{1}{2} \sin \theta e^{i\phi} \hat{S}_- + \cos \theta \hat{S}_z \right) |-a\rangle \\
&= \frac{\hbar}{2} \left[(d \sin \theta e^{-i\phi} + c \cos \theta) |+z\rangle + (c \sin \theta e^{i\phi} - d \cos \theta) |-z\rangle \right] \\
\Rightarrow |-a\rangle &= -(d \sin \theta e^{-i\phi} + c \cos \theta) |+z\rangle - (c \sin \theta e^{i\phi} - d \cos \theta) |-z\rangle \\
&\equiv c |+z\rangle + d |-z\rangle \\
\Rightarrow d \sin \theta e^{-i\phi} + c \cos \theta &= -c \\
\Rightarrow d &= -e^{i\phi} \frac{(1 + \cos \theta)}{\sin \theta} c = -e^{i\phi} \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} c
\end{aligned}$$

Using the normalization condition:

$$|c|^2 + |d|^2 = |c|^2 \left[1 + \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right] = |c|^2 \frac{1}{\sin^2 \frac{\theta}{2}} = 1 \Rightarrow |c| = \sin \frac{\theta}{2}$$

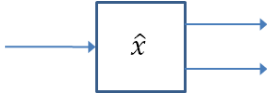
Conventionally, we choose

$$\begin{aligned}
c &= -e^{-i\phi} \sin \frac{\theta}{2} \\
\Rightarrow |-a\rangle &= -e^{-i\phi} \sin \frac{\theta}{2} |+z\rangle + \cos \frac{\theta}{2} |-z\rangle
\end{aligned}$$

■

EXAMPLES:

1. Suppose $|\chi\rangle = a|+z\rangle + b|-z\rangle$, $\mathcal{P}\left(+\frac{\hbar}{2}\right) = ?$ and $\mathcal{P}\left(-\frac{\hbar}{2}\right) = ?$



$$\begin{aligned}
|\chi\rangle &= a|+z\rangle + b|-z\rangle \\
&= a \left(\frac{1}{\sqrt{2}} |+x\rangle + \frac{1}{\sqrt{2}} |-x\rangle \right) + b \left(\frac{1}{\sqrt{2}} |+x\rangle - \frac{1}{\sqrt{2}} |-x\rangle \right) \\
&= \frac{a+b}{\sqrt{2}} |+x\rangle + \frac{a-b}{\sqrt{2}} |-x\rangle \\
\mathcal{P}\left(+\frac{\hbar}{2}\right) &= \frac{|a+b|^2}{2} \\
\mathcal{P}\left(-\frac{\hbar}{2}\right) &= \frac{|a-b|^2}{2}
\end{aligned}$$

2. Suppose that the first SG device is along the angles $\theta =$



$$|+a\rangle = \cos\frac{\pi}{3}|+z\rangle + \sin\frac{\pi}{3}|-z\rangle$$

$$= \frac{1}{2}|+z\rangle + \frac{\sqrt{3}}{2}|-z\rangle$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2}}|+x\rangle + \frac{1}{\sqrt{2}}|-x\rangle\right) + \frac{\sqrt{3}}{2}\left(\frac{1}{\sqrt{2}}|+x\rangle - \frac{1}{\sqrt{2}}|-x\rangle\right)$$

$$= \frac{1+\sqrt{3}}{2\sqrt{2}}|+x\rangle + \frac{1-\sqrt{3}}{2\sqrt{2}}|-x\rangle$$

$$\mathcal{P}\left(+\frac{\hbar}{2}\right) = \frac{|1+\sqrt{3}|^2}{8}$$

$$\mathcal{P}\left(-\frac{\hbar}{2}\right) = \frac{|1-\sqrt{3}|^2}{8}$$

3. Suppose $|\chi\rangle = \frac{1+i}{2}|+z\rangle + \frac{1-i}{2}|-z\rangle$

The expectation value of $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$

ANSWER:

- Method I: change of basis

$$\langle\chi|\hat{S}_z|\chi\rangle = \left(\frac{1-i}{2}\langle+z| + \frac{1+i}{2}\langle-z|\right)\hat{S}_z\left(\frac{1+i}{2}|+z\rangle + \frac{1-i}{2}|-z\rangle\right)$$

$$= \left(\frac{1-i}{2}\right)\left(\frac{1+i}{2}\right)\left(\frac{\hbar}{2}\right) + \left(\frac{1+i}{2}\right)\left(\frac{1-i}{2}\right)\left(-\frac{\hbar}{2}\right)$$

$$= 0$$

$$|\chi\rangle = \frac{1+i}{2}|+z\rangle + \frac{1-i}{2}|-z\rangle$$

$$= \frac{1+i}{2}\left(\frac{1}{\sqrt{2}}|+x\rangle + \frac{1}{\sqrt{2}}|-x\rangle\right) + \frac{1-i}{2}\left(\frac{1}{\sqrt{2}}|+x\rangle - \frac{1}{\sqrt{2}}|-x\rangle\right)$$

$$= \frac{1}{\sqrt{2}}|+x\rangle + \frac{i}{\sqrt{2}}|-x\rangle$$

$$\begin{aligned}
\langle \chi | \hat{S}_x | \chi \rangle &= \left(\frac{1}{\sqrt{2}} \langle +x | - \frac{i}{\sqrt{2}} \langle -x | \right) \hat{S}_x \left(\frac{1}{\sqrt{2}} | +x \rangle + \frac{i}{\sqrt{2}} | -x \rangle \right) \\
&= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} \left(\frac{\hbar}{2} \right) + \left(-\frac{i}{\sqrt{2}} \right) \left(\frac{i}{\sqrt{2}} \right) \left(-\frac{\hbar}{2} \right) \\
&= 0 \\
| \chi \rangle &= \frac{1+i}{2} | +z \rangle + \frac{1-i}{2} | -z \rangle \\
&= \frac{1+i}{2} \left(\frac{1}{\sqrt{2}} | +y \rangle + \frac{1}{\sqrt{2}} | -y \rangle \right) + \frac{1-i}{2} \left(\frac{-i}{\sqrt{2}} | +y \rangle + \frac{i}{\sqrt{2}} | -y \rangle \right) \\
&= 0 | +y \rangle + \frac{1+i}{\sqrt{2}} | -y \rangle \\
\langle \chi | \hat{S}_y | \chi \rangle &= \left(\frac{1-i}{\sqrt{2}} \langle -y | \right) \hat{S}_y \left(\frac{1+i}{\sqrt{2}} | -y \rangle \right) \\
&= \left(\frac{1-i}{\sqrt{2}} \right) \left(\frac{1+i}{\sqrt{2}} \right) \left(-\frac{\hbar}{2} \right) \\
&= -\frac{\hbar}{2}
\end{aligned}$$

- Method II: algebra

$$\begin{aligned}
\hat{S}_+ | -z \rangle &= \hbar | +z \rangle \\
\hat{S}_- | +z \rangle &= \hbar | -z \rangle \\
\langle \chi | \hat{S}_x | \chi \rangle &= \left(\frac{1-i}{2} \right) \left(\frac{1-i}{2} \right) \langle +z | \frac{1}{2} \hat{S}_+ | -z \rangle \\
&\quad + \left(\frac{1+i}{2} \right) \left(\frac{1+i}{2} \right) \langle -z | \frac{1}{2} \hat{S}_- | +z \rangle \\
&= -\frac{i}{2} \left(\frac{\hbar}{2} \right) + \frac{i}{2} \left(\frac{\hbar}{2} \right) \\
&= 0 \\
\langle \chi | \hat{S}_y | \chi \rangle &= \left(\frac{1-i}{2} \right) \left(\frac{1-i}{2} \right) \langle +z | \frac{1}{2i} \hat{S}_+ | -z \rangle \\
&\quad - \left(\frac{1+i}{2} \right) \left(\frac{1+i}{2} \right) \langle -z | \frac{1}{2i} \hat{S}_- | +z \rangle \\
&= -\frac{i}{2} \left(\frac{\hbar}{2i} \right) - \frac{i}{2} \left(\frac{\hbar}{2i} \right) \\
&= -\frac{\hbar}{2}
\end{aligned}$$

4. Suppose $\mathbf{x} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix}$

The expectation value of $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$

ANSWER:

$$\mathbf{x}^\dagger \cdot \mathbf{S}_z \cdot \mathbf{x} = \frac{1}{\sqrt{6}} (1-i \quad 2) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix} = -\frac{\hbar}{6}$$

$$\mathbf{x}^\dagger \cdot \mathbf{S}_x \cdot \mathbf{x} = \frac{1}{\sqrt{6}} (1-i \quad 2) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix} = \frac{\hbar}{3}$$

$$\mathbf{x}^\dagger \cdot \mathbf{S}_y \cdot \mathbf{x} = \frac{1}{\sqrt{6}} (1-i \quad 2) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix} = -\frac{\hbar}{3}$$

5. Suppose $\mathbf{x} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix}$

The expectation value of $\hat{S}$ along $\theta = 2\pi/3$ and $\phi = 0$

ANSWER:

The matrix form of $\mathbf{S}$ along $\vec{a}$:

$$\mathbf{S} = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$$

The eigenvalues and eigenvectors:

$$|\mathbf{S} - \lambda \mathbf{1}| = 0 \Rightarrow -(\cos^2 \theta + \sin^2 \theta) + \lambda^2 = 0 \Rightarrow \lambda = \pm 1$$

$$\lambda = 1 \Rightarrow \mathbf{v}_{+a} = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}$$

$$\lambda = -1 \Rightarrow \mathbf{v}_{-a} = \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\phi} \\ \cos \frac{\theta}{2} \end{pmatrix}$$

The unitary matrix has the eigenvectors of $\mathbf{v}_n$ as its columns,

$$\mathbf{U}_a = (\mathbf{v}_{+a} \quad \mathbf{v}_{-a}) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} e^{-i\phi} \\ \sin \frac{\theta}{2} e^{i\phi} & \cos \frac{\theta}{2} \end{pmatrix}$$

Along $\theta = 2\pi/3$ and $\phi = 0$ direction:

$$\mathbf{U}_a = \begin{pmatrix} \cos \frac{\pi}{3} & -\sin \frac{\pi}{3} \\ \sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$$

Change of vector:

$$\begin{aligned}\chi' &= \mathbf{U}_a^\dagger \cdot \chi \\ &= \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i \\ 2 \end{pmatrix} \\ &= \frac{1}{2\sqrt{6}} \begin{pmatrix} 1-2\sqrt{3}+i \\ 2+\sqrt{3}+i\sqrt{3} \end{pmatrix}\end{aligned}$$

The similarity transformation:

$$\begin{aligned}\mathbf{B} &= \mathbf{U}_a^\dagger \mathbf{S}_a \mathbf{U}_a = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \Rightarrow \chi'^\dagger \cdot \mathbf{B} \cdot \chi' &= \frac{1}{2\sqrt{6}} (1-2\sqrt{3}-i \quad 2+\sqrt{3}-i\sqrt{3}) \\ &\quad \times \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{2\sqrt{6}} \begin{pmatrix} 1-2\sqrt{3}+i \\ 2+\sqrt{3}+i\sqrt{3} \end{pmatrix} \\ &= \frac{14-4\sqrt{3}-10-4\sqrt{3}}{48} \hbar \\ &= \frac{1-\sqrt{3}}{12} \hbar\end{aligned}$$

6. Suppose $\hat{S}_x|\chi\rangle = \frac{\hbar}{2}|\chi\rangle$

The expectation value of $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$

ANSWER:

- Method I:

$$\begin{aligned}|\chi\rangle &= |+_x\rangle = \frac{1}{\sqrt{2}}|+_z\rangle + \frac{1}{\sqrt{2}}|-_z\rangle \\ \langle\chi|\hat{S}_x|\chi\rangle &= \frac{\hbar}{2} \\ |\chi\rangle &= \frac{1}{\sqrt{2}}|+_z\rangle + \frac{1}{\sqrt{2}}|-_z\rangle \\ &= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}|+_y\rangle + \frac{1}{\sqrt{2}}|-_y\rangle\right) + \frac{1}{\sqrt{2}}\left(\frac{-i}{\sqrt{2}}|+_y\rangle + \frac{i}{\sqrt{2}}|-_y\rangle\right) \\ &= \frac{1-i}{2}|+_y\rangle + \frac{1+i}{\sqrt{2}}|-_y\rangle \\ \langle\chi|\hat{S}_y|\chi\rangle &= \frac{\hbar}{2} - \frac{\hbar}{2} = 0 \\ \langle\chi|\hat{S}_z|\chi\rangle &= \frac{1}{2}\left(\frac{\hbar}{2}\right) + \frac{1}{2}\left(-\frac{\hbar}{2}\right) = 0\end{aligned}$$

- Method II:

The matrix form of $\mathbf{S}_x$:

$$\mathbf{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Diagonalize $\mathbf{S}_x$ and obtain the eigenvalues and eigenvectors:

$$|\mathbf{S}_x - \lambda \mathbf{1}| = 0 \Rightarrow \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$$

$$\lambda = 1 \Rightarrow \mathbf{v}_{+x} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda = -1 \Rightarrow \mathbf{v}_{-x} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The unitary matrix has the eigenvectors of $\mathbf{v}_n$ as its columns,

$$\mathbf{U}_x = (\mathbf{v}_{+x} \quad \mathbf{v}_{-x}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Change of vector:

$$\mathbf{x}' = \mathbf{U}_x^\dagger \cdot \mathbf{u}_{+x} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Expectation values:

$$\mathbf{x}'^\dagger \cdot \mathbf{S}_x \cdot \mathbf{x}' = \frac{1}{\sqrt{2}} (1 \quad 1) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{\hbar}{2}$$

$$\mathbf{x}'^\dagger \cdot \mathbf{S}_y \cdot \mathbf{x}' = \frac{1}{\sqrt{2}} (1 \quad 1) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

$$\mathbf{x}'^\dagger \cdot \mathbf{S}_z \cdot \mathbf{x}' = \frac{1}{\sqrt{2}} (1 \quad 1) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$$

(4) Spectral representation of Pauli matrix

$$\hat{\sigma}_x = |+\rangle\langle+| - |-\rangle\langle-| = |-z\rangle\langle+z| + |+z\rangle\langle-z|$$

$$\hat{\sigma}_y = |+\rangle\langle+| - |-\rangle\langle-| = i|-z\rangle\langle+z| - i|+z\rangle\langle-z|$$

$$\hat{\sigma}_z = |+z\rangle\langle+z| - |-z\rangle\langle-z|$$

PROOF:

$$\hat{\sigma}_x = |+\rangle\langle+| - |-\rangle\langle-|$$

$$= \left(\frac{1}{\sqrt{2}} |+\rangle + \frac{1}{\sqrt{2}} |-\rangle \right) \left(\frac{1}{\sqrt{2}} \langle+z| + \frac{1}{\sqrt{2}} \langle-z| \right)$$

$$- \left(\frac{1}{\sqrt{2}} |+\rangle - \frac{1}{\sqrt{2}} |-\rangle \right) \left(\frac{1}{\sqrt{2}} \langle+z| - \frac{1}{\sqrt{2}} \langle-z| \right)$$

$$= \frac{1}{2} |+\rangle\langle+z| + \frac{1}{2} |-\rangle\langle-z| + \frac{1}{2} |+\rangle\langle-z| + \frac{1}{2} |-\rangle\langle+z|$$

$$- \frac{1}{2} |+\rangle\langle+z| - \frac{1}{2} |-\rangle\langle-z| + \frac{1}{2} |+\rangle\langle-z| + \frac{1}{2} |-\rangle\langle+z|$$

$$= |-z\rangle\langle+z| + |+z\rangle\langle-z|$$

$$\begin{aligned}
\hat{\sigma}_y &= | +y \rangle \langle +y | - | -y \rangle \langle -y | \\
&= \left(\frac{1}{\sqrt{2}} | +z \rangle + \frac{i}{\sqrt{2}} | -z \rangle \right) \left(\frac{1}{\sqrt{2}} \langle +z | - \frac{i}{\sqrt{2}} \langle -z | \right) \\
&\quad - \left(\frac{1}{\sqrt{2}} | +z \rangle - \frac{i}{\sqrt{2}} | -z \rangle \right) \left(\frac{1}{\sqrt{2}} \langle +z | + \frac{i}{\sqrt{2}} \langle -z | \right) \\
&= \frac{1}{2} | +z \rangle \langle +z | - \frac{1}{2} | -z \rangle \langle -z | - \frac{i}{2} | +z \rangle \langle -z | + \frac{i}{2} | -z \rangle \langle +z | \\
&\quad - \frac{1}{2} | +z \rangle \langle +z | + \frac{1}{2} | -z \rangle \langle -z | - \frac{i}{2} | +z \rangle \langle -z | + \frac{i}{2} | -z \rangle \langle +z | \\
&= i | -z \rangle \langle +z | - i | +z \rangle \langle -z |
\end{aligned}$$

■

EXAMPLES:

1. $\hat{\sigma}_x | +z \rangle = ?$

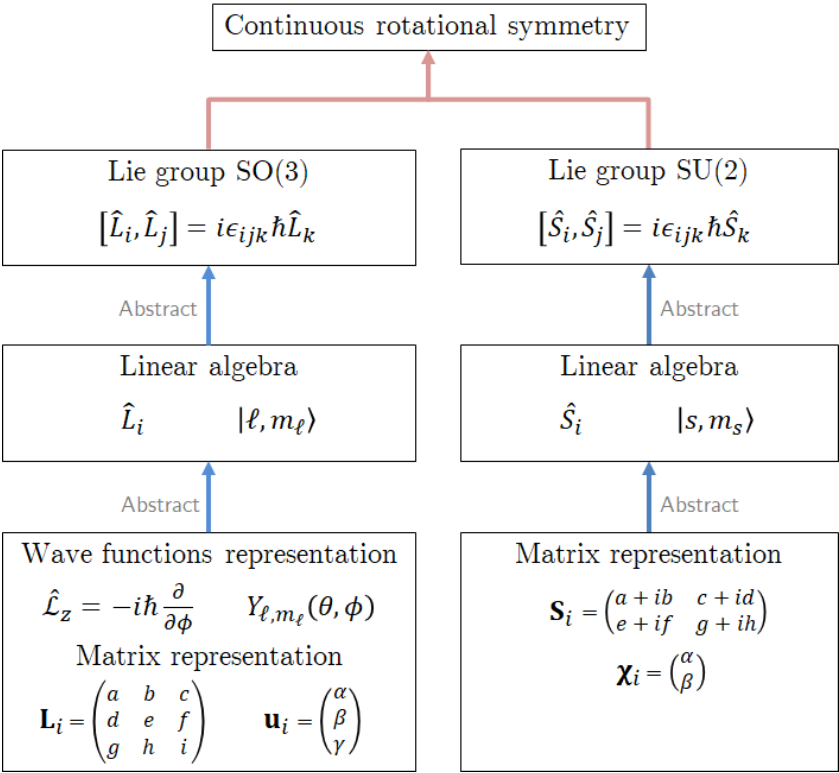
ANSWER:

$$\begin{aligned}
\hat{\sigma}_x | +z \rangle &= (| -z \rangle \langle +z | + | +z \rangle \langle -z |) | +z \rangle \\
&= | -z \rangle \langle +z | +z \rangle + | +z \rangle \langle -z | +z \rangle \\
&= | -z \rangle
\end{aligned}$$

or

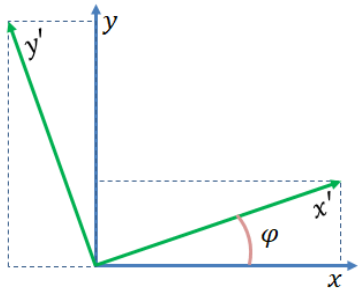
$$\begin{aligned}
\hat{\sigma}_x | +z \rangle &= \hat{\sigma}_x \left(\frac{1}{\sqrt{2}} | +x \rangle + \frac{1}{\sqrt{2}} | -x \rangle \right) \\
&= \frac{1}{\sqrt{2}} | +x \rangle - \frac{1}{\sqrt{2}} | -x \rangle \\
&= | -z \rangle
\end{aligned}$$

6-5 Continuous Group SO(3) and SU(2)



A. LIE GROUP SO(2)

(1) A two-dimensional geometric rotation through an angle φ



$$\begin{aligned} x' &= x \cos \varphi + y \sin \varphi \\ y' &= -x \sin \varphi + y \cos \varphi \end{aligned} \Rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \mathbf{R}(\varphi) = \begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix}$$

where $\mathbf{R}(\varphi + 2\pi) = \mathbf{R}(\varphi)$, $0 \leq \varphi < 2\pi$

- (2) The set of all functions $\mathbf{R}(\varphi)$ forms a group.

PROOF:

$$\mathbf{R}(\varphi_1)\mathbf{R}(\varphi_2) = \mathbf{R}(\varphi_1 + \varphi_2)$$

$$\mathbf{R}(0) = \mathbf{1}$$

$$\mathbf{R}^{-1}(\varphi) = \mathbf{R}(-\varphi)$$

$$(\mathbf{R}(\varphi_1)\mathbf{R}(\varphi_2))\mathbf{R}(\varphi_3) = \mathbf{R}(\varphi_1)(\mathbf{R}(\varphi_2)\mathbf{R}(\varphi_3))$$

- (3) The group is a $SO(2)$ group if and only if $\mathbf{R}$ is orthogonal and $\det \mathbf{R}(\varphi) = +1$.

PROOF:

$\mathbf{R}$ is orthogonal:

$$\mathbf{c}' = \mathbf{R} \cdot \mathbf{c}$$

Since $\mathbf{c}'^T \cdot \mathbf{c}' = \mathbf{c}^T \cdot \mathbf{R}^T \cdot \mathbf{R} \cdot \mathbf{c} = \mathbf{c}^T \cdot \mathbf{c}$, we obtain

$$\mathbf{R}^T \cdot \mathbf{R} = \mathbf{1}$$

$$\Rightarrow \mathbf{R}^T = \mathbf{R}^{-1}$$

$\mathbf{R}$ is special orthogonal, $\det \mathbf{R}(\varphi) = +1$:

$$\det \mathbf{R}^T \det \mathbf{R} = \det \mathbf{1} \Rightarrow |\det \mathbf{R}|^2 = 1 \Rightarrow \det \mathbf{R} = \pm 1$$

If $\det \mathbf{R} = +1$, it is a proper rotation which preserves handedness

If $\det \mathbf{R} = -1$, it is an improper rotation which do not preserve handedness

■

- (4) $SO(2)$ group is a Lie group.

PROOF:

For an infinitesimal rotation angle $d\varphi$

$$\mathbf{R}(\varphi) = \mathbf{R}(0) + \left. \frac{d\mathbf{R}(\varphi)}{d\varphi} \right|_{\varphi=0} \varphi + \frac{1}{2} \left. \frac{d^2\mathbf{R}(\varphi)}{d\varphi^2} \right|_{\varphi=0} \varphi^2 + \dots$$

where $\mathbf{R}(\varphi)$ is differentiable with respect to the parameter φ

$$\mathbf{R}(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{1}$$

$$\left. \frac{d\mathbf{R}(\varphi)}{d\varphi} \right|_{\varphi=0} = \begin{pmatrix} -\sin \varphi & \cos \varphi \\ -\cos \varphi & -\sin \varphi \end{pmatrix}_{\varphi=0} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Let $\mathbf{L} = i\hbar \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$\frac{d\mathbf{R}(\varphi)}{d\varphi} = -\frac{i}{\hbar} \mathbf{L}$$

$$\Rightarrow \mathbf{R}(\varphi) = \mathbf{1} - \frac{i}{\hbar} \mathbf{L} \varphi + \frac{1}{2} \left(-\frac{i}{\hbar} \mathbf{L} \right)^2 \varphi^2 + \dots$$

$$= \sum_n \frac{1}{n!} \left(-\frac{i}{\hbar} \varphi \mathbf{L} \right)^n$$

$$= e^{-\frac{i}{\hbar} \varphi \mathbf{L}}$$

is a rotation through angle φ about the z-axis.

$\Rightarrow$ The set of all functions $\mathbf{R}(\varphi)$ forms a SO(2) Lie group, where $\mathbf{L}$ is the generator of the group.

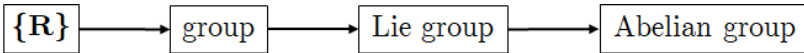
■

(5) SO(2) Lie group is an Abelian group.

PROOF:

The generators commute with one another, i.e., $[\mathbf{L}, \mathbf{L}] = 0$.

■



B. LIE GROUP SO(3)

(1) Matrix representation of rotation operators

$$\mathbf{R}_x(\varphi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & \sin \varphi \\ 0 & -\sin \varphi & \cos \varphi \end{pmatrix}$$

$$\mathbf{R}_y(\varphi) = \begin{pmatrix} \cos \varphi & 0 & -\sin \varphi \\ 0 & 1 & 0 \\ \sin \varphi & 0 & \cos \varphi \end{pmatrix}$$

$$\mathbf{R}_z(\varphi) = \begin{pmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

where $\mathbf{R}_i(\varphi + 2\pi) = \mathbf{R}_i(\varphi)$, $0 \leq \varphi < 2\pi$

Since $\mathbf{R}_i^T \cdot \mathbf{R}_i = \mathbf{1} \Rightarrow \mathbf{R}_i^T = \mathbf{R}_i^{-1}$ and $\det \mathbf{R}(\varphi) = +1$

$\Rightarrow \mathbf{R}(\varphi)$ forms a SO(3) group.

- (2) The set of all functions $\mathbf{R}(\varphi)$ forms a $\text{SO}(3)$ Lie group.

PROOF:

$\mathbf{R}_i(\varphi)$ is differentiable with respect to the parameter φ

$$\left. \frac{d\mathbf{R}_x(\varphi)}{d\varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = -\frac{i}{\hbar} \mathbf{L}_x$$

$$\left. \frac{d\mathbf{R}_y(\varphi)}{d\varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = -\frac{i}{\hbar} \mathbf{L}_y$$

$$\left. \frac{d\mathbf{R}_z(\varphi)}{d\varphi} \right|_{\varphi=0} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = -\frac{i}{\hbar} \mathbf{L}_z$$

$\Rightarrow \mathbf{R}(\varphi) = \mathbf{R}_x(\varphi)\mathbf{R}_y(\varphi)\mathbf{R}_z(\varphi) = e^{-\frac{i}{\hbar}\varphi\mathbf{L}\cdot\vec{a}}$ is a rotation through angle φ about an axis $\vec{a} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$.

$\Rightarrow$ The set of all functions $\mathbf{R}(\varphi)$ forms a $\text{SO}(3)$ Lie group, where $\mathbf{L}_i$ is the generator of the group.

■

- (3) $\text{SO}(3)$ Lie group is a non-Abelian group.

PROOF:

The generators do not commute with one another

$$\begin{aligned} [\mathbf{L}_x, \mathbf{L}_y] &= \mathbf{L}_x\mathbf{L}_y - \mathbf{L}_y\mathbf{L}_x \\ &= i\hbar \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} i\hbar \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \\ &\quad - i\hbar \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} i\hbar \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \\ &= \hbar^2 \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= i\hbar \mathbf{L}_z \end{aligned}$$

- (4) An operator that commutes with all generators of a Lie group is called a Casimir operator.

$$\mathbf{L}^2 = \mathbf{L}_x^2 + \mathbf{L}_y^2 + \mathbf{L}_z^2$$

C. LIE GROUP SU(2)

- (1) Matrix representation of spin rotation operators

$$\mathbf{U} = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} e^{-i\phi} \\ \sin \frac{\theta}{2} e^{i\phi} & \cos \frac{\theta}{2} \end{pmatrix}$$

The spin polarization matrix can be decomposed into 3 rotation matrices about the rectangular axes.

$$\mathbf{R}_x(\varphi) = \begin{pmatrix} \cos \frac{\varphi}{2} & -i \sin \frac{\varphi}{2} \\ -i \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{pmatrix}$$

$$\mathbf{R}_y(\varphi) = \begin{pmatrix} \cos \frac{\varphi}{2} & -\sin \frac{\varphi}{2} \\ \sin \frac{\varphi}{2} & \cos \frac{\varphi}{2} \end{pmatrix}$$

$$\mathbf{R}_z(\varphi) = \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix}$$

where $\mathbf{R}_i(\varphi + 2\pi) = -\mathbf{R}_i(\varphi)$, $\mathbf{R}_i(\varphi + 4\pi) = \mathbf{R}_i(\varphi)$, $0 \leq \varphi < 4\pi$

$$\mathbf{U} = \mathbf{R}_z(\phi) \mathbf{R}_y(\theta) \mathbf{R}_z(-\phi)$$

Since $\mathbf{R}_i^\dagger \cdot \mathbf{R}_i = \mathbf{1} \Rightarrow \mathbf{R}_i^\dagger = \mathbf{R}_i^{-1}$ and $\det \mathbf{R}_i(\varphi) = +1$
 $\Rightarrow \mathbf{R}(\varphi)$ forms a SU(2) group.

- (2) The set of all functions $\mathbf{R}(\varphi)$ forms a SU(2) Lie group.

PROOF:

$\mathbf{R}(\varphi)$ is differentiable with respect to the parameter φ

$$\left. \frac{d\mathbf{R}_x(\varphi)}{d\varphi} \right|_{\varphi=0} = -\frac{i}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -\frac{i}{2} \boldsymbol{\sigma}_x = -\frac{i}{\hbar} \mathbf{S}_x$$

$$\left. \frac{d\mathbf{R}_y(\varphi)}{d\varphi} \right|_{\varphi=0} = -\frac{i}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = -\frac{i}{2} \boldsymbol{\sigma}_y = -\frac{i}{\hbar} \mathbf{S}_y$$

$$\left. \frac{d\mathbf{R}_z(\varphi)}{d\varphi} \right|_{\varphi=0} = -\frac{i}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -\frac{i}{2} \boldsymbol{\sigma}_z = -\frac{i}{\hbar} \mathbf{S}_z$$

$\Rightarrow \mathbf{R}(\varphi) = \mathbf{R}_x(\varphi) \mathbf{R}_y(\varphi) \mathbf{R}_z(\varphi) = e^{-\frac{i}{\hbar} \boldsymbol{\varphi} \cdot \vec{\mathbf{S}}}$ is a rotation through angle φ
 about an axis $\vec{\mathbf{a}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$.

$\Rightarrow$ The set of all functions $\mathbf{R}(\varphi)$ forms a $SU(2)$ Lie group, where $\mathbf{S}_i$ is the generator of the group. ■

- (3) $SU(2)$ Lie group is a non-Abelian group.

PROOF:

The generators do not commute with one another

$$\begin{aligned} [\mathbf{S}_x, \mathbf{S}_y] &= \mathbf{S}_x \mathbf{S}_y - \mathbf{S}_y \mathbf{S}_x \\ &= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= i \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= i\hbar \mathbf{S}_z \end{aligned}$$
■

- (4) The Casimir operator is $\mathbf{S}^2 = \mathbf{S}_x^2 + \mathbf{S}_y^2 + \mathbf{S}_z^2$

D. CONNECTION BETWEEN $SU(2)$ AND $SO(3)$

- (1) Local mapping:

The Lie algebra structure (commutator of generators) determines the Lie group structure “locally”.

$$\begin{array}{ccc} [\mathbf{L}_x, \mathbf{L}_y] = i\hbar \mathbf{L}_z & & [\mathbf{S}_x, \mathbf{S}_y] = i\hbar \mathbf{S}_z \\ [\mathbf{L}_y, \mathbf{L}_z] = i\hbar \mathbf{L}_x & \xleftarrow{\text{isomorphic}} & [\mathbf{S}_y, \mathbf{S}_z] = i\hbar \mathbf{S}_x \\ [\mathbf{L}_z, \mathbf{L}_x] = i\hbar \mathbf{L}_y & & [\mathbf{S}_z, \mathbf{S}_x] = i\hbar \mathbf{S}_y \end{array}$$

$\Rightarrow SO(3)$ is locally isomorphic to $SU(2)$

$\Rightarrow$ There is a one-to-one local correspondence between $SO(3)$ and $SU(2)$.

- (2) Global mapping:

$SO(3)$ has a periodicity of 2π

$SU(2)$ has a periodicity of 4π

$\Rightarrow SO(3)$ is homomorphic to $SU(2)$

$\Rightarrow SO(3)$ is isomorphic to $SU(2)/Z_2$

$\Rightarrow$ There is a two-to-one correspondence between $SO(3)$ and $SU(2)$.

6-6 Addition of Two Spins

A. HILBERT SPACE OF TWO SPINS

- (1) Consider two individual spins. Each spin corresponds to a Hilbert space as

$$\begin{array}{cc} \boxed{\mathbb{V}_{s_1}} & \boxed{\mathbb{V}_{s_2}} \\ \{|s_1, m_{s_1}\rangle\} & \{|s_2, m_{s_2}\rangle\} \end{array}$$

$$\begin{array}{l} |s_1, m_{s_1}\rangle \in \mathbb{V}_{s_1} \text{ and } \hat{S}_1: \mathbb{V}_{s_1} \rightarrow \mathbb{V}_{s_1} \\ |s_2, m_{s_2}\rangle \in \mathbb{V}_{s_2} \text{ and } \hat{S}_2: \mathbb{V}_{s_2} \rightarrow \mathbb{V}_{s_2} \end{array}$$

where

$$[\hat{S}_1, \hat{S}_2] = 0$$

- (2) We now add two spins. The combined Hilbert space is

$$\mathbb{V}_{s_1} + \mathbb{V}_{s_2} = \mathbb{W}$$

$$\boxed{\mathbb{V}_{s_1}} + \boxed{\mathbb{V}_{s_2}} = \boxed{\mathbb{W}}$$

$$\{|s_1, m_{s_1}\rangle\} + \{|s_2, m_{s_2}\rangle\} = \{?, ?\}$$

- (3) Construct the basis states of $\mathbb{W}$

- Method I:

Consider the total spin operators $\hat{S} = \hat{S}_1 + \hat{S}_2$ and $\hat{S}_z = \hat{S}_{1z} + \hat{S}_{2z}$.

Since

$$[\hat{S}, \hat{S}_z] = (\hat{S}_1 + \hat{S}_2)(\hat{S}_{1z} + \hat{S}_{2z}) - (\hat{S}_{1z} + \hat{S}_{2z})(\hat{S}_1 + \hat{S}_2) \neq 0$$

thus, we can only consider the individual spins separately, i.e.,

$$\hat{S}_1: \mathbb{W} \rightarrow \mathbb{W}$$

$$\hat{S}_2: \mathbb{W} \rightarrow \mathbb{W}$$

Since

$$[\hat{S}_1, \hat{S}_2] = 0$$

the common eigenstates of $\hat{S}_1$ and $\hat{S}_2$ are the product of states belonging to $\mathbb{V}_{s_1}$ and $\mathbb{V}_{s_2}$, i.e.,

$$\left| s_1, m_{s_1}; s_2, m_{s_2} \right\rangle = \underbrace{\left| s_1, m_{s_1} \right\rangle}_{\in \mathbb{V}_{s_1}} \otimes \underbrace{\left| s_2, m_{s_2} \right\rangle}_{\in \mathbb{V}_{s_2}} \in \mathbb{W} = \mathbb{V}_{s_1} + \mathbb{V}_{s_2}$$

These basis states of $\mathbb{W}$ are called **tensor product states** and the number of basis states is $(2s_1 + 1) \times (2s_2 + 1)$.

- Method II:

Consider the square of the total spin operator $\hat{S}^2 = (\hat{S}_1 + \hat{S}_2)^2$.

Since

$$\begin{aligned} [\hat{S}^2, \hat{S}_z] &= [\hat{S}_x^2, \hat{S}_z] + [\hat{S}_y^2, \hat{S}_z] + [\hat{S}_z^2, \hat{S}_z] \\ &= \hat{S}_x [\hat{S}_x, \hat{S}_z] + [\hat{S}_x, \hat{S}_z] \hat{S}_x + \hat{S}_y [\hat{S}_y, \hat{S}_z] + [\hat{S}_y, \hat{S}_z] \hat{S}_y \\ &= i\hbar (-\hat{S}_x \hat{S}_y - \hat{S}_y \hat{S}_x + \hat{S}_y \hat{S}_x + \hat{S}_x \hat{S}_y) \\ &= 0 \end{aligned}$$

thus, we can consider the combined spins ($s = s_1 + s_2, \dots, |s_1 - s_2|$) as a whole, i.e.,

$$\hat{S}^2: \mathbb{W} \rightarrow \mathbb{W}$$

$$\hat{S}_z: \mathbb{W} \rightarrow \mathbb{W}$$

The common eigenstates of $\hat{S}^2$ and $\hat{S}_z$ are

$$\begin{aligned} |s, m_s\rangle &= \underbrace{|s_1 + s_2, m_{s_1+s_2}\rangle}_{\in \mathbb{V}_{s_1+s_2}} \oplus \dots \oplus \underbrace{||s_1 - s_2|, m_{|s_1-s_2|}\rangle}_{\in \mathbb{V}_{|s_1-s_2|}} \\ &\in \mathbb{W} = \mathbb{V}_{s_1+s_2} \cup \dots \cup \mathbb{V}_{|s_1-s_2|} \end{aligned}$$

where $s = s_1 + s_2, \dots, |s_1 - s_2|$ are the combined spins and $m_s = m_{s_1} + m_{s_2}$.

These basis states of $\mathbb{W}$ are called **direct sum states** and the number of basis states is

$$\begin{aligned} \sum_{s=|s_1-s_2|, s_1 \geq s_2}^{s_1+s_2} (2s+1) &= [2(s_1 + s_2) + 1] + \dots + [2(s_1 - s_2) + 1] \\ &= \sum_{n=0}^{2s_2} 2(s_1 - s_2 + n) + 1 \\ &= \frac{(4s_1 + 2)(2s_2 + 1)}{2} \\ &= (2s_1 + 1)(2s_2 + 1) \end{aligned}$$

EXAMPLES:

1. Find the tensor product states and direct sum states of the

addition of two spins:

$$s_1 = \frac{1}{2} \text{ and } s_2 = \frac{1}{2}$$

ANSWER:

- Tensor product states:

Number of basis states: $\left(2 \times \frac{1}{2} + 1\right) \left(2 \times \frac{1}{2} + 1\right) = 2 \times 2 = 4$

$$\left|\frac{1}{2}, +\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|\frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2}, +\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \left|\frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|\frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \left|\frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2}\right\rangle$$

- Direct sum states:

$$s = s_1 + s_2, \dots, |s_1 - s_2| = 1, 0$$

The number of basis states:

$$\sum_{s=0}^1 (2s + 1) = 3 + 1 = 4$$

The basis states are

$$\underbrace{|1, +1\rangle, |1, 0\rangle, |1, -1\rangle}_{s=1}, \underbrace{|0, 0\rangle}_{s=0}$$

where

$$\{|1, +1\rangle, |1, 0\rangle, |1, -1\rangle\} \in \mathbb{V}_1$$

$$\{|0, 0\rangle\} \in \mathbb{V}_0$$

Thus, we have

$$\{|1, +1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle\} \in \mathbb{W} = \mathbb{V}_1 \cup \mathbb{V}_0$$

2. Find the tensor product states and direct sum states of the addition of two spins:

$$s_1 = 1 \text{ and } s_2 = \frac{1}{2}$$

ANSWER:

- Tensor product states:

Number of basis states: $(2 \times 1 + 1) \left(2 \times \frac{1}{2} + 1\right) = 3 \times 2 = 6$

$$|1, +1\rangle \otimes \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|1, +1; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$|1, 0\rangle \otimes \left|\frac{1}{2}, +\frac{1}{2}\right\rangle = \left|1, 0; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$\begin{aligned}
|1, -1\rangle \otimes \left| \frac{1}{2}, +\frac{1}{2} \right\rangle &= \left| 1, -1; \frac{1}{2}, +\frac{1}{2} \right\rangle \\
|1, +1\rangle \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle &= \left| 1, +1; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
|1, 0\rangle \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle &= \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
|1, -1\rangle \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle &= \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle
\end{aligned}$$

- Direct sum states:

$$s = s_1 + s_2, \dots, |s_1 - s_2| = \frac{3}{2}, \frac{1}{2}$$

The number of basis states:

$$\sum_{s=1/2}^{3/2} (2s+1) = 4 + 2 = 6$$

The basis states are

$$\underbrace{\left| \frac{3}{2}, +\frac{3}{2} \right\rangle, \left| \frac{3}{2}, +\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{3}{2} \right\rangle}_{s=3/2}, \underbrace{\left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle}_{s=1/2}$$

where

$$\begin{aligned}
&\left\{ \left| \frac{3}{2}, +\frac{3}{2} \right\rangle, \left| \frac{3}{2}, +\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{3}{2} \right\rangle \right\} \in \mathbb{V}_{3/2} \\
&\left\{ \left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right\} \in \mathbb{V}_{1/2}
\end{aligned}$$

Thus, we have

$$\begin{aligned}
&\left\{ \left| \frac{3}{2}, +\frac{3}{2} \right\rangle, \left| \frac{3}{2}, +\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \left| \frac{3}{2}, -\frac{3}{2} \right\rangle, \left| \frac{1}{2}, +\frac{1}{2} \right\rangle, \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right\} \\
&\in \mathbb{W} = \mathbb{V}_{3/2} \cup \mathbb{V}_{1/2}
\end{aligned}$$

3. Consider a Hamiltonian $\hat{H} = J\hat{S}_1 \cdot \hat{S}_2$ where $s_1 = 1$ and $s_2 = \frac{1}{2}$.

Find the matrix representation of $\hat{H}$.

ANSWER:

The Hilbert space is

$$\mathbb{V}_1 + \mathbb{V}_{1/2} = \mathbb{W}$$

and the number of basis states of $\mathbb{W}$ is

$$(2s_1 + 1) \times (2s_2 + 1) = 3 \times 2 = 6$$

- Tensor product states:

$$\underbrace{\left| 1, +1; \frac{1}{2}, +\frac{1}{2} \right\rangle}_{|1\rangle}, \quad \underbrace{\left| 1, 0; \frac{1}{2}, +\frac{1}{2} \right\rangle}_{|2\rangle}, \quad \underbrace{\left| 1, -1; \frac{1}{2}, +\frac{1}{2} \right\rangle}_{|3\rangle}$$

$$\underbrace{\left|1, +1; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|4\rangle}, \quad \underbrace{\left|1, 0; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|5\rangle}, \quad \underbrace{\left|1, -1; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|6\rangle}$$

Since

$$\hat{S}_1 \cdot \hat{S}_2 = \frac{1}{2}(\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}) + \hat{S}_{1z}\hat{S}_{2z}$$

$$\hat{S}_1 \cdot \hat{S}_2 |1\rangle = \hat{S}_1 \cdot \hat{S}_2 \left|1, +1; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \frac{1}{2}(\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}) + \hat{S}_{1z}\hat{S}_{2z} \left|1, +1; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \hat{S}_{1z}\hat{S}_{2z} \left|1, +1; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= 1\hbar \cdot \frac{1}{2}\hbar \left|1, +1; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \frac{\hbar^2}{2} |1\rangle$$

$$\hat{S}_1 \cdot \hat{S}_2 |2\rangle = \hat{S}_1 \cdot \hat{S}_2 \left|1, 0; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \frac{1}{2}(\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}) + \hat{S}_{1z}\hat{S}_{2z} \left|1, 0; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \frac{1}{2}\hat{S}_{1+}\hat{S}_{2-} \left|1, 0; \frac{1}{2}, +\frac{1}{2}\right\rangle$$

$$= \frac{1}{2}\sqrt{1(1+1) - 0(0+1)}\hbar \cdot \sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right) - \frac{1}{2}\left(\frac{1}{2}-1\right)}\hbar \left|1, +1; \frac{1}{2}, -\frac{1}{2}\right\rangle$$

$$= \frac{\hbar^2}{2}\sqrt{2}|4\rangle$$

and so on, thus we obtain

$$\mathbf{H} = J \frac{\hbar^2}{2} \begin{array}{c} \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \end{array} \begin{array}{c} \backslash \\ \begin{array}{cccccc} |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \end{array} \end{array} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & -1 & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{pmatrix}$$

- Direct sum states:

The combined spins are $s = s_1 + s_2, \dots, |s_1 - s_2| = \frac{3}{2}, \frac{1}{2}$

$$\begin{array}{cccc} \underbrace{\left| \frac{3}{2}, +\frac{3}{2} \right\rangle}_{|1\rangle}, & \underbrace{\left| \frac{3}{2}, +\frac{1}{2} \right\rangle}_{|2\rangle}, & \underbrace{\left| \frac{3}{2}, -\frac{1}{2} \right\rangle}_{|3\rangle}, & \underbrace{\left| \frac{3}{2}, -\frac{3}{2} \right\rangle}_{|4\rangle} \\ \underbrace{\left| \frac{1}{2}, +\frac{1}{2} \right\rangle}_{|5\rangle}, & \underbrace{\left| \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|6\rangle} & & \end{array}$$

Since

$$\hat{S}^2 = (\hat{S}_1 + \hat{S}_2)^2 = \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1 \cdot \hat{S}_2$$

we obtain

$$\begin{aligned} \hat{S}_1 \cdot \hat{S}_2 &= \frac{\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2}{2} \\ \hat{S}_1 \cdot \hat{S}_2 |1\rangle &= \hat{S}_1 \cdot \hat{S}_2 \left| \frac{3}{2}, +\frac{3}{2} \right\rangle \\ &= \frac{\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2}{2} \left| \frac{3}{2}, +\frac{3}{2} \right\rangle \\ &= \frac{\frac{3}{2} \left(\frac{3}{2} + 1 \right) - 1(1+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right)}{2} \hbar^2 \left| \frac{3}{2}, +\frac{3}{2} \right\rangle \\ &= \frac{1}{2} \hbar^2 \left| \frac{3}{2}, +\frac{3}{2} \right\rangle \\ &= \frac{\hbar^2}{2} |1\rangle \end{aligned}$$

and so on, we obtain

$$\mathbf{H} = J \frac{\hbar^2}{2} \begin{array}{c} \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \end{array} \begin{pmatrix} & |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \end{pmatrix} \end{pmatrix} \end{array}$$

Since $\mathbb{W}$ can be decomposed into two subspaces $\mathbb{V}_{3/2}$ and $\mathbb{V}_{1/2}$, i.e., $\mathbb{W} = \mathbb{V}_{3/2} \cup \mathbb{V}_{1/2}$. Thus, we obtain two submatrices:

$$\begin{aligned} \mathbf{H}_{(s=3/2)} &= J \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}_{4 \times 4} \in \mathbb{V}_{3/2} \\ \mathbf{H}_{(s=1/2)} &= J \frac{\hbar^2}{2} \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}_{2 \times 2} \in \mathbb{V}_{1/2} \end{aligned}$$

B. THE CLEBSCH-GORDAN COEFFICIENT

- (1) Since the tensor product states $|s_1, m_{s_1}; s_2, m_{s_2}\rangle$ and the direct sum states $|s, m_s\rangle$ belong to the same Hilbert space $\mathbb{W}$, in general, we must be able to find the relationship between $|s_1, m_{s_1}; s_2, m_{s_2}\rangle$ and $|s, m_s\rangle$.

- (2) Method I: using ladder operators

Consider the square of the total spin operator $\hat{S}^2$

$$\begin{aligned}
 \hat{S}^2 &= (\hat{S}_1 + \hat{S}_2)^2 \\
 &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1 \cdot \hat{S}_2 \\
 &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1x}\hat{S}_{2x} + 2\hat{S}_{1y}\hat{S}_{2y} + 2\hat{S}_{1z}\hat{S}_{2z} \\
 &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z} \\
 &\quad + 2 \times \frac{1}{4} [(\hat{S}_{1+} + \hat{S}_{1-})(\hat{S}_{2+} + \hat{S}_{2-}) - (\hat{S}_{1+} - \hat{S}_{1-})(\hat{S}_{2+} - \hat{S}_{2-})] \\
 &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z} + (\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+})
 \end{aligned}$$

Since

$$\begin{aligned}
 &(\hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z}) |s_1, m_{s_1}; s_2, m_{s_2}\rangle \\
 &= (s_1(s_1 + 1)\hbar^2 + s_2(s_2 + 1)\hbar^2 + 2m_{s_1}m_{s_2}\hbar^2) |s_1, m_{s_1}; s_2, m_{s_2}\rangle \\
 &= (s_1(s_1 + 1) + s_2(s_2 + 1) + 2m_{s_1}m_{s_2}) \hbar^2 |s_1, m_{s_1}; s_2, m_{s_2}\rangle
 \end{aligned}$$

and

$$\begin{aligned}
 &(\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}) |s_1, m_{s_1}; s_2, m_{s_2}\rangle \\
 &= \sqrt{s_1(s_1 + 1) - m_{s_1}(m_{s_1} + 1)} \hbar \sqrt{s_2(s_2 + 1) - m_{s_2}(m_{s_2} - 1)} \hbar \\
 &\quad \times |s_1, m_{s_1} + 1; s_2, m_{s_2} - 1\rangle \\
 &\quad + \sqrt{s_1(s_1 + 1) - m_{s_1}(m_{s_1} - 1)} \hbar \sqrt{s_2(s_2 + 1) - m_{s_2}(m_{s_2} + 1)} \hbar \\
 &\quad \times |s_1, m_{s_1} - 1; s_2, m_{s_2} + 1\rangle
 \end{aligned}$$

thus, $\hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z}$ does not change m_s and $\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}$ changes m_s .

As $|s_1, m_{s_1}; s_2, m_{s_2}\rangle = |s_1, m_{s_1, \max}; s_2, m_{s_2, \max}\rangle$, we obtain

$$\hat{S}_{1+}\hat{S}_{2-} |s_1, m_{s_1, \max}; s_2, m_{s_2, \max}\rangle = 0$$

$$\hat{S}_{1-}\hat{S}_{2+} \left| s_1, m_{s_1, \max}; s_2, m_{s_2, \max} \right\rangle = 0$$

Thus, $\left| s_1, m_{s_1}; s_2, m_{s_2} \right\rangle = \left| s_1, m_{s_1, \max}; s_2, m_{s_2, \max} \right\rangle$ is the common eigenstates of $\hat{S}^2$ and $\hat{S}_z$. We can let one direct sum state as

$$\left| s_{\max}, m_{s, \max} \right\rangle = \left| s_1, m_{s_1, \max}; s_2, m_{s_2, \max} \right\rangle$$

Then iteratively using the total lowering operator

$$\hat{S}_- = \hat{S}_{1-} + \hat{S}_{2-}$$

$$\hat{S}_- \left| s_{\max}, m_{s, \max} \right\rangle = (\hat{S}_{1-} + \hat{S}_{2-}) \left| s_1, m_{s_1, \max}; s_2, m_{s_2, \max} \right\rangle$$

to find the states until reaches the state with $m_{s, \min}$.

EXAMPLES:

- Express the direct sum states in terms of the tensor product state of two spin- $\frac{1}{2}$ states.

ANSWER:

Consider one tensor product state $\left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$.

Since

$$\begin{aligned} \hat{S}^2 \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle &= \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z} + (\hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+}) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &= (\hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_{1z}\hat{S}_{2z}) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &= \left(\frac{3}{4}\hbar^2 + \frac{3}{4}\hbar^2 + 2 \times \frac{\hbar}{2} \times \frac{\hbar}{2} \right) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &= 2\hbar^2 \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \end{aligned}$$

and

$$\begin{aligned} \hat{S}_z \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle &= (\hat{S}_{1z} + \hat{S}_{2z}) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &= \left(\frac{\hbar}{2} + \frac{\hbar}{2} \right) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &= \hbar \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \end{aligned}$$

$\left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$ is the common eigenstate of $\hat{S}^2$ and $\hat{S}_z$.

Thus, we let

$$|1, +1\rangle = \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

Then, using the total lowering operator $\hat{S}_- = \hat{S}_{1-} + \hat{S}_{2-}$ to find the states $|1, 0\rangle$ and $|1, -1\rangle$:

For $|1, 0\rangle$,

$$\begin{aligned}\hat{S}_-|1, +1\rangle &= (\hat{S}_{1-} + \hat{S}_{2-}) \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ &\Rightarrow \sqrt{2}\hbar|1, 0\rangle = \hbar \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle + \hbar \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ &\Rightarrow |1, 0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle\end{aligned}$$

For $|1, -1\rangle$,

$$\begin{aligned}\hat{S}_-|1, 0\rangle &= (\hat{S}_{1-} + \hat{S}_{2-}) \left(\frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \right) \\ &\Rightarrow \sqrt{2}\hbar|1, -1\rangle = \frac{1}{\sqrt{2}}\hbar \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}}\hbar \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ &= \sqrt{2}\hbar \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ &\Rightarrow |1, -1\rangle = \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle\end{aligned}$$

For the state $|0,0\rangle$, it must be composed of those states that have opposite values of m_{s_1} and m_{s_2} .

$$|0,0\rangle = \alpha \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \beta \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

Apply the orthogonality:

$$\begin{aligned}\langle 1,0|0,0\rangle &= 0 \\ &\Rightarrow \left(\frac{1}{\sqrt{2}} \left\langle \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right| + \frac{1}{\sqrt{2}} \left\langle \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right| \right) \\ &\quad \times \left(\alpha \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \beta \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \right) = 0 \\ &\Rightarrow \frac{1}{\sqrt{2}}(\alpha + \beta) = 0 \\ &\Rightarrow \alpha = -\beta\end{aligned}$$

Using the normalization condition:

$$\begin{aligned}\langle 0,0|0,0\rangle &= |\alpha|^2 + |\beta|^2 = 2|\alpha|^2 = 1 \\ &\Rightarrow \alpha = \frac{1}{\sqrt{2}}, \quad \beta = -\frac{1}{\sqrt{2}}\end{aligned}$$

Thus, we obtain

$$|0,0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle - \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

The direct sum states are

(a) $s = 1$ and $m_s = 1, 0, -1$ (called triplet state)

$$|1, +1\rangle = \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

$$|1, -1\rangle = \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

(b) $s = 0$ and $m_s = 0$ (called singlet state)

$$|0, 0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle - \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

(3) Method II: using unitary operators

Consider two general angular momentum operators $\hat{J}_1$ and $\hat{J}_2$ and their corresponding basis states

$$|j_1, m_{j_1}\rangle \text{ and } |j_2, m_{j_2}\rangle$$

The tensor product states are

$$|j_1, m_{j_1}; j_2, m_{j_2}\rangle = |j_1, m_{j_1}\rangle \otimes |j_2, m_{j_2}\rangle$$

The direct sum states are

$$|j, m_j\rangle = |j_1 + j_2, m_{j_1+j_2}\rangle \oplus \cdots \oplus |j_1 - j_2, m_{|j_1-j_2|}\rangle$$

where $j = j_1 + j_2, \dots, |j_1 - j_2|$

The expansion of direct sum states in terms of the tensor product states is

$$\begin{aligned} \underbrace{|j, m_j\rangle}_{\text{direct sum states}} &= \sum_{m_{j_1}, m_{j_2}} \underbrace{|j_1, m_{j_1}; j_2, m_{j_2}\rangle}_{\text{tensor product states}} \underbrace{\langle j_1, m_{j_1}; j_2, m_{j_2} | j, m_j \rangle}_{\text{direct sum states}} \\ &= \sum_{m_{j_1}, m_{j_2}} \langle j_1, m_{j_1}; j_2, m_{j_2} | j, m_j \rangle \underbrace{|j_1, m_{j_1}; j_2, m_{j_2}\rangle}_{\text{tensor product states}} \\ &= \sum_{m_{j_1}, m_{j_2}} c_{m_{j_1}, m_{j_2}}^{j, m_j} \underbrace{|j_1, m_{j_1}; j_2, m_{j_2}\rangle}_{\text{tensor product states}} \end{aligned}$$

The coefficients of the expansion

$$c_{m_1, m_2}^{j, m} = c_{m_{j_1}, m_{j_2}}^{j, m_j} = \left\langle j_1, m_{j_1}; j_2, m_{j_2} \left| j, m_j \right\rangle \right.$$

are called Clebsch-Gordan (C.G.) coefficients. The C.G. coefficients are the matrix elements of the unitary operator.

EXAMPLES:

- Using the table of C.G. coefficients for $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$ to find the direct sum states

$m = 1$			$m = -1$			$m = 0$		
m_1, m_2		j	m_1, m_2		j	m_1, m_2		j
		1			1			1
$\frac{1}{2}, \frac{1}{2}$		1	$-\frac{1}{2}, -\frac{1}{2}$		1	$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
						$-\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

ANSWER:

$$|j, m\rangle = \sum_{m_1, m_2} c_{m_1, m_2}^{j, m} |j_1, m_1; j_2, m_2\rangle$$

For $|1, +1\rangle$,

$$c_{\frac{1}{2}, \frac{1}{2}}^{1, 1} = 1$$

$$|1, +1\rangle = \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

For $|1, 0\rangle$,

$$c_{\frac{1}{2}, -\frac{1}{2}}^{1, 0} = \frac{1}{\sqrt{2}}, \quad c_{-\frac{1}{2}, \frac{1}{2}}^{1, 0} = \frac{1}{\sqrt{2}}$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle + \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

For $|1, -1\rangle$,

$$c_{-\frac{1}{2}, -\frac{1}{2}}^{1, -1} = 1$$

$$|1, -1\rangle = \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

For $|0, 0\rangle$,

$$c_{\frac{1}{2}, -\frac{1}{2}}^{0, 0} = \frac{1}{\sqrt{2}}, \quad c_{-\frac{1}{2}, \frac{1}{2}}^{0, 0} = -\frac{1}{\sqrt{2}}$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle - \frac{1}{\sqrt{2}} \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle$$

Matrix representation of the unitary operator

$$\begin{pmatrix} |1, +1\rangle \\ |1, 0\rangle \\ |0, 0\rangle \\ |1, -1\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ \left| \frac{1}{2}, +\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, +\frac{1}{2} \right\rangle \\ \left| \frac{1}{2}, -\frac{1}{2}; \frac{1}{2}, -\frac{1}{2} \right\rangle \end{pmatrix}$$

—Chapter 7—

Time-Independent Perturbation Theory

7-1 Nondegenerate Perturbation Theory

A. FIRST-ORDER AND SECOND-ORDER CORRECTION

- (1) If the Hamiltonian of a system can be decomposed as

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

where $\hat{H}'$ is a small potential compared to $\hat{H}_0$. We shall then use the perturbation theory to find the approximated eigenvalues and eigenstates.

First step:

Suppose that we solved the Schrödinger equation of $\hat{H}_0$ and obtained a complete set of orthonormal eigenstates and corresponding eigenvalues:

$$\hat{H}_0 |\psi_n^{(0)}\rangle = E_n^{(0)} |\psi_n^{(0)}\rangle \text{ and } \langle \psi_n^{(0)} | \psi_m^{(0)} \rangle = \delta_{nm}$$

where $|\psi_n^{(0)}\rangle$ are non-degenerate states, i.e. each eigenstate corresponds to different eigenvalue.

Second step:

Parameterize the strength of the perturbation with a dimensionless quantity λ to keep track the order of the perturbation in a power series solution (in the end set $\lambda = 1$), i.e.,

$$\hat{H} = \hat{H}_0 + \lambda \hat{H}'$$

Thus, we obtain

$$(\hat{H}_0 + \lambda \hat{H}') |\psi_n\rangle = E_n |\psi_n\rangle \cdots \cdots \text{(a)}$$

and then we write eigenvalues and eigenstates as power series in λ :

$$E_n = \underbrace{E_n^{(0)}}_{\text{zeroth order}} + \lambda \underbrace{E_n^{(1)}}_{\text{first order correction}} + \lambda^2 \underbrace{E_n^{(2)}}_{\text{second order correction}} + \cdots$$
$$|\psi_n\rangle = \underbrace{|\psi_n^{(0)}\rangle}_{\text{zeroth order}} + \lambda \underbrace{|\psi_n^{(1)}\rangle}_{\text{first order correction}} + \lambda^2 \underbrace{|\psi_n^{(2)}\rangle}_{\text{second order correction}} + \cdots$$

Third step:

Substituting these series into equation (a), we have

$$\begin{aligned}
& (\hat{H}_0 + \lambda \hat{H}') \left(\left| \psi_n^{(0)} \right\rangle + \lambda \left| \psi_n^{(1)} \right\rangle + \lambda^2 \left| \psi_n^{(2)} \right\rangle + \dots \right) \\
& = \left(E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \right) \left(\left| \psi_n^{(0)} \right\rangle + \lambda \left| \psi_n^{(1)} \right\rangle + \lambda^2 \left| \psi_n^{(2)} \right\rangle + \dots \right)
\end{aligned}$$

and collect terms of the same order of the parameter λ on each side of the equation:

$$\begin{aligned}
O(\lambda^0): & \quad (\hat{H}_0 - E_n^{(0)}) \left| \psi_n^{(0)} \right\rangle = 0 \\
O(\lambda^1): & \quad (\hat{H}_0 - E_n^{(0)}) \left| \psi_n^{(1)} \right\rangle = (E_n^{(1)} - \hat{H}') \left| \psi_n^{(0)} \right\rangle \\
O(\lambda^2): & \quad (\hat{H}_0 - E_n^{(0)}) \left| \psi_n^{(2)} \right\rangle = (E_n^{(1)} - \hat{H}') \left| \psi_n^{(1)} \right\rangle + E_n^{(2)} \left| \psi_n^{(0)} \right\rangle \\
& \quad \vdots
\end{aligned}$$

(2) The first-order correction

From the first-order equation

$$(\hat{H}_0 - E_n^{(0)}) \left| \psi_n^{(1)} \right\rangle = (E_n^{(1)} - \hat{H}') \left| \psi_n^{(0)} \right\rangle$$

Multiplying $\left\langle \psi_n^{(0)} \right|$ on both sides, we obtain

$$\left\langle \psi_n^{(0)} \right| \hat{H}_0 \left| \psi_n^{(1)} \right\rangle - E_n^{(0)} \left\langle \psi_n^{(0)} \right| \left| \psi_n^{(1)} \right\rangle = E_n^{(1)} \underbrace{\left\langle \psi_n^{(0)} \right| \left| \psi_n^{(0)} \right\rangle}_{=1} - \left\langle \psi_n^{(0)} \right| \hat{H}' \left| \psi_n^{(0)} \right\rangle$$

Since

$$\left\langle \psi_n^{(0)} \right| \hat{H}_0 = \left\langle \hat{H}_0^\dagger \psi_n^{(0)} \right| = \left\langle \hat{H}_0 \psi_n^{(0)} \right| = \left\langle \psi_n^{(0)} \right| E_n^{(0)}$$

we obtain

$$\begin{aligned}
E_n^{(0)} \left\langle \psi_n^{(0)} \right| \left| \psi_n^{(1)} \right\rangle - E_n^{(0)} \left\langle \psi_n^{(0)} \right| \left| \psi_n^{(1)} \right\rangle &= E_n^{(1)} - \left\langle \psi_n^{(0)} \right| \hat{H}' \left| \psi_n^{(0)} \right\rangle \\
\Rightarrow E_n^{(1)} &= \left\langle \psi_n^{(0)} \right| \hat{H}' \left| \psi_n^{(0)} \right\rangle
\end{aligned}$$

The first-order correction to the energy is the expectation value of the perturbation in the unperturbed state.

Suppose that

$$\left| \psi_n^{(1)} \right\rangle = \sum_m c_{mn}^{(1)} \left| \psi_m^{(0)} \right\rangle = \sum_{m=n} c_{nn}^{(1)} \left| \psi_n^{(0)} \right\rangle + \sum_{m \neq n} c_{mn}^{(1)} \left| \psi_m^{(0)} \right\rangle$$

From the power series expansions of the eigenstate

$$\left| \psi_n \right\rangle = \left| \psi_n^{(0)} \right\rangle + \left| \psi_n^{(1)} \right\rangle + \dots = \left| \psi_n^{(0)} \right\rangle + \sum_m c_{mn}^{(1)} \left| \psi_m^{(0)} \right\rangle + \dots$$

Normalize the state to determine $c_{nn}^{(1)}$

$$\begin{aligned}
\langle \psi_n | \psi_n \rangle &= \left(\langle \psi_n^{(0)} | + \sum_m c_{mn}^{(1)*} \langle \psi_m^{(0)} | + \dots \right) \left(| \psi_n^{(0)} \rangle + \sum_m c_{mn}^{(1)} | \psi_m^{(0)} \rangle + \dots \right) \\
&= \langle \psi_n^{(0)} | \psi_n^{(0)} \rangle + \sum_m c_{mn}^{(1)} \langle \psi_n^{(0)} | \psi_m^{(0)} \rangle + \sum_m c_{mn}^{(1)*} \langle \psi_m^{(0)} | \psi_n^{(0)} \rangle + \dots \\
&= 1 + \sum_m c_{mn}^{(1)} \delta_{nm} + \sum_m c_{mn}^{(1)*} \delta_{mn} + \dots \\
&= 1 + c_{nn}^{(1)} + c_{nn}^{(1)*} + \dots
\end{aligned}$$

Drop the second-order terms in the normalization

$$\langle \psi_n | \psi_n \rangle = 1 + c_{nn}^{(1)} + c_{nn}^{(1)*} = 1 + 2\text{Re}\{c_{nn}^{(1)}\} = 1 \Rightarrow c_{nn}^{(1)} = 0$$

Thus, we obtain

$$| \psi_n^{(1)} \rangle = \sum_{m \neq n} c_{mn}^{(1)} | \psi_m^{(0)} \rangle$$

We therefore have

$$| \psi_n \rangle = | \psi_n^{(0)} \rangle + \sum_{m \neq n} c_{mn}^{(1)} | \psi_m^{(0)} \rangle + \dots$$

From the first-order equation

$$(\hat{H}_0 - E_n^{(0)}) | \psi_n^{(1)} \rangle = (E_n^{(1)} - \hat{H}') | \psi_n^{(0)} \rangle$$

Multiplying $\langle \psi_m^{(0)} |$ and $m \neq n$ on both sides, we obtain

$$\langle \psi_m^{(0)} | \hat{H}_0 | \psi_n^{(1)} \rangle - E_n^{(0)} \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle = E_n^{(1)} \underbrace{\langle \psi_m^{(0)} | \psi_n^{(0)} \rangle}_{=0} - \langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle$$

Since

$$\langle \psi_m^{(0)} | \hat{H}_0 = \langle \hat{H}_0^\dagger \psi_m^{(0)} | = \langle \psi_m^{(0)} | E_m^{(0)}$$

we obtain

$$\begin{aligned}
E_m^{(0)} \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle - E_n^{(0)} \langle \psi_m^{(0)} | \psi_n^{(1)} \rangle &= - \langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle \\
\Rightarrow (E_m^{(0)} - E_n^{(0)}) c_{mn}^{(1)} &= - \langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle \\
\Rightarrow c_{mn}^{(1)} &= \frac{\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}}
\end{aligned}$$

Thus, we obtain

$$|\psi_n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

NOTE:

For the perturbation approach to be valid, the matrix elements of the perturbation Hamiltonian are smaller than the unperturbed energy *differences*, i.e.,

$$\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle < E_n^{(0)} - E_m^{(0)}$$

(3) The second-order correction

From the second-order equation

$$(\hat{H}_0 - E_n^{(0)}) |\psi_n^{(2)}\rangle = (E_n^{(1)} - \hat{H}') |\psi_n^{(1)}\rangle + E_n^{(2)} |\psi_n^{(0)}\rangle$$

Multiplying $\langle \psi_n^{(0)} |$ on both sides, we obtain

$$\begin{aligned} \langle \psi_n^{(0)} | \hat{H}_0 | \psi_n^{(2)} \rangle - E_n^{(0)} \langle \psi_n^{(0)} | \psi_n^{(2)} \rangle \\ = E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle - \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(1)} \rangle - E_n^{(2)} \underbrace{\langle \psi_n^{(0)} | \psi_n^{(0)} \rangle}_{=1} \end{aligned}$$

$$\begin{aligned} \Rightarrow E_n^{(0)} \langle \psi_n^{(0)} | \psi_n^{(2)} \rangle - E_n^{(0)} \langle \psi_n^{(0)} | \psi_n^{(2)} \rangle \\ = E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle - \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(1)} \rangle - E_n^{(2)} \\ \Rightarrow 0 = E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle - \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(1)} \rangle + E_n^{(2)} \\ \Rightarrow E_n^{(2)} = \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(1)} \rangle - E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle \end{aligned}$$

Since

$$\langle \psi_n^{(0)} | \psi_n^{(1)} \rangle = \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle}{\underbrace{E_n^{(0)} - E_m^{(0)}}_{c_{mn}^{(1)}}} \langle \psi_n^{(0)} | \psi_m^{(0)} \rangle = 0$$

we obtain

$$\begin{aligned}
E_n^{(2)} &= \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_n^{(1)} \right\rangle \\
&= \left\langle \psi_n^{(0)} \left| \hat{H}' \sum_{m \neq n} \underbrace{\frac{\left\langle \psi_m^{(0)} \left| \hat{H}' \right| \psi_n^{(0)} \right\rangle}{E_n^{(0)} - E_m^{(0)}}}_{c_{mn}^{(1)}} \right| \psi_n^{(0)} \right\rangle \\
&= \sum_{m \neq n} \frac{\left| \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_m^{(0)} \right\rangle \right|^2}{E_n^{(0)} - E_m^{(0)}}
\end{aligned}$$

OS:

$$E_n^{(2)} = \sum_{m < n} \frac{\left| \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_m^{(0)} \right\rangle \right|^2}{E_n^{(0)} - E_m^{(0)}} - \sum_{m > n} \frac{\left| \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_m^{(0)} \right\rangle \right|^2}{E_m^{(0)} - E_n^{(0)}}$$

Energy levels m that lie below the state n give a positive contribution and push the energy level n up. Energy levels m that lie above the state n give a negative contribution and push the energy level n down.

(4) Thus, we obtain

$$\begin{aligned}
E_n &= E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \dots \\
&= \left\langle \psi_n^{(0)} \left| \hat{H}_0 \right| \psi_n^{(0)} \right\rangle + \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_n^{(0)} \right\rangle + \sum_{m \neq n} \frac{\left| \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_m^{(0)} \right\rangle \right|^2}{E_n^{(0)} - E_m^{(0)}} + \dots \\
|\psi_n\rangle &= |\psi_n^{(0)}\rangle + |\psi_n^{(1)}\rangle + \dots \\
&= |\psi_n^{(0)}\rangle + \sum_{m \neq n} \frac{\left\langle \psi_m^{(0)} \left| \hat{H}' \right| \psi_n^{(0)} \right\rangle}{E_n^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle + \dots
\end{aligned}$$

B. MATRIX REPRESENTATION

- (1) Choose the unperturbed eigenstates $|\psi_n^{(0)}\rangle$ as a basis, and then obtain the matrix form of the Schrödinger equation

$$\begin{pmatrix} E_1^{(0)} + a_{11} & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & E_2^{(0)} + a_{22} & a_{23} & a_{24} & \cdots \\ a_{31} & a_{32} & E_3^{(0)} + a_{33} & a_{34} & \cdots \\ a_{41} & a_{42} & a_{43} & E_4^{(0)} + a_{44} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix} = E \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix}$$

where $a_{nm} = \langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle$

(2) Since

$$E_n = \langle \psi_n^{(0)} | \hat{H}_0 | \psi_n^{(0)} \rangle + \langle \psi_n^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle + \sum_{m \neq n} \frac{|\langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} + \dots$$

The zeroth-order energy is a diagonal matrix of the unperturbed Hamiltonian:

$$\mathbf{H}_0 = \begin{pmatrix} E_1^{(0)} & 0 & 0 & 0 & \cdots \\ 0 & E_2^{(0)} & 0 & 0 & \cdots \\ 0 & 0 & E_3^{(0)} & 0 & \cdots \\ 0 & 0 & 0 & E_4^{(0)} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

The first-order correction to the energy is the diagonal elements of $\mathbf{H}'$:

$$\mathbf{H}' = \begin{pmatrix} a_{11} & 0 & 0 & 0 & \cdots \\ 0 & a_{22} & 0 & 0 & \cdots \\ 0 & 0 & a_{33} & 0 & \cdots \\ 0 & 0 & 0 & a_{44} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} + \begin{pmatrix} 0 & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & 0 & a_{23} & a_{24} & \cdots \\ a_{31} & a_{32} & 0 & a_{34} & \cdots \\ a_{41} & a_{42} & a_{43} & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$E_1^{(1)} = a_{11}$$

$$E_2^{(1)} = a_{22}$$

$\vdots$

The second-order correction to the energy is the sum of off-diagonal elements of $\mathbf{H}'$:

$$E_1^{(2)} = \frac{|a_{12}|^2}{E_1^{(0)} - E_2^{(0)}} + \frac{|a_{13}|^2}{E_1^{(0)} - E_3^{(0)}} + \dots$$

$$E_2^{(2)} = \frac{|a_{21}|^2}{E_2^{(0)} - E_1^{(0)}} + \frac{|a_{23}|^2}{E_2^{(0)} - E_3^{(0)}} + \dots$$

$\vdots$

C. EXAMPLES

- (1) A sodium nucleus has spin $\frac{3}{2}$ is placed in a constant magnetic field in the z -direction $\vec{B} = B_0 \vec{z}$

1. The zeroth-order Hamiltonian

$$\mathbf{H}_0 = -\boldsymbol{\mu} \cdot \mathbf{B} = \omega_0 \mathbf{S}_z = \begin{pmatrix} \frac{3}{2}\hbar\omega_0 & 0 & 0 & 0 \\ 0 & \frac{1}{2}\hbar\omega_0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2}\hbar\omega_0 & 0 \\ 0 & 0 & 0 & -\frac{3}{2}\hbar\omega_0 \end{pmatrix}$$

where the Larmor frequency $\omega_0 = eB_0/2m$

2. If a perturbative magnetic field $\vec{B}' = B_1 \vec{z}$ is applied to the system, the perturbed Hamiltonian.

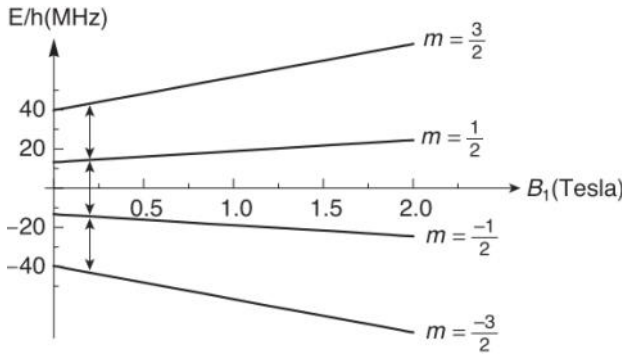
$$\mathbf{H}' = -\boldsymbol{\mu} \cdot \mathbf{B}' = \omega_1 \mathbf{S}_z = \begin{pmatrix} \frac{3}{2}\hbar\omega_1 & 0 & 0 & 0 \\ 0 & \frac{1}{2}\hbar\omega_1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2}\hbar\omega_1 & 0 \\ 0 & 0 & 0 & -\frac{3}{2}\hbar\omega_1 \end{pmatrix}$$

where the Larmor frequency $\omega_1 = eB_1/2m$

The energy shift due to the first-order perturbation:

$$\begin{aligned} E_1 &= E_1^{(0)} + E_1^{(1)} = \frac{3}{2}\hbar\omega_0 + \frac{3}{2}\hbar\omega_1 \\ E_2 &= E_2^{(0)} + E_2^{(1)} = \frac{1}{2}\hbar\omega_0 + \frac{1}{2}\hbar\omega_1 \\ E_3 &= E_3^{(0)} + E_3^{(1)} = -\frac{1}{2}\hbar\omega_0 - \frac{1}{2}\hbar\omega_1 \\ E_4 &= E_4^{(0)} + E_4^{(1)} = -\frac{3}{2}\hbar\omega_0 - \frac{3}{2}\hbar\omega_1 \end{aligned}$$

The new energies exhibit the linear dependence.



3. If a perturbative magnetic field $\vec{B}'' = B_2 \vec{x}$ is applied to the system, the perturbed Hamiltonian.
- Method I:

$$\mathbf{H}'' = -\vec{\mu} \cdot \mathbf{B}'' = \omega_2 \mathbf{S}_x = \begin{pmatrix} 0 & \frac{\sqrt{3}}{2} \hbar \omega_2 & 0 & 0 \\ \frac{\sqrt{3}}{2} \hbar \omega_2 & 0 & \hbar \omega_2 & 0 \\ 0 & \hbar \omega_2 & 0 & \frac{\sqrt{3}}{2} \hbar \omega_2 \\ 0 & 0 & \frac{\sqrt{3}}{2} \hbar \omega_2 & 0 \end{pmatrix}$$

There are no first-order energy corrections. The second-order energy corrections are

$$\begin{aligned} E_1^{(2)} &= \frac{\left| \langle \psi_1^{(0)} | \hat{H}'' | \psi_2^{(0)} \rangle \right|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{\left| \frac{\sqrt{3}}{2} \hbar \omega_2 \right|^2}{\frac{3}{2} \hbar \omega_0 - \frac{1}{2} \hbar \omega_0} = \frac{3}{4} \frac{\omega_2}{\omega_0} \hbar \omega_2 \\ E_2^{(2)} &= \frac{\left| \langle \psi_2^{(0)} | \hat{H}'' | \psi_1^{(0)} \rangle \right|^2}{E_2^{(0)} - E_1^{(0)}} + \frac{\left| \langle \psi_2^{(0)} | \hat{H}'' | \psi_3^{(0)} \rangle \right|^2}{E_2^{(0)} - E_3^{(0)}} = \frac{1}{4} \frac{\omega_2}{\omega_0} \hbar \omega_2 \\ E_3^{(2)} &= \frac{\left| \langle \psi_3^{(0)} | \hat{H}'' | \psi_2^{(0)} \rangle \right|^2}{E_3^{(0)} - E_2^{(0)}} + \frac{\left| \langle \psi_3^{(0)} | \hat{H}'' | \psi_4^{(0)} \rangle \right|^2}{E_3^{(0)} - E_4^{(0)}} = -\frac{1}{4} \frac{\omega_2}{\omega_0} \hbar \omega_2 \\ E_4^{(2)} &= \frac{\left| \langle \psi_4^{(0)} | \hat{H}'' | \psi_3^{(0)} \rangle \right|^2}{E_4^{(0)} - E_3^{(0)}} = -\frac{3}{4} \frac{\omega_2}{\omega_0} \hbar \omega_2 \end{aligned}$$

The energy shift due to the second-order perturbation:

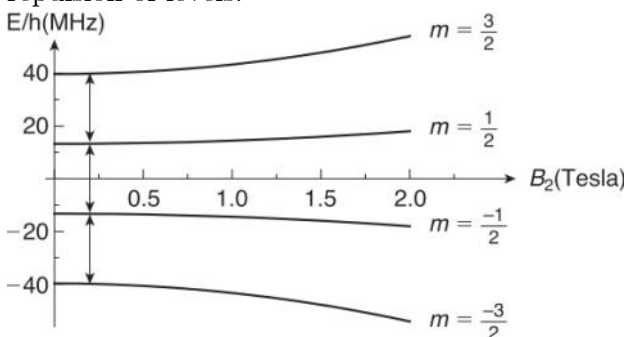
$$E_1 = \frac{3}{2}\hbar\omega_0 + \frac{3}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2$$

$$E_2 = \frac{1}{2}\hbar\omega_0 + \frac{1}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2$$

$$E_3 = -\frac{1}{2}\hbar\omega_0 - \frac{1}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2$$

$$E_4 = -\frac{3}{2}\hbar\omega_0 - \frac{3}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2$$

The quadratic dependence of the corrected energies illustrate the repulsion of levels.



- Method II:

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{H}'' = \begin{pmatrix} \frac{3}{2}\hbar\omega_0 & \frac{\sqrt{3}}{2}\hbar\omega_2 & 0 & 0 \\ \frac{\sqrt{3}}{2}\hbar\omega_2 & \frac{1}{2}\hbar\omega_0 & \hbar\omega_2 & 0 \\ 0 & \hbar\omega_2 & -\frac{1}{2}\hbar\omega_0 & \frac{\sqrt{3}}{2}\hbar\omega_2 \\ 0 & 0 & \frac{\sqrt{3}}{2}\hbar\omega_2 & -\frac{3}{2}\hbar\omega_0 \end{pmatrix}$$

The eigenvalues are

$$\lambda = \pm \frac{3}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2}, \quad \pm \frac{1}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2}$$

As $\omega_0 \ll \omega_2$

$$\sqrt{\omega_0^2 + \omega_2^2} \approx \omega_0 \left(1 + \frac{1}{2}\frac{\omega_2^2}{\omega_0^2} - \dots \right) = \omega_0 + \frac{1}{2}\frac{\omega_2}{\omega_0}\omega_2$$

$$\lambda_1 = \frac{3}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2} = \frac{3}{2}\hbar\omega_0 + \frac{3}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2$$

$$\begin{aligned}\lambda_2 &= \frac{1}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2} = \frac{1}{2}\hbar\omega_0 + \frac{1}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2 \\ \lambda_3 &= -\frac{1}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2} = -\frac{1}{2}\hbar\omega_0 - \frac{1}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2 \\ \lambda_4 &= -\frac{3}{2}\hbar\sqrt{\omega_0^2 + \omega_2^2} = -\frac{3}{2}\hbar\omega_0 - \frac{3}{4}\frac{\omega_2}{\omega_0}\hbar\omega_2\end{aligned}$$

(2) Considering a harmonic oscillator in an applied electric field

The Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - e\hat{x}\mathcal{E} = \hat{H}_0 + \hat{H}'$$

where $\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \Rightarrow \hat{H}_0|\psi_n^{(0)}\rangle = \hbar\omega\left(n + \frac{1}{2}\right)|\psi_n^{(0)}\rangle$

$$\hat{H}' = -e\hat{x}\mathcal{E}$$

$\mathcal{E}$ is the electric field coupled to a dipole operator $e\hat{x}$.

The induced dipole moment of the polarization of a harmonic oscillator is given by $d = \langle\psi|e\hat{x}|\psi\rangle$, where $|\psi\rangle$ is the eigenstate of $\hat{H}$.

The eigenstate up to the first order correction

$$|\psi_n\rangle = |\psi_n^{(0)}\rangle + \sum_{m \neq n} \frac{\langle\psi_m^{(0)}|\hat{H}'|\psi_n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

where $\langle\psi_m^{(0)}|\hat{H}'|\psi_n^{(0)}\rangle \ll |E_n^{(0)} - E_m^{(0)}|$

The dipole moment up to the linear order is

$$\begin{aligned}
d &= \langle \psi | e\hat{x} | \psi \rangle \\
&= \left(\langle \psi_n^{(0)} | + \sum_{m \neq n} \frac{\langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \langle \psi_m^{(0)} | \right) e\hat{x} \\
&\quad \times \left(| \psi_n^{(0)} \rangle + \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} | \psi_m^{(0)} \rangle \right) \\
&= \langle \psi_n^{(0)} | e\hat{x} | \psi_n^{(0)} \rangle + \sum_{m \neq n} \frac{\langle \psi_m^{(0)} | -e\hat{x}\mathcal{E} | \psi_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \langle \psi_n^{(0)} | e\hat{x} | \psi_m^{(0)} \rangle \\
&\quad + \sum_{m \neq n} \frac{\langle \psi_n^{(0)} | -e\hat{x}\mathcal{E} | \psi_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \langle \psi_m^{(0)} | e\hat{x} | \psi_n^{(0)} \rangle + \dots \\
&= \langle \psi_n^{(0)} | e\hat{x} | \psi_n^{(0)} \rangle - 2\mathcal{E} \sum_{m \neq n} \frac{\langle \psi_n^{(0)} | e\hat{x} | \psi_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \langle \psi_m^{(0)} | e\hat{x} | \psi_n^{(0)} \rangle
\end{aligned}$$

Since $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a})$

we obtain

$$\begin{aligned}
\langle \psi_n^{(0)} | \hat{x} | \psi_n^{(0)} \rangle &= \sqrt{\frac{\hbar}{2m\omega}} \langle \psi_n^{(0)} | \hat{a}^\dagger + \hat{a} | \psi_n^{(0)} \rangle = 0 \\
\Rightarrow d &= -2\mathcal{E} \sum_{m \neq n} \frac{\langle \psi_n^{(0)} | e\hat{x} | \psi_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \langle \psi_m^{(0)} | e\hat{x} | \psi_n^{(0)} \rangle
\end{aligned}$$

For the ground state, $n = 0, E_0^{(0)} - E_m^{(0)} = -\hbar\omega$

$$d = \frac{2e^2\mathcal{E}}{\hbar\omega} \sum_{m=0} \langle \psi_0^{(0)} | \hat{x} | \psi_m^{(0)} \rangle \langle \psi_m^{(0)} | \hat{x} | \psi_0^{(0)} \rangle = \frac{2e^2\mathcal{E}}{\hbar\omega} \langle \psi_0^{(0)} | \hat{x}^2 | \psi_0^{(0)} \rangle$$

Since

$$\begin{aligned}
\hat{x}^2 &= \frac{\hbar}{2m\omega} (\hat{a}^\dagger \hat{a}^\dagger + \hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}) \\
&= \frac{\hbar}{2m\omega} (\hat{a}^\dagger \hat{a}^\dagger + \hat{a} \hat{a} + 2\hat{a}^\dagger \hat{a} + 1)
\end{aligned}$$

we obtain

$$\langle \psi_0^{(0)} | \hat{x}^2 | \psi_0^{(0)} \rangle = \frac{\hbar}{2m\omega} \langle \psi_0^{(0)} | \hat{a}^\dagger \hat{a}^\dagger + \hat{a} \hat{a} + 2\hat{a}^\dagger \hat{a} + 1 | \psi_0^{(0)} \rangle = \frac{\hbar}{2m\omega}$$

$$d = \frac{2e^2\mathcal{E}}{\hbar\omega} \times \frac{\hbar}{2m\omega} = \frac{e^2\mathcal{E}}{m\omega^2}$$

7-2 Degenerate Perturbation Theory

A. DEGENERATE STATES

- (1) Degenerate states: few independent eigenstates share the same eigenvalue

$$\hat{H}_0 \left| \psi_m^{(0)} \right\rangle = E_m^{(0)} \left| \psi_m^{(0)} \right\rangle, \quad m \in [1, l]$$

where $E_1^{(0)} = \dots = E_l^{(0)}$ and $\left| \psi_1^{(0)} \right\rangle \neq \dots \neq \left| \psi_l^{(0)} \right\rangle$

It is a basic rule in quantum mechanics that degeneracies do not occur unless there is some symmetry to make them happen.

1. Accidental degeneracy due to the accidental values of some parameters .
2. Natural degeneracy due to the symmetries.

From the non-degenerate perturbation theory, we obtain

$$\begin{aligned} E_n &= E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \dots \\ &= E_n^{(0)} + \left\langle \psi_n^{(0)} \right| \hat{H}' \left| \psi_n^{(0)} \right\rangle + \sum_{m \neq n} \frac{\left| \left\langle \psi_n^{(0)} \right| \hat{H}' \left| \psi_m^{(0)} \right\rangle \right|^2}{E_n^{(0)} - E_m^{(0)}} + \dots \end{aligned}$$

The denominator in the above sum can vanish if m can be found such that

$$E_n^{(0)} = E_m^{(0)}$$

In this case, the calculation fails, and we must develop degenerate perturbation theory.

- (2) Assume that l energy states are degenerate, i.e., $E_1^{(0)} = \dots = E_l^{(0)}$ and $c_1 \neq \dots \neq c_l$. The matrix form of the Schrödinger equation on the basis $\left\{ \left| \psi_n^{(0)} \right\rangle \right\}$ is

$$\begin{pmatrix} E_1^{(0)} + a_{11} & a_{12} & a_{13} & a_{14} & \dots \\ a_{21} & E_1^{(0)} + a_{22} & a_{23} & a_{24} & \dots \\ a_{31} & a_{32} & E_3^{(0)} + a_{33} & a_{34} & \dots \\ a_{41} & a_{42} & a_{43} & E_4^{(0)} + a_{44} & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix} = E \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix}$$

where $a_{nm} = \left\langle \psi_n^{(0)} \left| \hat{H}' \right| \psi_m^{(0)} \right\rangle$

For l degenerate energy states, the Schrödinger equation can be expressed as

$$\underbrace{E_n^{(0)}}_{\text{blue}} c_n + \sum_{m=1}^l \underbrace{a_{nm}}_{\text{red}} c_m + \sum_{j \neq 1 \cdots l} \underbrace{a_{nj}}_{\text{green}} c_j = E_n c_n$$

For $j \neq 1 \cdots l$, we can substitute c_j with the first-order state correction and we obtain

$$E_n^{(0)} c_n + \sum_{m=1}^l a_{nm} c_m + \sum_{j \neq 1 \cdots l} a_{nj} \sum_{i \neq j} \frac{a_{ij}}{E_j^{(0)} - E_i^{(0)}} = E_n c_n$$

Considering the leading order in a_{nm} , i.e., the only significant a_{nm} of l degenerate states are those with $m = 1, \dots, l$. Thus, we obtain

$$E_n^{(0)} c_n + \sum_{m=1}^l a_{nm} c_m = E_n c_n$$

The above matrix can be approximated as a block diagonal matrix:

$$\begin{pmatrix} E_1^{(0)} + a_{11} & a_{12} & a_{13} & a_{14} & \cdots \\ a_{21} & E_1^{(0)} + a_{22} & a_{23} & a_{24} & \cdots \\ a_{31} & a_{32} & E_3^{(0)} + a_{33} & a_{34} & \cdots \\ a_{41} & a_{42} & a_{43} & E_4^{(0)} + a_{44} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix} = E \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix}$$

The eigenvalues of degenerate states up to the first-order are obtained by solving the subset of equations:

$$\begin{pmatrix} E_1^{(0)} + a_{11} & a_{12} \\ a_{21} & E_1^{(0)} + a_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

or solving the secular equation:

$$\begin{vmatrix} E_1^{(0)} + a_{11} - E & a_{12} \\ a_{21} & E_1^{(0)} + a_{22} - E \end{vmatrix} = 0$$

B. DEGENERATE SUBSPACE

- (1) We adopt the first-order equation [c.f.7-1-A]

$$\left(\hat{H}_0 - E_m^{(0)} \right) \left| \psi_m^{(1)} \right\rangle = \left(E_m^{(1)} - \hat{H}' \right) \left| \psi_m^{(0)} \right\rangle$$

Multiplying $\left\langle \psi_n^{(0)} \right|$ on both sides, we obtain

$$\begin{aligned} \langle \psi_n^{(0)} | \hat{H}_0 - E_m^{(0)} | \psi_m^{(1)} \rangle &= \langle \psi_n^{(0)} | E_m^{(1)} - \hat{H}' | \psi_m^{(0)} \rangle \\ \Rightarrow E_n^{(0)} \langle \psi_n^{(0)} | \psi_m^{(1)} \rangle - E_m^{(0)} \langle \psi_n^{(0)} | \psi_m^{(1)} \rangle &= E_m^{(1)} \langle \psi_n^{(0)} | \psi_m^{(0)} \rangle - \langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle \end{aligned}$$

Since $E_n^{(0)} = E_m^{(0)}$ as $n, m \in [1, l]$, we obtain

$$\langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle = E_m^{(1)} \langle \psi_n^{(0)} | \psi_m^{(0)} \rangle$$

- (2) Since $\langle \psi_n^{(0)} | \psi_m^{(0)} \rangle = 0$ for the distinct states $|\psi_n^{(0)}\rangle$ and $|\psi_m^{(0)}\rangle$, we obtain
- $$\langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle = 0$$

Thus, the ordinary perturbation theory fails. We write the unperturbed states in generic form:

$$|\psi_m^{(0)}\rangle \rightarrow \sum_m c_m |\psi_m^{(0)}\rangle$$

Any linear combination of the degenerate states is still an eigenstate of $\hat{H}_0$ with the same eigenvalue $E_j^{(0)}$:

$$\hat{H}_0 \sum_m c_m |\psi_m^{(0)}\rangle = E_m^{(0)} \sum_m c_m |\psi_m^{(0)}\rangle$$

- (3) Thus, we obtain

$$\begin{aligned} \langle \psi_n^{(0)} | \hat{H}' \sum_m c_m |\psi_m^{(0)}\rangle &= \langle \psi_n^{(0)} | \sum_m E_m^{(1)} c_m |\psi_m^{(0)}\rangle \\ \sum_m \langle \psi_n^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle c_m &= E_n^{(1)} c_n \Rightarrow \sum_m a_{nm} c_m = E_n^{(1)} c_n \end{aligned}$$

The first-order equation now exhibits a block diagonal matrix as

$$\begin{pmatrix} a_{11} & a_{12} & 0 & 0 & \cdots \\ a_{21} & a_{22} & 0 & 0 & \cdots \\ 0 & 0 & a_{33} & a_{34} & \cdots \\ 0 & 0 & a_{43} & a_{44} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix} = E^{(1)} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \end{pmatrix} \text{ where } E_1^{(0)} = E_2^{(0)}$$

The matrix corresponding to the m degenerate states are referred to as the degenerate submatrix. Diagonalize the degenerate submatrix by the secular equation:

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0$$

The eigenvalues of the submatrix are the first-order correction to the

energy:

$$E_1^{(1)} = \lambda_1$$

$$E_2^{(1)} = \lambda_2$$

OS:

Schmidt orthogonalization method

Let a degenerate set of eigenstates be given by $|\psi_1^{(0)}\rangle, |\psi_2^{(0)}\rangle, \dots$.

1. Normalize $|\psi_1^{(0)}\rangle$ and let it be the first state of the new set $|\tilde{\psi}_1^{(0)}\rangle$.

2. Define the inner product

$$\langle \tilde{\psi}_1^{(0)} | \psi_2^{(0)} \rangle = a_{12}$$

Choose

$$|\tilde{\psi}_2^{(0)}\rangle = |\psi_2^{(0)}\rangle - a_{12} |\tilde{\psi}_1^{(0)}\rangle$$

Test orthogonality

$$\begin{aligned} \langle \tilde{\psi}_1^{(0)} | \tilde{\psi}_2^{(0)} \rangle &= \langle \tilde{\psi}_1^{(0)} | \psi_2^{(0)} \rangle - a_{12} \langle \tilde{\psi}_1^{(0)} | \tilde{\psi}_1^{(0)} \rangle \\ &= a_{12} - a_{12} \\ &= 0 \end{aligned}$$

Normalize

$$\langle \tilde{\psi}_2^{(0)} | \tilde{\psi}_2^{(0)} \rangle = 1$$

3. Define the inner product

$$\langle \tilde{\psi}_1^{(0)} | \psi_3^{(0)} \rangle = a_{13}$$

$$\langle \tilde{\psi}_2^{(0)} | \psi_3^{(0)} \rangle = a_{23}$$

Choose

$$|\tilde{\psi}_3^{(0)}\rangle = |\psi_3^{(0)}\rangle - a_{13} |\tilde{\psi}_1^{(0)}\rangle - a_{23} |\tilde{\psi}_2^{(0)}\rangle$$

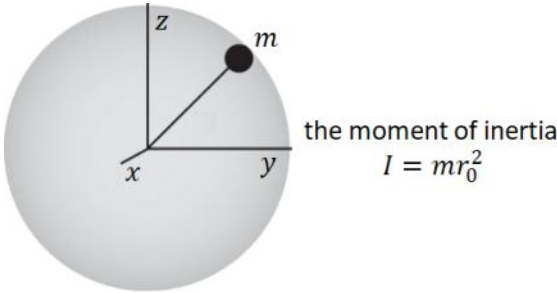
Test orthogonality

$$\begin{aligned} \langle \tilde{\psi}_1^{(0)} | \tilde{\psi}_3^{(0)} \rangle &= \langle \tilde{\psi}_1^{(0)} | \psi_3^{(0)} \rangle - a_{13} \langle \tilde{\psi}_1^{(0)} | \tilde{\psi}_1^{(0)} \rangle - a_{23} \langle \tilde{\psi}_1^{(0)} | \tilde{\psi}_2^{(0)} \rangle \\ &= a_{13} - a_{13} - a_{23} \times 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \langle \tilde{\psi}_2^{(0)} | \tilde{\psi}_3^{(0)} \rangle &= \langle \tilde{\psi}_2^{(0)} | \psi_3^{(0)} \rangle - a_{13} \langle \tilde{\psi}_2^{(0)} | \tilde{\psi}_1^{(0)} \rangle - a_{23} \langle \tilde{\psi}_2^{(0)} | \tilde{\psi}_2^{(0)} \rangle \\ &= a_{23} - a_{13} \times 0 - a_{23} \\ &= 0 \end{aligned}$$

EXAMPLES:

1. An electron is bound to move on the surface of a sphere and a small external magnetic field $\vec{B}' = B_1 \vec{x}$ is applied to the system.



The Hamiltonian of the system is

$$\hat{H} = \frac{\hat{L}^2}{2I} - \hat{\mu} \cdot \vec{B}' = \underbrace{\frac{\hat{L}^2}{2I}}_{\hat{H}_0} - \underbrace{\hat{\mu} \cdot B_1 \vec{x}}_{\hat{H}'}$$

Since

$$\hat{\mu} = -\frac{e}{2m} \hat{L} \quad [\text{c.f.6-2-B}]$$

we obtain

$$\hat{H} = \frac{\hat{L}^2}{2I} + \omega_1 \hat{L}_x$$

where the Larmor frequency is $\omega_1 = eB_1/2m$

Now, choose the basis such that

$$\hat{H}_0 |\ell, m_\ell\rangle = \frac{\hat{L}^2}{2I} |\ell, m_\ell\rangle = E_\ell^{(0)} |\ell, m_\ell\rangle, \quad E_\ell^{(0)} = \frac{\ell(\ell+1)\hbar^2}{2I}$$

$$|\ell, m_\ell\rangle = \begin{cases} |0, 0\rangle \\ |1, 1\rangle \\ |1, 0\rangle \\ |1, -1\rangle \\ \vdots \end{cases}$$

The matrix form of the Hamiltonian is

$$\mathbf{H} = \begin{matrix} & \backslash & |0, 0\rangle & |1, 1\rangle & |1, 0\rangle & |1, -1\rangle & \cdots & \cdots \\ \begin{matrix} \langle 0, 0| \\ \langle 1, 1| \\ \langle 1, 0| \\ \langle 1, -1| \\ \vdots \\ \vdots \end{matrix} & \left(\begin{array}{cccccc} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \hbar^2/I & \hbar\omega_1/\sqrt{2} & 0 & 0 & 0 \\ 0 & \hbar\omega_1/\sqrt{2} & \hbar^2/I & \hbar\omega_1/\sqrt{2} & 0 & 0 \\ 0 & 0 & \hbar\omega_1/\sqrt{2} & \hbar^2/I & 0 & 0 \\ 0 & 0 & 0 & 0 & \ddots & \cdots \\ 0 & 0 & 0 & 0 & \vdots & \ddots \end{array} \right) \end{matrix}$$

- (a) For $\ell = 0$, there is no energy correction.

(b) For $\ell = 1$:

$$\mathbf{H}_0 = \begin{array}{c} \backslash \quad |1,1\rangle \quad |1,0\rangle \quad |1,-1\rangle \\ \begin{array}{l} \langle 1, \quad 1| \\ \langle 1, \quad 0| \\ \langle 1, -1| \end{array} \end{array} \begin{pmatrix} \hbar^2/I & 0 & 0 \\ 0 & \hbar^2/I & 0 \\ 0 & 0 & \hbar^2/I \end{pmatrix}$$

$\Rightarrow$ There are 3 degenerate states for $\ell = 1$ ($m_\ell = 1, 0, -1$).

$$\mathbf{H}' = \omega_1 \mathbf{L}_x = \begin{array}{c} \backslash \quad |1,1\rangle \quad |1,0\rangle \quad |1,-1\rangle \\ \begin{array}{l} \langle 1, \quad 1| \\ \langle 1, \quad 0| \\ \langle 1, -1| \end{array} \end{array} \begin{pmatrix} 0 & \hbar\omega_1/\sqrt{2} & 0 \\ \hbar\omega_1/\sqrt{2} & 0 & \hbar\omega_1/\sqrt{2} \\ 0 & \hbar\omega_1/\sqrt{2} & 0 \end{pmatrix}$$

The secular equation is

$$\begin{vmatrix} -\lambda & \hbar\omega_1/\sqrt{2} & 0 \\ \hbar\omega_1/\sqrt{2} & -\lambda & \hbar\omega_1/\sqrt{2} \\ 0 & \hbar\omega_1/\sqrt{2} & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-\lambda) \left(\lambda^2 - \frac{\hbar^2 \omega_1^2}{2} \right) - \frac{\hbar\omega_1}{\sqrt{2}} \left(-\lambda \frac{\hbar\omega_1}{\sqrt{2}} \right) = 0$$

$$\Rightarrow -\lambda^3 + 2\lambda \frac{\hbar^2 \omega_1^2}{2} = 0$$

$$\Rightarrow \lambda = 0, \pm \hbar\omega_1$$

$$E_1^{(1a)} = \hbar\omega_1$$

$$E_1^{(1b)} = 0$$

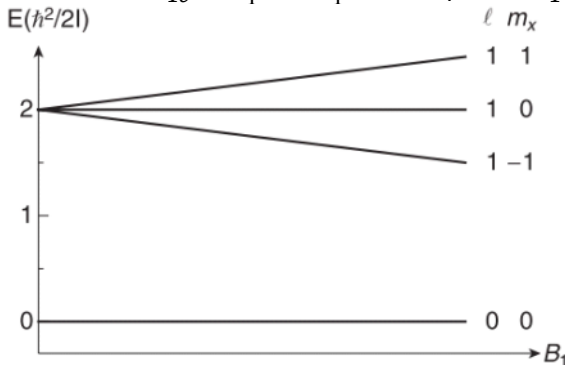
$$E_1^{(1c)} = -\hbar\omega_1$$

The perturbed energies are

$$E_{1a} = E_1^{(0)} + E_1^{(1a)} = \hbar^2/I + \hbar\omega_1$$

$$E_{1b} = E_1^{(0)} + E_1^{(1b)} = \hbar^2/I$$

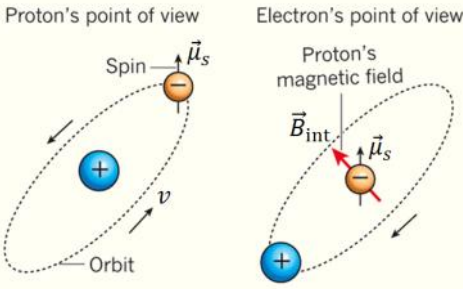
$$E_{1c} = E_1^{(0)} + E_1^{(1c)} = \hbar^2/I - \hbar\omega_1$$



7-3 Spin-Orbit Coupling

A. SPIN-ORBIT COUPLING IN AN ATOM

(1) Spin-orbit coupling



An electron moving in an electric field sees an effective magnetic field

$$\vec{B}_{\text{eff}} = \frac{\mu_0}{4\pi} \frac{I \vec{\ell} \times \vec{r}}{r^3} = -\frac{\mu_0}{4\pi} \frac{Ze \vec{v} \times \vec{r}}{r^3} = -\frac{\vec{v}}{c^2} \times \frac{Ze}{4\pi\epsilon_0} \frac{\vec{r}}{r^3} = \frac{\vec{E} \times \vec{v}}{c^2} = \frac{\vec{E} \times \vec{p}}{mc^2}$$

The relativistic correction with Llewellyn Thomas precession:

$$\vec{B}_{\text{eff}} = \frac{\vec{E} \times \vec{p}}{2mc^2}$$

The interaction between an electron spin magnetic moment and the induced magnetic field is

$$U = -\vec{\mu}_s \cdot \vec{B}_{\text{eff}}$$

Physically it can be understood as the Zeeman coupling between the magnetic moment carried by electron spin and the effective magnetic field seen by the electron.

Replace the spin magnetic moment with a spin operator

$$\hat{\mu}_s = -g_s \mu_B \frac{\hat{S}}{\hbar} = -g_s \frac{e\hbar}{2m} \frac{\hat{S}}{\hbar} = -g_s \frac{e}{2m} \hat{S}$$

We then obtain the Hamiltonian:

$$\hat{H}_{\text{SO}} = -\hat{\mu}_s \cdot \vec{B}_{\text{eff}} = -\left(-g_s \frac{e}{2m} \hat{S}\right) \cdot \frac{\vec{E} \times \hat{p}}{2mc^2} = g_s \frac{e}{4m^2 c^2} \hat{S} \cdot \vec{E} \times \hat{p}$$

Since

$$\vec{E} = \frac{\vec{F}}{-e} = \frac{\nabla U(\vec{r})}{e}$$

we obtain

$$\hat{H}_{\text{SO}} = \frac{g_s}{4m^2 c^2} \hat{S} \cdot \nabla U(\vec{r}) \times \hat{p}$$

- (2) If we take a central field, $U(\vec{r}) = U(r)\vec{r}$, such that

$$\nabla U(\vec{r}) = \frac{dU(r)}{dr} \frac{\vec{r}}{r}$$

we have

$$\hat{H}_{\text{LS}} = \frac{g_s}{4m^2c^2} \hat{S} \cdot \left[\frac{1}{r} \frac{dU(r)}{dr} \hat{r} \times \hat{p} \right] = \frac{g_s}{4m^2c^2} \frac{1}{r} \frac{dU(r)}{dr} \hat{S} \cdot \hat{L}$$

For hydrogen or hydrogen-like atoms, the electrostatic potential is spherically symmetric, i.e.,

$$U(r) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$$

Thus, we obtain

$$\hat{H}_{\text{LS}} = \frac{g_s}{4m^2c^2} \frac{e^2}{4\pi\epsilon_0} \frac{1}{r^3} \hat{L} \cdot \hat{S}$$

For electron spin, we have the gyromagnetic ratio $g_s = 2$.

$$\hat{H}_{\text{LS}} = \frac{1}{2m^2c^2} \frac{e^2}{4\pi\epsilon_0} \frac{1}{r^3} \hat{L} \cdot \hat{S} = f(r) \hat{L} \cdot \hat{S}$$

where $f(r) = \frac{1}{2m^2c^2} \frac{e^2}{4\pi\epsilon_0} \frac{1}{r^3}$

- (3) Consider the case where the spin-orbit coupling are much smaller than the Coulomb interaction. It is more appropriate to treat the spin-orbit coupling as a perturbation.

$$\hat{H} = \underbrace{\frac{\hat{p}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r}}_{\hat{H}_0} + \underbrace{f(r) \hat{L} \cdot \hat{S}}_{\hat{H}'}$$

The zeroth-order energies are Bohr energy levels.

$$E_n^{(0)} = -\frac{1}{2}m \left(\frac{e^2}{4\pi\epsilon_0 \hbar} \right)^2 \frac{1}{n^2} = -\frac{13.6}{n^2} = -\epsilon_n$$

B. TENSOR PRODUCT STATES (UNCOUPLED STATES)

- (1) Mathematical concepts of spin-orbit coupling

Hilbert subspace of a hydrogen atom $\mathbb{U}_{n,\ell}$:

$$\mathbb{H} = \mathbb{U}_{1,0} \cup \mathbb{U}_{2,1} \cup \mathbb{U}_{2,0} \cup \mathbb{U}_{3,2} \cup \mathbb{U}_{3,1} \cup \mathbb{U}_{3,0} \cdots$$

A vector space of a spin- $\frac{1}{2}$ particle $\mathbb{V}_{1/2}$:

The Hilbert space is

$$\mathbb{W} = \mathbb{H} + \mathbb{V}_{1/2}$$

EXAMPLES:

1. The space of $n = 2$ and $s = \frac{1}{2}$:

$$\mathbb{W} = (\mathbb{U}_{2,1} \cup \mathbb{U}_{2,0}) + \mathbb{V}_{1/2}$$

The number of basis states: $(3 + 1) \times 2 = 8$.

- (2) Choose the tensor product states $|\ell, m_\ell; s, m_s\rangle = |\ell, m_\ell\rangle \otimes |s, m_s\rangle$ as the basis

$$\begin{array}{ccc} \underbrace{\left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle}_{|1\rangle} & \underbrace{\left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle}_{|2\rangle} & \underbrace{\left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle}_{|3\rangle} \\ \underbrace{\left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|4\rangle} & \underbrace{\left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|5\rangle} & \underbrace{\left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|6\rangle} \\ \underbrace{\left| 0, 0; \frac{1}{2}, \frac{1}{2} \right\rangle}_{|7\rangle} & \underbrace{\left| 0, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|8\rangle} & \end{array}$$

The matrix representation of $\hat{H}_0$

$$\mathbf{H}_0 = \begin{array}{c} \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \\ \langle 7| \\ \langle 8| \end{array} \begin{array}{c} \backslash \\ \begin{pmatrix} -\varepsilon_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\varepsilon_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\varepsilon_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\varepsilon_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\varepsilon_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\varepsilon_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\varepsilon_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\varepsilon_2 \end{pmatrix} \end{array} \end{array}$$

where

$$\varepsilon_2 = \frac{13.6}{2^2} = 3.4 \text{ eV}$$

There are 8 degenerate states.

- (3) Since

$$\begin{aligned} \hat{L} \cdot \hat{S} &= \frac{1}{2} (\hat{L}_+ \hat{S}_- + \hat{L}_- \hat{S}_+) + \hat{L}_z \hat{S}_z \\ \hat{L}_z \hat{S}_z |\ell, m_\ell; s, m_s\rangle &= m_\ell m_s \hbar^2 |\ell, m_\ell; s, m_s\rangle \end{aligned}$$

$$\begin{aligned}
& \hat{L}_+ \hat{S}_- |\ell, m_\ell; s, m_s\rangle \\
&= \sqrt{\ell(\ell+1) - m_\ell(m_\ell+1)} \sqrt{s(s+1) - m_s(m_s-1)} \hbar^2 |\ell, m_\ell+1; s, m_s-1\rangle \\
& \hat{L}_- \hat{S}_+ |\ell, m_\ell; s, m_s\rangle \\
&= \sqrt{\ell(\ell+1) - m_\ell(m_\ell-1)} \sqrt{s(s+1) - m_s(m_s+1)} \hbar^2 |\ell, m_\ell-1; s, m_s+1\rangle
\end{aligned}$$

Therefore

$$\begin{aligned}
\hat{L} \cdot \hat{S} \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle &= \frac{\hbar^2}{2} \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle &= \frac{1}{2} \sqrt{2} \hbar^2 \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle &= -\frac{\hbar^2}{2} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{2} \sqrt{2} \hbar^2 \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{1}{2} \sqrt{2} \hbar^2 \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{\hbar^2}{2} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{1}{2} \sqrt{2} \hbar^2 \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{\hbar^2}{2} \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle \\
\hat{L} \cdot \hat{S} \left| 0, 0; \frac{1}{2}, \frac{1}{2} \right\rangle &= 0 \\
\hat{L} \cdot \hat{S} \left| 0, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle &= 0
\end{aligned}$$

$$\mathbf{H}' = \langle f(r) \rangle \frac{\hbar^2}{2} \begin{array}{c} \begin{array}{c} \backslash \\ \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \\ \langle 7| \\ \langle 8| \end{array} \begin{pmatrix} \begin{array}{cccccccc} |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle & |7\rangle & |8\rangle \\ \begin{array}{cccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & \sqrt{2} & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \end{pmatrix} \end{array} \end{array}$$

The 8×8 degenerate subspace can be separated into two subspaces. One is 2×2 and the other one is 6×6 .

Since there is no orbital angular momentum for $\ell = 0$ to create a magnetic field to yield the spin-orbit coupling, i.e., there is no energy correction. We are still not able to diagonalize the left 6×6 subspaces for $\ell = 1$.

C. DIRECT SUM STATES (COUPLED STATES)

(1) Choose the direct sum states as the zeroth-order state.

Define the total angular momentum operator $\hat{f} = \hat{L} + \hat{S}$ and $\hat{f}^2, \hat{f}_z, \hat{L}^2$, and $\hat{S}^2$ are mutually commuted.

The direct sum states $|j, m_j\rangle$ are common eigenstates of the operators $\hat{f}^2, \hat{f}_z, \hat{L}^2$, and $\hat{S}^2$, i.e.,

$$\hat{f}^2 |j, m_j\rangle = j(j+1)\hbar^2 |j, m_j\rangle, \quad \ell + s = \frac{3}{2} \geq j \geq |\ell - s| = \frac{1}{2}$$

$$\hat{f}_z |j, m_j\rangle = m_j \hbar |j, m_j\rangle$$

$$\hat{L}^2 |j, m_j\rangle = \ell(\ell+1)\hbar^2 |j, m_j\rangle$$

$$\hat{S}^2 |j, m_j\rangle = s(s+1)\hbar^2 |j, m_j\rangle$$

The basis states are

$$\begin{array}{cccc} \underbrace{\left| \frac{3}{2}, \frac{3}{2} \right\rangle}_{|1\rangle} & \underbrace{\left| \frac{3}{2}, \frac{1}{2} \right\rangle}_{|2\rangle} & \underbrace{\left| \frac{3}{2}, -\frac{1}{2} \right\rangle}_{|3\rangle} & \underbrace{\left| \frac{3}{2}, -\frac{3}{2} \right\rangle}_{|4\rangle} \\ \underbrace{\left| \frac{1}{2}, \frac{1}{2} \right\rangle}_{|5\rangle} & \underbrace{\left| \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|6\rangle} & & \end{array}$$

The spin-orbit coupling term can be written in terms of $\hat{f}$ as

$$\hat{f}^2 = (\hat{L} + \hat{S})^2 = \hat{L}^2 + 2\hat{L} \cdot \hat{S} + \hat{S}^2 \Rightarrow \hat{L} \cdot \hat{S} = \frac{\hat{f}^2 - \hat{L}^2 - \hat{S}^2}{2}$$

Acts on the direct sum states

$$\hat{L} \cdot \hat{S} |j, m_j\rangle = \frac{j(j+1) - \ell(\ell+1) - s(s+1)}{2} \hbar^2 |j, m_j\rangle$$

The matrix elements of $\hat{H}'$ in the direct sum basis is

$$\langle j, m_j | \hat{H}' | j, m_j \rangle = \langle f(r) \rangle \left[\frac{j(j+1) - \ell(\ell+1) - s(s+1)}{2} \hbar^2 \right]$$

$$\mathbf{H}' = \langle f(r) \rangle \frac{\hbar^2}{2} \begin{array}{c} \begin{array}{c} \backslash \\ \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \end{array} \begin{pmatrix} \begin{array}{cccccc} |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 \end{pmatrix} \end{array} \end{pmatrix} \end{array}$$

OS:

$$\begin{aligned}
\langle 1|\hat{H}'|1\rangle &= \langle f(r)\rangle \frac{\hbar^2}{2} \times \left[\frac{3}{2} \left(\frac{3}{2} + 1 \right) - 1(1+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right] \\
&= \langle f(r)\rangle \frac{\hbar^2}{2} \times \left[\frac{15}{4} - 2 - \frac{3}{4} \right] \\
&= \langle f(r)\rangle \frac{\hbar^2}{2} \times 1 \\
\langle 1|\hat{H}'|1\rangle &= \langle 2|\hat{H}'|2\rangle = \langle 3|\hat{H}'|3\rangle = \langle 4|\hat{H}'|4\rangle \\
\langle 5|\hat{H}'|5\rangle &= \langle f(r)\rangle \frac{\hbar^2}{2} \times \left[\frac{1}{2} \left(\frac{1}{2} + 1 \right) - 1(1+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right] \\
&= \langle f(r)\rangle \frac{\hbar^2}{2} \times \left[\frac{4}{4} - 2 - \frac{3}{4} \right] \\
&= \langle f(r)\rangle \frac{\hbar^2}{2} \times (-2) \\
\langle 5|\hat{H}'|5\rangle &= \langle 6|\hat{H}'|6\rangle
\end{aligned}$$

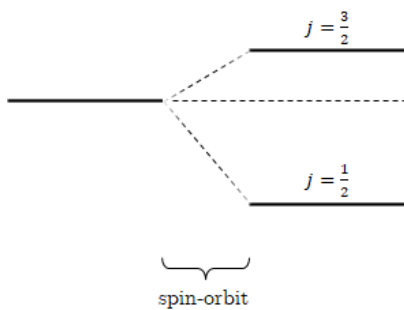
The spin-orbit coupling can only lift the degeneracy of the $j = \frac{3}{2}$ and $j = \frac{1}{2}$ states.

(2) First-order energy correction

$$\begin{aligned}
\left\langle \frac{1}{r^3} \right\rangle_{n\ell} &= \frac{1}{a_0^3 n^3 \ell(\ell+1/2)(\ell+1)} \\
\langle f(r) \rangle &= \frac{1}{2m^2 c^2} \frac{e^2}{4\pi\epsilon_0} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} \\
&= \frac{\epsilon_1 \alpha^2 a_0^3}{\hbar^2} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} \\
&= \frac{\epsilon_n \alpha^2}{n\ell(\ell+1/2)(\ell+1)\hbar^2}
\end{aligned}$$

where

$$\begin{aligned}
\alpha &= \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx \frac{1}{137} \text{ is the fine-structure constant.} \\
a_0 &= \frac{4\pi\epsilon_0 \hbar^2}{me^2} = \frac{\hbar}{m c \alpha} \approx 0.05 \text{ nm is the Bohr radius.} \\
\epsilon_1 &= \frac{mc^2}{2} \alpha^2 = 13.6
\end{aligned}$$



Thus, we obtain

$$\begin{aligned}
 \langle f(r) \rangle \frac{\hbar^2}{2} &= \frac{\varepsilon_2 \alpha^2}{2 \times 1(1 + 1/2)(1 + 1)\hbar^2} \\
 &= \frac{\varepsilon_2 \alpha^2 \hbar^2}{6\hbar^2 2} \\
 &= \frac{\varepsilon_2 \alpha^2}{12} \\
 &\approx 1.5 \times 10^{-5} \text{ eV}
 \end{aligned}$$

$$E_{3/2}^{(1)} = \frac{\varepsilon_2 \alpha^2}{12} \times 1$$

$$E_{1/2}^{(1)} = \frac{\varepsilon_2 \alpha^2}{12} \times (-2)$$

$$\Delta E_{\text{LS}} = E_{3/2}^{(1)} - E_{1/2}^{(1)} = \frac{\varepsilon_2 \alpha^2}{12} (1 - (-2)) = 4.5 \times 10^{-5} \text{ eV}$$

7-4 Zeeman Effect

A. ZEEMAN INTERACTION

- (1) For a single electron, the interaction between the applied magnetic field $\vec{B}$ and the magnetic moments in the atom is

$$\begin{aligned}\hat{H}_{\text{Zeeman}} &= -(\hat{\mu}_l + \hat{\mu}_s) \cdot \vec{B} \\ &= -\left(-\mu_B \frac{\hat{L}}{\hbar} - 2\mu_B \frac{\hat{S}}{\hbar}\right) \cdot \vec{B} \\ &= \frac{\mu_B \vec{B}}{\hbar} \cdot (\hat{L} + 2\hat{S})\end{aligned}$$

where $\mu_B = \frac{e\hbar}{2mc}$ is the Bohr magneton.

- (2) The total Hamiltonian of an electron in the presence of a magnetic field

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} + f(r)\hat{L} \cdot \hat{S} + \frac{\mu_B \vec{B}}{\hbar} \cdot (\hat{L} + 2\hat{S})$$

B. WEAK MAGNETIC FIELD

- (1) In this case, the Zeeman effect are much smaller than the spin-orbit coupling. Therefore, it is more appropriate to include the spin-orbit Hamiltonian in zeroth-order and treat the Zeeman interaction as a perturbation.

$$\hat{H} = \underbrace{\frac{\hat{p}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} + f(r)\hat{L} \cdot \hat{S}}_{\hat{H}_0} + \underbrace{\frac{\mu_B \vec{B}}{\hbar} \cdot (\hat{L} + 2\hat{S})}_{\hat{H}'}$$

- (2) Since $\hat{H}_0$ has included the spin-orbit interaction, we choose the direct sum states $|j, m_j\rangle$ as the zeroth-order states

$$\begin{array}{cccc}\underbrace{\left|\frac{3}{2}, \frac{3}{2}\right\rangle}_{|1\rangle} & \underbrace{\left|\frac{3}{2}, \frac{1}{2}\right\rangle}_{|2\rangle} & \underbrace{\left|\frac{3}{2}, -\frac{1}{2}\right\rangle}_{|3\rangle} & \underbrace{\left|\frac{3}{2}, -\frac{3}{2}\right\rangle}_{|4\rangle} \\ \underbrace{\left|\frac{1}{2}, \frac{1}{2}\right\rangle}_{|5\rangle} & \underbrace{\left|\frac{1}{2}, -\frac{1}{2}\right\rangle}_{|6\rangle} & & \end{array}$$

The zeroth-order energies are

$$\mathbf{H}_0 = \begin{matrix} & \begin{matrix} |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \end{matrix} \\ \begin{matrix} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \end{matrix} & \begin{pmatrix} E_{3/2}^{(0)} & 0 & 0 & 0 & 0 & 0 \\ 0 & E_{3/2}^{(0)} & 0 & 0 & 0 & 0 \\ 0 & 0 & E_{3/2}^{(0)} & 0 & 0 & 0 \\ 0 & 0 & 0 & E_{3/2}^{(0)} & 0 & 0 \\ 0 & 0 & 0 & 0 & E_{1/2}^{(0)} & 0 \\ 0 & 0 & 0 & 0 & 0 & E_{1/2}^{(0)} \end{pmatrix} \end{matrix}$$

$$\begin{aligned} E_j^{(0)} &= -\frac{13.6}{n^2} + \frac{\alpha^2 \varepsilon_1 a_0^3}{\hbar^2} \langle f(r) \rangle \langle j, m_j | \frac{\hat{f}^2 - \hat{L}^2 - \hat{S}^2}{2} | j, m_j \rangle \\ &= -\frac{13.6}{n^2} + \frac{\varepsilon_n \alpha^2}{n\ell(\ell+1/2)(\ell+1)} \frac{j(j+1) - \ell(\ell+1) - s(s+1)}{2} \end{aligned}$$

$$E_{3/2}^{(0)} = -3.4 + \frac{\varepsilon_2 \alpha^2}{12} \left(\frac{3}{2} \left(\frac{3}{2} + 1 \right) - 1(1+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right)$$

$$= -3.4 + \epsilon$$

$$E_{1/2}^{(0)} = -3.4 + \frac{\varepsilon_2 \alpha^2}{12} \left(\frac{1}{2} \left(\frac{1}{2} + 1 \right) - 1(1+1) - \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right)$$

$$= -3.4 - 2\epsilon$$

$$\text{where } \epsilon = \frac{\varepsilon_2 \alpha^2}{12} \approx 1.5 \times 10^{-5} \text{ eV.}$$

(3) Assuming $\vec{B} = B\vec{z}$, the perturbed Hamiltonian is

$$\hat{H}' = \frac{\mu_B B}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

The eigenstates of $\hat{L}_z$ and $\hat{S}_z$ are the tensor product states $|\ell, m_\ell; s, m_s\rangle$, i.e.,

$$\hat{H}' |\ell, m_\ell; s, m_s\rangle = \mu_B B (m_\ell + 2m_s) |\ell, m_\ell; s, m_s\rangle$$

(4) Matrix representation (method I)

$$\mathbf{H}' = \frac{\mu_B B}{\hbar} (\mathbf{L}_z + 2\mathbf{S}_z)$$

Using Clebsch-Gordan coefficients, the basis states are

$$\underbrace{\left| \frac{3}{2}, \frac{3}{2} \right\rangle}_{|1\rangle} = \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, \frac{1}{2} \right\rangle}_{|2\rangle} = \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, -\frac{1}{2} \right\rangle}_{|3\rangle} = \frac{1}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, -\frac{3}{2} \right\rangle}_{|4\rangle} = \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{1}{2}, \frac{1}{2} \right\rangle}_{|5\rangle} = -\frac{1}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|6\rangle} = -\frac{\sqrt{2}}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\mathbf{L}_z = \hbar \begin{matrix} & \begin{matrix} |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \end{matrix} \\ \begin{matrix} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \\ \langle 5| \\ \langle 6| \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 & \sqrt{2}/3 & 0 \\ 0 & 0 & -1/3 & 0 & 0 & \sqrt{2}/3 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & \sqrt{2}/3 & 0 & 0 & 2/3 & 0 \\ 0 & 0 & \sqrt{2}/3 & 0 & 0 & -2/3 \end{pmatrix} \end{matrix}$$

OS:

$$\hat{L}_z |1\rangle = \hbar \times 1 \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\hat{L}_z |2\rangle = \hbar \times \frac{1}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\hat{L}_z |3\rangle = \hbar \times \frac{1}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\hat{L}_z |4\rangle = \hbar \times (-1) \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\hat{L}_z |5\rangle = \hbar \times \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\hat{L}_z |6\rangle = \hbar \times \frac{\sqrt{2}}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\mathbf{S}_z = \hbar \begin{pmatrix} \langle 1| & |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \\ \langle 1| & 1/2 & 0 & 0 & 0 & 0 & 0 \\ \langle 2| & 0 & 1/6 & 0 & 0 & -\sqrt{2}/3 & 0 \\ \langle 3| & 0 & 0 & -1/6 & 0 & 0 & -\sqrt{2}/3 \\ \langle 4| & 0 & 0 & 0 & -1/2 & 0 & 0 \\ \langle 5| & 0 & -\sqrt{2}/3 & 0 & 0 & -1/6 & 0 \\ \langle 6| & 0 & 0 & -\sqrt{2}/3 & 0 & 0 & 1/6 \end{pmatrix}$$

OS:

$$\begin{aligned} \hat{S}_z|1\rangle &= \hbar \times \frac{1}{2} \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle \\ \hat{S}_z|2\rangle &= \hbar \times \left(\frac{\sqrt{2}}{2\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{1}{2\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle \right) \\ \hat{S}_z|3\rangle &= \hbar \times \left(\frac{1}{2\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{\sqrt{2}}{2\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle \right) \\ \hat{S}_z|4\rangle &= \hbar \times \left(-\frac{1}{2} \right) \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ \hat{S}_z|5\rangle &= \hbar \times \left(-\frac{1}{2\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{\sqrt{2}}{2\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle \right) \\ \hat{S}_z|6\rangle &= \hbar \times \left(-\frac{\sqrt{2}}{2\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{1}{2\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle \right) \\ \langle 2|\hat{S}_z|2\rangle &= \hbar \left(\frac{\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{2}}{2\sqrt{3}} - \frac{1}{\sqrt{3}} \times \frac{1}{2\sqrt{3}} \right) = \hbar \times \frac{1}{6} \\ \langle 5|\hat{S}_z|2\rangle &= \hbar \left(-\frac{1}{\sqrt{3}} \times \frac{\sqrt{2}}{2\sqrt{3}} - \frac{\sqrt{2}}{\sqrt{3}} \times \frac{1}{2\sqrt{3}} \right) = \hbar \times \left(-\frac{\sqrt{2}}{3} \right) \end{aligned}$$

Thus, we obtain

$$\mathbf{H}' = \mu_B B \begin{pmatrix} \langle 1| & |1\rangle & |2\rangle & |3\rangle & |4\rangle & |5\rangle & |6\rangle \\ \langle 1| & 2 & 0 & 0 & 0 & 0 & 0 \\ \langle 2| & 0 & 2/3 & 0 & 0 & -\sqrt{2}/3 & 0 \\ \langle 3| & 0 & 0 & -2/3 & 0 & 0 & -\sqrt{2}/3 \\ \langle 4| & 0 & 0 & 0 & -2 & 0 & 0 \\ \langle 5| & 0 & -\sqrt{2}/3 & 0 & 0 & 1/3 & 0 \\ \langle 6| & 0 & 0 & -\sqrt{2}/3 & 0 & 0 & -1/3 \end{pmatrix}$$

1. For $j = \frac{3}{2}$

$$\mathbf{H}'_{3/2} = \mu_B B \begin{array}{c} \backslash \\ \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \end{array} \begin{array}{c} |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \end{array} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2/3 & 0 & 0 \\ 0 & 0 & -2/3 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

Lucky! In the basis of the direct sum states, the degenerate subspace is already diagonal.

The first-order energies corrections are,

$$\begin{aligned} E_{3/2}^{(1a)} &= \left\langle \frac{3}{2}, \frac{3}{2} \left| \hat{H}' \right| \frac{3}{2}, \frac{3}{2} \right\rangle = 2\mu_B B \\ E_{3/2}^{(1b)} &= \left\langle \frac{3}{2}, \frac{1}{2} \left| \hat{H}' \right| \frac{3}{2}, \frac{1}{2} \right\rangle = \frac{2}{3}\mu_B B \\ E_{3/2}^{(1c)} &= \left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}' \right| \frac{3}{2}, -\frac{1}{2} \right\rangle = -\frac{2}{3}\mu_B B \\ E_{3/2}^{(1d)} &= \left\langle \frac{3}{2}, -\frac{3}{2} \left| \hat{H}' \right| \frac{3}{2}, -\frac{3}{2} \right\rangle = -2\mu_B B \end{aligned}$$

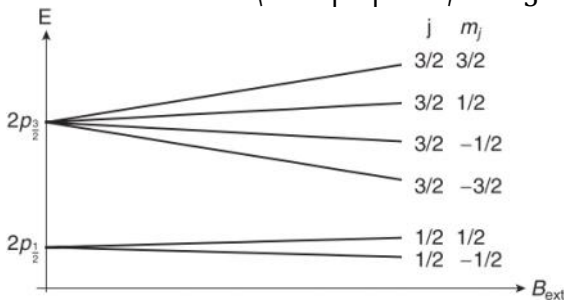
2. For $j = \frac{1}{2}$

$$\mathbf{H}'_{1/2} = \mu_B B \begin{array}{c} \backslash \\ \langle 5| \\ \langle 6| \end{array} \begin{array}{c} |5\rangle \\ |6\rangle \end{array} \begin{pmatrix} 1/3 & 0 \\ 0 & -1/3 \end{pmatrix}$$

Again! In the basis of the direct sum states, the degenerate subspace is already diagonal.

The first-order energies corrections are,

$$\begin{aligned} E_{1/2}^{(1a)} &= \left\langle \frac{1}{2}, \frac{1}{2} \left| \hat{H}' \right| \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{1}{3}\mu_B B \\ E_{1/2}^{(1b)} &= \left\langle \frac{1}{2}, -\frac{1}{2} \left| \hat{H}' \right| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{3}\mu_B B \end{aligned}$$



(5) Matrix representation (method II)

$$\mathbf{H}' = \frac{\mu_B B}{\hbar} (\mathbf{L}_z + 2\mathbf{S}_z)$$

1. For $j = \frac{3}{2}$

Using Clebsch-Gordan coefficients, the basis states are

$$\underbrace{\left| \frac{3}{2}, \frac{3}{2} \right\rangle}_{|1\rangle} = \left| 1, 1; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, \frac{1}{2} \right\rangle}_{|2\rangle} = \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, -\frac{1}{2} \right\rangle}_{|3\rangle} = \frac{1}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{3}{2}, -\frac{3}{2} \right\rangle}_{|4\rangle} = \left| 1, -1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\mathbf{L}_z = \hbar \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \end{array} \begin{array}{c} \backslash \\ |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \end{array} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/3 & 0 & 0 \\ 0 & 0 & -1/3 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\mathbf{S}_z = \hbar \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \end{array} \begin{array}{c} \backslash \\ |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \end{array} \begin{pmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/6 & 0 & 0 \\ 0 & 0 & -1/6 & 0 \\ 0 & 0 & 0 & -1/2 \end{pmatrix}$$

$$\mathbf{H}' = \mu_B B \begin{array}{c} \langle 1| \\ \langle 2| \\ \langle 3| \\ \langle 4| \end{array} \begin{array}{c} \backslash \\ |1\rangle \\ |2\rangle \\ |3\rangle \\ |4\rangle \end{array} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2/3 & 0 & 0 \\ 0 & 0 & -2/3 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

$$E_{3/2}^{(1a)} = \left\langle \frac{3}{2}, \frac{3}{2} \right| \hat{H}' \left| \frac{3}{2}, \frac{3}{2} \right\rangle = 2\mu_B B$$

$$E_{3/2}^{(1b)} = \left\langle \frac{3}{2}, \frac{1}{2} \right| \hat{H}' \left| \frac{3}{2}, \frac{1}{2} \right\rangle = \frac{2}{3}\mu_B B$$

$$E_{3/2}^{(1c)} = \left\langle \frac{3}{2}, -\frac{1}{2} \right| \hat{H}' \left| \frac{3}{2}, -\frac{1}{2} \right\rangle = -\frac{2}{3}\mu_B B$$

$$E_{3/2}^{(1d)} = \left\langle \frac{3}{2}, -\frac{3}{2} \right| \hat{H}' \left| \frac{3}{2}, -\frac{3}{2} \right\rangle = -2\mu_B B$$

2. For $j = \frac{1}{2}$

$$\underbrace{\left| \frac{1}{2}, \frac{1}{2} \right\rangle}_{|5\rangle} = -\frac{1}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{\sqrt{2}}{\sqrt{3}} \left| 1, 1; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\underbrace{\left| \frac{1}{2}, -\frac{1}{2} \right\rangle}_{|6\rangle} = -\frac{\sqrt{2}}{\sqrt{3}} \left| 1, -1; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| 1, 0; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

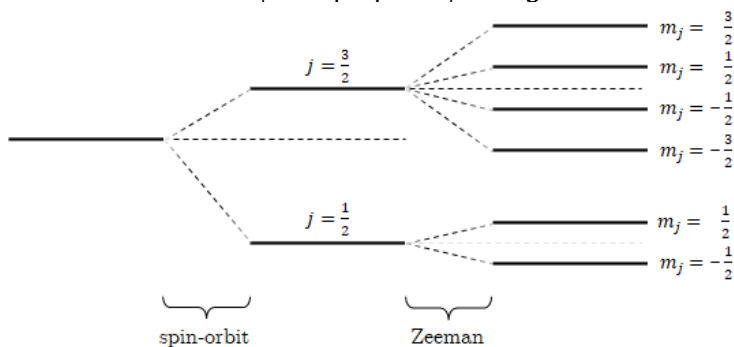
$$\mathbf{L}_z = \hbar \begin{array}{c} \backslash \quad |5\rangle \quad |6\rangle \\ \langle 5| \quad \langle 6| \end{array} \begin{pmatrix} 2/3 & 0 \\ 0 & -2/3 \end{pmatrix}$$

$$\mathbf{S}_z = \hbar \begin{array}{c} \backslash \quad |5\rangle \quad |6\rangle \\ \langle 5| \quad \langle 6| \end{array} \begin{pmatrix} -1/6 & 0 \\ 0 & 1/6 \end{pmatrix}$$

$$\mathbf{H}' = \mu_B B \begin{array}{c} \backslash \quad |5\rangle \quad |6\rangle \\ \langle 5| \quad \langle 6| \end{array} \begin{pmatrix} 1/3 & 0 \\ 0 & -1/3 \end{pmatrix}$$

$$E_{1/2}^{(1a)} = \left\langle \frac{1}{2}, \frac{1}{2} \right| \hat{H}' \left| \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{1}{3} \mu_B B$$

$$E_{1/2}^{(1b)} = \left\langle \frac{1}{2}, -\frac{1}{2} \right| \hat{H}' \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = -\frac{1}{3} \mu_B B$$



(6) Landé g factor (method III)

According to the Wigner-Eckhart theorem, the relations between the matrices in the direct sum basis that represent $\hat{L}_z$ or $\hat{S}_z$ and the matrix that represents $\hat{J}_z$ within a given j subspace are,

$$\langle j, m_j | \hat{L}_z | j, m_j' \rangle = \frac{\langle j, m_j | \hat{L} \cdot \hat{J} | j, m_j \rangle}{j(j+1)\hbar^2} \langle j, m_j | \hat{J}_z | j, m_j' \rangle$$

$$\langle j, m_j | \hat{S}_z | j, m'_j \rangle = \frac{\langle j, m_j | \hat{S} \cdot \hat{f} | j, m_j \rangle}{j(j+1)\hbar^2} \langle j, m_j | \hat{f}_z | j, m'_j \rangle$$

NOTES:

The equations above are the projection $\hat{L}_z$ or $\hat{S}_z$ to $\hat{f}_z$ direction in the space of $\mathbb{W}$. The lengths of the projection are given as

$$\frac{\langle j, m_j | \hat{L} \cdot \hat{f} | j, m_j \rangle}{j(j+1)\hbar^2} \text{ or } \frac{\langle j, m_j | \hat{S} \cdot \hat{f} | j, m_j \rangle}{j(j+1)\hbar^2}$$

Since $\hat{f} = \hat{L} + \hat{S}$, we obtain

$$\begin{aligned} \hat{S} = \hat{f} - \hat{L} &\Rightarrow \hat{S}^2 = \hat{f}^2 + \hat{L}^2 - 2\hat{L} \cdot \hat{f} \Rightarrow \hat{L} \cdot \hat{f} = \frac{\hat{f}^2 + \hat{L}^2 - \hat{S}^2}{2} \\ \hat{L} = \hat{f} - \hat{S} &\Rightarrow \hat{L}^2 = \hat{f}^2 + \hat{S}^2 - 2\hat{S} \cdot \hat{f} \Rightarrow \hat{S} \cdot \hat{f} = \frac{\hat{f}^2 + \hat{S}^2 - \hat{L}^2}{2} \\ \langle j, m_j | \hat{L} \cdot \hat{f} | j, m_j \rangle &= \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2} \hbar^2 \\ \langle j, m_j | \hat{S} \cdot \hat{f} | j, m_j \rangle &= \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2} \hbar^2 \end{aligned}$$

The $\hat{L}_z$ and $\hat{S}_z$ matrix elements are

$$\begin{aligned} \langle j, m_j | \hat{L}_z | j, m'_j \rangle &= \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} \langle j, m_j | \hat{f}_z | j, m'_j \rangle \\ \langle j, m_j | \hat{S}_z | j, m'_j \rangle &= \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \langle j, m_j | \hat{f}_z | j, m'_j \rangle \end{aligned}$$

Since $\hat{f}_z$ is diagonal in the direct sum basis, $\hat{L}_z$ and $\hat{S}_z$ are also diagonal within a given j subspace. Thus the theorem has allowed us to find the matrix representations, within a specific subspace, of operators that are not diagonal in the direct sum basis.

The first-order Zeeman energy correction:

$$\begin{aligned}
E_j^{(1)} &= \langle j, m_j | \hat{H}'_z | j, m_j \rangle \\
&= \frac{\mu_B B}{\hbar} \langle j, m_j | \hat{L}_z + 2\hat{S}_z | j, m_j \rangle \\
&= \frac{\mu_B B}{\hbar} \left[\frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} \right. \\
&\quad \left. + 2 \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \right] \langle j, m_j | \hat{J}_z | j, m_j \rangle \\
&= \frac{\mu_B B}{\hbar} \left[1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \right] m_j \hbar \\
&= g_j \mu_B B m_j
\end{aligned}$$

where g_j is Landé g factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$$

$$j = \frac{3}{2}:$$

$$g_{3/2} = 1 + \frac{3/2(3/2+1) + 1/2(1/2+1) - 1(1+1)}{2 \times 3/2(3/2+1)} = \frac{4}{3}$$

$$E_{3/2}^{(1a)} = \frac{4}{3} \mu_B B \times \frac{3}{2} = 2\mu_B B$$

$$E_{3/2}^{(1b)} = \frac{4}{3} \mu_B B \times \frac{1}{2} = \frac{2}{3} \mu_B B$$

$$E_{3/2}^{(1c)} = \frac{4}{3} \mu_B B \times \left(-\frac{1}{2}\right) = -\frac{2}{3} \mu_B B$$

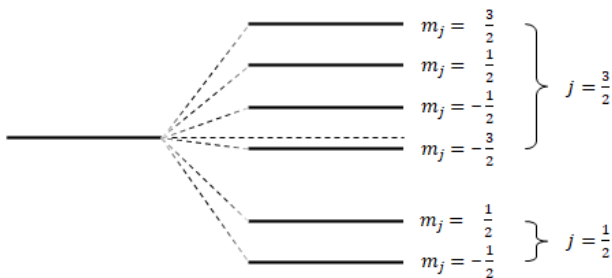
$$E_{3/2}^{(1d)} = \frac{4}{3} \mu_B B \times \left(-\frac{3}{2}\right) = -2\mu_B B$$

$$j = \frac{1}{2}:$$

$$g_{1/2} = 1 + \frac{1/2(1/2+1) + 1/2(1/2+1) - 1(1+1)}{2 \times 1/2(1/2+1)} = \frac{2}{3}$$

$$E_{1/2}^{(1e)} = \frac{2}{3} \mu_B B \times \frac{1}{2} = \frac{1}{3} \mu_B B$$

$$E_{1/2}^{(1f)} = \frac{2}{3} \mu_B B \times \left(-\frac{1}{2}\right) = -\frac{1}{3} \mu_B B$$



C. STRONG MAGNETIC FIELD

- (1) In this case, the Zeeman effect are much larger than the spin-orbit coupling. It is more appropriate to include the Zeeman Hamiltonian in zeroth-order and treat the spin-orbit coupling as a perturbation.

$$\hat{H} = \underbrace{\frac{\hat{p}^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} + \frac{\mu_B B}{\hbar}(\hat{L}_z + 2\hat{S}_z)}_{\hat{H}_0} + \underbrace{f(r)\hat{L} \cdot \hat{S}}_{\hat{H}'}$$

- (2) The zeroth-order state is the tensor product basis $|\ell, m_\ell; s, m_s\rangle$

$$\begin{array}{ccc} \underbrace{\left|1, \mathbf{1}; \frac{1}{2}, \frac{1}{2}\right\rangle}_{|1\uparrow\rangle} & \underbrace{\left|1, \mathbf{0}; \frac{1}{2}, \frac{1}{2}\right\rangle}_{|0\uparrow\rangle} & \underbrace{\left|1, -\mathbf{1}; \frac{1}{2}, \frac{1}{2}\right\rangle}_{|-1\uparrow\rangle} \\ \underbrace{\left|1, \mathbf{1}; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|1\downarrow\rangle} & \underbrace{\left|1, \mathbf{0}; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|0\downarrow\rangle} & \underbrace{\left|1, -\mathbf{1}; \frac{1}{2}, -\frac{1}{2}\right\rangle}_{|-1\downarrow\rangle} \end{array}$$

The zeroth-order energies are

$$\mathbf{H}_0 = \begin{array}{c} \begin{array}{c} \langle 1\uparrow| \\ \langle 0\uparrow| \\ \langle -1\uparrow| \\ \langle 1\downarrow| \\ \langle 0\downarrow| \\ \langle -1\downarrow| \end{array} \begin{pmatrix} E_{1\uparrow}^{(0)} & 0 & 0 & 0 & 0 & 0 \\ 0 & E_{0\uparrow}^{(0)} & 0 & 0 & 0 & 0 \\ 0 & 0 & E_{-1\uparrow}^{(0)} & 0 & 0 & 0 \\ 0 & 0 & 0 & E_{1\downarrow}^{(0)} & 0 & 0 \\ 0 & 0 & 0 & 0 & E_{0\downarrow}^{(0)} & 0 \\ 0 & 0 & 0 & 0 & 0 & E_{-1\downarrow}^{(0)} \end{pmatrix} \end{array}$$

$$\begin{aligned}
E_{n,m_\ell,m_s}^{(0)} &= -\frac{13.6}{n^2} + \langle \ell, m_\ell; s, m_s | \hat{H}_z | \ell, m_\ell; s, m_s \rangle \\
&= -\frac{13.6}{n^2} + \frac{\mu_B B}{\hbar} (m_\ell + 2m_s) \hbar \\
&= -\frac{13.6}{n^2} + \mu_B B (m_\ell + 2m_s) \\
E_{1\uparrow}^{(0)} &= -3.4 + 2\mu_B B \\
E_{0\uparrow}^{(0)} &= -3.4 + \mu_B B \\
\left. \begin{aligned} E_{-1\uparrow}^{(0)} &= -3.4 \\ E_{1\downarrow}^{(0)} &= -3.4 \end{aligned} \right\} &2 \text{ degenerate states} \\
E_{0\downarrow}^{(0)} &= -3.4 - \mu_B B \\
E_{-1\downarrow}^{(0)} &= -3.4 - 2\mu_B B
\end{aligned}$$

(3) Assuming $\vec{B} = B\vec{z}$, the perturbed Hamiltonian is

$$\hat{H}' = f(r)\hat{L} \cdot \hat{S}$$

The eigenstates are the direct sum states $|j, m_j, \ell, s\rangle$.

(4) Matrix representation (method I)

$$\text{Since } \hat{L} \cdot \hat{S} = \frac{1}{2}(\hat{L}_+ \hat{S}_- + \hat{L}_- \hat{S}_+) + \hat{L}_z \hat{S}_z$$

we obtain

$$\hat{L}_+ \hat{S}_- |\ell, m_\ell; s, m_s\rangle = \frac{\hbar^2}{2} \sqrt{\ell(\ell+1) - m_\ell(m_\ell+1)} \sqrt{s(s+1) - m_s(m_s-1)} \\ \times |\ell, m_\ell+1; s, m_s-1\rangle$$

$$\hat{L}_- \hat{S}_+ |\ell, m_\ell; s, m_s\rangle = \frac{\hbar^2}{2} \sqrt{\ell(\ell+1) - m_\ell(m_\ell-1)} \sqrt{s(s+1) - m_s(m_s+1)} \\ \times |\ell, m_\ell-1; s, m_s+1\rangle$$

$$\hat{L}_z \hat{S}_z |\ell, m_\ell; s, m_s\rangle = m_\ell m_s \hbar^2 |\ell, m_\ell; s, m_s\rangle$$

Therefore

$$\hat{L} \cdot \hat{S} \left| 1, \mathbf{1}; \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{\hbar^2}{2} \left| 1, \mathbf{1}; \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$\hat{L} \cdot \hat{S} \left| 1, \mathbf{0}; \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{\hbar^2}{2} \sqrt{2} \left| 1, \mathbf{1}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\hat{L} \cdot \hat{S} \left| 1, -\mathbf{1}; \frac{1}{2}, \frac{1}{2} \right\rangle = -\frac{\hbar^2}{2} \left| 1, -\mathbf{1}; \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{\hbar^2}{2} \sqrt{2} \left| 1, \mathbf{0}; \frac{1}{2}, -\frac{1}{2} \right\rangle$$

$$\begin{aligned}\hat{L} \cdot \hat{S} \left| 1, \mathbf{1}; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{\hbar^2}{2} \sqrt{2} \left| 1, \mathbf{0}; \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{\hbar^2}{2} \left| 1, \mathbf{1}; \frac{1}{2}, -\frac{1}{2} \right\rangle \\ \hat{L} \cdot \hat{S} \left| 1, \mathbf{0}; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{\hbar^2}{2} \sqrt{2} \left| 1, -\mathbf{1}; \frac{1}{2}, \frac{1}{2} \right\rangle \\ \hat{L} \cdot \hat{S} \left| 1, -\mathbf{1}; \frac{1}{2}, -\frac{1}{2} \right\rangle &= \frac{\hbar^2}{2} \left| 1, -\mathbf{1}; \frac{1}{2}, -\frac{1}{2} \right\rangle\end{aligned}$$

Thus, we obtain

$$\mathbf{H}' = \langle f(r) \rangle \frac{\hbar^2}{2} \begin{array}{c} \begin{array}{c} \backslash \\ \langle \mathbf{1} \uparrow | \\ \langle \mathbf{0} \uparrow | \\ \langle -\mathbf{1} \uparrow | \\ \langle \mathbf{1} \downarrow | \\ \langle \mathbf{0} \downarrow | \\ \langle -\mathbf{1} \downarrow | \end{array} \end{array} \begin{pmatrix} | \mathbf{1} \uparrow \rangle & | \mathbf{0} \uparrow \rangle & | -\mathbf{1} \uparrow \rangle & | \mathbf{1} \downarrow \rangle & | \mathbf{0} \downarrow \rangle & | -\mathbf{1} \downarrow \rangle \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & -1 & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{pmatrix}$$

The degenerate subspace is

$$\mathbf{H}' = \langle f(r) \rangle \frac{\hbar^2}{2} \begin{array}{c} \backslash \\ \langle -\mathbf{1} \uparrow | \\ \langle \mathbf{1} \downarrow | \end{array} \begin{pmatrix} | -\mathbf{1} \uparrow \rangle & | \mathbf{1} \downarrow \rangle \\ \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \end{pmatrix}$$

Since the degenerate subspace is already diagonal and has the same value, the degenerate states are still degenerate without lifting.

The first-order energies corrections are,

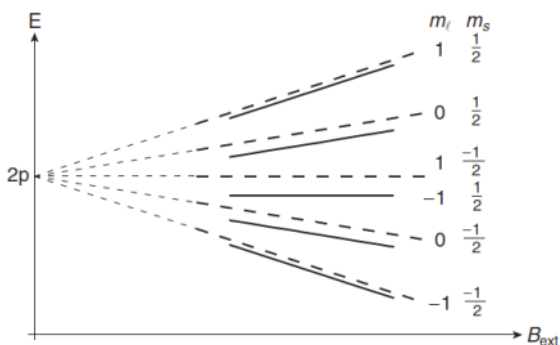
$$\begin{aligned}E_{-1\uparrow}^{(1)} &= -\epsilon \\ E_{1\downarrow}^{(1)} &= -\epsilon\end{aligned}$$

$$\text{where } \epsilon = \langle f(r) \rangle \frac{\hbar^2}{2} = \frac{\varepsilon_2 \alpha^2}{12} \approx 1.5 \times 10^{-5} \text{ eV}$$

The perturbed energies are

$$\begin{aligned}E_{1\uparrow}^{(0)} + E_{1\uparrow}^{(1)} &= -3.4 + 2\mu_B B + \epsilon + \eta \\ E_{0\uparrow}^{(0)} + E_{0\uparrow}^{(1)} &= -3.4 + \mu_B B + 0 + \eta \\ E_{-1\uparrow}^{(0)} + E_{-1\uparrow}^{(1)} &= -3.4 - \epsilon + \eta \\ E_{1\downarrow}^{(0)} + E_{1\downarrow}^{(1)} &= -3.4 - \epsilon + \eta \\ E_{0\downarrow}^{(0)} + E_{0\downarrow}^{(1)} &= -3.4 - \mu_B B + 0 + \eta \\ E_{-1\downarrow}^{(0)} + E_{-1\downarrow}^{(1)} &= -3.4 - 2\mu_B B + \epsilon + \eta\end{aligned}$$

$$\text{where } \eta = -\frac{\varepsilon_2 \alpha^2}{12} \times \frac{7}{4} = -2.6 \times 10^{-5} \text{ eV is the relativistic correction.}$$



where solid lines show the fine-structure (relativistic and spin-orbit) corrections to the Zeeman levels (dashed lines).

D. INTERMEDIATE MAGNETIC FIELD

- (1) In the intermediate magnetic field regime where the Zeeman corrections and the spin-orbit corrections are comparable, we have to treat both effects as perturbations to the zeroth-order Bohr energy levels.

$$\hat{H}' = \hat{H}_{\text{SO}} + \hat{H}_{\text{Zeeman}}$$

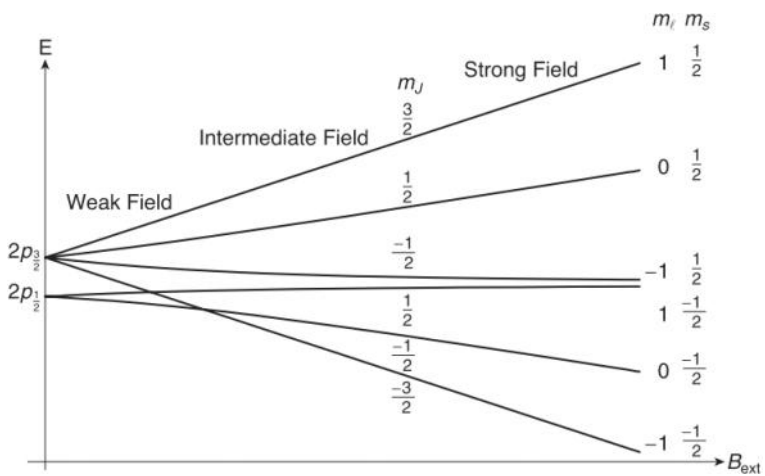
There is no obvious preferred basis.

- (2) The matrix $\mathbf{H}'$ represents the perturbation Hamiltonian $\hat{H}'$ in the direct sum basis

$$\mathbf{H}' = \begin{pmatrix} -a + 2b & 0 & 0 & 0 & 0 & 0 \\ 0 & -a + \frac{2b}{3} & 0 & 0 & -\frac{\sqrt{2}}{3}b & 0 \\ 0 & 0 & -a - \frac{2b}{3} & 0 & 0 & -\frac{\sqrt{2}}{3}b \\ 0 & 0 & 0 & -a - 2b & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{3}b & 0 & 0 & -5a + \frac{b}{3} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{3}b & 0 & 0 & -5a - \frac{b}{3} \end{pmatrix}$$

where $a = \frac{1}{128}\alpha^4 mc^2$ and $b = \mu_B B$

The off-diagonal terms come from the Zeeman interaction. Diagonalization of the matrix yields the energy shifts.



—Chapter 8—

Time-Dependent Perturbation Theory

8-1 First-Order Perturbation Theory

A. TIME-DEPENDENT HAMILTONIAN IN SCHRÖDINGER PICTURE

- (1) The time-dependent Hamiltonian

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$$

where $\hat{H}_0$ is time-independent and satisfies

$$\hat{H}_0 |\psi_n\rangle = E_n |\psi_n\rangle$$

Suppose $|\Psi(t)\rangle$ is the eigenstate of $\hat{H}(t)$, and can be expanded in terms of $|\psi_n\rangle$, i.e.,

$$|\Psi(t)\rangle = \sum_n c_n(t) e^{-\frac{iE_n t}{\hbar}} |\psi_n\rangle$$

- (2) Find $c_n(t)$ through the Schrödinger equation

$$\frac{d}{dt} |\Psi(t)\rangle = -\frac{i}{\hbar} \hat{H}(t) |\Psi(t)\rangle = -\frac{i}{\hbar} (\hat{H}_0 + \hat{V}(t)) |\Psi(t)\rangle$$

OS:

$$\frac{d}{dt} \Psi(x, t) = -\frac{i}{\hbar} \hat{\mathcal{H}}(t) \Psi(x, t)$$

$$\frac{d}{dt} \langle x | \Psi(t) \rangle = -\frac{i}{\hbar} \langle x | \hat{H}(t) | \Psi(t) \rangle$$

$$\frac{d}{dt} |\Psi(t)\rangle = -\frac{i}{\hbar} \hat{H}(t) |\Psi(t)\rangle$$

$$\Rightarrow \frac{d}{dt} \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} |\psi_j\rangle = -\frac{i}{\hbar} (\hat{H}_0 + \hat{V}(t)) \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} |\psi_j\rangle$$

$$\Rightarrow \sum_j \frac{dc_j(t)}{dt} e^{-\frac{iE_j t}{\hbar}} |\psi_j\rangle - \frac{i}{\hbar} \sum_j c_j(t) E_j e^{-\frac{iE_j t}{\hbar}} |\psi_j\rangle$$

$$= -\frac{i}{\hbar} \left[\sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} E_j |\psi_j\rangle + \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} \hat{V}(t) |\psi_j\rangle \right]$$

$$\Rightarrow \sum_j \frac{dc_j(t)}{dt} e^{-\frac{iE_j t}{\hbar}} |\psi_j\rangle = -\frac{i}{\hbar} \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} \hat{V}(t) |\psi_j\rangle$$

Multiplying both side by $\langle \psi_n |$

$$\begin{aligned}\langle \psi_n | \sum_j \frac{dc_j(t)}{dt} e^{-\frac{iE_j t}{\hbar}} | \psi_m \rangle &= -\frac{i}{\hbar} \langle \psi_n | \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} \hat{V}(t) | \psi_j \rangle \\ \sum_j \frac{dc_j(t)}{dt} e^{-\frac{iE_j t}{\hbar}} \underbrace{\langle \psi_n | \psi_j \rangle}_{=\delta_{nj}} &= -\frac{i}{\hbar} \sum_j c_j(t) e^{-\frac{iE_j t}{\hbar}} \langle \psi_n | \hat{V}(t) | \psi_j \rangle \\ \frac{dc_n(t)}{dt} &= -\frac{i}{\hbar} \sum_j c_j(t) e^{\frac{i(E_n - E_j)t}{\hbar}} \langle \psi_n | \hat{V}(t) | \psi_j \rangle\end{aligned}$$

Usually it is difficult to solve the full Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}(t)$. Here we just need to solve the equation with $\hat{V}(t)$ only.

(3) Physical interpretation of the time evolution of the system

1. Before application of $\hat{V}(t)$:

Time-independent, stationary state solutions satisfy $\hat{H}_0 | \psi_n \rangle = E_n | \psi_n \rangle$.

The probability of finding the state in the initial state $| \psi_m \rangle$ is assumed to be unit, i.e.,

$$| \Psi(0) \rangle = | \psi_m \rangle$$

2. During application of $\hat{V}(t)$:

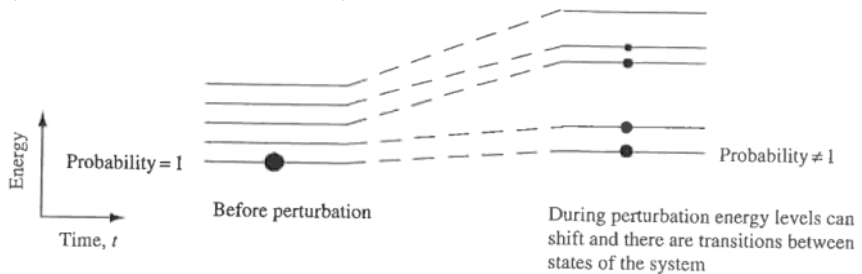
The state of the system evolves according to

$$| \Psi(t) \rangle = \sum_n c_n(t) e^{-\frac{iE_n t}{\hbar}} | \psi_n \rangle$$

and satisfies

$$\frac{dc_n(t)}{dt} = -\frac{i}{\hbar} \sum_j c_j(t) e^{\frac{i(E_n - E_j)t}{\hbar}} \langle \psi_n | \hat{V}(t) | \psi_j \rangle$$

The time-dependent change in potential causes transitions between eigenstates of the initial system (off-diagonal matrix elements) and can shift energy levels of the initial eigenstates (diagonal matrix elements).



3. After application of $\hat{V}(t)$:

Any time after $\hat{V}(t)$ is turned off, the final state of the system may be a stationary state

$$|\Psi(t)\rangle = e^{-iE_n t/\hbar} |\psi_n\rangle$$

B. TRANSITION PROBABILITY

(1) Assume a dimensionless parameter λ , such that

$$\hat{H}(t) = \hat{H}_0 + \lambda \hat{V}(t)$$

Expand the coefficient $c_n(t)$

$$c_n(t) = c_n^{(0)} + \lambda c_n^{(1)} + \dots$$

$$\frac{d}{dt} (c_n^{(0)} + \lambda c_n^{(1)} + \dots)$$

$$= -\frac{i}{\hbar} \sum_j (c_j^{(0)} + \lambda c_j^{(1)} + \dots) \langle \psi_n | \lambda \hat{V}(t) | \psi_j \rangle e^{\frac{i(E_n - E_j)t}{\hbar}}$$

$$= -\lambda \frac{i}{\hbar} \sum_j c_j^{(0)} \langle \psi_n | \hat{V}(t) | \psi_j \rangle e^{\frac{i(E_n - E_j)t}{\hbar}} + O(\lambda^2) + \dots$$

(2) The zeroth-order term $O(\lambda^0)$

$$\frac{dc_n^{(0)}(t)}{dt} = 0$$

$\Rightarrow c_n$ have no time dependence when there is no perturbation.

$$\text{Initial conditions: } |\Psi(0)\rangle = \sum_n c_n(0) e^{-\frac{iE_n \cdot 0}{\hbar}} |\psi_n\rangle = \sum_n c_n(0) |\psi_n\rangle$$

Since $|\Psi(0)\rangle = |\psi_m\rangle$, we obtain

$$c_n(0) = \delta_{nm}$$

$$\Rightarrow c_n^{(0)}(0) = c_n(0) = c_n^{(0)}(t) = \delta_{nm}$$

(3) The first-order term $O(\lambda^1)$

$$\begin{aligned}
\frac{dc_n^{(1)}(t)}{dt} &= -\frac{i}{\hbar} \sum_j c_j^{(0)}(t) \langle \psi_n | \hat{V}(t) | \psi_j \rangle e^{\frac{i(E_n - E_j)t}{\hbar}} \\
&= -\frac{i}{\hbar} \sum_j \delta_{mj}(t) \langle \psi_n | \hat{V}(t) | \psi_j \rangle e^{\frac{i(E_n - E_j)t}{\hbar}} \\
&= -\frac{i}{\hbar} \langle \psi_n | \hat{V}(t) | \psi_m \rangle e^{\frac{i(E_n - E_m)t}{\hbar}} \\
c_n^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t \langle \psi_n | \hat{V}(t') | \psi_m \rangle e^{\frac{i(E_n - E_m)t'}{\hbar}} dt'
\end{aligned}$$

(4) The first-order state vector correction

$$|\Psi(t)\rangle = \sum_n \left(c_n^{(0)}(t) + c_n^{(1)}(t) + \dots \right) e^{-\frac{iE_n t}{\hbar}} |\psi_n\rangle$$

The transition probability that the system can be found in a stationary state $|\psi_n\rangle$ after the perturbation is turned off at time t is approximated as the linear response, i.e., up to the first-order,

$$\mathcal{P}_{m \rightarrow n}(t) = \left| c_n^{(1)}(t) \right|^2 = \left| \frac{1}{\hbar} \int_0^t \langle \psi_n | \hat{V}(t') | \psi_m \rangle e^{\frac{i(E_n - E_m)t'}{\hbar}} dt' \right|^2$$

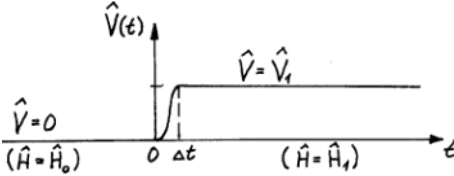
The transition rate, i.e., the transition probability per unit time

$$W = \frac{d}{dt} \mathcal{P} = \frac{d}{dt} \left| c_n^{(1)}(t) \right|^2$$

8-2 Adiabatic Process

A. FAST AND SLOW PERTURBATION

- (1) Suppose that the system starts with some Hamiltonian $\hat{H}_0$, but then *very quickly* changes to another Hamiltonian $\hat{H}$.

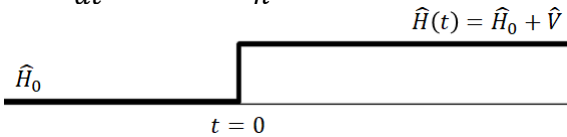


This occurs over a small timescale Δt , i.e., $\Delta t \ll \hbar/\Delta E$.

- (2) The change happens so quickly that the state does not have a chance to respond. After time Δt , the system still sits in the same state $|\psi_n\rangle$. The only difference is that the time dynamics is now governed by $\hat{H}$ rather than $\hat{H}_0$ (The process of the sudden change of the Hamiltonian is called a *quantum quench*).

PROOF:

$$\frac{d}{dt}|\Psi(t)\rangle = -\frac{i}{\hbar}\hat{H}(t)|\Psi(t)\rangle$$



$$d|\Psi(t)\rangle = -\frac{i}{\hbar}\hat{H}(t)|\Psi(t)\rangle dt$$

$$\int_{-\tau}^{\tau} d|\Psi(t)\rangle = -\frac{i}{\hbar} \int_{-\tau}^{\tau} \hat{H}(t)|\Psi(t)\rangle dt$$

$$|\Psi(\tau)\rangle - |\Psi(-\tau)\rangle = -\frac{i}{\hbar} \int_{-\tau}^0 \hat{H}_0 |\Psi(t)\rangle dt - \frac{i}{\hbar} \int_0^{\tau} (\hat{H}_0 + \hat{V}) |\Psi(t)\rangle dt$$

According to the mean value theorem

$$\begin{aligned} |\Psi(\tau)\rangle - |\Psi(-\tau)\rangle &= -\frac{i}{\hbar} \hat{H}_0 \tau |\Psi(\tau_-)\rangle - \frac{i}{\hbar} (\hat{H}_0 + \hat{V}) \tau |\Psi(\tau_+)\rangle \\ \lim_{\tau \rightarrow 0} [|\Psi(\tau)\rangle - |\Psi(-\tau)\rangle] &= 0 \end{aligned}$$

If the system is initially in an eigenstate $|\psi_k\rangle$ of $\hat{H}_0$, it will be in that state immediately after the change in the Hamiltonian has occurred. The transition amplitude to find it, after the change, in an eigenstate

$|\Psi_n(t)\rangle$ of $\hat{H}_0 + \hat{V}$ is then just the inner product between the eigenstates of these two different Hamiltonians, i.e., $\langle \Psi_n(t) | \psi_k \rangle$.

■

EXAMPLES:

1. Constant perturbation makes the transition from the state $|\psi_m\rangle$ to the state $|\psi_n\rangle$

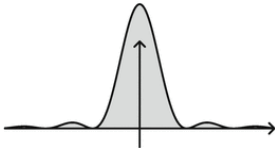
$$\begin{aligned} t = 0, & \quad \hat{V}(t) = 0 \\ t > 0, & \quad \hat{V}(t) = \hat{V} \end{aligned}$$



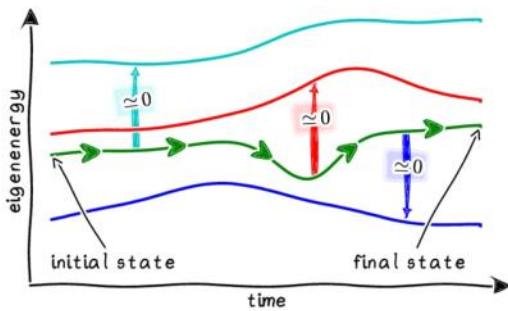
$$\text{Let } \omega_{nm} = \frac{E_n - E_m}{\hbar} \quad t = 0$$

$$\begin{aligned} c_n^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t \langle \psi_n | \hat{V}(t') | \psi_m \rangle e^{i\omega_{nm}t'} dt' \\ &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \int_0^t e^{i\omega_{nm}t'} dt' \\ &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \frac{e^{i\omega_{nm}t} - 1}{i\omega_{nm}} \\ &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \frac{\sin(\omega_{nm}t/2)}{\omega_{nm}/2} e^{i\omega_{nm}t/2} \end{aligned}$$

$$|c_n^{(1)}(t)|^2 = \frac{|\langle \psi_n | \hat{V} | \psi_m \rangle|^2}{\hbar^2} \text{sinc}^2(\omega_{nm}t/2)$$



- (3) Suppose that the Hamiltonian $\hat{H}(t)$ changes slowly in time, then there will be no transition from the eigenstate $|\psi_m\rangle$ of $\hat{H}(0)$ to a different eigenstate $|\psi_n\rangle$ of $\hat{H}(t)$.



PROOF:



$$\frac{d}{dt}|\Psi(t)\rangle = -\frac{i}{\hbar}\hat{H}(t)|\Psi(t)\rangle$$

$$|\Psi(t)\rangle = \sum_j c_j(t) e^{-\frac{i}{\hbar} \int_0^t E_j(t') dt'} |\psi_j(t)\rangle$$

where $\hat{H}(t)|\psi_j(t)\rangle = E_j(t)|\psi_j(t)\rangle \dots (a)$

Let $\theta_j(t) = \int_0^t E_j(t') dt'$

$$\sum_j \left[\frac{dc_j(t)}{dt} e^{-\frac{i\theta_j(t)}{\hbar}} |\psi_j(t)\rangle + c_j(t) \frac{d}{dt} e^{-\frac{i\theta_j(t)}{\hbar}} |\psi_j(t)\rangle + c_j(t) e^{-\frac{i\theta_j(t)}{\hbar}} \frac{d}{dt} |\psi_j(t)\rangle \right] = -\frac{i}{\hbar} \sum_j c_j(t) e^{-\frac{i\theta_j(t)}{\hbar}} \hat{H}(t) |\psi_j(t)\rangle$$

Since $\frac{d}{dt} e^{-\frac{i\theta_j(t)}{\hbar}} = -\frac{i}{\hbar} E_j(t) e^{-\frac{i\theta_j(t)}{\hbar}}$

we obtain

$$\sum_j \left[\frac{dc_j(t)}{dt} e^{-\frac{i\theta_j(t)}{\hbar}} |\psi_j(t)\rangle + c_j(t) e^{-\frac{i\theta_j(t)}{\hbar}} \frac{d}{dt} |\psi_j(t)\rangle \right] = 0$$

Multiplying both side by $\langle \psi_n(t) |$

$$\sum_j \left[\frac{dc_j(t)}{dt} e^{-\frac{i\theta_j(t)}{\hbar}} \underbrace{\langle \psi_n(t) | \psi_j(t) \rangle}_{=\delta_{nj}} + c_j(t) e^{-\frac{i\theta_j(t)}{\hbar}} \langle \psi_n(t) | \frac{d}{dt} |\psi_j(t)\rangle \right] = 0$$

$$\frac{dc_n(t)}{dt} = -\frac{i}{\hbar} \sum_j c_j(t) e^{-\frac{i(\theta_j(t) - \theta_n(t))}{\hbar}} \langle \psi_n(t) | \frac{d}{dt} |\psi_j(t)\rangle$$

Taking the time derivative of the eigenvalue equation (a)

$$\frac{d\hat{H}(t)}{dt}|\psi_j(t)\rangle + \hat{H}(t)\frac{d}{dt}|\psi_j(t)\rangle = \frac{dE_j(t)}{dt}|\psi_j(t)\rangle + E_j(t)\frac{d}{dt}|\psi_j(t)\rangle$$

Multiplying both side by $\langle\psi_n|$

$$\begin{aligned} \langle\psi_n(t)|\frac{d\hat{H}}{dt}|\psi_j(t)\rangle + E_n(t)\langle\psi_n(t)|\frac{d}{dt}|\psi_j(t)\rangle \\ = \frac{dE_j(t)}{dt}\underbrace{\langle\psi_n(t)|\psi_j(t)\rangle}_{\delta_{nj}} + E_j(t)\langle\psi_n(t)|\frac{d}{dt}|\psi_j(t)\rangle \end{aligned}$$

$$\text{For } n \neq j, \quad \langle\psi_n(t)|\frac{d}{dt}|\psi_j(t)\rangle = \frac{\langle\psi_n(t)|\frac{d\hat{H}}{dt}|\psi_j(t)\rangle}{E_j - E_n}$$

$$\begin{aligned} \frac{dc_n(t)}{dt} = -\frac{i}{\hbar}c_n(t)\langle\psi_n(t)|\frac{d}{dt}|\psi_n(t)\rangle \\ - \frac{i}{\hbar}\sum_{j \neq n} c_j(t) e^{-\frac{i(\theta_j(t)-\theta_n(t))}{\hbar}} \frac{\langle\psi_n(t)|\frac{d\hat{H}}{dt}|\psi_j(t)\rangle}{E_j - E_n} \end{aligned}$$

For the adiabatic approximation, i.e.,

$$\frac{d\hat{H}}{dt} \ll |E_j - E_n|$$

we obtain

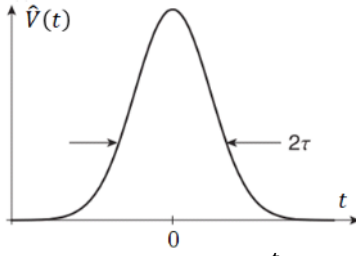
$$\begin{aligned} \frac{dc_n(t)}{dt} &\approx -\frac{i}{\hbar}c_n(t)\langle\psi_n(t)|\frac{d}{dt}|\psi_n(t)\rangle \\ c_n(t) &= c_n(0) \exp\left[-\frac{i}{\hbar}\int_0^t \langle\psi_n(t')|\frac{d}{dt}|\psi_n(t')\rangle dt'\right] = c_n(0)e^{-\frac{i\alpha_n(t)}{\hbar}} \\ \mathcal{P}_n &= |c_n(t)|^2 = |c_n(0)|^2 \end{aligned}$$

In the adiabatic limit the eigenstates of the Hamiltonian evolve independently of each other. ■

EXAMPLES:

1. Gaussian perturbation

$$\hat{V}(t) = \hat{V}e^{-t^2/\tau^2}$$



$$\begin{aligned}
 c_n^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t \langle \psi_n | \hat{V}(t') | \psi_m \rangle e^{i\omega_{nm}t'} dt' \\
 &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \int_0^t e^{i\omega_{nm}t'} e^{-t'^2/\tau^2} dt'
 \end{aligned}$$

As $t \rightarrow \infty$

$$\begin{aligned}
 c_n^{(1)}(t) &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \int_0^\infty e^{i\omega_{nm}t'} e^{-t'^2/\tau^2} dt' \\
 &= -\frac{i}{\hbar} \langle \psi_n | \hat{V} | \psi_m \rangle \sqrt{\pi} \tau e^{-\omega_{nm}^2 \tau^2 / 4} \\
 \left| c_n^{(1)}(\tau) \right|^2 &= \frac{\pi}{\hbar^2} \left| \langle \psi_n | \hat{V} | \psi_m \rangle \right|^2 \tau^2 e^{-\omega_{nm}^2 \tau^2 / 2}
 \end{aligned}$$

As the time constant τ becomes large enough, the probability $\left| c_n^{(1)}(\tau) \right|^2$ approaches zero, meaning that the system does not change states.

B. ADIABATIC AND DIABATIC PROCESSES

(1) Adiabatic processes

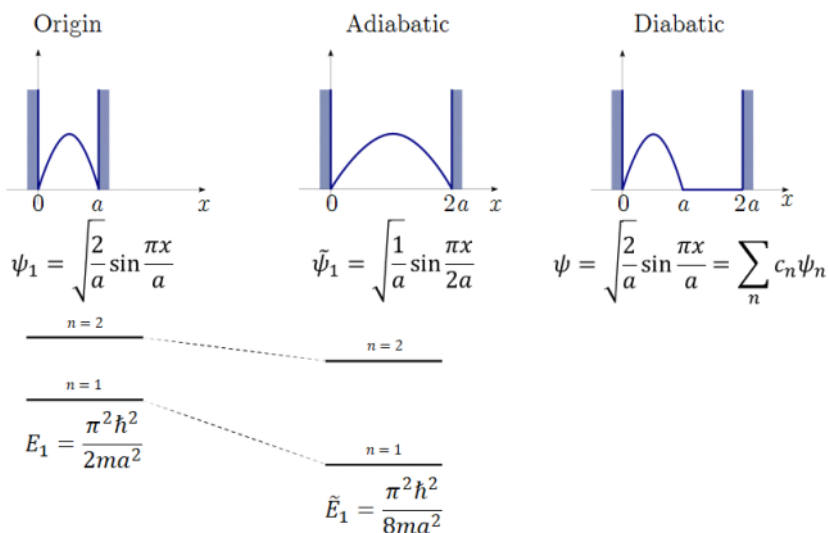
The system remains in the ground state, which is adjusted also smoothly to the new potential shape. Its state is always an eigenstate of the Hamiltonian at each time.

(2) Diabatic processes

The system remains in the original state, which is not an eigenstate of the new Hamiltonian. It is a superposition of several eigenstates of the new Hamiltonian.

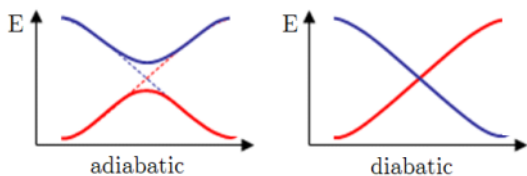
EXAMPLES:

1. Infinite potential well



- Initial wavefunction of particle-in-a-box.
- The wavefunction when we slowly change the width of box.
- The wavefunction right after we suddenly change the width of box.

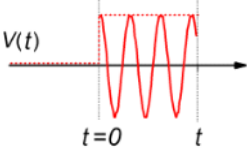
(3) Energy levels crossing and non-crossing



8-3 Fermi Golden Rule

A. HARMONIC PERTURBATION

- (1) The interaction of a system with an oscillating perturbation turned on at time $t = 0$



$$\hat{V}(t) = 2\hat{V}_0 \cos \omega t = \hat{V}_0(e^{i\omega t} + e^{-i\omega t})$$

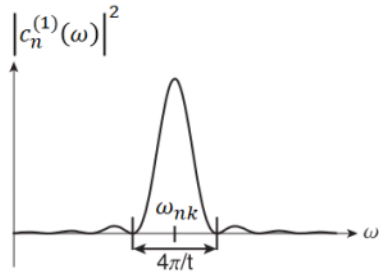
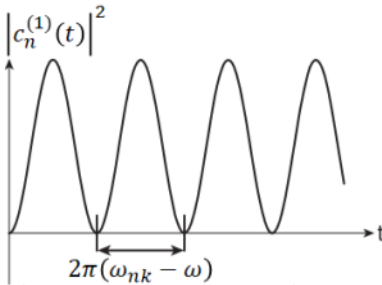
- (2) The transition from an initial state $|\psi_m\rangle$ to a final state $|\psi_n\rangle$

$$\begin{aligned} c_n^{(1)}(t) &= -\frac{i}{\hbar} \int_0^t \langle \psi_n | \hat{V}(t') | \psi_m \rangle e^{i\omega_{nm}t'} dt' \\ &= -\frac{i}{\hbar} \langle \psi_n | \hat{V}_0 | \psi_m \rangle \int_0^t (e^{i\omega t'} + e^{-i\omega t'}) e^{i\omega_{nm}t'} dt' \\ &= -\frac{iV_{nm}}{\hbar} \int_0^t (e^{i(\omega_{nm}-\omega)t'} + e^{i(\omega_{nm}+\omega)t'}) dt' \\ &= -\frac{iV_{nm}}{\hbar} \left[\frac{e^{i(\omega_{nm}-\omega)t} - 1}{i(\omega_{nm} - \omega)} + \frac{e^{i(\omega_{nm}+\omega)t} - 1}{i(\omega_{nm} + \omega)} \right] \\ &= -\frac{iV_{nm}}{\hbar} \left[e^{\frac{i(\omega_{nm}-\omega)t}{2}} \frac{\sin\left(\frac{\omega_{nm}-\omega}{2}t\right)}{\frac{\omega_{nm}-\omega}{2}} \right. \\ &\quad \left. + e^{\frac{i(\omega_{nm}+\omega)t}{2}} \frac{\sin\left(\frac{\omega_{nm}+\omega}{2}t\right)}{\frac{\omega_{nm}+\omega}{2}} \right] \end{aligned}$$

The transition probability

$$\begin{aligned}
|c_n^{(1)}(t)|^2 &= \frac{V_{nm}^2}{\hbar^2} \left[\frac{\sin^2\left(\frac{\omega_{nm}-\omega}{2}\right)t}{\left(\frac{\omega_{nm}-\omega}{2}\right)^2} + \frac{\sin^2\left(\frac{\omega_{nm}+\omega}{2}\right)t}{\left(\frac{\omega_{nm}+\omega}{2}\right)^2} \right. \\
&\quad \left. + (e^{i\omega t} + e^{-i\omega t}) \frac{\sin\left(\frac{\omega_{nm}-\omega}{2}\right)t}{\frac{\omega_{nm}-\omega}{2}} \frac{\sin\left(\frac{\omega_{nm}+\omega}{2}\right)t}{\frac{\omega_{nm}+\omega}{2}} \right] \\
&= \frac{V_{nm}^2}{\hbar^2} \left[\frac{\sin^2\left(\frac{\omega_{nm}-\omega}{2}\right)t}{\left(\frac{\omega_{nm}-\omega}{2}\right)^2} + \frac{\sin^2\left(\frac{\omega_{nm}+\omega}{2}\right)t}{\left(\frac{\omega_{nm}+\omega}{2}\right)^2} \right. \\
&\quad \left. + 4 \frac{\cos^2 \omega t - \cos \omega_{nm} t \cos \omega t}{\omega_{nm}^2 - \omega^2} \right]
\end{aligned}$$

- (3) $|c_n^{(1)}(t)|^2$ shows an oscillatory dependence. $|c_n^{(1)}(\omega)|^2$ shows that the resonance peak resembles a Dirac delta function in the limit that the frequency width $\Delta\omega = 4\pi/t \rightarrow 0$, which implies $t \rightarrow \infty$.



For larger $t \rightarrow \infty$

$$\lim_{t \rightarrow \infty} \frac{\sin^2\left(\frac{\omega_{nm}-\omega}{2}\right)t}{\left(\frac{\omega_{nm}-\omega}{2}\right)^2} = \lim_{t \rightarrow \infty} \text{sinc}^2\left(\frac{\omega_{nm}-\omega}{2}\right)t = 2\pi t \delta(\omega_{nm} - \omega)$$

OS:

Let $x = (\omega_{nk} - \omega)/2$

$$\begin{aligned}
\int_{-\infty}^{\infty} \text{sinc}^2 xt \, dx &= \int_{-\infty}^{\infty} \left(\frac{\sin xt}{x}\right)^2 dx = \pi t \int_{-\infty}^{\infty} \frac{\sin^2 xt}{\pi x^2 t} dx = \pi t \\
&\Rightarrow \lim_{t \rightarrow \infty} \frac{\sin^2 xt}{\pi x^2 t} = \pi t \delta(x)
\end{aligned}$$

$$\begin{aligned}
\left|c_n^{(1)}(\infty)\right|^2 &= \frac{V_{nm}^2}{\hbar^2} [2\pi t \delta(\omega_{nm} - \omega) + 2\pi t \delta(\omega_{nm} + \omega)] \\
&= \frac{2\pi V_{nm}^2 t}{\hbar^2} [\delta(\omega_{nm} - \omega) + \delta(\omega_{nm} + \omega)] \\
&= \frac{2\pi V_{nm}^2 t}{\hbar} [\delta(E_{nm} - E) + \delta(E_{nm} + E)]
\end{aligned}$$

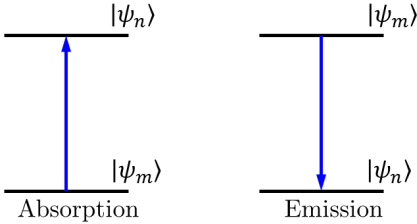
OS:

$$\frac{\delta(\omega)}{\hbar} = \delta(\hbar\omega) = \delta(E)$$

The transition rate

$$W_{m \rightarrow n} = \frac{d}{dt} \left| c_n^{(1)}(\infty) \right|^2 = \frac{2\pi V_{nm}^2}{\hbar} [\delta(E_{nm} - E) + \delta(E_{nm} + E)]$$

(4) Physical interpretation for $\delta(E_{nm} - E)$ and $\delta(E_{nm} + E)$ terms:



For $\delta(E_{nm} - E)$:

$$E_{nm} - E = 0 \Rightarrow E = E_n - E_m > 0.$$

Transition from lower energy level E_m to higher energy level E_n , i.e., absorption.

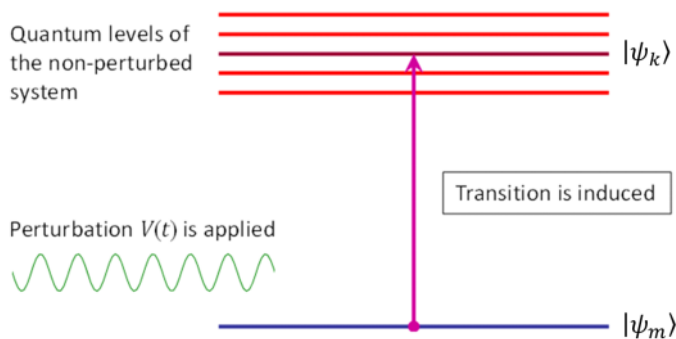
For $\delta(E_{nm} + E)$:

$$E_{nm} + E = 0 \Rightarrow E = E_m - E_n > 0.$$

Transition from higher energy level E_m to lower energy level E_n , i.e., emission.

B. FERMI GOLDEN RULE

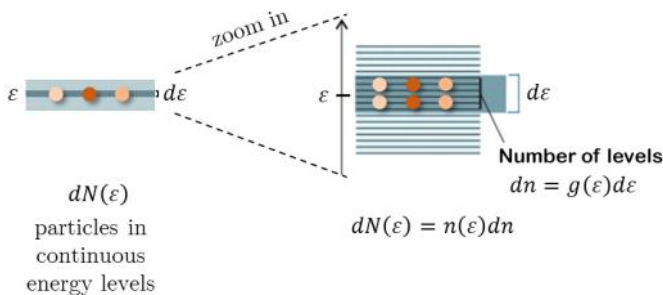
(1) Consider the transition from an initial state $|\psi_m\rangle$ to a *continuum of states* $\{|\psi_k\rangle\}$:



The total transition probability is obtained by summing the contributions of all the final states between ε_k and $\varepsilon_k + d\varepsilon$:

$$\bar{W} = \frac{1}{\mathcal{V}} \sum_k W_{km} = \frac{2\pi V_{km}^2}{\hbar} \frac{1}{\mathcal{V}} \sum_k \delta(\varepsilon - \varepsilon_k)$$

(2) Introduce the density of states



For some physical quantities, we need to evaluate

$$\sum_k f(\varepsilon_k)$$

The summations over k can be converted to integrals as follows:

$$\sum_k f(\varepsilon_k) = \mathcal{V} \int f(\varepsilon_k) \frac{d^3k}{(2\pi)^3} = \mathcal{V} \int f(\varepsilon) g(\varepsilon) d\varepsilon$$

Since

$$\sum_k f(\varepsilon_k) = \sum_k \int f(\varepsilon) \delta(\varepsilon - \varepsilon_k) d\varepsilon = \int f(\varepsilon) \sum_k \delta(\varepsilon - \varepsilon_k) d\varepsilon$$

Thus, we obtain

$$g(\varepsilon) = \frac{1}{\mathcal{V}} \sum_k \delta(\varepsilon - \varepsilon_k) = \int \delta(\varepsilon - \varepsilon_k) \frac{d^3k}{(2\pi)^3}$$

Since the volume contains one $\vec{k}$ -point is $\frac{(2\pi)^3}{\mathcal{V}}$. The number of states in the interval dk is

$$dn(\varepsilon_k) = \frac{d^3k}{\frac{(2\pi)^3}{\mathcal{V}}} = \mathcal{V} \frac{d^3k}{(2\pi)^3}$$

Thus, we obtain

$$\begin{aligned} g(\varepsilon) &= \int \delta(\varepsilon - \varepsilon_k) \frac{1}{\mathcal{V}} dn(\varepsilon_k) = \frac{1}{\mathcal{V}} \int \delta(\varepsilon - \varepsilon_k) \frac{dn(\varepsilon_k)}{d\varepsilon_k} d\varepsilon_k \\ &= \frac{1}{\mathcal{V}} \frac{dn(\varepsilon)}{d\varepsilon} \end{aligned}$$

$\Rightarrow g(\varepsilon)$ is the number of states per volume per energy and called the **density of states (DOS)**.

$$\begin{aligned} g(\varepsilon) &= \frac{1}{\mathcal{V}} \frac{dn(\varepsilon)}{d\varepsilon} = \frac{1}{\mathcal{V}} \sum_k \delta(\varepsilon - \varepsilon_k) \\ \Rightarrow \bar{W} &= \frac{2\pi V_{km}^2}{\hbar} g(\varepsilon) \end{aligned}$$

This is so called **Fermi golden rule**.

C. DIPOLE TRANSITION SELECTION RULE

- (1) The power of the electromagnetic wave is

$$P = \int \vec{S} \cdot d\vec{A} = \int (\vec{E} \times \vec{H}) \cdot d\vec{A} = \frac{q^2 \dot{v}^2}{6c^3 \pi \epsilon_0}$$

where $\vec{S}$ is the Poynting vector. In most cases of interest, the wavelength of the photon emitted or absorbed by an atom is on the order of several 1000 Å. An atom has dimensions on the order of Å.

We can therefore use the approximation that $\vec{E}$ is constant in space (known as electric dipole approximation).

$x = A \sin \omega t$ (simple harmonic motion)

$$\dot{v} = \frac{d^2x}{dt^2} = -\omega^2 x$$

$$P = \frac{2q^2 \omega^4 x^2}{3c^3 4\pi \epsilon_0}$$

$$\langle P \rangle = \frac{2\omega^4 q^2 \langle x^2 \rangle}{3c^3 4\pi \epsilon_0} = \frac{\omega^4 q^2 A^2}{3c^3 4\pi \epsilon_0} = \frac{\omega^4}{12c^3 \pi \epsilon_0} \langle q\vec{r} \rangle^2$$

- (2) Consider the transition between two states of an atom in an

electromagnetic field:

$$V(t) = q\hat{r}\vec{E}(t)$$

where $q\hat{r}$ is the dipole operator. Thus, we obtain

$$\langle \hat{P} \rangle = \frac{\omega^4}{12c^3\pi\epsilon_0} \langle q\hat{r} \rangle^2$$

The dipole transition rate is

$$\bar{W} = \frac{2\pi}{\hbar} \left| \int \psi_f^*(r, \theta, \phi) V(t) \psi_i(r, \theta, \phi) d\tau \right|^2 g(\varepsilon)$$

The transition probability rate is obtained by applying the time-dependent perturbation theory to a system that undergoes a transition from an initial state $\psi_i(r, \theta, \phi)$ to a final state $\psi_f(r, \theta, \phi)$ that is part of a continuum of states. Thus, we obtain

$$\langle \hat{P} \rangle = \frac{\omega^4}{12c^3\pi\epsilon_0} \langle q\hat{r} \rangle^2 \propto \bar{W}$$

If $W_{fi} = 0$, the transition is forbidden.

If $W_{fi} \neq 0$, the transition is allowed.

(3) Determine Δm

1. Radial integral

$$\begin{aligned} \langle \hat{r} \rangle &= \int \psi_f^*(r, \theta, \phi) \hat{r} \psi_i(r, \theta, \phi) d^3r \\ &= \int_0^\infty R^*(r) r R(r) r^2 dr \\ &\quad \times \int_0^\pi \int_0^{2\pi} Y^*(\theta, \phi) f(\theta, \phi) Y(\theta, \phi) \sin \theta d\theta d\phi \end{aligned}$$

The radial integral equals some constant. Hence it gives no selection rule.

2. Angular integral

Consider $\Phi_m(\phi) = e^{im\phi}$ only

(a) z-axis

$$\begin{aligned}
\langle \hat{r}_z \rangle &= \int \psi_f^*(r, \theta, \phi) \hat{z} \psi_i(r, \theta, \phi) d^3r \\
&= \int \psi_f^*(r, \theta, \phi) r \cos \theta \psi_i(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi \\
&= C_{n,\ell} \int_0^{2\pi} e^{-im_f\phi} e^{im_i\phi} d\phi \\
&= C_{n,\ell} \int_0^{2\pi} e^{(-i\Delta m\phi)} d\phi
\end{aligned}$$

If $\Delta m \neq 0$, then $\langle r_z \rangle = 0$

If $\Delta m = 0$, then $\langle r_z \rangle \neq 0$

(b) x -axis

$$\begin{aligned}
\langle \hat{r}_x \rangle &= \int \psi_f^*(r, \theta, \phi) r \sin \theta \cos \phi \psi_i(r, \theta, \phi) r^2 \sin \theta dr d\theta d\phi \\
&= C_{n,\ell} \int_0^{2\pi} e^{-im_f\phi} \cos \phi e^{im_i\phi} d\phi \\
&= C_{n,\ell} \frac{1}{2} \int_0^{2\pi} \left(e^{i(m_i-m_f+1)\phi} + e^{i(m_i-m_f-1)\phi} \right) d\phi
\end{aligned}$$

$$\text{only if } \begin{cases} m_i - m_f + 1 = 0, & \Delta m = 1 \\ m_i - m_f - 1 = 0, & \Delta m = -1 \end{cases} \Rightarrow \langle r_x \rangle \neq 0$$

Thus $\Delta m = m_f - m_i = \pm 1$

(c) y -axis

$$\begin{aligned}
\langle \hat{r}_y \rangle &= C_{n,\ell} \int_0^{2\pi} e^{-im_f\phi} \sin \phi e^{im_i\phi} d\phi \\
&= C_{n,\ell} \frac{1}{2i} \int_0^{2\pi} \left(e^{i(m_i-m_f+1)\phi} - e^{i(m_i-m_f-1)\phi} \right) d\phi \\
&\Rightarrow \Delta m = \pm 1
\end{aligned}$$

Conclusion: The transition occurs only if $\Delta m = 0, \pm 1$.

(4) Determine $\Delta \ell$

1. Conservation of parity

In quantum mechanics, the physical observables are related to the probability density $|\psi(x)|^2$. When the potential is symmetric under the reflection transformation, i.e., $V(-x) = V(x)$, the probability density must follow $|\psi(-x)|^2 = |\psi(x)|^2$.

The wavefunction could be either $\psi(-x) = \psi(x)$ or $\psi(-x) = -\psi(x)$ to satisfy the condition.

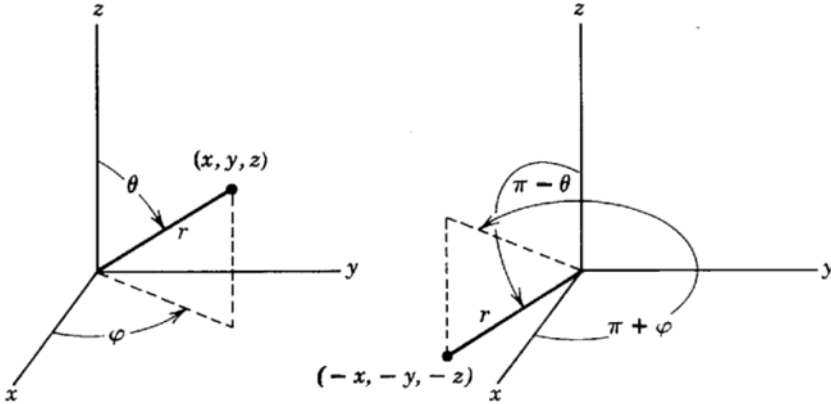
2. The wavefunction, therefore, has definite "parity":

Even parity: $\psi(-x) = \psi(x)$ (even/symmetric function)

Odd parity: $\psi(-x) = -\psi(x)$ (odd/antisymmetric function)

The Coulomb potential is symmetric about the origin point. The wavefunctions have definite parity.

In spherical coordinate $\psi_{n\ell m}(-r, -\theta, -\phi) = \psi_{n\ell m}(r, \pi - \theta, \pi + \phi)$



$$R_{n\ell}(-r) = R_{n\ell}(r)$$

$$P_{\ell m}(-\theta) = P_{\ell m}(\pi - \theta) = (-1)^{\ell+|m|}P(\theta)$$

$$\Phi_m(-\phi) = \Phi_m(\pi + \phi) = (-1)^{|m|}\Phi(\phi)$$

$$Y_{\ell m}(-\theta, -\phi) = Y_{\ell m}(\pi - \theta, \pi + \phi)$$

$$= (-1)^{\ell+2|m|}Y(\theta, \phi)$$

$$= (-1)^{\ell}Y(\theta, \phi)$$

$$\psi_{n\ell m}(-r, -\theta, -\phi) = (-1)^{\ell}\psi_{n\ell m}(r, \theta, \phi)$$

In order to obtain

$$\langle \hat{r} \rangle = \int \psi_f^*(r, \theta, \phi) \hat{r} \psi_i(r, \theta, \phi) d^3r \neq 0$$

The integrand must be even. Since r has odd parity, therefore we require the product $\psi_f^*(r, \theta, \phi) \psi_i(r, \theta, \phi)$ to be odd and therefore the initial and final state have to be of opposite parity.

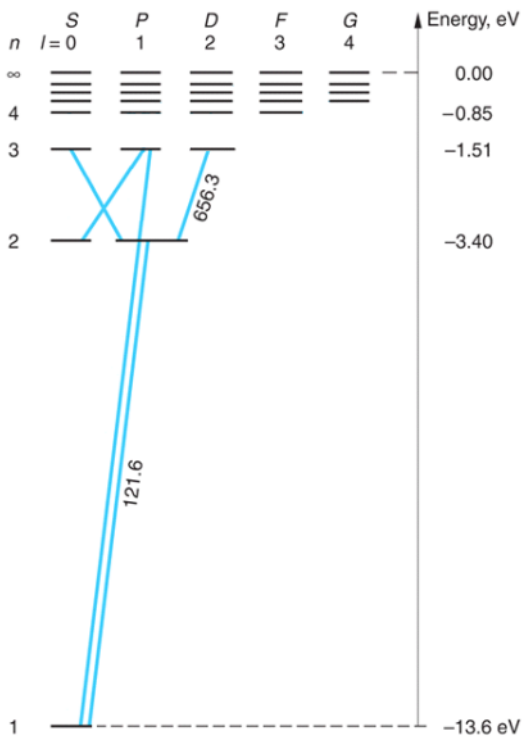
$$\Delta\ell = \ell_f - \ell_i = \pm 1$$

(5) Transitions are governed by selection rules.

$$\Delta\ell = \ell_f - \ell_i = \pm 1$$

$$\Delta m = m_f - m_i = 0, \pm 1$$

n	S	P	D	F	G	Energy, eV
$l=0$	1	2	3	4		



—Chapter 9—

Quantum Entanglement

9-1 Quantum States of Multi-Particle System

A. QUANTUM STATE OF SINGLE-PARTICLE

- (1) Suppose experimentalist A prepares a stream of electrons such that each electron is in the spin state $|\psi\rangle_A$:

$$|\psi\rangle_A = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$$

Meanwhile, experimentalist B prepares a stream of electrons which is a mixture: half the electrons are in the state $|\uparrow\rangle$ and half are in the state $|\downarrow\rangle$.

Can we distinguish the A stream from the B stream?

1. Measuring the spin in the z -direction:

Evidently, both will give 50% $|\uparrow\rangle$ and 50% $|\downarrow\rangle$. Thus, the stream A and the stream B are indistinguishable.

2. Measuring the spin in the x -direction:

The quantum state $|\psi\rangle_A$ is in fact just that of a spin in the x -direction, i.e., $|\psi\rangle_A = |\rightarrow\rangle$. Thus, we obtain 100% $|\rightarrow\rangle$.

The state $|\uparrow\rangle$ and $|\downarrow\rangle$ are linear combination of $|\rightarrow\rangle$ and $|\leftarrow\rangle$:

$$|\uparrow\rangle = \frac{1}{\sqrt{2}}(|\rightarrow\rangle + |\leftarrow\rangle)$$

$$|\downarrow\rangle = \frac{1}{\sqrt{2}}(|\rightarrow\rangle - |\leftarrow\rangle)$$

Thus, we obtain 50% $|\rightarrow\rangle$ and 50% $|\leftarrow\rangle$.

- (2) The state $|\psi\rangle_A = |\rightarrow\rangle$ is called a pure state and the expectation value of an operator $\hat{A}$ is

$$\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$$

The stream B, in contrast, is in a mixed state: different pure quantum states occurring with different probabilities, but with no phase coherence between them. In other words, these relative probabilities in B of different quantum states do not derive from probability amplitudes: the probabilities of the different quantum states in the mixed state B are exactly like classical probabilities.

OS:

That being said, though, to find the probability of measuring spin

up in some such mixed state, one first uses the classical-type probability for each component state, then for each quantum state in the mix, one finds the probability of spin up in that state by the standard quantum technique.

Therefore, for a mixed state in which the system is in the state $|\psi_n\rangle$ with probability p_n ,

$$\sum_n p_n = 1$$

the expectation value of an operator $\hat{A}$ is

$$\langle \hat{A} \rangle = \sum_n p_n \langle \psi_n | \hat{A} | \psi_n \rangle$$

- (3) Mathematically, the measurement postulate of quantum mechanics (4-3-A-V) is

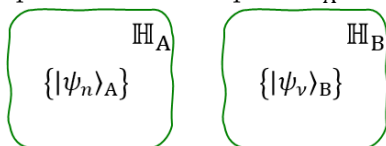
$$|\psi_n\rangle = \frac{\hat{P}_n}{\sqrt{\langle \psi | \hat{P}_n | \psi \rangle}} |\psi\rangle$$

where $\hat{P}_n = |\psi_n\rangle\langle\psi_n|$ is a projection operator.

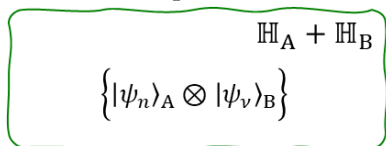
B. QUANTUM STATE OF TWO-PARTICLE

- (1) Tensor product

Suppose we have two particles labeled A and B. The basis state of each individual particle is described by $\{|\psi_n\rangle_A\}$ and $\{|\psi_\nu\rangle_B\}$ respectively in a single-particle Hilbert space $\mathbb{H}_A$ and $\mathbb{H}_B$.



Then the Hilbert space for the combined system of two particles is



where $\mathbb{W} = \mathbb{H}_A + \mathbb{H}_B$ is the Hilbert space and the basis of $\mathbb{W}$ is the

tensor product of $\{|\psi_n\rangle_A\}$ and $\{|\psi_\nu\rangle_B\}$:

$$|\psi_n\rangle_A \otimes |\psi_\nu\rangle_B \equiv |\psi_n\rangle_A |\psi_\nu\rangle_B \text{ for all } \begin{cases} |\psi_n\rangle_A \in \mathbb{H}_A \\ |\psi_\nu\rangle_B \in \mathbb{H}_B \end{cases}$$

EXAMPLES:

1. Consider a system of two spin-1/2 particles A and B:
 - (a) The basis states of the single-particle Hilbert space $\mathbb{H}_A$ is

$$\{|+z\rangle, |-z\rangle\}$$
 The basis states of the single-particle Hilbert space $\mathbb{H}_B$ is

$$\{|+z\rangle, |-z\rangle\}$$
 - (b) The basis states of the combined Hilbert space $\mathbb{W} = \mathbb{H}_A + \mathbb{H}_B$ is

$$\{|+z\rangle|+z\rangle, |+z\rangle|-z\rangle, |-z\rangle|+z\rangle, |-z\rangle|-z\rangle\}$$
 - (c) The number of basis states of $\mathbb{W}$ is $2 \times 2 = 2^2 = 4$
2. Suppose a quantum system contains 20 particles. What is the number of basis states?

ANSWER:

The combined Hilbert space is

$$\mathbb{W} = \mathbb{H}_1 + \mathbb{H}_2 + \cdots + \mathbb{H}_{20}$$

The number of basis states of $\mathbb{W}$ is

$$\underbrace{2 \times 2 \cdots 2}_{20\text{-particle}} = 2^{20} = 1,048,576$$

Thus, even in quantum systems with a modest number of particles, the quantum state can carry huge amounts of information.

(2) Entanglement

Any two-particle state can be written as a superposition of these basis vectors:

$$|\psi\rangle = \sum_{m\mu} c_{m\mu} |\psi_m\rangle_A \otimes |\psi_\mu\rangle_B \equiv \sum_{m\mu} c_{m\mu} |\psi_m\rangle_A |\psi_\mu\rangle_B$$

where $\sum_{m\mu} |c_{m\mu}|^2 = 1$

If $|\psi\rangle$ can be written in terms of definite individual-particle states, i.e.,

$$|\psi\rangle = |\psi\rangle_A |\psi\rangle_B$$

The state is said to be **unentangled**.

If $|\psi\rangle$ can **not** be written in terms of definite individual-particle states, i.e.,

$$|\psi\rangle \neq |\psi\rangle_A |\psi\rangle_B$$

The state is said to be **entangled**.

EXAMPLES:

1. Verify the entanglement of the following quantum states of two spin-1/2 particles:

$$(a) \quad |\psi\rangle = \frac{1}{\sqrt{2}} (|+z\rangle |-z\rangle - |-z\rangle |+z\rangle)$$

$$(b) \quad |\psi\rangle = \frac{1}{\sqrt{2}} (|+z\rangle |-z\rangle + |-z\rangle |-z\rangle)$$

ANSWER:

$$(a): \quad |\psi\rangle \neq |\psi\rangle_A |\psi\rangle_B$$

Thus, the state is entangled.

$$\begin{aligned} (b): \quad |\psi\rangle &= \frac{1}{\sqrt{2}} (|+z\rangle |-z\rangle + |-z\rangle |-z\rangle) \\ &= \frac{1}{\sqrt{2}} (|+z\rangle + |-z\rangle) |-z\rangle \\ &= |+x\rangle_A |-z\rangle_B \end{aligned}$$

Thus, the state is unentangled.

(3) Inner product and outer product

Consider a system of two particles A and B:

The inner product between the tensor product basis states is defined as follows:

$$\begin{aligned} & \left(\langle \psi_n |_A \otimes \langle \psi_\nu |_B \right) \cdot \left(|\psi_m\rangle_A \otimes |\psi_\mu\rangle_B \right) \equiv \langle \psi_n | \psi_m \rangle_A \langle \psi_\nu | \psi_\mu \rangle_B \\ & = \delta_{nm} \delta_{\nu\mu} \end{aligned}$$

We calculate the inner product for A, calculate the inner product for B, and then multiply the two resulting numbers.

The partial projection operator for A written in terms of the outer product is

$$\hat{P}_n^A = |\psi_n\rangle \langle \psi_n| \otimes \hat{1}$$

(4) Partial measurement

An arbitrary two-particle state is

$$|\psi\rangle = \sum_{m\mu} c_{m\mu} |\psi_m\rangle_A |\psi_\mu\rangle_B$$

We perform a measurement on particle A and obtain the collapse state

$$\begin{aligned} |\psi'\rangle &= \frac{\hat{P}_n^A}{\sqrt{\langle\psi|\hat{P}_n^A|\psi\rangle}} |\psi\rangle \\ &= \frac{(|\psi_n\rangle\langle\psi_n| \otimes \hat{1}) \sum_{m\mu} c_{m\mu} (|\psi_m\rangle \otimes |\psi_\mu\rangle)}{\sqrt{\sum_{iv} c_{iv}^* (\langle\psi_i| \otimes \langle\psi_v|) (|\psi_n\rangle\langle\psi_n| \otimes \hat{1}) \sum_{j\mu} c_{j\mu} (|\psi_j\rangle \otimes |\psi_\mu\rangle)}} \\ &= \frac{\sum_{m\mu} c_{m\mu} (|\psi_n\rangle\langle\psi_n||\psi_m\rangle \otimes \hat{1}|\psi_\mu\rangle)}{\sqrt{\sum_{ivj\mu} c_{iv}^* c_{j\mu} (\langle\psi_i||\psi_n\rangle\langle\psi_n||\psi_j\rangle \otimes \langle\psi_v|\hat{1}|\psi_\mu\rangle)}} \\ &= \frac{1}{\sqrt{\sum_{ivj\mu} c_{iv}^* c_{j\mu} \delta_{in} \delta_{nj} \delta_{v\mu}}} \sum_{m\mu} c_{m\mu} \delta_{nm} |\psi_m\rangle |\psi_\mu\rangle \\ &= \frac{1}{\sqrt{\sum_v c_{nv}^* c_{nv}}} \sum_{\mu} c_{n\mu} |\psi_n\rangle |\psi_\mu\rangle \\ &= \frac{1}{\sqrt{\sum_v |c_{nv}|^2}} \sum_{\mu} c_{n\mu} |\psi_n\rangle_A |\psi_\mu\rangle_B \end{aligned}$$

Thus, for the chosen state n of A, we sum over all the associated states of B:

$$|\psi\rangle \xrightarrow{\text{collapse}} |\psi'\rangle = \frac{1}{\sqrt{\sum_v |c_{nv}|^2}} \sum_{\mu} c_{n\mu} |\psi_\mu\rangle_B \otimes |\psi_n\rangle_A = |\psi\rangle_A |\psi\rangle_B$$

$\Rightarrow$ After the collapse, the two-particle state becomes unentangled.

EXAMPLES:

1. A system of two spin-1/2 particles is in the "singlet state"

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle_A |-\rangle_B - |-\rangle_A |+\rangle_B \right)$$

Suppose we measure $\hat{S}_z$ on particle A. What are the probabilities of the possible outcomes and the associated "collapse" states?

ANSWER:

For the outcome is $+\frac{\hbar}{2}$, we choose the only state $|\psi_n\rangle_A = |+\rangle_A$.

Thus, we obtain the collapse state

$$|\psi'\rangle = \frac{1}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2}} \frac{1}{\sqrt{2}} |+\rangle_A |-\rangle_B = |+\rangle |-\rangle$$

and the outcome probability

$$|\langle\psi'|\psi\rangle|^2 = \frac{1}{2}$$

For the outcome is $-\frac{\hbar}{2}$, we choose the only state $|\psi_n\rangle_A = |-\rangle_A$.

Thus, we obtain the collapse state

$$|\psi'\rangle = \frac{1}{\sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2}} \frac{-1}{\sqrt{2}} |-\rangle_A |+\rangle_B = -|-\rangle |+\rangle$$

and the outcome probability

$$|\langle\psi'|\psi\rangle|^2 = \frac{1}{2}$$

$\Rightarrow$ The two possible outcomes $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$, occur with equal probability. In either case, the two-particle state collapses so that A is in the observed spin eigenstate, and B has the opposite spin. After the collapse, the two-particle state is no longer entangled.

9-2 Bell's Inequality

A. THE EINSTEIN-PODOLSKY-ROSEN "PARADOX"

OS:

In 1935, Einstein, Podolsky and Rosen tried to use the property of entanglement to argue that quantum mechanics is incomplete.

- (1) Consider an entangled state of two spin-1/2 particles:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(| +z \rangle_A | -z \rangle_B - | -z \rangle_A | +z \rangle_B \right)$$

Measure $\hat{S}_z$ on particle A.

If the measurement outcome is $+\frac{\hbar}{2}$, the new state is $| +z \rangle | -z \rangle$.

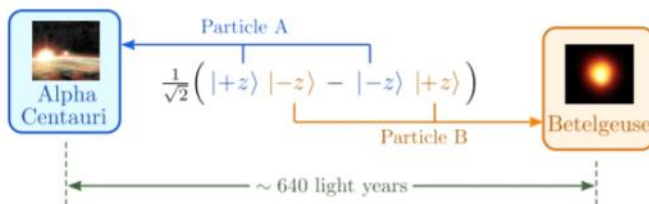
If the measurement outcome is $-\frac{\hbar}{2}$, the new state is $| -z \rangle | +z \rangle$.

After the collapse, the two-particle state is unentangled and each particle has a definite spin.

- (2) The measurement postulate of quantum mechanics

The collapse happens instantaneously, regardless of the distance separating the particles.

COMMENT:



1. Imagine that we prepare the two-particle state in a laboratory on Earth. Particle A is then transported to the laboratory of in the Alpha Centauri star system, and particle B is transported to the laboratory in the Betelgeuse star system, separated by about 640 light years.
2. We measure $\hat{S}_z$ on particle A, which induces an instantaneous collapse of the two-particle state. The outcome of A is either $| +z \rangle$ or $| -z \rangle$ with probability 1/2.
3. Immediately afterwards, we measure $\hat{S}_z$ on particle B, and

obtain, with 100% certainty, the opposite spin.

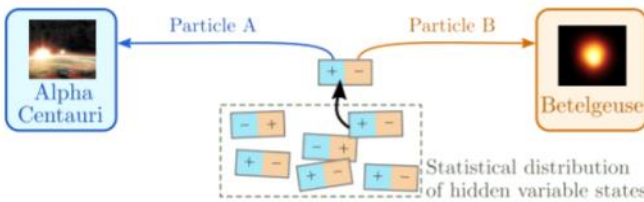
⇒ State collapse is *nonlocal*.

⇒ The instantaneous collapse of the wavefunction is nonlocal.

(3) The arguments of EPR

1. The outcome of A with a probability is due to the ignorance of some information from the beginning, i.e., the outcome of any measurement should be determined if we already knew the "hidden variable".
2. According to the special relativity, no way to transmit information faster than light. That means that the measurement of particle A does not affect the measurement of particle B instantaneously. In other words, if state collapse is locality, the measurement of A will take some time to travel to affect the measurement of B.

EPR suggested *a statistical distribution of hidden variable states* to make quantum theory deterministic and local.



B. BELL'S INEQUALITY (1964)

- (1) We repeatedly prepare the particle pairs in the singlet state

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(| +z \rangle_A | -z \rangle_B - | -z \rangle_A | +z \rangle_B \right)$$

Many round of the experiment are conducted; for each round, both experimentalists' random choices of spin axis S_1 , S_2 , or S_3 are recorded, along with their measurement results.

	 Alpha Centauri	$\frac{1}{\sqrt{2}}(+\rangle -\rangle - -\rangle +\rangle)$		 Betelgeuse
		Particle A	Particle B	
Experiment 1	S_1	-	+	S_2
Experiment 2	S_3	+	+	S_2
Experiment 3	S_2	+	-	S_1
Experiment 4	S_3	-	+	S_3
Experiment 5	S_3	-	-	S_1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- (2) Hidden variable theory predicts that the two experimentalists choose different spin axes, the probability that their results have opposite signs is $\mathcal{P}(\text{oppo}) \geq 1/3$ which is called Bell's inequality.

COMMENT:

There are 8 distinct possible outcomes of A along each direction.

$(+, +, +) \cdots \cdots$ (a)

$(+, +, -)(+, -, +)(-, +, +) \cdots \cdots$ (b)

$(+, -, -)(-, +, -)(-, -, +) \cdots \cdots$ (c)

$(-, -, -) \cdots \cdots$ (d)

1. If two experimentalists happened to choose the same spin axes, the probability for the two measurement results to have opposite signs is 1.
2. If two experimentalists happened to choose different spin axes, what is the probability for the two measurement results to have opposite signs?

For case (a) and (d): The probability for opposite signs is 1.

For case (b) and (c):

Consider $\underbrace{(+, +, -)}_A \underbrace{(-, -, +)}_B$

A	B	result
S_1	S_2	opposite
S_1	S_3	same
S_2	S_1	opposite
S_2	S_3	same
S_3	S_1	same
S_3	S_2	same

The probability for opposite signs is $1/3$.

$\Rightarrow$ The two experimentalists choose different spin axes, the

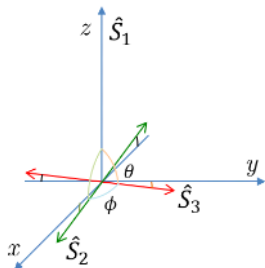
probability that their results have opposite signs is

$$\mathcal{P} = \frac{6 + 2 \times 6 + 6}{6 \times 8} = \frac{1}{2} \geq \frac{1}{3} \dots \text{Bell's inequality}$$

(3) Quantum theory predicts a probability $\mathcal{P}(\text{oppo}) < 1/3$

COMMENT:

Consider a case:



where $\hat{S}_1$ along the z -axis, $\hat{S}_2$ along $\theta = 2\pi/3$ and $\phi = 0$, and $\hat{S}_3$ along $\theta = 2\pi/3$ and $\phi = \pi/2$.

Suppose we choose $\hat{S}_1$ for A and obtain $+\hbar/2$. The state of A collapses to $|+z\rangle$ and the state of B collapses to $|-z\rangle$. If we choose $\hat{S}_2$ for B, we obtain

$$\begin{aligned} |+\theta\rangle &= \cos\frac{\theta}{2}|+z\rangle + \sin\frac{\theta}{2}e^{i\phi}|-z\rangle = \frac{1}{2}|+z\rangle + \frac{\sqrt{3}}{2}|-z\rangle \\ |-\theta\rangle &= -\sin\frac{\theta}{2}e^{-i\phi}|+z\rangle + \cos\frac{\theta}{2}|-z\rangle = -\frac{\sqrt{3}}{2}|+z\rangle + \frac{1}{2}|-z\rangle \\ \Rightarrow |-z\rangle &= \frac{\sqrt{3}}{2}|+\theta\rangle + \frac{1}{2}|-\theta\rangle \end{aligned}$$

The probability of measuring $-\hbar/2$ along $\hat{S}_2$ is $1/4$. All the other possible scenarios are worked out similarly. The result is that the overall probability of the two experimentalists obtaining opposite results (in the cases where we choose different measurement axis) is $1/4 < 1/3$. Bell's inequality is violated!

9-3 Density Matrix

A. PURE STATE AND MIXED STATE

- (1) The expectation value of an arbitrary operator $\hat{A}$

For a pure state:

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle \text{ where } |\psi\rangle = \sum_n c_n |\psi_n\rangle, \text{ and } \sum_n c_n^2 = 1$$

For a mixed state:

$$\langle \hat{A} \rangle = \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle \text{ where } \sum_i p_i = 1$$

The equations are different to obtain $\langle \hat{A} \rangle$ in a pure state and in a mixed state.

- (2) Define the density matrix

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

where $|\psi_i\rangle$ are (not necessarily orthogonal) normalized pure states.

Thus, for a pure state:

$$\hat{\rho} = |\psi\rangle \langle \psi|$$

is a projection operator, i.e., $\hat{\rho}^2 = \hat{\rho}$. Thus, we obtain

$$\begin{aligned} \langle \hat{A} \rangle &= \langle \psi | \hat{A} | \psi \rangle \\ &= \sum_n \langle \psi | \hat{A} | \psi_n \rangle \underbrace{\langle \psi_n |}_{\hat{1}} | \psi \rangle \\ &= \sum_n \langle \psi_n | \underbrace{|\psi\rangle \langle \psi|}_{=\hat{\rho}} \hat{A} | \psi_n \rangle \\ &= \text{Tr}(\hat{\rho} \hat{A}) \end{aligned}$$

For a mixed state:

$$\langle \hat{A} \rangle = \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle$$

Choose an orthonormal basis $|\phi_n\rangle$ and then

$$\begin{aligned}
\langle \hat{A} \rangle &= \sum_i p_i \langle \psi_i | \hat{A} \sum_n \underbrace{|\phi_n\rangle \langle \phi_n|}_{\hat{1}} | \psi_i \rangle \\
&= \sum_n \sum_i p_i \langle \phi_n | \psi_i \rangle \langle \psi_i | \hat{A} | \phi_n \rangle \\
&= \text{Tr} \left(\sum_i p_i | \psi_i \rangle \langle \psi_i | \hat{A} \right) \\
&= \text{Tr}(\hat{\rho} \hat{A})
\end{aligned}$$

Now, we have the same equation to obtain $\langle \hat{A} \rangle$. We need to determine a density matrix for a pure state or a mixed state.

EXAMPLES:

1. For all spins in the state $|\rightarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$, in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ the density matrix is

- (1) Using Dirac notation:

$$\begin{aligned}
\hat{\rho} &= |\rightarrow\rangle \langle \rightarrow| \\
&= \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \frac{1}{\sqrt{2}}(\langle \uparrow| + \langle \downarrow|) \\
&= \frac{1}{2}(|\uparrow\rangle \langle \uparrow| + |\uparrow\rangle \langle \downarrow| + |\downarrow\rangle \langle \uparrow| + |\downarrow\rangle \langle \downarrow|)
\end{aligned}$$

- (2) Using matrix:

$$\begin{aligned}
\rho &= \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 1 \\ \sqrt{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\
\rho^2 &= \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \rho
\end{aligned}$$

and

$$\begin{aligned}
\langle \mathbf{s}_x \rangle &= \frac{\hbar}{4} \text{Tr} \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) = \frac{\hbar}{4} \text{Tr} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{\hbar}{2} \\
\langle \mathbf{s}_y \rangle &= \frac{\hbar}{4} \text{Tr} \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right) = \frac{\hbar}{4} \text{Tr} \begin{pmatrix} i & -i \\ i & -i \end{pmatrix} = 0 \\
\langle \mathbf{s}_z \rangle &= \frac{\hbar}{4} \text{Tr} \left(\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) = \frac{\hbar}{4} \text{Tr} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} = 0
\end{aligned}$$

2. 50% of spins in the state $|\uparrow\rangle$ and 50% of spins in the state $|\downarrow\rangle$, in

the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis, the density matrix is

(1) Using Dirac notation:

$$\hat{\rho} = \frac{1}{2}|\uparrow\rangle\langle\uparrow| + \frac{1}{2}|\downarrow\rangle\langle\downarrow| = \frac{1}{2}\hat{1}$$

$$\hat{\rho}^2 = \frac{1}{4}\hat{1} \neq \hat{\rho}$$

and

$$\langle\hat{s}_x\rangle = \text{Tr}(\hat{\rho}\hat{s}_x) = \frac{\hbar}{4}\text{Tr}(\hat{s}_x) = 0$$

$$\langle\hat{s}_y\rangle = \text{Tr}(\hat{\rho}\hat{s}_y) = \frac{\hbar}{4}\text{Tr}(\hat{s}_y) = 0$$

$$\langle\hat{s}_z\rangle = \text{Tr}(\hat{\rho}\hat{s}_z) = \frac{\hbar}{4}\text{Tr}(\hat{s}_z) = 0$$

$\Rightarrow$ The Pauli matrices are all traceless.

(2) Using matrix:

$$\boldsymbol{\rho} = \frac{1}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\mathbf{1}$$

$$\boldsymbol{\rho}^2 = \frac{1}{2}\mathbf{1} \cdot \frac{1}{2}\mathbf{1} = \frac{1}{4}\mathbf{1} \neq \boldsymbol{\rho}$$

and

$$\langle\mathbf{s}_x\rangle = \frac{\hbar}{4}\text{Tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = \frac{\hbar}{4}\text{Tr}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = 0$$

$$\langle\mathbf{s}_y\rangle = \frac{\hbar}{4}\text{Tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right) = \frac{\hbar}{4}\text{Tr}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = 0$$

$$\langle\mathbf{s}_z\rangle = \frac{\hbar}{4}\text{Tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right) = \frac{\hbar}{4}\text{Tr}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 0$$

If 50% of spins in the state $|\rightarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ and 50% of spins in the state $|\leftarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$, the density matrix is

$$\hat{\rho} = \frac{1}{2}|\rightarrow\rangle\langle\rightarrow| + \frac{1}{2}|\leftarrow\rangle\langle\leftarrow|$$

$$\boldsymbol{\rho} = \frac{1}{2} \times \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \times \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2}\mathbf{1}$$

We obtain the same density matrix as the previous one. The reason is that the spins are completely random, all directions are equally likely. Thus, the density matrix describing such a state cannot depend on the direction we choose for our axes.

B. THE BLOCH SPHERE

(1) Density matrix for spin-1/2

Generally, this will be a 2×2 matrix that can be written as linear combination of the identity $\hat{1}$ and the Pauli matrices $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$, as

$$\begin{aligned}\rho &= \frac{1}{2}(\mathbf{1} + \vec{a} \cdot \vec{\sigma}) \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{a_x}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{a_y}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \frac{a_z}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 + a_z & a_x - ia_y \\ a_x + ia_y & 1 - a_z \end{pmatrix}\end{aligned}$$

↓matrix form

$$\hat{\rho} = \frac{1}{2}(\hat{1} + \vec{a} \cdot \vec{\sigma})$$

The coefficient $\vec{a}$ can be calculated as the expectation value of the Pauli matrices

$$\begin{aligned}\text{Tr}(\hat{\rho}\hat{\sigma}_x) &= \text{Tr}\left(\frac{1}{2}(\hat{1} + \vec{a} \cdot \vec{\sigma})\hat{\sigma}_x\vec{x}\right) \\ &= \frac{1}{2}\text{Tr}(\hat{\sigma}_x) + \frac{\vec{a}}{2} \cdot \text{Tr}(\vec{\sigma}\hat{\sigma}_x)\vec{x} \\ &= 0 + \frac{\vec{a}}{2} \cdot \text{Tr}(\hat{\sigma}_x^2 - i\hat{\sigma}_z + i\hat{\sigma}_y)\vec{x} \\ &= \vec{a} \cdot \vec{x} \\ \Rightarrow \vec{a} &= \text{Tr}(\hat{\rho}\vec{\sigma})\end{aligned}$$

(2) Bloch vector and Bloch sphere

The coefficient $\vec{a}$ is named the Bloch vector.

All spin 1/2 density matrices lie on or within the so-called Bloch sphere (with radius $|\vec{a}| = 1$) and are determined by the Bloch vector $\vec{a}$. The length of the Bloch vector thus tells us something about the mixedness, the polarization of an ensemble, i.e. of a beam of spin 1/2 particles, e.g. electrons. We say that the beam is polarized if $\vec{a}_i = 1$ and completely unpolarized if $\vec{a}_i = 0$, for all i . This means that pure and mixed states can be characterized via the Bloch vector in the following way

$$\text{pure state} \quad \hat{\rho}^2 = \hat{\rho} \quad \Rightarrow \quad |\vec{a}| = 1$$

$$\text{mixed state} \quad \hat{\rho}^2 \neq \hat{\rho} \quad \Rightarrow \quad |\vec{a}| < 1$$

A totally mixed state ($\vec{a}_i = 0$, for all i) can then be written as

$$\hat{\rho}_{\text{mix}} = \frac{1}{2}\hat{1}$$

such that

$$\begin{aligned}\mathrm{Tr}(\hat{\rho}_{\mathrm{mix}}) &= 1 \\ \mathrm{Tr}(\hat{\rho}_{\mathrm{mix}}^2) &= \frac{1}{4}\mathrm{Tr}(\hat{1}) = \frac{1}{2}\end{aligned}$$

(3) Purity

Choose an orthonormal basis $|\phi_n\rangle$ and then

$$\begin{aligned}\mathrm{Tr}(\hat{\rho}) &= \mathrm{Tr}\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right) \\ &= \sum_n \langle\phi_n|\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|\right)|\phi_n\rangle \\ &= \sum_i p_i \sum_n \langle\psi_i|\phi_n\rangle\langle\phi_n|\psi_i\rangle \\ &= \sum_i p_i \langle\psi_i|\psi_i\rangle \\ &= \sum_i p_i \\ &= 1\end{aligned}$$

The trace of $\hat{\rho}$ is equal to 1 for pure states and for mixed states. We then try to obtain

$$\begin{aligned}\mathrm{Tr}(\hat{\rho}^2) &= \sum_n \langle\phi_n|\sum_{ij} p_i p_j |\psi_i\rangle\langle\psi_i|\psi_j\rangle\langle\psi_j|\phi_n\rangle \\ &= \sum_{ij} p_i p_j \langle\psi_i|\psi_j\rangle\langle\psi_j|\psi_i\rangle \underbrace{\sum_n |\phi_n\rangle\langle\phi_n|}_{=\hat{1}} \\ &= \sum_{ij} p_i p_j \underbrace{|\langle\psi_i|\psi_j\rangle|^2}_{=\delta_{ij}} \\ &= \sum_i p_i^2\end{aligned}$$

Since $0 \leq p_i \leq 1$ and therefore $p_i^2 \leq p_i$. Thus, the trace of $\hat{\rho}^2$ is a good measure for the mixedness of a density matrix, i.e., it is equal to 1 for pure states and strictly smaller than 1 for mixed states.

Since

$$\rho^2 = \frac{1}{4} \begin{pmatrix} 1 + 2a_z + a_x^2 + a_y^2 + a_z^2 & 2(a_x - ia_y) \\ 2(a_x + ia_y) & 1 - 2a_z + a_x^2 + a_y^2 + a_z^2 \end{pmatrix}$$

If $\rho^2 = \rho$, we have a pure state, i.e.,

$\Rightarrow$ From the diagonal terms, we have $a_x^2 + a_y^2 + a_z^2 = 1$

$\Rightarrow$ For a Bloch vector of norm 1, $|\vec{a}|^2 = a_x^2 + a_y^2 + a_z^2 = 1$, we have a pure state, i.e., from the density matrix ρ ,

$$\rho = \frac{1}{2} \begin{pmatrix} 1 + a_z & a_x - ia_y \\ a_x + ia_y & 1 - a_z \end{pmatrix}$$

we obtain

$$\rho_{11}\rho_{22} = (1 + a_z)(1 - a_z) = 1 - a_z^2$$

$$\rho_{12}\rho_{21} = (a_x - ia_y)(a_x + ia_y) = a_x^2 + a_y^2$$

$$\Rightarrow \rho_{12}\rho_{21} = \rho_{11}\rho_{22}$$

EXAMPLES:

- Which of the following density matrices correspond to a pure state?

$$\rho_1 = \begin{pmatrix} \frac{2}{7} & 0 \\ 0 & \frac{5}{7} \end{pmatrix}, \rho_2 = \begin{pmatrix} \frac{1}{4} & \frac{i\sqrt{3}}{4} \\ -\frac{i\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}, \rho_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \rho_4 = \begin{pmatrix} \frac{1}{5} & \frac{\sqrt{2}}{5} \\ \frac{\sqrt{2}}{5} & \frac{4}{5} \end{pmatrix}$$

ANSWER:

A pure state must satisfy $\rho_{mn}\rho_{nm} = \rho_{mm}\rho_{nn}$ for every m and n .

$$\rho_1: 0 \times 0 \neq \frac{2}{7} \times \frac{5}{7} \quad \boxed{\times}$$

$$\rho_2: \frac{i\sqrt{3}}{4} \times \left(-\frac{i\sqrt{3}}{4}\right) = \frac{1}{4} \times \frac{3}{4} \quad \boxed{\checkmark}$$

$$\rho_3: 0 \times 0 = 0 \times 1 \quad \boxed{\checkmark}$$

$$\rho_4: \frac{\sqrt{2}}{5} \times \frac{\sqrt{2}}{5} \neq \frac{1}{5} \times \frac{4}{5} \quad \boxed{\times}$$

C. PARTIAL TRACE

- Consider two subsystems A and B, with the Hilbert space $\mathbb{W} = \mathbb{H}_A + \mathbb{H}_B$. In general, the density matrix is

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

and

$$|\psi_i\rangle = |\psi_i\rangle_A |\psi_i\rangle_B = \sum_{m\mu} c_m^i c_\mu^i |\psi_m\rangle_A |\psi_\mu\rangle_B$$

where

$$\sum_m |c_m^i|^2 = \sum_\mu |c_\mu^i|^2 = 1$$

Thus, we have

$$\begin{aligned}\hat{\rho} &= \sum_i p_i |\psi_i\rangle_A \langle\psi_i|_A \otimes |\psi_i\rangle_B \langle\psi_i|_B \\ &= \sum_i \sum_{m\mu} p_i |c_m^i|^2 |c_\mu^i|^2 |\psi_m\rangle_A \langle\psi_m|_A \otimes |\psi_\mu\rangle_B \langle\psi_\mu|_B\end{aligned}$$

(2) Suppose that we are interested only in the properties of subsystem A.

To do this, the observable quantities we are interested in have the following form

$$\hat{A} \otimes \hat{1}_B$$

where $\hat{A}$ is an Hermitian operator pertaining to system A, and $\hat{1}_B$ is the identity operator for system B. The expectation value is:

$$\begin{aligned}\langle \hat{A} \otimes \hat{1}_B \rangle &= \text{Tr}(\hat{\rho}(\hat{A} \otimes \hat{1}_B)) \\ &= \text{Tr}\left(\left(\sum_i \sum_{m\mu} p_i |c_m^i|^2 |c_\mu^i|^2 |\psi_m\rangle_A \langle\psi_m|_A \otimes |\psi_\mu\rangle_B \langle\psi_\mu|_B\right)(\hat{A} \otimes \hat{1}_B)\right) \\ &= \sum_n \sum_{im} p_i |c_m^i|^2 \langle\phi_n|\psi_m\rangle_A \langle\psi_m|\hat{A}|\phi_n\rangle_A \sum_{\nu\mu} |c_\mu^i|^2 \langle\phi_\nu|\psi_\mu\rangle_B \langle\psi_\mu|\hat{1}|\phi_\nu\rangle_B\end{aligned}$$

Since

$$\begin{aligned}\sum_{\nu\mu} |c_\mu^i|^2 \langle\phi_\nu|\psi_\mu\rangle_B \langle\psi_\mu|\hat{1}|\phi_\nu\rangle_B &= \sum_{\nu\mu} |c_\mu^i|^2 \langle\psi_\mu|\phi_\nu\rangle_B \langle\phi_\nu|\psi_\mu\rangle_B \\ &= \sum_\mu |c_\mu^i|^2 \langle\psi_\mu|\psi_\mu\rangle_B \\ &= 1\end{aligned}$$

We obtain

$$\begin{aligned}\langle \hat{A} \otimes \hat{1}_B \rangle &= \sum_n \sum_{im} p_i |c_m^i|^2 \langle \phi_n | \psi_m \rangle_A \langle \psi_m | \hat{A} | \phi_n \rangle_A \\ &= \text{Tr} \left(\sum_{im} p_i |c_m^i|^2 | \psi_m \rangle \langle \psi_m | \hat{A} \right)\end{aligned}$$

We then define a reduced density matrix

$$\hat{\rho}_A = \text{Tr}_B[\hat{\rho}] = \sum_v \langle \phi_v | \hat{\rho} | \phi_v \rangle_B = \sum_{im} p_i |c_m^i|^2 | \psi_m \rangle \langle \psi_m |$$

where $\text{Tr}_B[\dots]$ denotes a partial trace which means tracing out the B part of the Hilbert space.

The expectation value can be written as

$$\langle \hat{A} \otimes \hat{1}_B \rangle = \text{Tr}(\hat{\rho}_A \hat{A})$$

$\Rightarrow$ When computing physical predictions regarding subsystem A alone, neglecting B, it is not necessary to consider the full density matrix $\hat{\rho}$; it is sufficient to consider the smaller reduced density matrix $\hat{\rho}_A$. In other words, $\hat{\rho}_A$ contains all physical information about subsystem A, when we are not interested in the properties of subsystem B, which is expressed by the mathematical fact that the observables have the form $\hat{A} \otimes \hat{1}_B$.

EXAMPLES:

1. Consider an entangled state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)$.

The density matrix is

$$\begin{aligned}\hat{\rho} &= |\psi\rangle\langle\psi| \\ &= \frac{1}{2} (|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle)(\langle\uparrow|\langle\downarrow| - \langle\downarrow|\langle\uparrow|) \\ &= \frac{1}{2} (|\uparrow\rangle|\downarrow\rangle\langle\uparrow|\langle\downarrow| - |\downarrow\rangle|\uparrow\rangle\langle\uparrow|\langle\downarrow| + |\downarrow\rangle|\uparrow\rangle\langle\downarrow|\langle\uparrow| - |\uparrow\rangle|\downarrow\rangle\langle\downarrow|\langle\uparrow|)\end{aligned}$$

Tracing over system B yields the reduced density matrix

$$\hat{\rho}_A = \text{Tr}_B[\hat{\rho}] = \langle\uparrow|\hat{\rho}|\uparrow\rangle_B + \langle\downarrow|\hat{\rho}|\downarrow\rangle = \frac{1}{2} (|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|)$$

Express $\hat{\rho}_A$ as a maatrix in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis:

$$\begin{aligned}
\rho_A &= \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) + \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} (0 \ 1) \\
&= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
&\Rightarrow \rho_A^2 \neq \rho_A
\end{aligned}$$

$\Rightarrow$ The density matrix corresponds to a pure state, while the reduced density matrix corresponds to a mixed state. This shows that the partial trace can transform pure states into statistical mixtures.

2. Consider the state of the composite system consisting of independent subsystems.

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_A + |\downarrow\rangle_A) \otimes |\uparrow\rangle_B = |\rightarrow\rangle_A \otimes |\uparrow\rangle_B$$

The density matrix is

$$\begin{aligned}
\hat{\rho} &= |\psi\rangle\langle\psi| \\
&= (|\rightarrow\rangle_A \otimes |\uparrow\rangle_B)(\langle\rightarrow|_A \otimes \langle\uparrow|_B) \\
&= (|\rightarrow\rangle_A\langle\rightarrow|_A) \otimes (|\uparrow\rangle_B\langle\uparrow|_B) \\
&= \hat{\rho}_A \otimes \hat{\rho}_B
\end{aligned}$$

The reduced density matrices are

$$\begin{aligned}
\hat{\rho}_A &= \text{Tr}_B[\hat{\rho}] = \text{Tr}_B[\hat{\rho}_A \otimes \hat{\rho}_B] = \hat{\rho}_A \text{Tr}_B[\hat{\rho}_B] = \hat{\rho}_A = |\rightarrow\rangle_A\langle\rightarrow|_A \\
\hat{\rho}_B &= \text{Tr}_A[\hat{\rho}] = \text{Tr}_A[\hat{\rho}_A \otimes \hat{\rho}_B] = \hat{\rho}_B \text{Tr}_A[\hat{\rho}_A] = \hat{\rho}_B = |\uparrow\rangle_B\langle\uparrow|_B
\end{aligned}$$

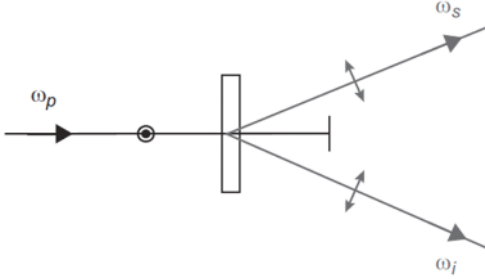
The matrix form in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis:

$$\begin{aligned}
\Psi &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \\
\rho &= \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
&= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}
\end{aligned}$$

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\rho_B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

3. In order to describe the polarization state of the two-photon system, the polarization of each photon must be specified.



This polarization state is

$$|x, x\rangle = |x_s\rangle \otimes |x_i\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$|y, y\rangle = |y_s\rangle \otimes |y_i\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

- (a) The density matrix that corresponds to a pure state that the superposition consists of equal parts $|x, x\rangle$ and $|y, y\rangle$, and a mixture consists of equal parts $|x, x\rangle$ and $|y, y\rangle$.
- (b) For a two-photon system prepared in an equal mixture of states $|x, x\rangle$ and $|y, y\rangle$ determine the probability that the signal photon is measured to be polarized along 45° , given that the idler photon is found to be polarized along this same direction. The density operator corresponding to this state is given by

$$\hat{\rho} = \frac{1}{2} |x, x\rangle \langle x, x| + \frac{1}{2} |y, y\rangle \langle y, y| \rightarrow \rho = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We want to find

$$\mathcal{P}(45_s | 45_i) = \frac{\mathcal{P}(45_s, 45_i)}{\mathcal{P}(45_i)} = \frac{\mathcal{P}(45_s) \mathcal{P}(45_i | 45_s)}{\mathcal{P}(45_i)} = ?$$

ANSWER:

- (a) The state and density matrix corresponding to a pure state is

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|x, x\rangle + |y, y\rangle) = \frac{1}{\sqrt{2}}(|x\rangle \otimes |x\rangle + |y\rangle \otimes |y\rangle)$$
$$\hat{\rho} = |\psi\rangle\langle\psi|$$

In the matrix form:

$$\begin{aligned}\Psi &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} \left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ \rho &= \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} (1 \quad 0 \quad 0 \quad 1) = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}\end{aligned}$$

The density matrix corresponding to a mixed state is

$$\hat{\rho} = \frac{1}{2}|x, x\rangle\langle x, x| + \frac{1}{2}|y, y\rangle\langle y, y|$$

In the matrix form:

$$\begin{aligned}\rho &= \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} (1 \quad 0 \quad 0 \quad 0) + \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} (0 \quad 0 \quad 0 \quad 1) \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\end{aligned}$$

- (b) $\mathcal{P}(45_s^\circ, 45_i^\circ) = \text{Tr}(\hat{\rho}|45_s^\circ, 45_i^\circ\rangle\langle 45_s^\circ, 45_i^\circ|)$

where

$$|45_s^\circ, 45_i^\circ\rangle \xrightarrow{\text{matrix form}} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$|45_s^\circ, 45_i^\circ\rangle\langle 45_s^\circ, 45_i^\circ| \xrightarrow{\text{matrix form}} \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Thus, we obtain

$$\begin{aligned} \mathcal{P}(45_s^\circ, 45_i^\circ) &= \text{Tr}(|45_s^\circ, 45_i^\circ\rangle\langle 45_s^\circ, 45_i^\circ|) \\ &\xrightarrow{\text{matrix form}} \text{Tr} \left(\frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} \right) \\ &= \frac{1}{8} \text{Tr} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} \\ &= \frac{1}{4} \end{aligned}$$

We also have

$$|45_i^\circ\rangle\langle 45_i^\circ| \xrightarrow{\text{matrix form}} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\hat{1}_s \otimes |45_i^\circ\rangle \xrightarrow{\text{matrix form}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\begin{aligned} \mathcal{P}(45_i^\circ) &= \text{Tr}(\hat{1}_s \otimes \hat{\rho} |45_i^\circ\rangle\langle 45_i^\circ|) \\ &\xrightarrow{\text{matrix form}} \text{Tr} \left(\frac{1}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \right) \\ &= \frac{1}{4} \text{Tr} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \\ &= \frac{1}{2} \end{aligned}$$

Thus, we obtain

$$\mathcal{P}(45_s^\circ|45_i^\circ) = \frac{\mathcal{P}(45_s^\circ, 45_i^\circ)}{\mathcal{P}(45_i^\circ)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$